

Please enter below

What will be the output for the code snippet given below when the following stack is passed to the given method?

stack (top-bottom): 75312

```
public static Deque<Integer> updatestack (Deque<Integer> stack) (  
  
    int val = stack.size()-1;  
  
    int value1 = 1;  
  
    while (val>0)  
  
        int topElement = stack.pop();  
  
        Deque<Integer> tempStack = new Deque<Integer>();  
  
        while(value1 <= val) (  
  
            int element = stack.pop();  
  
            tempStack.add(element+topElement);  
  
            value1 += 1;  
  
            while(tempStack.peek()!=null) (  
  
                stack.add(tempStack.pop());  
  
                stack.add(topElement);  
  
                value1 += 1;  
  
            value1 = 1;  
  
        return stack;  
    )
```

- 7 33 24 15 9
- 7 12 3 1 2
- 7 26 17 8 2
- 7 33 26 17 2

1.What will be the worst case time complexity for the below method?

```
public void demo (int n, boolean val) (  
  
    if (val) {  
  
        for (int index1=0;index1<n;index1++) { for (int index2=n;index2>index1; index2--)  
        { System.out.println(index1+index2);  
  
        }  
  
        }  
  
    else (  
  
        for (int index1=0;index1<n;index1++) { for (int index2=0;index2<index1; index2++)  
        { System.out.println(index1+index2);  
  
        }  
  
        }  
  
    )  
    )
```

```
for (int indexl=0;indexl<n;index++) { System.out.println(indexl);
```

- $O(n^2+n)$
- $O(n\log n)$
- $O(n^2)$
- $OO(n)$

2. Multiple Choice (Select 1 out of 4 options, for the question below)

Questionid: c6006

What is product backlog?

- A prioritized list of work for the development team that is derived from the roadmap and its requirements.
- A list of tasks identified by the Scrum team to be completed during the Scrum sprint.
- A list of tasks identified by team for promotions
- A list of team members.

3 Multiple Choice (Select 1 out of 4 options, for the question below)

Questionid, af085

Please refer below

Given the below code:

```
interface A()
```

```
class superclass implements A
```

```
class TestQuestion extends superclass { public static void main(String[]) { superclass superob = new  
superclass(); TestQuestion subob = new TestQuestion(); System.out.println(superob instanceof A);  
System.out.println(superob instanceof TestQuestion); System.out.println(subob instanceof A);
```

What will be the output when the preceding program is compiled and executed?

A The program will display the output as:

true
false
true

(B)

The program will generate compilation error

(C)

The program will display the output as:

false
true
false

(D) The program will throw a runtime exception

Multiple Choice (Select 1 out of 4 options, for the question below)

4 Alter the end of each sprint, your team will hold two meetings. What are those meetings called?

- **Sprint Review**
- Sprint Planning
- Sprint Release
- Release Roadmap

5 Which of the following scope is used only when the dependency is of pom type?

- **import**
- system
- test
- Runtime

6 Multiple Choice Select one of 4 options. The question below

QuestionId: 20005

Your current project has several branches like master, beta and push-notification. You have just finished the notification feature in the push-notification branch and you want to commit it to the beta branch. How can you accomplish this?

- Checkout the push-notification branch and run `git merge beta`
- Checkout the master branch and run `git merge beta --no-commit`
- Delete the push-notification branch and it will be committed to the master branch automatically
- **checkout the beta branch and run `git merge push-notification`**

7 . public class Rainbow {

```

public enum MyColor (
    RED (0xff0000), GREEN (0x00ff00), BLUE (0x0000ff):
private final int rgb;
15. MyColor (int rgb) (this.rgb = rgb; ) 16. public int getRGB () (return rgb;)
17. ):
18. public static void main(String[] args)
19. // insert code here
20.
21.)

```

- MyColor skyColor = BLUE;
- MyColor treeColor = MyColor.GREEN;
- if (RED.getRGB() < BLUE.getRGB()) {}
- Compilation fails due to other error(s) in the code.
- MyColor purple = new MyColor(0xff00ff);
- MyColor purple = MyColor.BLUE + MyColor.RED;

8.QuestionId Oc050

Please refer below

What is the output expected from the following code?

```

import java.io.Serializable;

public class Account implements Serializable

private String accountName;

private transient String accountNo

public String bankName
public Account(String accountName, String accountNo) (

this.accountName = accountName;

this.accountNo = accountNo;

bankName = "BankIndia";

1

@Override

public String toString() ( return "Account [accountName = " + accountName + ", bankName = " + bankName
+ "]; accountNo = " + accountNo + " ]"; )

import java.io.*;

public class FileStream (

```

```

public static void main(String[] args) throws Exception {

Account account = new Account ("Finance India", 130345678");

ObjectOutputStream

objOutputStream new ObjectOutputStream( new FileOutputStream("account.dat"));

objOutputStream.writeObject (account);

= new ObjectInputStream( FileInputStream("account.dat")); new

ObjectInputStream objInputStream

Account accountRead (Account)objInputStream.readObject();

System.out.println(accountRead); objOutputStream.close();

objInputStream.close();

Account [accountName=FinanceIndia, accountNo=null, bankName=null]

Account [accountName=FinanceIndia, accountNo=20345678, bankName=BankIndia]

Account [accountName=null, accountNo=null, bankName=BankIndia]

Account [accountName=FinanceIndia, accountNo=null, bankName=BankIndia]

```

Questionid: 68078

Please refer below

What will be the output for the following code?

```

public static void main(String[] args)

int num1 = 07

int num2 = 0;

for (int var= 0; var < 5 ; var++) if ((++num1 > 2) && (++num2 > 2)) {

num1++;

System.out.println(num1+"and"+num2);

```

- 5 and 2
- 8 and 5
- 6 and 3
- 6 and 4

Question No: 12 | Max Mark(s): 01 |

Multiple Choice (Select 1 out of 4 options, for the question below)

QuestionId: bc008

is a command used to get files from the remote repository to the local repository, but not into the working directory

- **git fetch**
- git pull
- git clone
- git remote add

Question No: 15 Max Muks): 01

Questionid: 05024

A company has a business application the provides its users with many different reports: receivables reports, payables reports, revehus projects, and so on. The company has just purchased some new, state-of-the-art, wireless printers, and a programmer has been assigned the task of enhancing all of the reports to use not only the company's old printers, but the now wireless printers as well. When the programmer starts looking into the application, the programmer discovers that because of the design of the application, it is necessary to make changes to each report to support the new printers. Which two design concepts most likely explain this situation?

- Inheritance
- **Low cohesion**
- **Tight coupling**
- High cohesion
- Loose coupling
- Object immutability

Multiple Choice (Solect 1 out of 4 options, for the question below.)

QuestionId: 14002

What does "Timeboxed" means in agile terms?

- Flexible
- Frequent

- Fixed

- Fast

Multiple Choice (Select 1 out of 4 options, for the question below)

QuestionId: 23042

There are two general uses of plugins. The first is build and the second is

- reporting

- servers

- dependencies

- distribution management

Multiple Choice (Select 1 out of 4 options, for the question below)

QuestionId 17043

What will the following command do?

mvn dependency tree

- Verify the dependencies defined in the POM.xml

- Download all the transitive dependencies

- Print out a report of your projects dependencies in a tree format

- Prune unused dependencies from your dependency tree

QuestionId 69051

Please refer below

Consider the following class:

```
public class Deer {
```

```
    private String name;
```

```
    private int age;
```

```
    private int height;
```

```
    Deer(String name, int age, int height) {
```

```
        this.name = name;
```

```
        this.age = age;
```

```
        this.height = height;
```

//Getters and Setters.

What will be the output for the following code?

```
class TestDeso
```

```
public static void main (String[] args) throws java.lang.Exception
```

```
int[] abc=new User("Rock", 11,155), new User ("Jill",15,165),new User("John", 13,159), new  
User("James",14,169); for (int i=0;i<4; i++){
```

```
if (i%2==0) System.out.println(abc[i].getName());
```

```
else
```

```
System.out.println(abc[i].getAge());
```

- 155 15 159 14
- **Rock 15 John 14**
- Rock 11 Jill 15 John 13 James 14
- 11 Jill 13 James

QuestionId c5021

Please refer below

Given the below code. Which statement is true?

11. class PingPong2 (

12. synchronized void hit (long n) (

13. for(int i=1; i < 3; i++)

14. System.out.print(n+"")++

15.)

16.)

17. public class Tester implements Runnable{

18. static PingPong2 pp2 new PingPong2(); 19. public static void main(String[] args) {

20. new Thread(new Tester()).start(); 21. new Thread(new Tester()).start();

22.)

23. public void run() { pp2.hit (Thread.currentThread().getId());}

24. }

- The output could be 5-1 6-1 6-25-2
- The output could be 6-1 6-2 5-1 5-2
- The output could be 6-1 5-2 6-2 5-1
- The output could be 6-1 6-2 5-1 7-1

Multiple Choice (Select 1 out of 4 options, for the question below)

Questionid ed084

Please refer below

What will get console out during execution?

```
public class TestLocalDeno
```

```
public static void main(String[] args) { Date startUtilDate new Date(); //Assume that the obtained date
is Thu Nov 03 20:40:45 IST 2016 Date tempUtilDate startUtilDate: startUtilDate.setDate(26)
System.out.println(tempUtilDate);
```

```
LocalDate startLocalDate LocalDate.of (2016, Month, JUNE, 01): LocalDate tempLocalDate
startLocalDate: startLocalDate startLocalDate plusDays (10): System.out.println(tempLocalDate):
```

- Sat Nov 26 20:40:45 IST 2016 2016-06-01
- Sat Nov 03 20:40:45 IST 2016 2016-06-01
- Sat Nov 25 20:40:45 IST 2016 2016-06-10
- Sat Nov 03 20:40:45 IST 2016 2016-06-10

Multiple Choice (Select 1 out of 4 options, for the question below.)

QuestionId: 05056

Consider a Java program being executed to read gigabytes of a file. For each line it reads, a Java object corresponding to the line is created. After some time, JVM is not able to allocate memory to create object. In this scenario, choose the correct statement from below

- JVM will throw an Exception which programmer can handle and program can continue to read the next line.

- JVM will throw an Error which programmer can handle and program can continue to read the next line.
- JVM will throw an Error and program will abort as this situation cannot be handled by programming
- JVM will throw an Exception and program will abort as this situation cannot be handled by programming

1)What will the mvn dependency: tree command do

[1] Verify the dependencies defined in the POM file.

[2] Print out a report of your project's dependencies:

[3] Download all the transitive dependencies.

(4) Prune unused dependencies from your dependency tree

2)

```
public interface Deduction {  
    static void deduct() {  
        System.out.println("deduct");  
    }  
}  
  
@SuppressWarnings("")  
public class Customer implements Deduction {  
    public void serviceCharge(){  
        //deduction functionality being invoked  
        deduct();  
    }  
}  
  
public class TechSol {  
  
    public static void main(String[] args){  
        new Customer().serviceCharge();  
    }  
}
```

What will happen to the above code when compiled and executed?

a)The code will not get compiled because the static method of interface is not accessed using interface name

b)The code will not get compiled as the method deduct() is undefined for the Customer class

c)Will get executed successfully leaving the output "deduct"

d)Will lead to run time ambiguity as the static method is not accessed using Interface name

3) What is true about 'mvn clean'?

A) It generates site documentation of the project,

B) It is used to migrate the application.

C) It cleans up artifacts created by prior builds.

d) This is used to build the application.

4) Which command is used to compare the working directory with staging area?

gita state

git add

Scorching heat

git diff

5) What will be the output for the following code?

```
class Employee{
```

```
String name:
```

```
double salary:
```

```
Employee (String name, double salary) {
```

```
this.name=name;
```

```
this.salary=salary;
```

```
}
```

```
}
```

```
class Testers
```

```
public static void main(String[] args) {
```

```

Employee employee1=new Employee("John",30000.00);
Employee employee2=new Employee("Sam", 40000.00);
Employee employee3=new Employee ("Jon", 35000.00); Map<Integer, Employee> map=new
HashMap<Integer, Employee>();
map.put(1, employee1);
map.put(2, employee2);
map.put(1, new Employee ("Jon", 30000.00));
for (Map. EntryInteger, Employees> entry:map.entrySet())
).salary))
{
System.out.println("EmployeeId: "tentry.getKey()+" Name: "tantry.getValue().name+" Salary:
"+entry.getValm().salary);
}
}
}

```

EmployeeId: 1 Name: John Salary: 30000.0 EmployeeId: 2 Name: Sam Salary: 40000.0

The code will throw runtime exception as the same key cannot be used for 2 values

Set cannot be converted to Map. Entry

EmployeeId: 1 Name: Jon Salary: 35000.0 EmployeeId: 1 Name: John Salary: 30000.0

EmployeeId: 1 Name: Jon Salary: 35000.0 EmployeeId: 2 Name: Sar Salary: 40000.0

6) Given that the current directory is empty, and that the user has read and write permissions, with the following code snippet:

```

11. import java.io.*;
12. public class DOS (
13. public static void main(String[] args) {
14. File dir = new File("dir");
15. dir.mkdir();
16. File fl= new File(dir, "fl.txt");
17. try {
18. f1.createNewFile();
19.) catch (IOException e) (;)

```

```

20. File newDir = new File("newDir");
21. dir.renameTo (newDir);
22. }
23. }

```

Which statement is true?

Compilation falls.

The file system has a new empty directory named dir.

The file system has a new empty directory named newDir.

The file system has a directory named dir, containing a file 11.txt

The file system has a directory named newDir, containing a file f1.txt.

7)

The below swapAdjacent() method should swap the data of two adjacent nodes in a linked list.

E.g. For the linked list, 12->15->21->45, the modified linked list should be 15->12->45->21.

Which line of code should you place instead of //Missing code to make the method work correctly?

```

public void swapadjacent (Node head) (
    int tempData;
    Node tempshead;
    while(temp.getNext()!=null) (
        tempData=temp.getData();1111
        //Missing code
        temp.getNext().setData (tempbata) temp-tamp.getNext().getNext();
    }
}

```

```
temp=temp.getNext().getData()
```

```
temp.setData(temp.getData());
```

```
Dtemp.setData(temp.getNext().getData()); Chatgpt answered
```

```
temp.setData(tempData)
```

8)

Consider the array given below.

arr (35,49,56,65,67,78,82,85,89,92,96,99)

Which values of arr can be searched during the third iteration of binary search?

49, 65, 85 and 96

35, 65, 82 and 92

49, 65, 82 and 92 chatgpt answered

35, 65, 82 and 96

9) Please refer below

What will be the output for the code snippet given below when the following stack is passed to the given method?

stack (top-bottom): 2 1 9 6 7

```
public static Queue<Integer> queuePunction(Queue<Integer> stack)
```

```
1 Queue<Integer> outQueue = new ArrayDeque<Integer>();
```

```
Queue<Integer> tempQueue = new ArrayDeque<Integer>();
```

```
while (stack.peek()!=null) {
```

```
Integer value = stack.pop();
```

```
if (values2 == 0)
```

```
outQueue.add(value);
```

```
as e
```

```
tempQueue.add(value);
```

```
Integer temp = 0;
```

```
while (tempQueue.peek()!=null) {
```

```
temp + tempQueue.poll();
```

```
outQueue.add(temp);
```

```
return outQueue;
```

Note: Order of displaying the elements of the queue is from front to rear.

100% sure answer: 2 6 1 1 0 1 7

9) What will be the output for the following code?

```
public class Employee {  
    private String name; private int empld;  
    private String city;  
    Employee()  
    {  
        this.city "New York"  
    }  
    Employee (String name, int empld) (  
    this():  
    thie.name name;  
    this.empld = mapld;  
    public static void main(String[] args) { Employee employael = new Employee ("John", 101); Employse  
    employee2 = new Employee ();  
    System.out.println(employeeel.name + " " + employeeel.empld+ + employeeel.city):  
    System.out.println(employee2.name + " " + employee2.empld + + employee2.city);  
    }  
    John 101 New York John 101 New York  
    John 101 null null 0 New York  
    John 101 New York null 0 New York  
    John 101 null John 101 New York
```

10)

Consider the code snippet given below. How many threads are active at line numbers 1, 2, and 3 respectively?

```
public class MyThreadDemo implements Runnable {  
    public void run() { System.out.println("Thread is running");  
    public static void main(String[] args) {  
        MyThreadDemo obj = new MyThreadDemo ();  
        Thread threadObj = new Thread(obj);  
        // 1 threadObj.start();  
        // 2. threadObj.run();// 3
```


Code will provide a run time error

Code will provide an error on compilation

main thread, 2, 3 respectively

main thread, 2 respectively

11) Which of the three framework activities are present in the Adaptive Software Development(ASD)?

Requirements gathering, adaptive cycle planning, iterative development

speculation, collaboration, learning

analysis, design, coding

iterative development, coding, design

12)

What will be the output for the following code?

```
class Demo 1
```

```
String test;
```

```
public Demo (String test) { this.test = test;
```

```
@Override
```

```
public int hashCode() { return 1;
```

```
public class Tester {
```

```
public static void main(String args[]){
```

```
Map<Demo, String> tests = new HashMap<Demo, String>(); tests.put(new Demo ("testKey"),  
"testValue");
```

```
tests.put(new Demo ("test2Key"), "test2Value");
```

```
tests.put(new Demo("testKey"), "testValue overridden"); System.out.print(tests.size());
```

```
System.out.print(" "+tests.get(new Demo ("testKey")));
```

```
}}
```

```
}
```

Ans -A) 3 null

13) Given the stack: (Top Bottom) 7->12->15->4->2

What will be the content of the returned stack when the given stack is passed as input to the below method?

```
public static Deque<Integer> calculate (Deque<Integer> stack) {  
    Deque<Integer> outStack = new ArrayDeque<Integer>(); if (stack.peek()==null) {  
        return outStack;  
    } else {  
        while (stack.peek() !=null) {  
            outStack.push(stack.pop());  
            outStack.push(stack.pop());  
            outStack.push(outStack.pop()*outStack.pop());  
        }  
        return outStack;  
    }  
}
```

84-85

5->84->8

84->5->8

8->5>84

14)

What defines the Agile Scrum approach?

System that relies on incremental development

System that relies on no development

System that relies on ad hoc development

System that relies on conditional development

Question No: 38 | Max Mark(s): 01 |

Multiple Choice (Select 1 out of 4 options, for the question below)

QuestionId: 13068

Please refer below

Given the code fragments:

-

```
public class Book implements Comparator<Book> {  
    String name;  
    double price;  
    public Book () {}  
    public Book (String name, double price) {  
        this.name = name;  
        this.price = price;
```

```
        public int compare (Book b1, Book b2) {  
            return b1.name. compareTo(b2.name) ;
```

```
        public String toString() {  
            return name + ":" + price;
```

and

```
List<Book>books = Arrays.asList (  
    new Book ("Beginning with Java", 2),  
    new Book ("A Guide to Java Tour", 3)
```

```
    Collections.sort (books, new Book());
```

```
    System.out.print (books) ;
```

What will be the output for the following code?

[A Guide to Java Tour:3.0, Beginning with Java:2.0]

[Beginning with Java:2.0, A Guide to Java Tour:3.0]

A compilation error occurs because the Book class does not override the abstract method compare To().

An Exception is thrown at run time.

Question No: 37 | Max Mark(s): 01 |

Multiple Choice (Select 1 out of 4 options, for the question below.)

What will the following command print to the terminal?

git remote -v

A list of remote repositories and their URL's

The current git version your running

An inline editor for modifying remote repositories

The last 5 git versions you have installed

Question 38

```
public Demo (String test) {
```

```
this. test = test;
```

```

@Override

public int hashCode () {
return 1;

public class Tester {
public static void main (String args[]) {

Map<Demo, String> tests = new HashMap<Demo, String>();
tests.put(new Demo("test1Key"), "test1Value") ;
tests.put(new Demo ("test2Key"), "test2Value") ;
tests.put(new Demo("test1Key"), "test1Value overridden") ;
System.out.print(tests.size());
System.out.print(" "+tests.get(new Demo("test1Key")));

-

```

3 null

2 test1Value overridden

3 test1Value overridden

2 null

Question No: 40 | Max Mark(s): 01 |

Multiple Choice (Select 1 out of 4 options, for the question below.)

Which Maven phase copies a project's artifact into the local repository to use as a dependency?

Package

install

Compile

Deploy

What will be the output for the following code?

```
public Deer (int age) { System.out.print ("DeerAge"); }  
private boolean hasHorns () { return false; }
```

```
public static void main(String[] args) {  
    Deer deer = new Reindeer (5) ;  
    System.out.println(", "+deer.hasHorns ());
```

11:

12:

```
1: public class Deer {  
    : public Deer() { System.out.print("Deer"); }
```

3:

4:

5:

6:

7:

8:

9: 1

```
10: class Reindeer extends Deer {  
    public Reindeer (int age) { System.out.print("Reindeer"); }  
    public boolean hasHorns () { return true; }  
13: }
```

DeerReindeer,false

ReindeerDeer,false

DeerAgeReindeer,false

DeerAgeReindeer,true

The code will not compile because of line 7

Given the below code. What can be the possible outputs?

```
1. public class Threads2 implements Runnable {  
2.  
3. public void run() {  
4. System.out.println("run.");  
5. throw new RuntimeException("Problem") ;  
6. }  
7. public static void main (String[] args) {  
8. Thread t = new Thread(new Threads2()) ;  
9. t.start ();  
10. System.out.println("End of method. ") ;  
11. }  
12. }
```

java.lang.RuntimeException: Problem

run

java.lang.RuntimeException: Problem

End of method.

java.lang.RuntimeException: Problem

End of method.

run.

java.lang.RuntimeException: Problem

Answer:

End of method.

run.

java.lang.RuntimeException: Problem

run.

java.lang.RuntimeException: Problem

End of method.

Please refer below

Consider the array given below.

arr= {125, 126, 177, 185, 196, 199, 201, 208, 329, 431, 565, 589, 618, 679, 818, 903, 913}

How many iterations will be required to find the element 818 using binary search algorithm?

Auto Nerd

Answer

3

4

2

1

Question No: 48

Multiple Choice (Select 1 out of 4 options, for the question below.)

QuestionId: 4d003

How does the tool like FIT and Fitness compliment agile?

Continuous Integration

Unit Testing

Code coverage metrics

Automated Acceptance Tests

What is the result for the below given code?

1. public class KungFu {
2. public static void main (String[] args) {
3. Integer x = 400;
4. Integer y = x;
5. x++;

```
6. StringBuilder sb1 = new StringBuilder ("123");
7. StringBuilder sb2 = sb1;
8. sb1.append ("5") ;
9. System.out.println((x == y) + " " + (sb1 == sb2)) ;
10. }
11. }
```

true true

false true

true false

false false

Compilation fails.

An exception is thrown at runtime.

Identify the issue in the below code fragment.

```
public class Example {
    public String formatInput (String i) {
        if(i.trim().length() == 9) { //Line 3
            StringBuilder sb = new StringBuilder();
            sb=sb.insert(0,"+1("); //Line 5
            sb=sb.insert(6,")"); //Line 6
```

```
sb=sb.insert(10,"-"); //Line 7
return sb.toString();
return null;
public static void main (String[] args) {
Example obj = new Example();
String i;
obj.formatInput(i);
```

Compilation Fails at Line 3

Compilation Fails at Line 6

NullPointerException will be thrown if the value of i is null

Compilation Fails at Line 7

StringIndexOutOfBoundsException will be thrown if the length is 9

Question No: 45 | Max Mark(s): 01 |

Multiple Choice (There may be more than 1 correct answer)

QuestionId: 66004

Which of the following techniques are used for estimating effort?

Role Play

Observation

Affinity estimation

Planning poker

Discussion

1. Consider the following form and servlet code:

Test.html:

```
<form action="Demo?vall=Second" method="post">
<input type="hidden" name="vall" value="First"/>
X:<input type="radio" name="val2" value="Third"/>
Y:<input type="radio" name="val2" value="Four"/>
<input type="submit"/>
</form>
```

Demo Servlet:

```
protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    response.setContentType("text/plain");
    PrintWriter Out=response.getWriter();
    Out.write("<html> <body>");
    String[] vall=request.getParameterValues ("vall");
    for (int i=0;i<vall.length;i++)
    {
        Out.write(vall[i]+":");
    }
    String val2= request.getParameter("val2");
    Out.write(val2+":");
    Out.write("</body> </html>");
}
```

Assuming the user changes none of the default settings and presses SUBMIT, what will be the servlet output in the response?

Second: First:Third:

Second:First:Four:

Second:First:null:

First:second:second:first

2. The below code snippet shows the minimum required details of a hibernate.cfg.xml file where Driver1 is the driver class for a database and URL1 is the URL of the database to connect. The mapping details of the Employee entity is defined in emp.hbm.xml.

Identify the mandatory statement which is missing from the below configuration file?

```
<hibernate-configuration>
<session-factory >
<property name="hibernate.connection.driver_class"> Driver1</property>
<property name="hibernate.connection.url">URL1</property>
<property name="hibernate.connection.username">username1</property> <property
name="hibernate.connection.password">password1</property>
<property name="hibernate.dialect"> Dialect1</property>
</session-factory>
</hibernate-configuration>
```

give the correct answer from below

1.<property name="hibernate.hbm2ddlauto">create-drop</property>

2.<mapping resource="emp.hbm.xml"/>

3.<property name="hibernate.hbm2ddlauto">drop</property>

4.<property name="hibernate.show_sql">true</property>

3. Please refer below FoxTravels maintains two classes Driving License and Citizen. They have one to one relationship between them. Assume that these two classes are annotated correctly with Cascade Type ALL and there are no records currently available in the database. What will be the output of the code given below?

```
DrivingLicense d1=new DrivingLicense();
d1.setLicenceNumber(101);
Calendar issueDate =Calendar.getInstance();
issueDate.set (12, 12, 2005);
Calendar expiry= Calendar.getInstance();
expiry.set (12, 12, 2025);
d1.setLicenceIssued(issueDate);
d1.setLicence Expiry(expiry);
d1.setVehicleType("LMV")
Citizen citzn = new Citizen();
citzn. setcustId (5001) ;
citzn.setcustName("Ram");
citzn.setDriving License (d1);
session.getTransaction().begin();
session.persist (citzn);
session.getTransaction().commit();
option
```

1. One record will be inserted in Driving License Table and one record will be inserted in Citizen Table
2. One record will be inserted in Driving License Table but no record will be inserted in Citizen Table
3. No record will be inserted in Driving License Table but one record will be inserted in Citizen Table
4. No record will be inserted in Driving License Table and no record will be inserted in Citizen Table

5. Please refer below Consider a MobileService class in the package com.test with appropriate properties and methods. A Spring configuration file defining a bean of Mobile Service class is shown below:

```
<bean id="mobileService1" class="com.test.MobileService"/>
```

```
<bean id="mobileService2" class="com.test.MobileService"/>
```

Select the CORRECT statement from the following options.

options

1. Spring does not allow multiple bean definitions of same class

2. Creates two beans mobileService1 and mobileService2 of type MobileService class, which can be accessed in application by using the id of the respective
3. Creates only one bean in the container of type MobileService class
4. Spring does not allow more than one bean definition in the configuration

6. Please refer below Consider the following JSP code:

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
pageEncoding="ISO-8859-1"%>
```

```
<html>
```

```
<body>
```

```
<%! int p;%>
```

```
<%! public void jspDestroy ()
```

```
{
```

```
out.print(Integer.parseInt(p));
```

```
} %>
```

```
System.out.println("self-destructing");
```

```
<%! public void jsplnit() {
```

```
System.out.println(<%= p %>);
```

```
}
```

```
%>
```

```
</body>
```

```
</html>
```

Which of the following are valid reasons for the above JSP failing to translate and compile?

1. **Expression embedded in declaration**
2. **Data member p not initialized before use**
3. Local variable x not initialized before use
4. **Placement of jspDestroy() method before jsplnit() method**
5. The page actually complies and translate without any problem
6. Integer type is not applicable for parcing arguments

7. Multiple Choice (Select 1 out of 4 options, for the question below.)

Which MySQL function can be used to work with the current date?

_TODAY

CURRDATE()

CURDATE()

NOW()

8. What is the return type for ServletContext.getResource() and ServletContext.getResourceAsStream() methods?

1. java.io.Resource and java.io.InputStream
2. java.io.Resource and java.io.BufferedReader
3. java.net.URL and java.io.InputStream
4. java.io.File and java.io.InputStream
5. java.net.URL and java.io.FileInputStream

9. Please refer below

What will be the output for the following code?

DemoServlet:

```
ServletConfig config=getServletConfig();
Enumeration<String> e=config.getInitParameterNames();
String str="";
while(e.hasMoreElements()) {
str=e.nextElement();
out.print("<br>Name: "+str);
out.print(" value: "+config.getInitParameter(str));
}
```

Web.xml:

```
<servlet>
<servlet-name>DemoServlet</servlet-name>
<servlet-class>com.test. DemoServlet</servlet-class>
<init-param>
<param-name>username</param-name>
<param-value>system</param-value>
</init-param>
<init-param>
<param-name>password</param-name>
<param-value>oracle</param-value>
</init-param>
</servlet>
```

(A)

Name: password value: oracle

Name: username value: system

10.

Which techniques deliver a stream that simply provides the raw bytes from the database without any conversion?

getCharacterStream

getBinaryStream

getAsciiStream

getUnicodeStream

11

Given the HTML form:

1. <html>
2. <body>
3. <form action="submit.jsp">
4. Name: <input type="text" name="i1">

5. Price: <input type="text" name="i2">

6. <input type="submit">
7. </form>
8. </body>
9. </html>

Product Class:

```
package com.example;

public class Product {
    private String name;
    private double price;

    public Product() {
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice (double price) {
        this.price = price;
    }
}

} Assume the product attribute does not exist in any scope. Which code snippet, in
submit.jsp, instantiates an instance of com.example.Product that contains the results of the
form submission?

(A) <jsp:useBean id="com.example. Product" /> <jsp: setProperty name="product"
property="*" />
```

(B) `<jsp:useBean class="product1" id="com.test. Product" /> ${product.name = param.il}
$(product.price = param.i2)`

(C) `<jsp:useBean id="product" class="com.example. Product">`

`<jsp:setProperty name="product" property="name" param="il" />`

`<jsp:setProperty name="product" property="price" param="12" /> </jsp:useBean>`

(D) `<jsp:useBean id="product" type="com.example.Product"> <jsp:setProperty
name="product" property="name" value="<%= request.getParameter("il") %>" />
<jsp:setProperty name="product" property="price" value="<%= request.getParameter("12")
%>" />`

12

Not sure

Which of the following statements about a column are true?

1. You cannot decrease the width of a CHAR column
2. You cannot convert a CHAR data type column to the VARCHAR data type.
3. You cannot increase the width of a VARCHAR column.
4. You cannot modify the data type of a column if the column contains non-null data.
5. You cannot specify the column's position when adding a new column to a table

13

What is the benefit of PreparedStatement?

1. Slow performance
2. Encourages SQL injection
3. PreparedStatement is best suited with repetitive kind of queries
4. More memory usage

14

Please refer below Evaluate this SQL statement:

```
SELECT manufacturer_id, COUNT(*), order_date  
FROM inventory  
WHERE price > 5.00
```

```
GROUP BY order_date, manufacturer_id
```

```
HAVING COUNT(*) > 10
```

```
ORDER BY order date DESC;
```

Which clause specifies the rows to be returned from the inventory table?

1. HAVING COUNT(*) > 10
2. GROUP BY order_date, manufacturer_id
3. SELECT manufacturer_id, COUNT(*), order_date
4. ORDER BY order date DESC

5. **WHERE price > 5.00**

15

Identify the different stages of lifecycle in Hibernate.

1. **Transient Persistent Detached**
2. Persistent Transient Detached
3. Transient Detached Persistent
4. Persistent Detached Transient

16

In an web page how would you write the JSP standard action code to import a JSP segment that generates user preference specific content?

1. **<jsp:include page="user-pref.jsp"> <jsp:param name="userPref" value="\${user.preference}" /> </jsp:include>**
2. <jsp:import page="user-pref.jsp"> <jsp:param name="userPref" value="\${user.preference}" /> </jsp:include>
3. <jsp:import file="user-pref.jsp"> <jsp:param name="userPref" value="\${user.preference}" /> </jsp:import>
4. <jsp:include path="user-pref.jsp"> <jsp:param name="userPref" value="\${user.preference}" /> </jsp:include>

17

Please refer below What would be the result of the following Delete statement in MySQL?
Delete from customer Where customer_id IN (100, 102, 105)

1. **Customers 100, 102 and 105 would be deleted**
2. All customers not between 100 and 105 would be deleted.
3. All customers except 100, 102 and 105 would be deleted.
4. Customers between 100 and 105 would be deleted.

18.

Test.html

```
<form method="get" action="Demo">  
Name: <input type="text" name="user" /> <br/>  
<input type="submit" value="submit">  
</form>
```

Demo Servlet:

```
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws  
ServletException, IOException {
```

```
PrintWriter out = response.getWriter();
```

```
request.setAttribute("User", "David");
```

```
String user = (String) request.getAttribute("user");
out.println("<html>");
out.println("<body>");
    out.println("<form action='Second'>");
out.println("<input type='hidden' name='user' value='"+user+"'>");
out.println("<input type='submit' value='submit' >");
out.println("</form>");
out.println("</body>");
out.println("</html>");
}
```

Second Servlet:

```
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    // TODO Auto-generated method stub
    PrintWriter out = response.getWriter();
    String user = request.getParameter("user");
    out.println("Welcome "+user);
}
```

1. Welcome John
2. Welcome David
3. Compilation Error
4. Welcome null

19.

Gen has written a JSP code in which he is calling the getMessage() function which is returning a value "hello world". Identify which code he should use to get this.

(A) **<html> <!-- String message="hello world"; String getMessage() { return message; } %> Message for you: <%= getMessage() %> </html>**

(B) <html> <% String message="hello world"; String getMessage() { return message; } %> Message for you: <%= getmessage() %> </html>

(C) <html> <%= String message="hello world"; String getMessage() { return message; } %> Message for you: <%= getMessage() %> </html>

(D) <html> <!-- String message="hello world"; String getMessage() { return message; } %> Message for you: <!-- getMessage() %> </html>

20.

Assume that we have two entity classes created for validating the Citizen's voter details. As per the rules, a citizen is supposed to have only one voter identity with him. Which among the following mappings are true with respect to One to One Mapping?

- (A) `@Entity @Table (name "Citizen")`
`public class Citizen {`
`@Id private int citizenId;`
`private String citizenName;`
`@OneToOne(cascade CascadeType.ALL)`
`@JoinColumn(name="citizen VoterId")`
`private Voter citizenVoter;`
`//getter and setters`
`@Entity @Table(name = "Voter")`
`public class Voter {`
`@Id private int citizenVoter Id;`
`private String location; //getters and setters`
`}`
- (B) `@Entity @Table(name = "Citizen")`
`public class Citizen( @OneToOne(cascade = CascadeType.ALL)`
`private Voter citizenVoter;`
`@Id private int citizen Id;`
`private String citizenName;`
`//getters and setters`
`}`
`@Entity @Table(name "Voter")`
`public class Voter {`
`name = "citizenVoterId")`
`@Id @JoinColumn (`
`private int citizenVoterId;`
`private String location;`
`//getters and setter`
`}`
- (C) `@Entity @Table(name public class Citizen "Citizen")`
`{`
`@Id private int citizen Id;`
`private String citizenName;`
`@OneOne (cascade = CascadeType.ALL)`
`@JoinedColumn(name="citizenVoterId")`
`private Voter citizenVoter;`
`//getters and setters`
`@Entity @Table(name public class Voter "Voter")`
`{`
`@Id private int citizenVoter Id;`
`private String location;`
`//getters and setters`
`}`
- (D) `@Entity @Table(name = "Citizen")`
`public class Citizen {`
`@Id`
`private int citizenId;`
`private String citizenName;`
`@OneToOne (cascade = CascadeType.ALL)`
`@JoinColumn (name=" 'citizenVoterId", unique = = true)`

```
private Voter citizenVoter;  
//getters and setters  
} @Entity  
@Table(name public class Voter "Voter") {  
    @Id private int citizenVoterId;  
    private String location;  
    //getters and setters  
}
```

21.

Multiple Choice (There may be more than 1 correct answer)

Please refer below

Consider the following code:

```
<html>  
<body>  
<jsp:useBean id="address" class="AddressBean" scope="session" />  
state = <jsp:getProperty name="address" property="state" />  
</body>  
</html>
```

Which of the following are equivalent to the third line ?

1. <% state = address.getState(); %>
2. <% out.write("state = "); out.print(address.getState()); %>
3. <% out.print("state = "); out.print(address.getState()); %>
4. <% out.print("state = " + address.getState()); %>
5. state = <%= address.getState() %>

22

How many jsp implicit objects are there which are created by web container that are available to all the jsp pages?

1. 8
2. 9
3. 10
4. 7

23

Which tag is used to enable annotation support in Spring?

1. <annotation-config /> to bean configuration.
2. <annotation-context:config /> to bean configuration.
3. <context:configuration>
4. <context:annotation-config/> to bean configuration

24

What would be the result of the following statement in MySQL if there are duplicate values in each table?

```
SELECT actor_id, first_name, last_name
```

```
FROM actor
```

```
UNION
```

```
SELECT actor_id, first_name, last_name
```

```
FROM actor_copy
```

```
ORDER BY actor_id
```

1. All records from both tables will be returned.
2. Only records from the actor_copy table will be returned.
3. Only records from the actor table will be returned.
4. **Only unique records from both tables will be returned.**
5. No records will be returned.

25

Which defines the welcome files in a web application deployment descriptor?

1. (A) <welcome>
 <welcome- file > / welcome.jsp</welcome-file>
 </welcome>
 <welcome> <welcome-file>/index.html</welcome-file>
 </welcome>
2. (B) <welcome-file-list> <welcome-file> welcome.jsp</welcome-file> <welcome-file>index.html</welcome-file> </welcome-file-list>
3. (C) <welcome> <welcome-file>welcome.jsp</welcome </welcome > </welcome >
 <welcome-file>index.html</welcome-file> </welcome >
4. (D) <welcome-file-list> <welcome-file> welcome.jsp</welcome-file> <welcome-file>/index.html</welcome-file> </welcome-file-list> <welcome> <welcome-file>
 <welcome-name> welcome </welcome-name> <location> welcome.jsp </location>
 </welcome-file>
5. E) <welcome-file> <welcome-name>index</welcome-name> <location> index
 </location> </welcome-file>

26 [Not sure- Acc to Chatgpt and Gbard - answer: 1 & 2](#)

there may be more than 1 correct answers

Please refer below What are the possible outcomes by executing the doGet method in ClubServlet given below?

```
public class ClubServlet extends HttpServlet
```

```
{
```

```
public void doGet(HttpServletRequest req, HttpServletResponse res) throws  
ServletException, IOException
```

```
{
```

```
RequestDispatcher rd=getServletContext().getRequestDispatcher ("PlayServlet");
```

```
rd. forward (req, res);
```

1. HTTP 500 error
2. NullPointerException
3. An illegalArgumentException
4. HTTP 255 error
5. HTTP 404 error

27

Your web application requires the ability to load and remove web files dynamically to the web containers file system. Which HTTP methods are used to perform these actions?

- ✓ 1. PUT
2. POST
3. SEND
- ✓ 4. DELETE
5. REMOVE
6. DESTROY

28

Please refer below What will be the result of running the following jsp file taking into account that the Web server has just been started and this is the first page loaded by the server?

```
<html>
<body>
<%= request.getSession(false).getId() %>
</body>
</html>
```

1. It will not compile
2. It will print the session id
3. It will produce a NullPointerException as the getSession(false) method's call returns null, cause no session had been created.
4. It will produce an empty page

29

Which of the following statements correctly store an object associated with a name at a place where all the servlets/JSPs of the same webapp participating in a session can use it? (Assume that request, response, name, value etc. are references to objects of appropriate types.)

1. request.setAttribute(name, value)
- ✓ 2. response.setAttribute(name, value)
- ✗ 3. request.getSession().setAttribute(name, value)

4. `servlet.getServletContext().setAttribute(name, value)`

5. `request.setParameter(name, value)`

why we use Request Dispatcher to forward a request to another resource, instead of using a `sendRedirect`?

1. Redirects are no longer supported in the current servlet API
2. Redirects are not a cross-platform portable mechanism
3. The RequestDispatcher does not use the reflection API
4. The RequestDispatcher does not require a round trip to the client, and thus is more efficient and allows the server to maintain request state.

30

Which of the following is incorrect about dependency injection in Spring?

1. It helps in achieving loose coupling between objects.
2. It helps in testing the module by injecting the dependent Mock Objects.
3. It increases the scope of code reusability.
4. It helps in managing database connections.

31

There's a Mobile class which has `mobilenno`, `model` and `price` have properties. The corresponding `Mobile.hbm.xml` is written and the location is specified in `hibernate.cfg.xml`. In the main file, the `SessionFactory` object (`sf`) is created from which a session is opened as below:

1. `SessionFactory factory= cfg.buildSessionFactory (serviceRegistry);`
2. `Session session= factory.openSession();`
3. `Transaction tx = session.beginTransaction();`
4. `Mobile m1=new Mobile();`
5. `m1.setMno (102);` //what is the state
6. `m1.setModel("Motorola");`
7. `m1.setPrice (12000);`
8. `session.save(m1);` //what is the state
9. `tx.commit();`
10. `session.close();`
11. `m1.setModel ("Samsung");` //what is the state
12. `m1.setPrice (13000);`
13. `Session s1= factory.openSession();`
14. `tx= session.beginTransaction();`
15. `m1.setModel ("Apple");` //what is the state
16. `session.update(m1);` //what is the state
17. `.commit();`
18. `tx =null;`

Identify what are the object states in 4,8,11,15 & 16.

1. Transient, Persistent, Detached, Persistent, Persistent
2. Transient, Persistent, Persistent, Detached, Persistent
3. Transient, Persistent, Detached, Detached, Detached
4. Transient, Persistent, Detached, Detached, Persistent

32

Consider the following code snippet:

Servlet:

```
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
```

```
    ServletContext sc = this.getServletContext();
```

```
    RequestDispatcher dis = sc.getRequestDispatcher("/Menu.html");
```

```
    PrintWriter out = response.getWriter();
```

```
    if (dis != null)
```

```
        dis.forward(request, response);
```

```
    else
```

```
        out.print("Content not Available");
```

```
        out.print("Ending..");
```

```
}
```

Menu.html:

```
<html>
```

```
<body>
```

```
    <p>Content from Menu</p>
```

```
</body>
```

```
</html>
```

Which one of the following will be the correct outcome when the above code is executed?

1. Content from Menu
2. Content from Menu Ending..
3. Content not Available Ending..
4. Content not Available

33. Following are some of the rules framed for writing a java bean. Select correct option from the following that gives the combination of correct rules.

1. It must have a public no-arg constructor
It should have private accessor and mutator methods
It should have private member variables
The return type of getter method should be of same type of setter type argument
It must implement Comparable interface
2. It must have a parameterised constructor
It should override toString method. It should have private member variables

The return type of getter method should be of same type of setter type argument
It must implement comparable interface

3. It must have a parameterised constructor

It should override toString method.

It should have public member variables

The return type of getter method should be of same type of setter type argument

It must implement serializable interface

4. It must have a public no-arg constructor

It should have public accessor and mutator methods

It should have private member variables

The return type of getter method should be of same type of setter type argument

It must implement serializable interface

34. please refer below Consider the following scenario:

Statement A: <jsp:include/> standard action inserts the response of a jsp page at runtime.

Statement B: include directive inserts the source of a jsp page at translation time.

Which of the following statement is correct with respect to the above?

1. Statement A is true and B is false
2. Statement A is false and B is false
3. Statement A is true and B is true
4. Statement A is false and B is true

35

Consider the following JSP page sources:

<%-- file one.jsp begins here-->

<%@page import="java.util.*" contentType="text/html" session="true"%>

<html>

<body>

<ArrayList al=new ArrayList(0);

a1.add("Jack Russell");

a1.add("Labrador");

a1.add("Great Dane");

%>

<@include file="two.jsp"%>

</body>

</html>

<%--file two.jsp begins-->

<%@ page import="java.util.*" contentType="text/html"%>

<table>

<%for (int i=0;i<al.size(0); i++) {%>

```
<r><td> <%=al.get(i) %></td></tr>
```

```
<%} %>
```

```
</table>
```

```
<%-- two.jsp ends here-->
```

Which of the following statements are true about the above two JSP page sources, given that client is requesting for one.jsp?

1. Translation will fail because the import attribute of the page directive is repeated across the page source.
2. The local variable al in two.jsp will not be recognized
3. The order of the import and contentType attributes in both JSP page sources is immaterial.
4. Removing the Session attribute from one.jsp will make no difference to the generated servlet code
5. The import, contentType, and session attributes should appear in separate page directives.

36

Which condition must be satisfied by a query in order to use it as a sub-query after the IN keyword?

1. The outer query must share a column with the same name as the inner query
2. It should return only a single row
3. It should return only a single column
4. The output of the query must be ordered

37

Which of the following are examples of cross-cutting concerns?

1. Logging
2. Security
3. Debugging
4. Wiring

39

Please refer below Consider the following code for address.jsp:

```
<html>
```

```
<body>
```

```
<jsp:useBean id="address" class="AddressBean" />
```

```
<jsp:setProperty name="address" property="city" value="Sydney" />
```

```
<jsp:setProperty name="address" property="city" />
```

```
<jsp:getProperty name="address" property="city" />
```

```
</body>
```

```
</html>
```

Also consider the URL

<http://localhost:8080/guest/address.jsp?city=Melbourne&city=Canberra>

(Assume that the city property is not an indexed property.)

Which one of the following is the output if the above page is accessed via the mentioned URL?

1. Sydney
 2. ☒ Melbourne
 3. Canberra
 4. SydneyMelbourneCanberra
- ☒ MelbourneCanberra

40

APIs for dynamically registering servlets and filters can be called at?

1. Context initialization time
2. There is no way to dynamically register servlets and filters
3. Context destruction time
4. Injected into a servlet context

41

Select the different ways to configure metadata in Spring.

1. By XML Based configuration file
2. By Annotation based configuration.
3. By Java based configuration
4. Only XML or Annotation based is possible but not Java based

42

Refer the following database schema.

Table Courses

Id

associatedStudents (Many-to-one relationship with Students)

Table Students

Id

optedCourses (Many-to-one relationship with Courses)

Which of the following options implements this relationship correctly?

1. #Student.java
@Entity
public class Student {
@Id Long id;
@ManyToMany

```

Set<Courses> optedCourses;
}
#Courses.java
@Entity
public class Courses {
    @Id
    Long id;
    @ManyToMany
    Set<Student> associatedStudents;
}

```

```

2. #Student.java
@Entity
public class Student {
    @Id
    Long id;
    @JoinColumn
    Set<Courses> optedCourses;
}
#Courses.java
@Entity
public class Courses
{
    @Id
    Long id;
    @ManyToMany
    Set<Student> associatedStudents;
}

```

```

3. #Student.java
@Entity
public class Student {
    @Id
    Long id;
    @JoinColumn (mappedBy=id)
    Set<Courses> optedCourses;
}
#Courses.java
@Entity

public class Courses {
    @Id
    Long id;

    @JoinColumn (mappedBy='id')
    Set<Student> associatedStudents}

```

```

4. #Student.java
@Entity

public class Student {
@Id
Long id;
@OneToMany
Set<Courses> optedCourses;
}
# Courses.java
@Entity
public class Courses {
@Id
Long id;
@OneToMany

Set<Student> associatedStudents;
}

```

43

Multiple Choice (Select 1 out of 4 options, for the question below)

What sequence should be followed when closing a database resource?

- ☒ 1. Connection, Statements, and then ResultSet
- ☐ 2. Statements, ResultSet, and then Connection
- ☐ 3. Connection, ResultSet, and then Statements
- ☒ 4. ResultSet, Statements, and then Connection

44

What will the following code produce as a result?

```

static ({
try
{ Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306: test", "db",
"mysql");
} catch (SQLException e)
{
e.printStackTrace();
}
public static void main(String[] args)
{ try
{
Statement st con.createStatement();
System.out.println("Database Connected");
st.close();
}
}
}

```

```
con.close();  
} catch (Exception e) {}  
}  
}
```

1. ✓ Compile Time Error Due to con variable is not accessible outside the block scope.
2. ClassNotFoundException occurs.
3. ✗ Database Connected.
4. SQL exception occurs when run.

45. Users of your web application have requested that they should be able to set the duration of their sessions. So for example, one user might want a webapp to stay connected for an hour rather than the webapp's default of fifteen minutes; another user might want to stay connected for a whole day. Furthermore, you have a special login servlet that performs user authentication and retrieves the object from the database. You want to augment this code to set up the user's specified session duration. Which code snippet in the login servlet will accomplish this goal?

1. User user = // retrieves the user objects object from the database session.setDurationInterval (user.getSessionDuration());

B. User user = // retrieves the User objects from the database session.setInactiveInterval (user.getSessionDuration());

C. User user = // retrieves the user objects object from the database session.setInactiveInterval (user.get.SessionDuration());

D. User user = // retrieves the user objects object from the database session.setDuration (user.getSessionDuration());

E. User user = // retrieves the user objects object from the database session.setMaxInactiveInterval (user.getSessionDuration());

F. User user = // retrieves the user objects object from the database session.setMaxDurationInterval (user.getSessionDuration());

46. Ronald has observed that his Web application has some HTML tags or features that all the Web browsers do not support. If the client's browser does not support it, the application may or may not run or may generate undesired output in the Web browser of the client. Therefore Ronald wants to identify the type of browser and other specific information about the client that sends the request to the Web application before invoking the servlet. Which of the following option will help Ronald provide this functionality?

1. By programming filters
2. By creating security constraints
3. By modifying the web deployment descriptor
4. By implementing the Listeners

47. In what order should a database connection be created?

1. Import JDBC packages.
2. Open a connection to the database.

3. Load and register the JDBC driver.
48. Execute the statement object and returns query resultset.
5. Create a statement object to perform a query.
6. Close the resultset and statement objects.
7. Process the resultset.
8. Close the connection.

1,2,3,5,4,7,8,6

1,3,2,5,4,7,6,8

2,1,3,4,8,7,5,6

1,3,2,4,5,6,7,8

49. Which three EL expressions, inserted at line 15, are valid and evaluate to "3"? (Choose three)

1. A. `${vals.2}`
2. B. `${vals ["2"] }`
3. C. `${vals.index}`
4. D. `${vals[index] }`
5. E. `${vals} [index]`
6. F. `${vals. (vals.index) }`
7. G. `${vals [vals[index-1]] }`

50. Given below is the code snippet. What will be the current state of an entity, which was in managed state, and delete method has been invoked on by the corresponding transaction is still not committed?

```
CustomerEntity entity = new (CustomerEntity) session.get(CustomerEntity.class, 1002);
```

```
session.beginTransaction();
```

```
session.delete(entity);
```

1. New State
2. Managed State
3. Removed State
4. State Undefined

MainTest

1. Consider Mobile Service class with two properties mobile Number and model as shown below. <pre>package com.test; public class MobileService(private long mobileNumber; private String model; //parameterized constructor public MobileService (long mobileNumber, String model) { this.mobileNumber = mobileNumber; this.model = model; config.xml Configuration file as below: <bean id="mobileService" class="com.test. MobileService"> <constructor-arg value="Nokia"/> <constructor-arg value="1234567892"/> </bean></pre> The main method of the application is shown below: <pre>ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml"); MobileService m = (MobileService) ac. getBean ("mobileService");</pre>	What will be the output for the above code? Throws Exception as the constructor arguments are not passed in appropriate order in the bean definition mobileService bean of MobileService class type is created with values as "1234567892" for mobile Number and "Nokia" for model mobileService bean of MobileService class type is created with values as "0" for mobile Number and "1234567892" for model mobileService bean of MobileService class type is created with values as "0" for mobile Number and "null" for model
2. A servlet sets a sessionn scoped attribute product with an instance of com.example. product an forward to a JSP. Which of the following options output the name of the product in response? <pre>package com.example; public class Product { private String name; private double price; public Product() { public String getName () { return name; } } public void setName(String name) { this.name = name; } public double getPrice() { return price; } }</pre>	<pre><%=product.getName()%></pre> <pre><jsp:useBean id="product" class="com.example.Prod uct" /> <%= product.getName() %></pre> <pre><jsp:useBean class="com.example.Prod uct" id="product" scope="page"> <%= product.getName() %> </jsp:useBean></pre> <pre><jsp:useBean id="product" type="com.example.Prod uct" scope="page" /></pre>

<pre> public void setPrice (double price) { this price = price; } } %= product.getName() %> </pre>	<pre> <%= product.getName() %> <jsp:useBean id="product" type="com.example.Prod uct"> <%= product.getName() %> </pre>
3. Read the following code of the Hibernate configuration file and identify the right tag to refer the POJO class Locker. <pre> <?xml version="1.0" encoding="UTF-8"?> <!DOCTYPE hibernate-configuration PUBLIC "-//Hibernate/Hibernate Configuration DTD 3.0//EN" "http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd"> <hibernate-configuration> <session-factory> < property name="hibernate.dialect">org.hibernate. dialect. MySQL5Dialect</property> property name="hibernate.connection.driver_class">com.mysql.jdbc.Driver</p roperty> < <property name="hibernate.connection.url">jdbc:mysql://localhost:3306/hiber natedb</property> <property name="hibernate.connection.username">root</property> <property name="hibernate.connection.password">root</property> <property name="hibernate.show_sql">true</property> --- ="com.mybank. Locker"/> </session-factory> </hibernate-configuration> </pre>	mapping resource mapping class mapping file load mappingfile
4. Consider the following JSP page sources: <pre> <-- file one.jsp begins here--> <%@page import="java.util.*" contentType="text/html" session="true"> <html> <body> </pre>	Which of the following statements are true about the above two JSP page sources, given that client is requesting for one.jsp? Translation will fail because the import

<pre> <%ArrayList al=new ArrayList(0); al. add ("Jack Russell"); al. add ("Labrador"); al. add("Great Dane");> <@include file="two.jsp"> </body> </html> <!--file two.jsp begins--> <%@page import="java.util.*" contentType="text/html"> <table> <%for (int i=0;i<al.size(0); i++) {%> <r><td> <%=al.get(i) %></td></tr> <88> </table> <8--two.jsp ends here--> </pre>	attribute of the page directive is repeated across the page source. The local variable al in two.jsp will not be recognized The order of the import and contentType attributes in both JSP page sources is immaterial. Removing the Session attribute from one.jsp will make no difference to the generated servlet code The import, contentType, and session attributes should appear in separate page directives
5. Consider the following JSP code snippet: <pre> Employee Bean john if (john == null) (EmployeeBean)request.getAttribute("john"); harry = new EmployeeBean(); request.setAttribute("john", john); } </pre> Which one of the following is an equivalent action code for the above code?	<pre> <jsp:declareBean id = "john" class = "EmployeeBean" scope= "request" /> <jsp:createBean id = "john" class = "EmployeeBean" /> <jsp:useBean id = "john" class = "EmployeeBean" scope = "request" /> <jsp:useBean id = "john" class = "EmployeeBean" /> </pre>
6. What will be the outcome of running the following jsp code? <pre> <body> int i = 1, j = 10 do { </pre>	<pre> i = 6 and j = 5 i = 5 and j = 5 i = 6 and j = 4 </pre>

<pre> if(i > i) * ibreak; j = - i } while (i < 5) ; System.out.println I ^ prime prime i=^ prime prime +i+^ prime prime and j =^ prime prime +j) : </body> </pre>	i = 5 and j = 6
7. What will be the output for the following code? <pre> static (try (} catch (SQLException e) (e.printStackTrace(); Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306: test", "db", "mysql"); public static void main(String[] args) (try { Statement st con.createStatement(); System.out.println("Database Connected"); st.close(); con.close(); } catch (Exception e) () </pre>	Compile Time Error Due to con variable is not accessible outside the block scope. ClassNotFoundException occurs. Database Connected. SQL exception occurs when run.
8. What will be the output of the following code snippet? <pre> <% page language="java"> <html> <head></head> </pre>	It will display the date as mm:dd:yy It will display the date and time It will display time as hh/mm/ss

<pre> <body> <java.util.Date now=new java.util.Date(); 8> <H1><%= now.getHours() %>: <%=now.getMinutes()>: <%=now.getSeconds () %></H1> </body> </html> </pre>	It will display the current time as hh:mm:ss
9. Consider the following deployment descriptor(web.xml)file: <pre> <session-config> <session-timeout>70</session-timeout> </session-config> </pre> Then, in the Servlet program code setMaxInactiveInterval (600) (seconds) for the session object is invoked. After how long would the session expire?	70 minutes 60 minutes Gives illegalStateException 70 seconds
10. You are creating a form to allow a user to upload files into a web application that will be handled by a servlet. Which of the following HTML snippet would you use?	<pre> <form action="UploadServlet" method="POST"> <form action="UploadServlet" enctype="multipart/form -data" method="POST"> <form action="uploadservlet.jsp " enctype="multipart/form -data" method="POST"> <form action="uploadservlet.jsp " method="GET"> </pre>
11. Consider the following classes defined in a Spring Application: <pre> package com.test.service; public class MobileService (private String mobileName; public MobileService() { //getter and setter methods package com.test.service; public class PersonService (private MobileService mob; //getter and setter methods </pre> The Spring configuration file includes the following declarations:	What will be the output for the above code? NoSuchBeanDefinitionException

<pre> <bean id="mobileService1" class="com.test.service. MobileService"> <property name="mobileName" value="Nokia"/> </bean> <bean id="mobileService2" class="com.test.service MobileService"> <property name="mobileName" value="Samsung"/> </bean> <bean id="personService" class="com.test.service. PersonService" autowire="byType" /> The main method for the application is defined as follows: ApplicationContext ac= new ClassPathXmlApplicationContext("config.xml"); PersonService pl= (PersonService) ac.getBean("personService"); System.out.println("I have: "+pl.getMob ().getMobileName()); </pre>	
12. What will be the output for the following code when the user enters the name as "Henry"? Demo.html: <pre> <form action="Demo" method="get"> <hr> Name: <input type="text" name="vall">

 User Id: <input type="text" name="val2" value="U1001">

 <input type="submit" value="login"> </form> </pre> Demo Servlet: <pre> protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException { PrintWriter out = response.getWriter(); String n=request.getParameter("vall"); String u=request.getParameter("val2"); if(u!=null) response.sendRedirect("Welcome"); else out.write(n+" "+ u); Welcome Servlet: protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException { String name = request.getParameter("vall"); System.out.println(name); } </pre>	The output will be a redirection to the "Henry" servlet

There's a Mobile class which has mobileno, model and price have properties. The corresponding Mobile.hbm.xml is written and the location is specified in hibernate.cfg.xml. In the main file, the SessionFactory object (sf) is created from which a session is opened as below: 1. SessionFactory factory = cfg.buildSessionFactory (serviceRegistry); 2. Session session = factory.openSession(); 3. Transaction tx = session.beginTransaction(); 4. Mobile ml = new Mobile(); //what is the state 5. ml.setMno(102); 6. ml.setModel("Motorola"); 7. ml.setPrice(12000); 8. session.save(ml); //what is the state 9. tx.commit(); 10. session.close(); 11. ml.setModel("Samsung"); //what is the state 12. ml.setPrice (13000); 13. Session s1 = factory.openSession(); 14. tx session.beginTransaction(); 15. ml.setModel("Apple"); //what is the state 16. session.update (ml); //what is the state 17. tx.commit(); 18. tx = null;	Identify what are the object states in 4,8,11,15 & 16. Transient, Persistent, Detached, Persistent, Persistent Transient, Persistent, Persistent, Detached, Persistent Transient, Persistent, Detached, Detached, Detached Transient, Persistent, Detached, Detached, Persistent
13. Consider the following JSP code: <pre><%@ page language="java" contentType="text/html; charset=ISO-8859-1" pageEncoding="ISO-8859-1"%> <html> <body> <%! int p;%> <%! public void jspDestroy () { print(Integer.parseInt(p)); System.out.println("self-destructing"); <%! public void jsplnit() { System.out.println(<%= p %>); </body> </html></pre>	Which of the following are valid reasons for the above JSP failing to translate and compile? ✓ Expression embedded in declaration Data member p not initialized before use Local variable x not initialized before use Placement of jspDestroy() method before jsplnit() method The page actually compiles and translates without any problem

	Integer Type is not applicable for parsing arguments
14. Consider the following code: <pre> public class StudentData { int studentId; String student Name; //Getter Setter Methods Servlet1: protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException { HttpSession session = request.getSession(true); StudentData student = new StudentData(); student.setStudent Id (1001); student.setStudentName ("Mary"); session.setAttribute("studentInfo", student); Servlet2: protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException { request.getSession(false); HttpSession session1 = request.getSession(true); PrintWriter out = response.getWriter(); StudentData student = (StudentData) session.getAttribute("studentInfo"); session1.setAttribute("studentInfo", student); StudentData student1 = (StudentData) out.print(student.studentName+" "+studentId); </pre>	What will be the output when Servlet2 is called? O Mary 1001 Onull null Onull 1001 Mary null
15. Let us first look at how to implement transaction using JDBC API. For the Employee Management scenario, the code given below is used to retrieve employee record based on the employee id from the database using plain JDBC API	In order to complete the above code rearrange the above statement. i,iii, ii ,iv iv,iii, ii

Now the development team would like to enable transaction support explicitly instead of the auto-commit mode that the JDBC driver uses by default. <pre> public void insert (Employee employee) { // Assume code to set database properties with appropriate details try{ Class.forName (driver);//register driver con = DriverManager.getConnection (url, username, password);//create connection //line 1 String query = "insert into employee values (?, ?, ?)"; // Create prepared statement ps = connection.prepareStatement (query); ps.setLong(1, employee.getEmpId()); ps.setString(2, employee.getEmpName()); ps.setString(3, employee.getDepartment()); // Execute query ps.executeUpdate(); //line 2 } catch (Exception e) { //line 3 printStackTrace(); finally { //Assume code to close prepared statement, resultset and database connection } 1. //turn off auto-commit con.setAutoCommit(false); ii. // If there is any error. conn.rollback(); iii. // If there is no error. conn.commit(); iv. //turn on auto-commit con.setAutoCommit(true); </pre>	iv, iii,i
16. Evaluate this SELECT statement: <pre> SELECT first_name, last_name FROM physician WHERE physician_id NOT IN (SELECT physician_id FROM physician WHERE license_no = 17852); </pre> Which one of the following SELECT statement would achieve the same result?	(A) SELECT first_name, last_name FROM physician WHERE physician_id IN (SELECT physician_id FROM physician WHERE license_no = 17852); (B) SELECT first_name, last_name FROM

	physician WHERE physician_id != ALL (SELECT physician_id FROM physician WHERE license no = 17852); (C) ✓ SELECT first_name, last_name FROM physician_id WHERE license no <> 17852 AND license_no IS NOT NULL; (D) SELECT first_name, last_name FROM physician WHERE physician_id = 17852;
17. What will be the output when we request two times to the following servlet? <pre> public class TestDemo implements Servlet { public TestDemo() { System.out.println("Servlet Constructor"); } public void init (ServletConfig config) throws ServletException { System.out.println("Init Method"); } public void destroy() { System.out.println("Destroy Method"); } public void service (ServletRequest request, ServletResponse response) throws ServletException, IOException { System.out.println("Service Method"); } public ServletConfig getServletConfig() { return null; } public String getServletInfo() { return null; } } </pre>	(A) Servlet Constructor Init Method Service Method Destroy Method (B) Servlet Constructor Servlet Constructor Init Method Service Method Destroy Method (C) Servlet Constructor Servlet Constructor Init Method Service Method Service Method Destroy Method (D) Servlet Constructor Init Method Service Method Service Method Destroy Method
18. In the case of stored procedures, which statements are true regarding IN parameters?	IN Parameter is the default parameter

	IN Parameter value should be passed to the procedure for processing The same parameter will serve as input and output The IN parameter is a variable whose value will change in the invoked procedure.
19. Assume that we have two entity classes created for validating the Cezary's voter details. As per the rule, a citizen is supposed to have only one voter identity with which among the following mappings is then with respect to One to One Mapping? (A) <pre> @Entity @Table (name = "Citizen") public class Citizen { @Id private int citizenId; private String citizenName; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Voter citizenVoter; } @Entity @Table(name = "Voter") public class Voter { @Id private int voterId; private String location; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Citizen citizen; } //getters and setters </pre> B) <pre> @Entity @Table(name = "Citizen") public class Citizen { @Id private int citizenId; private String citizenName; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Voter citizenVoter; } @Entity @Table(name = "Voter") public class Voter { @Id private int voterId; private String location; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Citizen citizen; } //getters and setters </pre> (C) <pre> @Entity @Table(name = "Citizen") public class Citizen { @Id private int citizenId; private String citizenName; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Voter citizenVoter; } @Entity @Table(name = "Voter") public class Voter { @Id private int voterId; private String location; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Citizen citizen; } //getters and setters </pre> (D) <pre> @Entity @Table(name = "Citizen") public class Citizen { @Id private int citizenId; private String citizenName; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId", unique = true) private Voter citizenVoter; } @Entity @Table(name = "Voter") public class Voter { @Id private int voterId; private String location; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Citizen citizen; } //getters and setters </pre>	(D) <pre> @Entity @Table(name = "Citizen") public class Citizen { @Id private int citizenId; private String citizenName; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId", unique = true) private Voter citizenVoter; } @Entity @Table(name = "Voter") public class Voter { @Id private int voterId; private String location; @OneToOne (cascade = CascadeType.ALL) @JoinColumn (name = "citizenVoterId") private Citizen citizen; } //getters and setters </pre>

@Id private int citizenVoter Id; private String location; //getters and setters	

1. Which of the following are the features of DevOps?(multi ans)(kk)

Automation

Collaboration

Integration

Deployment

2. What is the file name built by Gradle?

build.xml

config.gradle

pom.gradle

Build.gradle

3. Assume you have a class called Car and this has the following variables: id, make, model and price. Assume you want to configure a derived query method which will return the car with the id of 45.

Which of these methods of a CrudRepository will you invoke to get this car?

Find(45)

findAll().filterById(45)

findById(45)

findByMake().id(45)

4. Given the below code:

```
@RestController
```

```
@RequestMapping("/friendsfamily")
```

```
public class FriendsFamilyController
```

```
{
```

```
@GetMapping("/{phoneNumber}")
```

```
public FriendsFamilyDTO getFriendsFamily()
```

```
{
```

```
}
```

//code to search for Friends Family based on the phone number and return the same

} Choose the correct line of code that should occupy the parameter section of the request method.

```
@PathParam("phoneNumber") long phoneNumber
```

```
@RequestParam("phoneNumber") long phoneNumber
```

```
@PathVariable("phoneNumber") long phoneNumber
```

```
@RequestBody("phoneNumber") long phoneNumber
```

5. While selecting multiple fields using JPQL how are the field records returned?

- a. Using Object[] arrays for field values
- b. Using a special class which is used to hold multiple fields
- c. Using Pair<> and Triplet<> objects
- d. Using List<> collections for field values

6. What will be the output for the following piece of code?(run)

```
//The JUnit files are imported

public class TestClass {

    @Test

    public void testingMethod () {

        String message = "Test";

        assertEquals (3, message.length());

    }

}
```

- a. Compilation Error
- b. Runtime Error
- c. Test Ran with Success
- d. Test Ran with Failure

7. Please refer below(run)

Assuming that necessary imports are done, what is the output when following code is executed?

```
package com.test.service;

@Service

public class CustomerService {

    public CustomerService() {

        System.out.print("CustomerService constructor called ");

    }

}

package com.test.dao;

@Repository

public class CustomerDao {
```



```

public CustomerDao () {
    System.out.print("CustomerDao constructor called ");
}
}

package com.test

@SpringBootApplication

public class DemoSpringBootApplication {

    public static void main(String[] args) {
        SpringApplication.run(DemoSpringBootApplication.class, args);
    }
}

```

- a. CustomerService constructor called CustomerDao constructor called
- b. CustomerDao constructor called CustomerService constructor called
- c. Exception is Thrown at Runtime
- d. Nothing will be displayed in console.

8. Please refer below

Assume that the @Transactional annotation support is enabled appropriately with Spring configuration.

What is the behavior of the following code sample?

```

public class EmployeeServiceImpl {
    -----

    @Transactional

    public void updateEmployee (int empId, String baseLocation) {

        Employee employee = getEmployee (employee.getEmpId());
    }
}

```

```
}
```

```
@Transactional (propagation=Propagation.REQUIRES_NEW)
```

```
public Employee getEmployee (int empId) {
```

```
// IMPLEMENTATION
```

```
}
```

a. The getEmployee() method executes code in a new transaction

b. The getEmployee() method executes in the transaction scope of update Employee() method

c. Spring does not allow overriding the default values of @Transactional annotation

d. The getEmployee() method executes in non-transaction scope

9. The software engineering practice of merging all developer working copies to the shared mainline several times a day is known as

a. Continuous integration

b. Continuous deployment

c. Continuous delivery

d. DevOps

10. Please refer below(run)

Given the below code. (Assume necessary imports are done.)

```
@RestController
```

```
@RequestMapping("/employee")
```

```
public class EmployeeController
```

```
{
```

```
@PostMapping("/create")
```

```
public EmployeeDTO addEmployee (@RequestBody EmployeeDTO employeeDTO){
```

```
//code to add employee and return the same back
```

```
}
```

```
}
```

Choose the correct statement.

- a. @PostMapping is not supported when the controller class is annotated with @RequestMapping
- b. @PostMapping is not allowed to hold path matching info ("/create")
- c. Compilation Error
- d. This controller is perfectly fine and will work as expected

11.

Which Maven phase copies a project's artifact into the local repository to use as a dependency?

- a. Package
- b. Install
- c. Compile
- d. Deploy

12. What are the features of Jenkins?(multi ans)

- a. It is a free and open-source automation tool
- b. Jenkins provides a vast number of plugins
- c. It is difficult to set up and install on multiple operating systems
- d. Provides pipeline support

13. Assume you have a table in your database called Car which maps to a Car class in your Java application. There is a member variable in your class called "make" representing the brand/manufacturer of a car. There is also a "type" variable conveying whether the car is a sedan, SUV, hatchback etc. You have also set up a CrudRepository for the Car entity. Let's say you wish to perform a search such as "give me all cars of brand myBrand of type hatchback". Which of these is a method which you can set up in the CrudRepository to enable such a search?

(A) List<Car> findByMakePlusType (String make, String type);

(B)List<Car> findByMake (String make);

(C)List<Car> findByMakeAndType(String make, String type);

(D)Car findByMakeAndType();

14.Which represents a programming paradigm which can be applied to Spring applications which allows for new behavior to be added without modifying the existing code base?

Bean-based design

Aspect-oriented programming

Config-based design

Intrusion-free programming

Object-oriented programming

15. For an Entity class with a composite key, we can replace the use of the @Embeddedid annotation as long as we apply a specific annotation to the class pointing it to the composite key class. Which represents that annotation?

@IdClass

@Embed

@Composite

@Reference

16. Please refer below

Consider the following classes defined in a Spring Application:

```
package com.mphasis.service;
```

```
public class MobileService {
```

```
long mobileNumber;
```

```

String model;

public MobileService (long mobileNumber, String model) {

this.mobileNumber = mobileNumber;

this.model = model;

}

public MobileService() {

//getter and setter methods

}

package com.mphasis.service;

public class PersonService {

private Mobileservice mobile;

//getter and setter methods

}

```

The Spring configuration file includes the following declarations:

```

<bean id="m1" class="com.mphasis.service. MobileService">

    <constructor-arg value="93423" />

    <constructor-arg value="Nokia" />

</bean>

<bean id="personService" class="com.mphasis.service.PersonService" autowire="no"
/> The main method for the application is defined as follows:

```

```

ApplicationContext ac= new ClassPathXmlApplicationContext("config.xml");
PersonService p1= (PersonService) context.getBean("personService");

System.out.println(p1.getMobile().getModel());

```

What is the output of the above code execution?

Throws Exception

Display mobile name as "Nokia"

Display mobile name as null

Display model name as null

17)

```
public class OwnerController (
```

```
@RequestMapping(value = "/owner/{ownerId}", method RequestMethod. POST)
```

```
@ResponseBody
```

```
public Owner findowner (@PathVariable("ownerId") int ownerId) {
```

```
return nav Owner();
```

A. RequestMethod.GET method is more accurate than POST

B. @PathVariable should be replaced with the @PathParam annotation

C. @ResponseBody could be removed

D. Returning the 201 HTTP status code is better

Select multiple correct option

18)

TechData core team works on a project based on Spring Boot application development with JPA integration for ORM support. The team uses a class called DataUserDAOImpl to define the EntityManager instance, this class implements the DataUserDAO interface. The specification mentions that the Entity Manager created should be maintained by the container and there should not be any developer's side efforts to

manage the Entity Manager. Help the team identify which among the following can be a valid option to achieve this?

(A)

```
public class DataUserDAOImpl implements DataUserDAO
```

```
@PersistenceContext
```

```
private EntityManager entityManager;
```

```
//Defining required methods to perform database operations
```

B)

```
public class DataUserDAOImpl implements DataUserDAO
```

```
@PersistenceUnit
```

```
private EntityManager entityManager;
```

```
//Defining required methods to perform database operations
```

(C)

```
public class DataUserDAOImpl
```

```
@PersistenceContext
```

```
private EntityManager entityManager;
```

```
//Defining required methods to perform database operations
```

(D)

```
public class DataUserDAOImpl
```

```
@PersistenceUnit
```

```
private EntityManager entityManager;
```

```
//Defining required methods to perform database operations
```

20)

If you have configured a table that you want to use to generate primary keys using the TABLE generation strategy, which of the following statements are true?

- a) Each table has an auto-incremented column for primary keys
- b) While sharing a database tables every entity which uses the TABLE strategy needs to have an entry in the shared table
- c) Every entity which uses the TABLE strategy needs to have a separate table
- d) Multiple tables can share the same table for primary key generation (if single ans)

21)

Consider the below given Spring Application code.

```
package com.test.service;

public class EmployeeService {

    private int employeeNo;

    private String employeeName;

    @Autowired

    private Employee employee;

    //getter and setter methods for all properties

}

package com.test.model;

public class Employee {

    private int employeeId;

    private String name;

    private String role;

    //getter and setter methods for all properties
```


}

Assume required beans are available in the Spring configuration. Select

- a) Autowiring is generally advisable for workflow classes such as controller, service and repository classes.
- b) Code throws compilation error as @Autowired has been used in Employee Service class on domain reference(Employee employee).
- c) Domain object has to be created using new operator during database access to represent the data in a database.
- d) It is generally not advisable to make domain objects as Spring beans.

22)

Which of the following properties can be used to configure a column in the JPA @Column annotation?(multi ans)

- a) columnLength
- b) column Type
- c) columnDefinition
- d) precision
- e) scale

23)

What are the correct implementation classes of ApplicationContext?

- a) FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext

b) FileSystemXmlApplicationContext, WebXmlApplicationContext, ClassPathXmlApplicationContext

c) AdvancedApplicationContext, FileApplicationContext

d) FileSystemApplicationContext, ClassPathApplicationContext

24. Which of the following is true when you use the @IdClass annotation to specify composite keys?(multiple ans)

We do not need an additional class representing the composite key

We need an additional class representing the composite key

This annotation is to send up relationships and not to specify keys

We need to use @Id annotations on composite key fields in the entity class

25. Assume you have defined a CrudRepository in a package com.myapp. Which of these annotations can you use in a configuration class to scan for such repositories and activate them for your app?

@Bean(package = "com.myapp")

@EnableJpaRepositories (basePackages = {"com.myapp"})

@Scan(package = "com.myapp")

@Configuration("com.myapp")

26. What is a self-managed solution that provides source code collaboration for professional teams of any size, across any distance?

Bitbucket

Git

Github

docker hub

28. Cyber Data team is working on a project which uses Spring JDBC with Spring Boot. Help the team to identify, which of the below operations are supported and will be automatically handled by Spring Boot JDBC?(multiple ans)

Setting up loop for iterating over results.

Specifying SQL statements, declaring parameters and providing values.

Defining connection parameters like driver class name, database URL, username and password.

Opening connection, Preparing and executing statements, Processing exceptions, Handling Transactions, Closing connections, statement and result set.

29. Which of the following is NOT a validator annotation that Spring has built-in support for?

@Size

@NotDuplicateId

@Email

@NotNull

30. Which of the following is true about update and delete queries in JPQL?(multi)

You need to run these queries within a transaction

JPQL can perform these queries on only one record at a time

JPQL does not support these queries

You need to invoke `.executeUpdate()` to run these queries

31. Which statement best describes Spring Initializr?

A JDBC library which simplifies database operations

A Maven plugin to build Spring projects

The official reference documentation for Spring Boot

A utility to generate the starter kit for a Spring project

32. When deploying your web application as an external war file what are the first two details you need to specify in your app?(multi ans)

Have your configuration derive from `SpringBootServletInitializer`

Set the Maven packaging as "war"

Set the Maven packaging as external-war

Have your configuration have the annotation `@ExternalWar`

33. What annotation should be used in a JUnit test class to define a method to initialize any variables or classes before running the set of tests?

`@SpringBootTest`

`@initialize`

`@Constructor`

`@Before`

`@Test`

34. The development team is working on fund Transfer method of Banking application. The team wants to know the default rollback policy in Spring transaction management. Identify the Correct options from the given statements.(multi ans)

Rollback for any Exception

Rollback for RuntimeException

Rollback for checked exceptions

Always commit

commit for checked exceptions

35. Which annotations can be added to your Spring Boot application to enable auto-configuration?

@EnableAutoConfiguration

@EnableAutoConf

@SpringBootApplication

@EnableSpringAutoConfiguration

@SpringBootApplication

36. Consider Mobile Service class with two properties mobile Number and model as shown below.

```
package com.spring;
```

```
public class MobileService {
```

```
    private long mobileNumber;
```

```
    private String model;
```

```
    //default constructor
```

```
    //getter & setters
```

```
}
```

config.xml:

```
<bean id="mobileService" class="com.spring. MobileService">
```

```
<constructor-arg value="1234567891"/>
```

```
<constructor-arg value="Nokia"/>
```

```
</bean>
```

The main method:

```
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");  
MobileService m = (MobileService)ac.getBean("mobileService");
```

What is the error in the above scenario?

Attribute name is not provided in the constructor-arg tag of bean definition in the configuration file.

Compilation Error

Use of index/type attribute is mandatory in constructor-arg tag.

Use of scope attribute is mandatory in constructor-arg tag.

Parameterized constructor is not added in the POJO class.

37.If you are using starter templates for your application, what do you need to watch out for in the boilerplate HTML code?

Update the boilerplate code to use your code

Update the contents of the HTML to represent what you want in your site

Update references to the CSS and other assets to point to your location

Update the pom.xml file to include CSS dependencies

38.Assume you have implemented a one-to-one relationship in your JPA application using the @MapsId annotation in one of the entity class definitions. Which statement about the corresponding tables created is TRUE?

a. There will be two tables with foreign key references in both classes, referencing each other

b. No tables will be created

c .There will only be a single table created with one entity embedded in the other

d. There will be two tables with a foreign key reference from one class to another

39-----is the point at which we say a build is ready for deployment into the production environment.

Build

Test

Release

Deploy

40. Assume you have defined a CrudRepository in a package com.myapp. Which of these annotations can you use in a configuration class to scan for such repositories and activate them for your app?

@Bean(package = "com.myapp")

@EnableJpaRepositories (basePackages = {"com.myapp"})

@Scan(package = "com.myapp")

@Configuration("com.myapp")

41. Which of the following statements are true about entities?(multi)

Entity objects correspond to records in a database table

They map to relationships between tables in a database

They map to a table in the underlying database

They represent constraints on values in a database table

42.What is the property in Spring which helps you to change the port of your running application?

port

server.port

server.address.port

application.port

43.Please refer below

Identify the correct annotation to be placed in the blanks given below.

package mypack;

@Repository (value = "librarydaobean")

public class LibraryDAOImpl implements ILibraryDAO

{ private EntityManagerFactory emf;

public void setEntityManagerFactory (EntityManagerFactory emf)

{ this.emf = emf; }

}

@Persistence Unit

@PersistenceContext

@Persistence

@Autowired

Hale Tech team uses a Spring-based application to maintain their user details. The team has implemented a backend using the CRUDRepository of Spring Data commons. The HaleUser class is an entity class that is mapped to the table in the relational schema. The class is designed as shown below:(multi)

@Entity


```
public class HaleUser {
```

```
@Id
```

```
private Integer haleUserId;
```

```
private String haleUserName;
```

```
//getters and setters
```

```
} Help the team identify which among the following are valid methods supported by Spring?
```

```
findAll()
```

```
findById(Integer haleUserId)
```

```
save(Integer haleUserId)
```

```
deleteById(Integer haleUserId)
```

44. Please refer below

Which of the following are true regarding the below Spring controller?

```
@RestController
```

```
public class OwnerController {
```

```
@RequestMapping(value = "/owner/{ownerId}", method= RequestMethod. POST)
```

```
@ResponseBody
```

```
public Owner findOwner (@PathVariable("ownerId") int ownerId) {
```

```
return new Owner();
```

```
}
```

```
}
```

1. RequestMethod. GET method is more accurate than POST

2. @PathVariable should be replaced with the @PathParam annotation

3. @ResponseBody could be removed

4. Returning the 201 HTTP status code is better

45. Choose the components with which Jenkins is more closely integrated. (multi ans)

1. Build tools like Maven.

2. Version Control systems like Git and SVN (Apache Subversion).

3. Eclipse

4. VS Code

46. What must you add to your Spring Boot application in order to access and return results from a Spring Boot service?

1. A service dependency and an import statement for the service

2. A RestController object that takes the service as its end point

3. A bean in the Rest controller to call the respective service URL

4. A request mapping in the application Rest controller

47. Please refer below

Consider the following classes defined in a Spring application:

package com.demo.company;

public class Address {

private String city;

private String pincode;

public Address () { }

//getter and setter methods}

```
package com.demo.company;  
  
public class Employee {  
  
    private Address address;  
  
    //getter and setter methods)
```

Config.xml:

```
<bean id="address" class="com.demo.company. Address">  
    <property name="city" value="Los Angeles"/>  
    <property name="pincode" value="91167"/>  
</bean>  
  
<bean id="employee" class="com.demo.company.Employee" autowire="byName" />
```

Main class:

```
ApplicationContext ac= new ClassPathXmlApplicationContext("config.xml");  
  
Employee e1= (Employee) ac.getBean ("employee");  
  
System.out.println("City: "+e1.getAddress().getCity()+" Street:  
"+e1.getAddress().getPincode());
```

What is the output of the above code execution?

- 1.Throws NullPointerException
- 2.Display city as "Los Angeles" and pincode as null
3. Display city and pincode as null
- 4.No bean named 'address' available
- 5.Display city as "Los Angeles" and pincode as "91167"

48.Where can you add an external browser and change the default for the Spring Boot Dashboard?

1.Spring Tool Suite 4 -> Preferences -> General -> Web Browser

2.Spring -> Boot Dashboard -> Preferences -> Web Browser -> General

3.Spring Tool Suite 4 -> Boot Dashboard -> General

4.Boot Dashboard -> Show Properties-> Web Browser

49.MphasisRun application has been developed to maintain the history of employees participating in all the run events organized by the company. The team uses JpaRepository for its back end implementation. The entity class used for storing the employee details is coded as shown below

@Entity

```
public class MphasisRunners{
```

@Id

```
private Integer mphasisRunnerEmpId:
```

```
private String phasisRunner EmpName;
```

```
//getters and setters
```

```
}
```

Assume that mphasisRunners is an instance of MphasisRunners class. Help the team decide which among the following methods can be used to persist the MphasisRunners record?

1.save(mphasisRunners)

3.persist(mphasisRunners)

4.saveAndFlush(mphasisRunners)

5.save(mphasisRunnerEmpId)

50.For the code given below, choose the equivalent option.

```
package com.foo;
```

```
@Configuration
```

```
public class AcmeConfig {
```

```
@Bean
```

```
public AcmeService acmeService() {
```

```
    return new AcmeServiceImpl();
```

(A)

```
<beans> <bean id="acmeConfig" class="com.AcmeService"/> </beans>
```

(B)

```
<beans> <bean id="acmeService" class="com.AcmeServiceImpl"/>
```

```
</beans>
```

(C)

```
<beans>
```

```
<bean id="acmeConfig" class="AcmeServiceImpl"/>
```

```
</beans>
```

(D)

```
<beans>
```

```
<bean id="acmeConfig" className="com.foo.AcmeServiceImpl"/>
```

```
</beans>
```

(E)

```
<beans>
```

```
<bean id="acmeService" class="com.foo.AcmeServiceImpl"/>
```

```
</beans>
```

1. Questionid: c6045

Please refer below

Assume that the @Transactional annotation enabled appropriately with Spring configuration
What is the behavior of the following code sample?

```
public class EmployeeserviceImp
{
    -----
    @Transactional
    public void updateEmployee (int empld, String baseLocation)
    Employee employee =getEmployee (employee.getEmpld());
    -----
    @Transactional (propagation Propagat. REQUIRES_NEW)
    public Employee getEmployee (int empld)
{
    // IMPLEMENTATION
}
}
```

1. The getEmployee() method executes code in a new transaction
2. The getEmployee() method executes in the transaction scope of update Employee() method
3. Spring does not allow overriding the default values of @Transactional annotation
4. The getEmployee() method executes in non-transaction scope

2.

Questionid: 04005 Please refer below Consider the code given below:

```
public interface MusicianDAO {
//CRUD methods
}
public class MusicianDAOImpl extends MusicianDAO
{
//Implements CRUD operations using Spring JdbcTemplate
-----
}
```

Which among the following is translate checked exceptions valid annotation to be used at the class level to unchecked exceptions by eliminating the need for an explicit try-catch block. the repository code?

1. @Component
2. @Repository
3. @ExceptionHandler(value = Exception.class)
4. @Configuration

3. .

Multiple Choice (There may be more than 1 correct answer)

Questionid: 25026

Which of the following are advantages of modelling relationships using embedded objects?

- a. The same embeddable can be shared across multiple entities

- b. JPA providers are more performant with embedded objects
- c. Easier to use embeddables rather than individual fields
- d. Can represent relationships using composition in the object-oriented world

4.

Assume you have a table in your database called Car which maps to a Car class in your Java application. There is a member variable in your class called "make" representing the brand/manufacturer of a car. There is also a "type" variable conveying whether the car is a sedan, SUV, hatchback etc. You have also set up a CrudRepository for the Car entity. Let's say you wish to perform a search such as "give me all cars of brand myBrand of type hatchback. Which of these is a method which you can set up in the CrudRepository to enable such a search?

- (A)
List<Car> findByMakePlusType (String make, String type):
- (B)
List<Car> findByMake (String make);
- (C)
List<Car> findByMakeAndType (String make, String type);
- (D)
Car findByMakeAndType();

5.

Please refer below

Given the code below:

```
@Entity
public class Employee {
    @ManyToOne
    @JoinColumn (name="dept_id")
    private Department dept;
}
```

How do I retrieve employees who belong to a particular department

- (A)"select e from Employees e where e.dept =?1"
- (B)"select e from Employees e where e.dept_id =?1"
- (C)"select e from Employees e where e.dept.id = ?1"
- (D)"select e from Employees e where e.department_id=?1"

6.

Please refer below

Identify the correct annotation to be placed in the blanks given below.

```
package mypack;
@Repository(value "librarydaobean")
public class LibraryDAOImpl implements ILibraryDAO
{
    private EntityManagerFactory emf;
    _____
    public void setEntityManagerFactory (EntityManagerFactory emf)
    {
        this.emf= emf;
    }
}
```


@Persistence Unit
@PersistenceContext
@Persistence
@Autowired

7.

Refer the following code

```
@RestController  
public class ApiController{  
    //code goes here
```

```
@RequestMapping("/")  
public String index(){
```

```
    logger.debug("A DEBUG Message");  
    //code goes here
```

```
    return "Display set-up for logging configuration.";  
}  
}
```

Which of the following added 3 and Line 7 to log the message "An INFO message"?

- (A) `Logger logger = LoggerFactory.getLogger(ApiController.class);
logger.info("An INFO Message");`
- (B) `Logger logger = LoggerFactory.getLogger;
logger.info("An INFO Message");`
- (C) `Logger logger = new LoggerFactory (ApiController.class);
logger.info("An INFO Message");`
- (D) `Logger logger = LoggerFactory.getLogger (SpringBootApplication.class);
logger.info("An INFO Message");`

8.

Which are the embedded servers that the spring-boot-starter-web dependency supports?

- a. Undertow
- b. httpserver
- c. Jetty
- d. Tomcat
- e. IIS

9.

Which of the following is NOT a reason to use an Object Relational Mapping framework when working with relational databases using a high-level programming language such as Java?

- 1. Relational tables express relationships using constraints and programming languages use design patterns
- 2. Relational tables consist of rows and columns and objects consist of fields
- 3. There is no way to query data from a database without using a language such as Java
- 4. The fundamental unit of programming for relational databases and Java is different

10. Assume you are building an app for a car insurance company where a customer can own multiple cars, but a car can only belong to one customer. Which represents a valid relationship in their corresponding Java classes (Customer and Car)?

a. Use the @OneToMany annotation in Customer and @OneToOne in Car b. Use the @Many ToOne annotation in Customer and @OneToMany in Car c. Use the @Many ToOne annotation in both Car and Customer d. Use the @Many ToOne annotation in Car and @OneToMany in Customer	11. Which of the following is true for the generation strategy SEQUENCE for primary key generation? a. It uses an auto-incremented column to produce a sequence of primary keys b. It uses a database sequence generator to generate primary keys c. Multiple tables can use the same sequence generator d. It uses a shared table where sequential primary keys are stored
12. Consider the following code snippet. Which of the following gives the order given OutputTest s is executed? <pre> import org.junit.*; import static org.junit.Assert.*; import java.io.*; public class OutputTest{ private File output; @Before public void createOutputFile() { output= new File(...); } @After public void deleteOutputFile() { output.delete(); } @Test public void SomethingWithFile() { ... } } </pre> 1. createOutputFile() deleteOutputFile() testSomethingWithFile() 2.createOutputFile() testSomethingWithFile() delete OutputFile() 3.testSomethingWithFile() createOutputFile() deleteOutputFile() 4.deleteOutputFile() createOutputFile() testSomethingWithFile()	13. Can you have two methods mapped to the same path in your controller? a. Yes, the application will figure out which method to invoke based on the servlet context b. Yes, but each method has to respond to a different HTTP request c. Yes, but each method should have a different name d. No this is not possible, every method has to be mapped to a different path
14. . -----is a suite of plugins which supports implementing and integrating continuous delivery pipelines into Jenkins. a. Jenkins Pipeline b. Jenkins Job c. Jenkins build d. Jenkins procedure	

15. Which of the following aims at building, testing and releasing software faster and more frequently?
a. Continuous integration b. Continuous deployment c. Continuous delivery d. DevOps
16. What Java Spring Rest command would retrieve data from a restful API using RestTemplate ? An object called restTemplate has been created from the restTemplate class. (A) restTemplate.retrieveRest ("http://restwebsite", String.class, quote) (B) restTemplate.retrieveObject (String.class, "http://restwebsite") (C) restTemplate.getRestObject (String.class, "http://restwebsite") (D) restTemplate.getForObject ("http://restwebsite", String.class)
17. Assume you have a table in your database called Car which maps to a Car class in your Java application. There is a member variable in your class called "bhp" representing the maximum horsepower generated by the car's engine. Which represents a method signature in a CrudRepository which will return all cars whose engines can generate more than a specified horsepower, and ordered in the descending order of BHP? (A) List<Car> findByBhpGreaterThanBortByähpDesc (int horsepower); (B) List<Car> findByBhpGreaterThanOrderByBhp (int horsepower); (C) List<Car> findByBhpGreaterThan (int horsepower); (D) List<Car> findByBhpGreaterThanOrderByBhpDesc(int horsepower);
18. Which of the following are build automation tool? a. Gradle b. Maven c. Jenkins d. Docker
19. Which is the one annotation that you would apply to the class which contains the main() method for your Spring Boot app? a. @EnableAutoConfiguration b. @SpringBootApplication c. @ComponentScan d. SpringBootConfiguration
20. Assume you have configured your persistence.xml file with the property javax.persistence.schema- generation.database.action with a value of drop-and-create. You have also set properties for a create script, a load script and a drop script with the names of create.sql, drop.sql and load.sql. What is the order of execution of these scripts? a. create.sql, load.sql, drop.sql b. drop.sql, create.sql, load.sql c. load.sql, drop.sql, create.sql d. The order is randomly assigned
21. The DataCore team uses the Spring framework for its data access layer's implementations. Initially, the team had used Spring JDBC, later migrated to Spring ORM. As per the current requirements, now the client needs the data access layer to be implemented using Spring

Data JPA. Help the team identify, which among the following can be easily achieved using Spring Data JPA for the above implementation?

(multiple ans)

- a. Spring Data JPA provides common database operations support to an application, hence developers need not implement them.
- b. Spring Data JPA provides inconsistency in data access layer implementation by using its interfaces.
- c. Spring Data provides support for both SQL and NoSQL databases.
- d. Spring Data JPA reduces the number of artifacts in the data access layer as the only artifact required is the interface that extends JpaRepository

22.

What annotation should be used in a JUnit test class to define a method to initialize any variables or classes before running the set of tests?

- a. @Before
- b. @SpringBootTest
- c. @Test
- d. @Constructor
- e. @BeforeClass

23. The TechData core team works on a project based on S class called DataUserDAOimpi to define the EntityManager instance, this class implements the DataUserDAO interface. The specification mentions that the Entity Manager created should be maintained by the container and there should not be any developer's side efforts to manage the Entity Manager. Help the team identify which among the following can be a valid option to achieve this?

- a.

```
public DAOImpl implements DataUserDAO
{
    @PersistenceContext
    private EntityManager entityManager;

    //Defining required methods to perform database operations

}
```
- b.

```
public DataUserDAOImpl implements DataUserDAO
{
    @PersistenceUnit
    private EntityManager entityManager;

    //Defining required methods to perform database operations
}
```
- c.
- d.

24. #Component("user")

```
public class User
{
    @Value ("100")
    private int id;
```

```
//@value("John")
@value("#('john'.toUpperCase())")
private String FirstName;
@value("Doe")

private String lastName;
@Value("#{user.firerName.concat(' ').concate(user.lastName))")
private String fullName;
@Value("#{hobbies['1']}")
private String hobby;
@Value("#{400>300 || 150 < 100}") // true
private boolean flag;
// Getter Setter methods
}
Config.xml
<util: list list-class="java.util.ArrayList" id="hobbies">
<value>Singing</value>
<value>Dancing</value>
<value>Reading</value>
</util:list>
```

What will be when printed?

- a. id=100, fullName JOHN DOE, hobby-Dancing, flag=false
- b. id=100, fullName JOHN Doe, hobby-Singing, flag=true
- c. id=100, fullName= JOHN Doe, hobby Dancing, flag=true
- d. id=100, fullName JOHN Doe, hobby-Singing, flag=false

25. What annotation must be used when creating a Rest Template Application method using the RestTemplateBuilder to call REST services from other Spring Boot applications?

```
@BeanApplication
@Spring
@SpringBeanApplication
@Bean
```

26. Jenkins uses-----architecture to manage distributed builds.

```
Client-Server
MVC
Master-Slave
distributed
```

27. Please refer below

Consider a class "MyDataApplicant" properties given below.

```
public class MyDataApplicant{
private String applicantId;
private String applicant Name;
//gettere and setters
}
```

Assumption

-MyDataApplicant table exists -jdbcTemplate is database with instance of JdbcTemplate class propert

Which is a proper Spring JobeTemplate code from the options given below to fetch the total number of appli have submitted their requests to organization?

	(A)String sql "SELECT count(*) FROM MyDataApplicant"; int count=jdbcTemplate.queryForObject (sql, String.class); (B)String sql "SELECT count(*) FROM MyDataApplicant int count jdbcTemplate.queryForObject(sql); (C) String sql="SELECT count(*) FROM MyDataApplicant"; int count=jdbcTemplate.queryForObject(sql, Integer.class); (D) String sql = "SELECT * FROM MyDataApplicant"; int count=jdbcTemplate.queryForObject (sql, Integer.class);
28. Jenkins is a -----tool.	Continuous Integration Continuous Delivery Build Deployment Versioning
29. Questionid: 37038	Sweta is working on a Spring REST application. Sweta uses controller class annotatedwith @RestController that replaces the use of----- @Controller and @ResponseBody @Configuration and @ResponseBody. @Controller and @RequestMapping. @Controller and @Response
30. Which is not a phase of Maven?	validate verify run compile
31. FoxTravels maintains two classes Driving License and Citizen. They have one to one relationship between them. Assume that these two classes are annotated correctly with Cascade Type.ALL and there are no records currently available in the database.	What will be the output of the code given below? <pre>DrivingLicense dl = new DrivingLicense(); dl.setLicenceNumber(101); Calendar issueDate = Calendar.getInstance(); issueDate.set (12, 12, 2005); Calendar expiry = Calendar.getInstance(); expiry.set (12, 12, 2025); dl.setLicenceIssued(issueDate); dl.setLicenceExpiry(expiry); dl.setVehicleType("LMV") Citizen citzn = new Citizen(); citzn. setcustId (5001); citzn.setcustName("Ram"); citzn.setDrivingLicense (d1); session.getTransaction().begin(); session.persist (citzn); session.getTransaction().commit();</pre> a. One record will be inserted in Driving License Table and one record will be inserted in Citizen Table b. One record will be inserted in Driving License Table but no record will be inserted in Citizen Table c. No record will be inserted in Driving License Table but one record will be inserted in Citizen Table d. No record will be inserted in Driving License Table and no record will be inserted in Citizen Table

32.

In the absence of a Java config file, what is the name of the properties file in which a Spring Boot app looks for in the classpath for the application configurations?

- a. jpa.properties
- b. config.properties
- c. application.properties
- d. app.config

33. Please refer below

Consider the code given below:

```
public interface MusicianDAO {  
    //CRUD methods  
}  
  
public class MusicianDAOImpl extends MusicianDAO {  
    //Implements CRUD operations using Spring JdbcTemplate  
    -----  
}
```

Which among the following is a valid annotation to be used at the class level to translate checked exceptions to unchecked exceptions by eliminating the need for an explicit try-catch block in the repository code?

- a. @Component
- b. @Repository
- c. @ExceptionHandler(value = Exception.class)
- d. @Configuration

34. How can you select multiple fields using a Criteria API query?

- a. Using the criteriaQuery.selectAll() method
- b. Using the criteriaQuery.select() method
- c. Using the criteriaQuery.multiselect() method
- d. This is not possible using the criteria API

35. Which of the following is the correct sequence of Spring Bean life cycle?

- a. 1) Instantiation
2) Dependency Injection
3) Business Method Execution
4) Initialization
5) Destruction
- b. 1) Instantiation
2) Initialization
3) Dependency Injection
4) Business Method Execution
5) Destruction
- c. 1) Instantiation
2) Initialization
3) Business Method Execution
4) Dependency Injection
5) Destruction
- d. 1) Instantiation
2) Dependency Injection
3) Initialization
4) Business Method Execution
5) Destruction

36.

You retrieve the same set of fields over and over again from an underlying database table. How will you represent the field information within your query?

- a. Using a special Map<> class that JPQL provides
- b. Using a constructor for a custom object which holds those field values
- c. Using a List<Object> type
- d. Use Object[] arrays

37.

John works on a code implemented using @Transactional annotation. Help him identify, which among the following are correct usages of this annotation?

(A)

```
public interface DataStoreService {  
    @Transactional void updateDataStoreDetails (DataStore dataStore); void  
    insertDataStoreDetails (DataStore dataStore); }
```

(B)

```
@Transactional  
public class DataStoreServiceImpl {  
    public void updateDataStoreDetails (DataStore dataStore) { //code goes here  
    public void insertDataStoreDetails (DataStore dataStore) {  
        //code goes here  
    }  
}
```

(C)

```
public class DataStoreServiceImpl {  
    @Transactional  
    public void updateDataStoreDetails (DataStore dataStore) {  
        //code goes here  
    }  
}
```

(D)

```
public class DataStoreServiceImpl {  
    public void updateDataStoreDetails (DataStore datastore) {  
        @Transactional  
        //code goes here  
    }  
}
```

38.

Which correctly describes the relationship between dependency injection and inversion of control (IoC)?

- a. Dependency injection is one way to prevent IoC
- b. IoC and dependency injection are competing design patterns
- c. IoC is one way to prevent dependency injection
- d. Dependency injection is one way to implement IoC

39.

An application 'EasyRider' is added with the Employee Persistence module recently that works for persisting employee details. Some of the code snippets from the newly added module are given below for your observation.

```
//import statements  
@SpringBootApplication (scanBasePackages="com.mphasis.irs")  
public class IrsApplication{  
    public static void main(String[] args) throws Exception {  
        ApplicationContext context = SpringApplication.run(IrsApplication.class, args);
```



```

Employee Service employeeService = (EmployeeService)context.getBean
("employeeService");
System.out.println(employeeService.persistEmployee());
}
}
//import statements
public class EmployeeService{
public String persistEmployee () {
//call to employee repository is made
return "employee persisted successfully";
}
}

```

As you observe, NoSuchBeanDefinitionException will be thrown at run-time because of EmployeeService class. Identify the option that shall make the code work.

- a. Annotate Employee Service class either with @Component or @Service annotation
- b. Annotate Employee Service class with @Configuration annotation
- c. Annotate Employee Service class with @ComponentScan annotation
- d. Annotate Employee Service class with @ComponentScan annotation @Bean

40. Assume you have a class called Car and this has the following variables: id, make, model and price. You would like Spring to implement a method which returns all cars which have a price lower than a specified value. Which derived query method signatures will accomplish this?

- a. List<Car> findByPriceLess Than (double price);
- b. List<Car> find TopByPriceOrderByPriceAsc()(double price);
- c. Car findByPrice Lowest (double price);
- d. List<Car> findCarsCheaper Than (double price);

41.

In Spring MVC, which of the following are valid as per the use of @PathVariable annotation?(multiple ans)

- (A) @RequestMapping(value="/users/{userId}")
public String viewUser(@PathVariable("users") String user, Model m)
- (B) @RequestMapping(value="/users/{userId}/addresses")
public String viewUserAddress (@PathVariable String userId, @PathVariable String addressId, Model m)
- (C) @RequestMapping(value="/users/{userId}")
public String viewUser(@PathVariable("userId") String personnellId, Model m)
- (D) @RequestMapping(value="/users/{userId}")
public String viewUser(@PathVariable String userId, Model m)
- (E) @RequestMapping(value="/users/{userId}/addresses/{addressId}")
public String viewUserAddress (@PathVariable String userId, @PathVariable String addressId, Model m)

42.

Assume that we have two entity classes created for validating the Citizen's voter details. As per the rules, a citizen is supposed to have only one voter identity with him. Which among the following mappings are true with respect to One to One Mapping?

(A)

```
@Entity @Table (name "Citizen") public class Citizen
{
    @Id private int citizenId; private String citizenName;
    @OneToOne(cascade CascadeType.ALL) @JoinColumn (name="citizen VoterId")
    private Voter citizenVoter;
    //getter and setters
}
@Entity @Table (name = "Voter")
public class Voter
{ @Id private int citizenVoter Id;
  private String location;
  //getters and setters
}
```

(B)@Entity @Table (name "Citizen")

```
public class Citizen
{
    @Id private int citizenId;
    private String citizenName; @OneToOne (cascade CascadeType.ALL)
    private Voter citizenVoter;
    //getters and setters
}
@Entity @Table(name = "Voter")
public class Voter {
    @Id @JoinColumn(name = "citizenVoterId")
    private int citizenVoterId;
    private String location;
    //getters and setter
}
```

(C)

```
@Entity @Table (name "Citizen")
public class Citizen
{
    @Id private int citizenId;
    private String citizenName;
    @OneToOne(cascade = CascadeType.ALL)
    @JoinColumn (name="citizenVoterId", unique true)
    private Voter citizenVoter;
    //getters and setters
}
@Entity @Table(name = "Voter")
public class Voter
{
    @Id private int citizenVoter Id;
    private String location;
    //getters and setters
}
```

(D)

```
@Entity @Table(name = "Citizen")
public class Citizen
```

```

{ @Id private int citizen Id;
private String citizenName;
@OneToOne (cascade = CascadeType. ALL)
@JoinColumn (name="citizenVoterId", unique = true)
private Voter citizenVoter;
//getters and setters
}

@Entity @Table(name = "Voter")
public class Voter
{
@Id private int citizenVoter Id;
private String location;
//getters and setters
}

```

43.

Within your programming language you want to model composition i.e. one object is composed or nested within another. However, the fields of the nested object map to database table columns of the outer entity. What annotation will you use?

- a. @Flatten
- b. @Compose
- c. @Embedded
- d. @Nested
- e. @Composite Embedded

44.

. Please refer below

Consider the following Spring bean and configuration class:

```

package com.test.service;
public class BikeServiceImpl {
public BikeServiceImpl() {
System.out.println("Registration in progress...");
}
public void showStatus () {
System.out.println("Registered");
}
}

package com.test.configuration;
@Configuration
public class SpringConfig {
@Bean
public BikeServiceImpl myRegistration () {
return new BikeServiceImpl();
}
}

```

What is the output when the following code is executed?

```

package com.test;
public class Tester {
public static void main(String[] args) {
AbstractApplicationContext ctx = new
AnnotationConfigApplicationContext(SpringConfig.class);

```

```

BikeServiceImpl bikeService bikeService.showStatus(); (BikeServiceImpl)
ctx.getBean("myregistration");
ctx.close();
}
}

```

- a. Registration in progress... Registered
- b. Registered
- c. Registered Registration in progress...
- d. The code will throw RuntimeException exception

45.

For any build to run, the first step is to create a in Jenkins.

- a. job
- b. task
- c. pipeline
- d. process

46.

HighTechSolutions team uses Spring JDBC for performing database operations to the User entity. The User class is designed as below:

```

public class User{
private Integer userId;
private String userName;
private String userUnit;
}

```

The team needs to retrieve all the user records. Assume that the jdbcTemplate is an instance of JdbcTemplate object.

Which among the following Spring JDBC template code is a valid option to fetch the records?

(A)

```

public List<User> getUser() {
String sql = "select userId, userName, userUnit from User ";
return jdbcTemplate.query (sql, new RowMapper User >() { public User mapRow
(ResultSet rs, int rowNum)
throws SQLException {
User user = new User ();
//values fetched from rs and set to the User object
return user;
}
});
}

```

```

public List<User> getAll() {
String sql = "select userId, userName userUnit from User ";
return jdbcTemplate.queryForObject(sql, new RowMapper< User >() {
public User mapRow (ResultSet rs, int rowNum) throws SQLException { User user new
User (); //values fetched from rs and set to the User object return user;
});
}
}

```

(C)

```

public List<User> getAll() {

```

```
String sql = "select userId, userName return jdbcTemplate.query (sql, User.class);
userUnit from User ";
}
(D)
public List<User> getAll() {
String sql = "select u.userId, u.userName, u.userUnit from User u"; return
jdbcTemplate.executeQuery (sql, User.class);
}
(E)
public List<User> getAll() {
String sql = "select userId, userName, userUnit from User ";
return jdbcTemplate.query(sql, new RowMapper< User>() { public User mapRow
(ResultSet rs, int rowNum)
throws SQLException { User user = new User ();
//values fetched from rs and set to the User object
return user;
}
});
}
```

47.

Which is the one annotation that you would apply to the class which contains the main() method for your Spring Boot app?

- a. @EnableAutoConfiguration
- b. @SpringBootApplication
- c. @ComponentScan
- d. @SpringBootConfiguration

48.

In which phase DevOps really kicks in?

- a. Build
- b. Plan
- c. Code
- d. Release

49.

Given the below code:

```
1 public interface CustomerRepository extends JpaRepository<User, Long> (
2. // code goes here
```

```
3. Collection<User> findAllActiveUsers(); // complete this line
```

```
4. }
```

Which of the following options will allow users to get a list of all active users and sort them by name? Use following query to get the list of active users with JPA.

"SELECT * FROM USERS u WHERE u.status = 1"

(A)

```
@Query (value = "SELECT * FROM USERS u WHERE u.status = 1", nativeQuery = true)
Collection<User> findAllActiveUsers (Sort.by (Sort. Direction.ASC, "name"));
```

(B)

```
@nativeQuery(value = "SELECT * FROM USERS u WHERE u.status = 1",
nativeQuery=true)
```

```
Collection<User> findAllActiveUsers (Sort.by(Sort. Direction. ASC, "name")); true)
```

(C)

```
@Query (value = "SELECT * FROM USERS u WHERE u.status = 1", nativeQuery = true)
Collection<User> findAllActiveUsers().sort("name");
(D)
@Query (value = "SELECT * FROM USERS u WHERE u.status = 1")
Collection<User> findAllActiveUsers (Sort.by (Sort. Direction. ASC));
```

50.

Which of the following properties can be used to configure a column in the JPA @Column annotation?(multiple ans)

- a.columnLength
- b.column Type
- c.columnDefinition
- d.precision
- e.scale

51.

TechDataStore team uses Spring JDBC to perform database operations. The TechData User class is designed as shown below:

```
public class TechDataUser {
    private Integer techDataUserId;
    private String techDataUserName;
    private String techDataUserDept;
}
```

The team needs to delete a TechDataUser record based on the techDataUserId. that the jdbcTemplate is an instance of a JdbcTemplate object.

Assume Which among the following Spring JDBC Template code is a valid option to delete the record?

```
(A) public int delete(int techDataUserId) {
    return jdbcTemplate.remove("delete from TechDataUser where techDataUserId=?",
    techDataUserId);
}
```

```
(B)
public int delete(int techDataUserId) {
    return jdbcTemplate.update("delete from TechDataUser where techDataUserId=?",
    techDataUserId);
}
```

```
(C)
public int delete(int techDataUserId) {
    return jdbcTemplate.delete("delete from TechDataUser where techDataUserId=?",
    techDataUserId);
}
```

```
(D)
public int delete(int techDataUserId) {
    return jdbcTemplate.query("delete from TechDataUser where techDataUserId=?",
    techDataUserId);
}
```

52.

. What annotation must be used when creating a Rest Template Application method using the RestTemplateBuilder to call REST services from other Spring Boot applications?

- a. @BeanApplication

	b. @Spring c. @SpringBeanApplication d. @Bean
53.	Which of the following are advantages of modelling relationships using embedded objects?(multiple ans) a. The same embeddable can be shared across multiple entities b. JPA providers are more performant with embedded objects c. Easier to use embeddables rather than individual fields d. Can represent relationships using composition in the object-oriented world
54.	Which of the following are the advantages of Jenkins? a. Cost Savings b. Plugin EcoSystem c. Increased Productivity d. Increased Development Time
55.	Assume that a Library Repository must be created for the given Library class. Which of the following is the correct way of coding? (Assume all the import statements are provided correctly) <pre>@Entity public class Library{ @Id private String libraryId; private String libraryName; }</pre> (A)public interface LibraryRepository extends JpaRepository< Library, Integer> { } (B)public interface Library Repository extends JpaRepository< Library, String> { } (C)public interface LibraryRepository extends JpaRepository<String> { (D)public interface LibraryRepository extends JpaRepository<Library> { }
56.	Assume you have a table in your database called Car which maps to a Car class in your Java application. There is a member variable in your class called "make" representing the brand/manufacturer of a car. You have also set up a CrudRepository for the Car entity. Which represents a method in the CrudRepository which can be used to retrieve all Car instances of a specific make? (A)List<Car> selectWhereMake(String make); (B)List<Car> findByMake(String make); (C)List<Car> findById(String make); (D)List<Car> selectFromCarWhereMake(String make);
57.	Which one of the following allows a single team to handle the entire application lifecycle from development to testing, deployment, and operations? a. Jenkins b. DevOps c. Git d. Maven
58.	Which techniques in Spring will achieve inversion of control? Model-view-controller paradigm(multiple ans) a. Factory pattern

	b. Builder pattern c. Dependency injection d. Server locator pattern
59.	You have to create a request mapping for a messaging service that gets data from the user via the POST method. Which annotation should go with the function "public void addMsg(@RequestBody Msg message){ ... }"? (A) @RequestMapping (value="/message", method=RequestMethod. POST) (B) @RequestMapping (value="/message", request=RequestMethod.POST) (C) @RequestMapping (value="/message", request=FormVar.POST) (D) @RequestMapping (value="/message", method="POST")
60.	Consider the GreetingService class given below: <pre>package com.test.service; public class GreetingService { private String greeting; public String getGreeting() { return greeting; } public void setGreeting(String greeting) { this.greeting greeting; } }</pre> A Spring configuration file(config.xml) defining bean of the GreetingService class is shown below: <pre><bean id= "greetingService" class="com.test.service.GreetingService" /></pre> The main method of the application includes the following statements: <pre>ApplicationContext context new ClassPathXmlApplicationContext("config.xml"); GreetingService greetingService (GreetingService) context.getBean("greetingService"); greetingService.setGreeting("Hi, Welcome to Spring Course!!"); System.out.println(greetingService.getGreeting());</pre> What will happen when we execute the above main method? a. Displays the greeting message "Hi, Welcome to Spring Course!!" b. Displays null as the greeting property is not initialized c. POJO class setter method cannot be invoked on the bean d. Compilation Error
61.	A _____ is a runnable task that Jenkins controls to achieve a required objective. a.plugin b.job c.stage d.pipeline
62.	Consider a class "MyDataApplicant" with the properties given below: <pre>public class MyDataApplicant (//Id field private String applicantId; private String applicantName; //getters and setters</pre>

}

Assumptions:

-MyDataApplicant table exists in the database with the above properties

-jdbcTemplate is an instance of JdbcTemplate class

Which is a proper Spring JdbcTemplate code from the options given below to fetch the total number of applicants who have submitted their requests to the organization?

(A)

String sql= "SELECT count(*) FROM MyDataApplicant "; int count
jdbcTemplate.queryForObject(sql, String.class);

(B)

String sql ="SELECT count(*) FROM MyDataApplicant "; int count =
jdbcTemplate.queryForObject (sql);

(C)

String sql = "SELECT count(*) FROM MyDataApplicant "; int count =
jdbcTemplate.queryForObject(sql, Integer.class);

(D)

String sql = "SELECT FROM MyDataApplicant "; int count
jdbcTemplate.queryForObject (sql, Integer.class);

63.

Which of the following are DevOps tool?(multiple ans)

a. Git

b. Jenkins

c. Maven

d. Eclipse

64.

Which is not a phase of Maven?

a. validate

b. verify

c. run

d. compile

65.

Which of the following is NOT a reason to use an Object Relational Mapping framework when working with relational databases using a high-level programming language such as Java?

a. Relational tables express relationships using constraints and programming languages use design patterns

b. Relational tables consist of rows and columns and objects consist of fields

c. There is no way to query data from a database without using a language such as Java

d. The fundamental unit of programming for relational databases and Java is different

66.

For an Entity class with a composite key, we can replace the use of the @EmbeddedId annotation as long as we apply a specific annotation to the class pointing it to the composite key class. Which represents that annotation?

a. @IdClass

b. @Embed

c. @Composite

d. @Reference

67.

Refer the following code for a Java class.

```
1. package com.mphasis;  
2. import javax.persistence. Entity;  
3. import javax.persistence. GeneratedValue;  
4. import javax.persistence. GenerationType;  
5. import javax.persistence.Id;  
6. // code goes here  
7. public class Customer {  
8. // code goes here  
9. // code goes here  
10. private Long id;  
11. private String firstName; private String lastName;  
12.  
}
```

// Setters and Getters

Which of the following lines of code when added to Lines 6, 8, and 9 will enable mapping of the Java class with the database using JPA where the id field is managed by JPA during generation of records?

(A)

@Table

@Id

@GeneratedValue (strategy=GenerationType.AUTO)

(B)

@Entity

@Id

@GeneratedValue(strategy=GenerationType.AUTO)

(C)

@Table

@Identity

@GeneratedValue(strategy=GenerationType.AUTO)

(D)

@Table

@Id

@AutoGeneratedValue (strategy=GenerationType.AUTO)

68. What dependency artifact Id must be added to see the secure actuators in spring boot?

spring-boot-starter-security

spring-boot-dev-tools

spring-boot-start-web-services

spring-boot-secure-actuators

69.

Which of the following can be used to specify the scope of a bean?

1. In a XML based configuration file:

```
<bean id="a1" class="ConstructorInjection.TaxiService" scope="prototype"/>
```

2. In a Java based configuration file:

```
@Bean(name="a3")
```

```
@Scope("prototype")
```

```
public TaxiService getTaxi(){}
```

3. In the POJO class:

```
@Component("a2")
@Scope("prototype")
public class TaxiService{}
```

- a. Only 1
- b. 1,2
- c. 1,2,3
- d. 2,3

70.

Please refer below

HaleTech team uses a Spring-based application to maintain their user details. The team has implemented a backend using the CRUDRepository of Spring Data commons. The HaleUser class is an entity class that is mapped to the table in the relational schema. The class is designed as shown below:

```
@Entity
public class HaleUser{
    @Id
    private Integer haleUserId:
    private String haleUserName:
    //getters and and setters
```

Help the identify which among following are valid methods supported by Spring?

- a. findAll()
- b. findById(Integer haleUserId)
- c. save(Integer haleUserId)
- d. deleteById(Integer haleUserId)

71.

Given the below code:

RestController

```
@RequestMapping("/friendsfamily")
```

```
public class Friends FamilyController
```

```
{
```

```
@GetMapping("/{phoneNumber}")
```

```
public Friends FamilyDTO getFriendsFamily()
```

```
{
```

```
//code to search for Friends Family based on the phone number and return the same
```

```
}
```

```
}
```

Choose the correct line of code that should occupy the parameter section of the request method.

```
@PathParam("phoneNumber")long phoneNumber
```

```
@RequestParam("phoneNumber")long phoneNumber
```

```
@PathVariable("phoneNumber")long phone Number
```

```
@RequestBody("phoneNumber")long phoneNumber
```

72.

Select the correct statements about JPA and Hibernate

- a. JPA is an implementation of Hibernate
- b. Hibernate is a specification for persisting data
- c. JPA is a specification for persisting data
- d. Hibernate is an implementation of JPA

73.

Which scope in Spring ensures that object exists between every request in an HTTP session?

(a) Singleton

(b) Prototype

(c) Request

(d) Session

74. Which of the following is a repository hosting service tool that features collaboration and access control?

- a. Bitbucket
- b. Git
- c. Github
- d. docker hub

75. Given the below code. (Assume necessary imports are done.)

```
@RestController
@RequestMapping("/friendsfamily")
public class FriendsFamilyController
{
    @PostMapping (produces="application/json", consumes="application/json")
    public FriendsFamilyDTO addFriendsFamily(@RequestBody FriendsFamilyDTO friendsFamilyDTO)
    {
        //code to add friends family and return the same back
    }
}
```

Identify the things that optional from the following.

- a. @RequestBody
- b. produces="application/json"
- c. consumes="application/json"

options

Only a

Only b and c

Only a and c

a,b,c

76. What Spring Starter Project options are configurable from the New Spring Start Project popup?

- a. Port
- b. Name
- c. Service URL
- d. Java Version
- e. Package
- f. Annotation Class

77. Assume you have a class called Car and this has the following variables: id, make, model and price. You would like to search for cars matching a particular model value, but you are not sure about the case used in the model, and you want the search to be case-insensitive. Which derived query methods can you configure in your CrudRepository to serve this purpose?

- a. findByModelIgnoreCase(String model)
- b. findByMakeAndModel(String make, String model)
- c. findByModel(String model)
- d. findByModelLike(String make)

78. Select the method signatures in a CrudRepository which are equivalent

```
Car findTopOrderByPrice()
Car findFirstOrderByPrice()
Car findFirstOrderByPriceDesc()
```

Car findBySortByPriceLimit1()

79. Given the following configuration class, Choose the correct statements.

```
public class ApplicationConfig ( private DataSource dataSource;
    @Autowired

    public ApplicationConfig(DataSource dataSource) { this.dataSource=dataSource;

    @Bean(name="clientRepository")

    ClientRepository jpaClientRepository() { return new JpaClientRepository();
    }}
```

- a. @Configuration annotation is missing
- b. @Bean scope is prototype
- c. @Bean name is ambiguous
- d. Default or no-arg constructor is missing

80. What is the output of the following code?

```
public class TutorialExample {

    Test (groups = "group")

    public void testMethod1ForGroupA() {

        System.out.println("Running test method of group");

        @Test(groups = "groupA")

        public void testMethod2ForGroupA() {

            System.out.println("Running test method of group");

            Assert.fail();

            @Test(dependsOnGroups = "group")

            public void dependentTestOnGroupA() {

                System.out.println("Running the dependent test");

            }

        }

    }

}
```

- A.
- testMethod1ForGroupA-Pass
 - testMethod2ForGroupA-Fail
 - dependent TestOnGroupA-Pass

- B.
- testMethod1ForGroupA-Pass
 - testMethod2ForGroupA-Fail dependent

TestOnGroupA -Fail

C

testMethod1ForGroupA-Pass
testMethod2ForGroupA-Fail
dependent TestOnGroupA -Skipped

D.

testMethod1ForGroupA-Pass
testMethod2ForGroupA-Pass
dependent TestOnGroupA-Pass

Q#1. Which annotation is not used to inject dependencies into a Spring bean?

(a) @Inject

(b) @Bean

(c) @Autowired

(d) @Resource

What is Spring Core?

(a) A Java framework for creating user interfaces

(b) An IDE for developing Spring based applications

(c) A sub-module of Spring framework

(d) A scripting language of the Spring Framework

Q#3. Which of the following annotations is NOT used to specify that a class is a Spring bean?

(a) @Service

(b) @Qualifier

(c) @Component

(d) @Controller

Q#4. Which of the following is INCORRECT about dependency injection in Spring?

(a) It helps in achieving loose coupling between objects.

(b) It helps in testing the module by injecting the dependent Mock Objects.

(c) It increases the scope of code reusability.

(d) It helps in managing database connections.

.

Q#5. Consider the below @Value annotation, which is written using SpEL (Spring Expression Language):

```
@Value("#{myBean.myProperty != null ? myBean.myProperty : 'default'}")
```

Which is the shorter representation of the above annotation using Elvis operator?

- (a) @Value("#{myBean.myProperty ?: null}")
- (b) @Value("#{myBean.myProperty ? myBean.myProperty : 'default'}")
- (c) @Value("#{myBean.myProperty ? null}")
- (d) @Value("#{myBean.myProperty ?: 'default'}")

Q#6. In Spring's XML configuration file, which XML tag is used to define a bean?

- (a) <component>
- (b) <service>
- (c) <bean>
- (d) <dependency>

Q#7. What is the purpose of the @Autowired annotation in Spring?

- (a) It marks a bean for automatic injection
- (b) It configures the transaction management
- (c) It specifies the automatic ordering of bean initialization
- (d) It creates an object of a Spring bean

Q#8. Which annotation is used to enable Spring's aspect-oriented programming (AOP) support?

- (a) @Aspect
- (b) @AspectJ
- (c) @EnableAop
- (d) @EnableAspectJAutoProxy

Q#9. What is the purpose of the @Qualifier annotation in Spring?

- (a) It specifies the order of bean initialization among multiple beans

(b) It marks a bean for automatic injection

(c) It resolves ambiguity when multiple beans of the same type are present

(d) It marks a class to become a Spring bean

Q#10. Which annotation is used to declare a class as a Spring bean in a Java based configuration class?

(a) @Service

(b) @Bean

(c) @Component

(d) @Autowired

Q#11. Which of the following tag is required to enable component scanning in Spring XML configuration?

(a) <context:scan package="com.example" />

(b) <scan-components base-package="com.example" />

(c) <component-scan base-package="com.example" />

(d) <scan: component base-package="com.example" />

Q#12. Which is the incorrect purpose of the @Value annotation in Spring?

(a) It is generally used for injecting values into configuration variables

(b) It supports Spring Expression Language (SpEL)

(c) It is used to assign default values to variables and method arguments

(d) It is used to inject dependencies as part of autowiring

Q#13. Which annotation is used to enable transaction management in Spring?

(a) @Transactional

(b) @EnableTransactionManagement

(c) @ConfigureTransaction

(d) @EnableTransactions

Q#14. How can you retrieve a bean from the Spring ApplicationContext using Java code?

(a) Using the `ApplicationContext.getBean()` method

(b) Using the `@Bean` annotation

(c) Using the `@Autowired` annotation

(d) Using the `@Resource` annotation

Q#15. Which scope in Spring ensures that object exists between every request in an HTTP session?

(a) Singleton

(b) Prototype

(c) Request

(d) Session

Q#16. How can you define property placeholders in Spring XML configuration?

(a) Using the `<context:property-placeholder>` tag

(b) Using the `<context:value>` tag

(c) Using the `<context:property>` tag

(d) Using the `<context:placeholder>` tag

Q#17. In order to enable the Spring expression language (SpEL) in Spring beans, Which annotation is used?

(a) `@ExpressionLanguage`

(b) `@EnableSpEL`

(c) `@SpEL`

(d) `@Value`

Q#18. If you want to configure a bean for lazy initialization in Spring, how can you configure a bean to be initialized lazily?

(a) Using the `@Lazy` annotation

(b) Setting the "lazy-init" attribute to "true" in XML configuration

(c) Using the `@PostConstruct` annotation

(d) Setting the "scope" attribute to "lazy" in XML configuration

Q#19. To be eligible for Spring's event handling mechanism, which interface should a class implement?

(a) `EventListener`

(b) `ApplicationListener`

(c) `EventHandler`

(d) `EventSubscriber`

Q#20. How can you register an event listener in Spring XML configuration?

(a) Using the `<listener>` tag

(b) Using the `<event-listener>` tag

(c) Using the `<application-listener>` tag

(d) Using the `<bean>` tag with the `ApplicationListener` interface

Q#21. How can you enable asynchronous method execution in Spring?

(a) Using the `@Async` annotation

(b) Using the `<async-method-execution>` tag

(c) Using the `@EnableAsync` annotation

(d) Setting the "async" attribute to "true" in XML configuration

.

Q#22. What is the purpose of the `@Qualifier` annotation in Spring?

(a) It specifies the order of bean initialization

(b) It marks a bean for automatic injection

(c) It resolves ambiguity when multiple beans of the same type are present

(d) It creates an object for the annotated bean

Q#23. How can you enable Spring's support for method-level caching?

(a) Using the `@Cacheable` annotation

(b) Using the `<cacheable>` tag

(c) Using the `@EnableCaching` annotation

(d) Setting the "cacheable" attribute to "true" in XML configuration

Q#24. How can you define the order of bean initialization in Spring?

(a) Using the `<order>` tag

(b) Using the @Order annotation

(c) Using the @Priority annotation

(d) Using the <priority> tag

Q#25. Which is the correct option to enable Spring MVC in a Spring based application?

(a) Using the @EnableMvc annotation

(b) Using the <enable-mvc> tag

(c) Using the @EnableWebMvc annotation

(d) Using the @EnableSpringMvc annotation

Q#26. Which annotation is used to enable scheduling tasks in Spring?

(a) @Scheduled

(b) @EnableScheduling

(c) @TaskScheduler

(d) @Scheduling

Q#27. If you want to schedule a method to be executed at fixed intervals in Spring, Which annotation will you use?

(a) @FixedRate

(b) @FixedDelay

(c) @Cron

(d) @FixedInterval

Q#28. To access a bean property, which SpEL expression is correct?

(a) #{beanName.propertyName}

(b) \${beanName.propertyName}

(c) @beanName.propertyName

(d) %beanName.propertyName%

Q#29. Which annotation is used to mark a method as an exception handling method in Spring MVC?

(a) **@ExceptionHandler**

(b) @ExceptionHandler

(c) @HandleException

(d) @ExceptionHandlerResolver

Q#30. If you want to write global exception handling code that can be applied to multiple controllers in Spring MVC, which annotation will you use?

(a) @GlobalExceptionHandler

(b) @GlobalExceptionHandler

(c) **@ControllerAdvice**

(d) @EnableGlobalExceptionHandler

Q#31. In Spring JDBC, which method will you use to execute a SQL query that can map the results to a Java object?

(a) executeQueryForObject()

(b) fetchObject()

(c) executeForObject()

(d) **queryForObject()**

Q#32. What is the role of the DispatcherServlet in Spring MVC?

(a) It configures the Spring MVC dependencies

(b) It manages the application's database connections

(c) **It handles incoming requests and delegates them to controllers**

(d) It handles security authentication and authorization.

Q#33. Which of the following is helpful in passing data from a controller method to a view in Spring MVC?

(a) HttpRequest object

(b) HttpServletResponse object

(c) HttpSession object

(d) **ModelAttributes to the Model object**

Q#34. How can you enable cross-origin resource sharing (CORS) in Spring MVC?

- (a) Using the `@EnableCors` annotation
- (b) Using the `<enable-cors>` tag
- (c) Using the `@CrossOrigin` annotation
- (d) Using the `@EnableCrossOrigin` annotation

Q#35. Which of the following is used to hold the model data and view information in Spring MVC?

- (a) `ViewResolver`
- (b) `Model`
- (c) `ModelAttribute`
- (d) `ModelAndView`

Q#36. Which of the following is an incorrect use of Http GET request in a Spring MVC application?

- (a) `@GetMapping(value="/", produces = MediaType.TEXT_PLAIN_VALUE)`
- (b) `@RequestMapping(value = "/emp/{id}", method = GET)`
- (c) `@GetMapping(consumes = {"text/plain", "application/*"})`
- (d) `@GetMapping("/")`

Q#37. In Spring Security, which annotation is not enabled by `@EnableMethodSecurity` by default?

- (a) `@PreAuthorize`
- (b) `@PostAuthorize`
- (c) `@PreFilter`
- (d) `@Secured`

Q#38. Which advice type is executed regardless of the method execution outcome, including exceptions in Spring AOP?

- (a) Before advice
- (b) After advice
- (c) Around advice
- (d) After advice

Q#39. Which pointcut expression is correct for the below statement?
“Any method with zero or more parameters inside EmpDao class”

- (a) **public** * com.dev.dao.EmpDao.*()
- (b) **public** * com.dev.dao.*.*(..)
- (c) **public** * com.dev.dao.EmpDao.*(..)
- (d) **public** Integer com.dev.dao.EmpDao.*()

Q#40. In the context of Spring AOP, select the method which is the candidate for below Pointcut Expression?

public * *(Invoice)

- (a) **public** Integer saveInvoice(Integer id){....}
- (b) **public void** deleteInvoice(Integer id){....}
- (c) **public void** updateInvoice(Invoice inv){....}
- (d) **public** Invoice getInvoice(Integer id){....}

Q#41. Which one is the another way of writing below annotation in Spring Security?
@Secured(“ROLE_MANAGER”, “ROLE_ADMIN”)

- (a) @PreAuthorize(“ROLE_MANAGER”, “ROLE_ADMIN”)
- (b) @PreAuthorize(“hasRole(‘ROLE_MANAGER’) or hasRole(‘ROLE_ADMIN’)”)
- (c) @PreAuthorize(“hasRole(‘ROLE_MANAGER’) | hasRole(‘ROLE_ADMIN’)”)
- (d) @PreAuthorize(“role(‘ROLE_MANAGER’) or role(‘ROLE_ADMIN’)”)

Q#42. Which is a correct cron expression in Spring Scheduling that executes a task every year on Feb 14th 9:00:00 PM, if given day(14th) is Sunday or Tuesday only?

- (a) 0 0 9 2 14 1,3
- (b) 0 0 9 2 14 SUN,TUE
- (c) 0 0 21 14 2 SUN,TUE
- (d) 0 0 9 14 2 1,3

Q#43. As part of the cron expressions improvement in Spring 5.3, the below cron expression can also be written as:

0 0 0 * * 0

- (a) @monthly

(b) @weekly

(c) @yearly

(d) @midnight

Q#44. Which is a correct cron expression in Spring Scheduling that executes a task every year to wish happy birthday on 24th March.

(a) 0 0 0 3 24 *

(b) 0 0 24 3 0 *

(c) 0 0 0 24 3 *

(d) 0 0 0 24 3 0

Q#45. In Spring JDBC, what does the NamedParameterJdbcTemplate class offer?

(a) Support for executing SQL statements with positional parameters

(b) Support for executing SQL statements with named parameters

(c) Support for executing stored procedures

(d) Support for executing batch updates

Core Spring 4.2 Certification Mock Exam

Question

Container

Question 1

Given the following Spring configuration file, what is the correct answer:

```
<bean class="com.spring.service.MyServiceImpl">
    <property name="repository" ref="jpaDao"/>
</bean>

<bean class="com.spring.repository.JpaDao"/>
```

1. The first declared bean MyServiceImpl is missing an id must be named myService
2. The second declared bean JpaDao is missing an id must be named jpaDao
3. Answers 1 and 2 are both rights
4. Answers 1 and 2 are both wrong

Question 2

Given the Spring configuration file, which are the correct statements?

```
<bean class="com.spring.service.BankServiceImpl"
      p:bankName="NationalBank">
</bean>
```

1. The p namespace has to be declared
2. Bean id is bankServiceImpl
3. The BankServiceImpl references a NationalBank bean
4. NationalBank is a scalar value

Question 3

What the name of the bean defined in the following configuration class? Select a single answer.

```
@Configuration
public class ApplicationConfig {

    @Autowired
    private DataSource dataSource;

    @Bean
    ClientRepository clientRepository() {
        ClientRepository accountRepository = new JpaClientRepository();
        accountRepository.setDataSource(dataSource);
        return accountRepository;
    }
}
```



```
}  
}
```

1. JpaClientRepository
2. jpaClientRepository
3. clientRepository
4. Two beans are defined: a data source and a repository

Question 4

How could you externalize constants from a Spring configuration file or a Spring annotation into a `.properties` file? Select one or more answers

1. By using the `<util:constant />` tag
2. By declaring the `ConstantPlaceholderConfigurer` bean post processor
3. By using the `<context:property-placeholder />` tag
4. By using the `c:` namespace

Question 5

What statement is not correct in live environment? Select a unique answer.

1. Constructor and properties autowiring in the same bean are not compatible
2. A bean should have a default or a no-args constructor
3. The `<constructor-arg>` tag could take type, name and index to reduce ambiguity
4. None of the above
5. All of the above

Question 6

What are the right affirmations about the `@PostConstruct`, `@Resource` and the `@PreDestroy` annotations?

1. Those annotations are specified in the JSR-250
2. The Spring Framework embedded those annotations
3. The `<context:component-scan>` tag enable them
4. The `<context:annotation-config>` tag enable them
5. Declaring the `CommonAnnotationBeanPostProcessor` enable them

Question 7

What is/are typically case(s) where you usually need to manually instantiate an `ApplicationContext`?

1. In a web application
2. In an integration test running with the `SpringJUnit4ClassRunner`
3. In a standalone application started with a main method
4. None of the above

Question 8

Select the right statement about referring a Spring configuration file inside the package com.example.myapp in the below example?

```
ApplicationContext context = new
```

```
ClassPathXmlApplicationContext("classpath:/com.example.myapp.config.xml");
```

1. The classpath: prefix could be omitted
2. Package name using the dot character is not well formatted
3. The slash character preceding com.example could be omit
4. All of the above
5. None of the above

Question 9

How to auto-inject into a field a Spring bean by its name? Select one or more answer choices.

1. With the name attribute of the @Autowired annotation
2. By using the single @Qualifier annotation
3. By using both the @Autowired and the @Qualifier Spring annotations
4. By using the @Autowired annotation and naming the field with the bean name

Question 10

What are the main advantages of using interfaces when designing business services? Select one or more answer choices.

1. Mocking or stubbing the service
2. Be able to use the Spring auto-injection
3. Can do dependency checking
4. Loosely coupled code

Question 11

Select one or many correct answers about Spring bean life cycle.

1. The method annotated with @PostConstruct is called after bean instantiation and before properties setting of the bean
2. The method @PreDestroy of a prototype bean is called when the bean is garbage collected
3. The init() method declared in the init-method attribute of a bean is called before the afterPropertiesSet callback method of the InitializingBean interface
4. The method annotated with @PostConstruct is called before the afterPropertiesSet callback method of the InitializingBean interface

Question 12

Given the following configuration class, what are the correct affirmations? Select one or more answers.

```
public class ApplicationConfig {  
  
    private DataSource dataSource;  
  
    @Autowired  
    public ApplicationConfig(DataSource dataSource) {  
        this.dataSource = dataSource;  
    }  
  
    @Bean(name="clientRepository")  
    ClientRepository jpaClientRepository() {  
        return new JpaClientRepository();  
    }  
}
```

1. @Configuration annotation is missing
2. Default or no-arg constructor is missing
3. @Bean name is ambiguous
4. @Bean scope is prototype

Question 13

What are the features of the XML <context:namespace>? Select one or many answers.

1. @Transactional annotation scanning
2. @Aspect annotation detection enabling
3. @Autowired annotation enabling
4. @Component annotation scanning

Test

Question 14

Select one or more correct statements about developing integration test with Spring support.

1. A new Spring context is created for each test class
2. To get a reference on the bean you want to test, you have to call the getBean() method of the Spring context
3. Spring context configuration could be inherited from the super class
4. The Spring context configuration file has to be provided to the @ContextConfiguration annotation

Question 15

What are the main advantage(s) for using Spring when writing integration tests?

1. Reuse Spring configuration files of the application
2. Create mock or stub

3. Be able to use the rollback after the test pattern
4. Use dependency injection

Question 16

What are the main advantage(s) for using Spring when writing unit tests?

1. Reuse Spring configuration files of the application
2. Use dependency injection
3. Provide some mocks for servlet classes
4. All of the above
5. None of the above

Question 17

What is right about the Spring test module?

1. It provides an abstraction layer for the main open source mock frameworks
2. Provides the @Mock annotation
3. It dynamically generates mock objects
4. All of the above
5. None of the above

Question 18

Select correct statement(s) about transactional support of the Spring test module.

1. Transaction manager could be set within the @TransactionConfiguration annotation
2. Method annotated with @Before is executed outside of the test's transaction
3. Spring test may rollback the transaction of a service configured with the REQUIRES_NEW propagation
4. The transaction of a method annotated with the @Rollback annotation with its default values is rolled back after the method has completed

AOP

Question 19

Considering 2 classes AccountServiceImpl and ClientServiceImpl. Any of these 2 classes inherits from each other. What is the result of the following pointcut expression?

```
execution(* *..AccountServiceImpl.update(..))  
&& execution(* *..ClientServiceImpl.update(..))
```

1. Matches public update methods of the 2 classes, whatever the arguments
2. Matches any update methods of the 2 classes, whatever the arguments and method visibility
3. Matches any update methods of the 2 classes, with one more arguments and whatever method visibility
4. No joint point is defined

Question 20

Using the Spring AOP framework, what is the visibility of the method matches by the following join point?

```
@Pointcut("execution(* *(..))")  
private void anyOperation() {};
```

1. All methods, whereas their visibility
2. All methods, except private method
3. Protected and public methods
4. Public methods

Question 21

What are the 2 correct statements about AOP proxy?

1. AOP proxies are created by Spring in order to implement the aspect contracts
2. AOP proxies are always created with a JDK dynamic proxy
3. Only classes that implements a least one interface could be proxied
4. All methods could be proxied
5. Proxies are created by a BeanPostProcessor

Question 22

What is an after throwing advice? Select a unique answer.

1. Advice that could throw an exception
2. Advice to be executed if a method exits by throwing an exception
3. Advice that executes before a join point
4. Spring does not provide this type of advice

Question 23

What is an after returning advice? Select a unique answer.

1. Advice to be executed regardless of the means by which a join point exits
2. Advice that surrounds a method invocation and can perform custom behavior before and after the method invocation
3. Advice to be executed before method invocation
4. Advice to be executed after a join point completes without throwing an exception

Question 24

What is an advice? Select a unique answer.

1. An action taken by an aspect at a particular join point
2. A point during the execution of a program
3. An aspect and a pointcut
4. A predicate that matches join points

Question 25

What is a pointcut? Select the single answer.

1. Code to execute at a join point
2. An expression to identify joinpoints
3. An advice and a jointpoint
4. None of the above

Question 26

Select method's signatures that match with the following pointcut:

`execution(* com.test.service..*.*(*))`

1. `void com.test.service.MyServiceImpl#transfert(Money amount)`
2. `void com.test.service.MyServiceImpl#transfert(Account account, Money amount)`
3. `void com.test.service.account.MyServiceImpl#transfert(Money amount)`
4. `void com.test.service.account.MyServiceImpl#transfert(Account account, Money amount)`
5. None of the above

Question 27

What are the unique right answer about Spring AOP support?

1. An advice could proxied a constructor's class
2. A pointcut could select methods that have a custom annotation
3. Static initialization code could be targeted by a point cut
4. Combination of pointcuts by &&, || and the ! operators is not supported

Question 28

Using the Spring AOP framework, what are the joinpoint methods of the following pointcut expressions?

`execution(public * *.*(..))`

1. The execution of all public method
2. The execution of all public method returning a value

3. The execution of all public method having at least one parameter
4. The execution of all public method in class belonging to the default java package

Data Access

Question 29

Why is it a best practice to mark transaction as read-only when code does not write anything to the database? Select one or more answers.

1. It is mandatory for using Spring exception translation mechanism
2. May be improve performance when using Hibernate
3. Spring optimizes its transaction interceptor
4. Provides safeguards with Oracle and some other databases

Question 30

What data access technology is supported by the Spring framework? Select one or more answers.

1. JDBC
2. NoSQL
3. Hibernate
4. JPA

Question 31

What is not provided by the JdbcTemplate? Select a unique answer.

1. Data source access
2. Open/close data source connection
3. JDBC exception wrapping into DataAccess Exception
4. JDBC statement execution

Question 32

Using JdbcTemplate, what is the Spring provided class you will use for result set parsing and merging rows into a single object? Select a unique answer.

1. RowMapper
2. RowCallbackHandler
3. ResultSetExtractor
4. ResultSetMapper

Question 33

What configuration is supported by the LocalSessionFactoryBean which supports Hibernate 4 or higher? Select a unique answer.

1. Listing entity classes annotated with @Entity

2. Scanning a package to detect annotated entity classes (with @Entity)
3. Listing hibernate XML mapping configuration file (.hbm.xml)
4. All above

Transaction

Question 34

What is/are incorrect statements about XML declaration of the transaction manager bean? Select one or more answers.

1. The tx namespace provides JTA transaction manager declaration shortcut syntax
2. Id of the bean has to be *transactionManager*
3. Depending the application persistence technology, the HibernateTransactionManager or the DataSourceTransactionManager could be used as bean class
4. Default transaction timeout could be given

Question 35

Assuming @Transactional annotation support is enabled and the transferMoney method is called through a Spring AOP proxy, what is the behavior of the following code sample?

```
@Transactional(propagation=Propagation.REQUIRED)
public void transferMoney(Account src, Account target, double amount) {
    add(src, -amount);
    add(src, amount);
}

@Transactional(propagation=Propagation.REQUIRES_NEW)
public void add(Account account, Double amount) {
    // IMPLEMENTATION
}
```

1. The add() method executes code in a new transaction
2. The add() method uses the transaction of the transferMoney() method
3. When calling the add() method, an exception is thrown
4. Other behavior

Question 36

Does Spring provide programmatic transaction management? Select a unique answer.

1. Yes with the TransactionTemplate class
2. Yes with the TransactionService class
3. Yes using the @Transactional bean post processor
4. No

Question 37

What is the transaction behavior of the PROPAGATION_REQUIRES_NEW mode? Select a unique answer.

1. If a transaction exists, the current method should run within this transaction. Otherwise, it should start a new transaction and run within its own transaction.
2. If a transaction is in progress, the current method should run within the nested transaction of the existing transaction. Otherwise, a new transaction has to be started and run within its own transaction.
3. The current method must start a new transaction and run within its own transaction. If there is an existing transaction in progress, it is suspended.
4. None of the above

Question 38

What is the default rollback policy in transaction management?

1. Rollback for any Exception
2. Rollback for RuntimeException
3. Rollback for checked exceptions
4. Always commit

Spring @MVC

Question 39

What could not return a Spring MVC controller? Select a single answer.

1. An absolute path to the view
2. A logical view name
3. A new JstlView
4. void
5. null value

Question 40

Where do you cannot declare Spring MVC controller? Select one or more answers.

1. In a Spring application context XML configuration file
2. Into the web.xml file of the web application
3. Into the java code by using annotations
4. Into the JSP pages

Question 41

What is the easiest method to write a unit test?

1. `void displayAccount(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException`
2. `void displayAccount(HttpServletRequest req, HttpSession Session) throws ServletException, IOException`
3. `@RequestMapping("/displayAccount")
String displayAccount(@RequestParam("accountId") int id, Model model)`
4. `@RequestMapping("/displayAccount")
String displayAccount(@PathVariable("accountId") int id, Model model)`

Spring Security

Question 42

How could you secure MVC controller with Spring Security? Select a unique answer.

1. With the `@Secured` annotation
2. With the `@RolesAllowed` annotation
3. In a XML security configuration file
4. All of the above
5. None of the above

Question 43

What are the possible mechanisms provided by Spring Security to store user details? Select one or more correct answers.

1. Database
2. JAAS
3. LDAP
4. Properties file

Question 44

What is right about Spring Security configuration and the security namespace? Select one or more correct answers.

1. The access attribute of the intercept-url tag support both EL and constants together.
2. The patterns declared into the intercept-url tag are analyzed from up to bottom. Winning is the first that matches.
3. The patterns declared into the intercept-url tag use by default the java regex syntax.
4. Security rules may apply depending request parameter

REST

Question 45

Which of the following is true regarding the below Spring controller?

```
@RestController
public class OwnerController {

    @RequestMapping(value = "/owner/{ownerId}", method = RequestMethod.POST)
    @ResponseBody
    public Owner findOwner(@PathVariable("ownerId") int ownerId) {
        return new Owner();
    }
}
```

1. RequestMethod.GET method is more accurate than POST
2. @PathVariable should be replaced with the @PathParam annotation
3. Returning the 201 HTTP status code is better
4. @ResponseBody could be removed

Question 46

Which of the following statements is true regarding the @ResponseStatus annotation?

1. @ResponseStatus is detected on nested exceptions
2. TheExceptionHandlerResolver uses the @ResponseStatus annotation to map exception to HTTP status code
3. A controller handler is annotated with the @ResponseStatus, the response status set by RedirectView takes precedence over the annotation value.
4. The @ResponseStatus annotation can go on a @RequestMapping method or a @RestController class or a business exception class.

Microservice

Question 47

Compared to monolithic application, what are the advantage(s) of microservices?

1. The base code is easy to understand
2. Imply a simple distributed system
3. Easier deployment

4. Fine-grained scaling

Question 48

What Spring Cloud provides in a microservices architecture?

1. A Service Discovery implementation
2. A server for externalized configuration
3. A Dockerfile building an image that runs any Spring Boot application
4. Netflix OSS integration for Spring Boot

Spring Boot

Question 49

What provides Spring Boot?

1. Support for Jetty and Undertow as embedded containers
2. Java code generation
3. Auto-configuration of the Spring Framework and third libraries
4. Convenient dependency descriptors to load transitive dependencies
5. Support both Java-based and YAML for Spring application context configuration

Question 50

What is the name of the default environment configuration file of Spring Boot?

1. configuration.spring
2. configuration.yml
3. configuration.xml
4. application.properties
5. application.json

Q 1 - What is spring?

A - Spring is an open source development framework for enterprise Java.

B - Spring is a proprietary framework.

C - Spring is a development framework for .Net applications.

D - Spring is a development framework for PHP based applications.

Q 2 - Which of the following is correct assertion about spring?

A - Spring enables developers to develop enterprise-class applications using POJOs.

B - Spring is organized in a modular fashion.

C - Testing an application written with spring is simple because environment-dependent code is moved into this framework.

D - All of above.

Q 3 - What is Dependency Injection?

A - It is a design pattern which implements Inversion of Control for software applications.

B - It is one of the spring module.

C - It is a technique to get dependencies of any project.

D - It is used to promote tight coupling in code.

Q 4 - Which of the following is correct about dependency injection?

A - It helps in decoupling application objects from each other.

B - It helps in deciding the dependencies of objects.

C - It stores objects states in database.

D - It stores object states in file system.

Q 5 - What AOP stands for?

A - Aspect Oriented Programming

B - Any Object Programming

C - Asset Oriented Programming

D - Asset Oriented Protocol

Q 6 - What is true about cross-cutting concerns?

A - The functions that span multiple points of an application are called cross cutting concerns.

B - Cross-cutting concerns are conceptually separate from the application's business logic.

C - Logging is one of the examples of cross cutting concerns.

D - All of the above.

Q 7 - Which are the modules of core container?

A - Beans, Core, Context, SpEL

B - Core, Context, ORM, Web

C - Core, Context, Aspects, Test

D - Bean, Core, Context, Test

Q 8 - Which are the modules of Data Access/ integration layer?

A - JDBC, ORM, OXM, JMS, Transactions

B - JDBC, ORM, OXM, JMS

C - JDBC, ORM, Web, Beans

D - JDBC, ORM, OXM, JMS

Q 9 - Which are the modules of Web layer?

A - WebSocket, Servlet, Web, Portlet

B - WebSocket, Servlet, Web-MVC, Web

C - HTML, JSP, WEB, Portlet

D - HTML, Servlet, WEB, Portlet

Q 10 - Which of the statement is not correct?

A - Core and beans modules provide the fundamental parts of the framework, including Dependency Injection feature.

B - The SpEL module provides a powerful Expression Language for querying and manipulating an object graph at runtime.

C - Aspects module provides integration with AspectJ.

D - None of the above.

Q 11 - Which of the statement is correct?

A - The JDBC module provides a JDBC-abstraction layer that removes the need to do tedious JDBC related coding.

B - The ORM module provides integration layers for popular object-relational mapping APIs, including JPA, JDO, Hibernate, and iBatis.

C - The Java Messaging Service JMS module contains features for producing and consuming messages.

D - All of the above.

Q 12 - Which of the statement is correct?

A - The AOP module provides aspect-oriented programming implementation allowing you to define method-interceptors and pointcuts to cleanly decouple code that implements functionality that should be separated.

B - The Aspects module provides integration with AspectJ - Which is again a powerful and mature aspect oriented programming (AOP) framework.

C - The Instrumentation module provides class instrumentation support and class loader implementations to be used in certain application servers.

D - All of the above.

Q 13 - What types of Dependency injection does spring supports?

A - Constructor based, Setter based

B - Constructor based, Setter based, Getter Based

C - Setter based, Getter based, Properties based

D - Constructor based, Setter based, Properties based

Q 14 - Which are the IoC containers in Spring?

A - BeanFactory, ApplicationContext

- [B - BeanFactory, ApplicationContext, IoCContextFactory](#)
- [C - BeanFactory, BeanContext, IoCContextFactory](#)
- [D - BeanFactory, ApplicationContext, BeanContext](#)

Q 15 - Which is the correct implementation class of BeanFactory?

- [A - XmlBeanFactory](#)
- [B - ClassPathBeanFactory](#)
- [C - FileSystemBeanFactory](#)
- [D - AdvancedBeanFactory](#)

Q 16 - Which are the correct implementation classes of ApplicationContext?

- [A - FileSystemXmlApplicationContext, ClassPathXmlApplicationContext, WebXmlApplicationContext](#)
- [B - FileSystemApplicationContext, ClassPathApplicationContext, WebApplicationContext](#)
- [C - AdvancedApplicationContext, FileApplicationContext](#)
- [D - FileSystemApplicationContext, ClassPathApplicationContext](#)

Q 17 - Which of the following stands true for spring beans?

- [A - Spring beans are managed by the Spring IoC container.](#)
- [B - Spring beans are instantiated, assembled, and otherwise managed by a Spring IoC container.](#)
- [C - Spring beans are simple POJOs.](#)
- [D - All of the above.](#)

Q 18 - Which is the way to provide configuration metadata to spring?

- [A - XML Based configuration file.](#)
- [B - Annotation based configuration.](#)
- [C - Java based configuration.](#)
- [D - All of the above.](#)

Q 19 - What is bean scope?

- [A - Bean scope forces Spring to produce a new bean instance as per the scope defined.](#)
- [B - Bean scope defines the accessibility of bean in a java class.](#)
- [C - Bean scope defines the accessibility of bean in a java package.](#)
- [D - Bean scope defines the accessibility of bean in a web application.](#)

Q 20 - What is singleton scope?

- [A - This scopes the bean definition to a single instance per Spring IoC container.](#)
- [B - This scopes the bean definition to a single instance per HTTP Request.](#)
- [C - This scopes the bean definition to a single instance per HTTP Session.](#)
- [D - This scopes the bean definition to a single instance per HTTP Application/ Global session.](#)

Q 21 - What is prototype scope?

- [A - This scopes a single bean definition to have any number of object instances.](#)
- [B - This scopes the bean definition to a single instance per HTTP Request.](#)
- [C - This scopes the bean definition to a single instance per HTTP Session.](#)
- [D - This scopes the bean definition to a single instance per HTTP Application/ Global session.](#)

Q 22 - What is request scope?

- [A - This scopes a bean definition to an HTTP request.](#)
- [B - This scopes the bean definition to Spring IoC container.](#)
- [C - This scopes the bean definition to HTTP Session.](#)
- [D - This scopes the bean definition HTTP Application/ Global session.](#)

Q 23 - What is session scope?

- [A - This scopes a bean definition to an HTTP session.](#)
- [B - This scopes the bean definition to Spring IoC container.](#)
- [C - This scopes the bean definition to HTTP request.](#)
- [D - This scopes the bean definition to HTTP Application/ Global session.](#)

Q 24 - What is global-session scope?

- [A - This scopes a bean definition to an HTTP Application/ Global session.](#)
- [B - This scopes the bean definition to Spring IoC container.](#)
- [C - This scopes the bean definition to HTTP request.](#)
- [D - This scopes the bean definition to HTTP Session.](#)

Q 25 - What is default scope of bean in Spring framework?

- [A - singleton](#)
- [B - prototype](#)
- [C - request](#)
- [D - session](#)

Phase 4

Single Answer

1. Rest parameters are used when

- datatypes of parameters are not known
- **number of parameters are not known//Ans**
- function needs to be called recursively
- function needs to be called asynchronously

2. Assume you have an array `myArray = [3 4, 5, 6, 7]` How could you use that to produce another array `times Two = [6,8, 10, 12, 14]`?

- `var times Two = myArray*2`
- `var times Two = myArray.map(myArray*2);`
- `var times Two = [...myArray]*2`
- **`var times Two = myArray.map(num => num*2)//Ans`**

3. Typescript can be installed by using _

- typescript installer
- **npm //Ans**
- powershell
- typescript as standalone zip file

4. According to the styling rules of organization Zee, the font-weight of every first paragraph within the section should be bold. Which structured pseudoclass can be used by the developer working on the web development project for organization Zee?

- **p:first-of-type//Ans**
- section: first-child
- section p:nth-child(1)
- section: first-of-type

5. Lets say you want to capitalize the first letter of each word and transform the rest of the word to lower case Here the words are delimited by whitespace characters like space tab, or line-feed character. Which Pipe do you use to achieve this?

- UpperCase Pipe
- LowerCase Pipe
- keyvalue Pipe
- Json Pipe

- **TitleCase Pipe.//Ans**

6.Match each element description option to its HTML5 element.

- A. <main>
B. <aside>
C. <figure>
D. <nav>

- A:Defines self-contained content
B:Defines content other than the page content
C:Defines navigation links
D:Defines the main content of a document
- **A:Defines the main content of a document**
B:Defines content other than the page content
C:Defines self-contained content
D:Defines navigation links.....//Ans

- A:Defines the main content of a document
B:Defines self-contained content
C:Defines content other than the page content
D:Defines navigation links
- A:Defines self-contained content
B:Defines the main content of a document
C: Defines content other than the page content
D: Defines navigation links

7. _____option of Angular CLI indicates to use inline templates for the project.

- --inline
- --inline-templates
- --inline-temp
- **--inline-template//Ans**

8. Rank the elements of the CSS box model from most inner to outer layer.

- Padding
Border
Content
Margin
- Margin
Border
Padding
Content
- **Content**
Padding
Border
Margin..... //Ans
- Content
Border
Padding
Margin

9. What is the correct statement to bind required and maxlength validators to name field in a reactive form?

- `name: ['', Validators.required, Validators.maxLength(10)]`
- **`name: ['', [Validators.required, Validators.maxLength(10)]]//Ans`**
- `name: [Validators.required, Validators.maxLength(10)]`
- `name: Validators.required, Validators.maxLength(10)`

10. What is a proper syntax to construct a module?

- **`var variable_name = angular.module("app_name", []);//Ans`**
- `var variable_name = angular.module("app_name");`
- `var variable_name = angular.module();`
- `var variable_name = new module();`

11. _____ means that if we declare a variable (declared but not initialized) at the end of a function, the runtime will hoist it to the top and we will not have any error if we would have used that variable before being declared.

- **`Variable hoisting//Ans`**
- verloading
- Inheritance
- Static Typing

12. Which of the following options would correctly set the viewport to make your website look good on all devices?

- `<meta name="port" content="width=device-width, initial-scale=0" />`
- **`<meta name="viewport" content="width=device-width, initial-scale=1" />//Ans`**
- `<meta name="port" content="width=device-width, initial-scale=1" />`
- `<meta name="viewport" content="width=devicewidth, initial-scale=1" />`

13. Which component life cycle hook is called immediately before Angular destroys the directive or component?

- `ngOnFinalize()`
- `ngAfterDestory()`
- **`ngOnDestroy()//Ans`**
- `ngBefore Destroy()`

14. Which of the following attribute directive adds and removes CSS classes on an HTML element?

- **ngClass//Ans**
- ngStyle
- ngCSS
- ng-class

15. Rest parameters are used when _____

- datatypes of parameters are not known
- **number of parameters are not known//Ans**
- function needs to be called recursively
- function needs to be called asynchronously

16. Write a statement with <p> element to initialize two variables (empid and salary).

- <p ng-app="" ng-init="empid: 'E1001';salary:7000">
- <p ng-app="" ng-init="empid: 'E1001',salary:7000">
- <p ng-app="" ng-init="empid='E1001',salary=7000">
- **<p ng-app="" ng-init="empid='E1001';salary=7000">//Ans**

17. Which of the following JavaScript function executes asynchronously?

- **async functionName(); //Ans**
- setInterval
- clearInterval
- Javascript is not multi threaded, hence no support for async calls

18. Which function is used to encode a Uniform Resource Identifier (URI) by replacing each instance of certain characters by one, two or three escape sequences representing the UTF-8 encoding of the character?

- **encodeURIComponent() //Ans**
- encode()
- encode ToURI()
- encoding()

19.Mr. Cailin has developed a webpage. Cailin has included the border styling using the below code.

border: 5px blue solid dashed double;

The border is not rendered when the web page is viewed in the browser. What is the reason for border not being rendered?

- The border width, border color and border style values should be specified individually
- **The order of the values are incorrect**
- There is no property called border in css
- **Multiple border-styles cannot be specified to border short hand property.//Ans**

D(D)20.Which CSS code snippet sets the background top vertically and center horizontally in the container and repeat the background vertically?

- background: url('bg.jpg') repeat-x 0px center;
- **background: url('bg.jpg') repeat-y center 0px; //Ans**
- background: url('bg.jpg') repeat-y 0px center;
- background: url('bg.jpg') repeat-x center 0px;

21.Which of the following is a valid arrow function?

- **let sum =(x:number , y :number)=> x+y //Ans**
- **let sum() =(x:number , y :number): number => x+y**
- let sum=(x :number , y:number) => return x+y
- let sum : (x :number , y :number) =>x+y

22. Which of the following perform transformations on data?

- **Filter plugins//Ans**
- Logging Plugin
- Search Plugin
- Monitor Plugin

23. Which is the main node configuration file for ElasticSearch?

- **elasticsearch.yml //Ans**
- elasticsearch.config
- jvm.options
- log4j2 properties

24. There are two general uses of plugins. The first is build and the second is_____

- **Reporting //Ans**
- servers
- dependencies
- distribution management

25. _____ is a suite of plugins which supports implementing and integrating continuous delivery pipelines into Jenkins.

- **Jenkins Pipeline//Ans**
- Jenkins Job
- Jenkins Build
- Jenkins Procedure

26. Which of the following command format the response of request in json format?

- **curl -XGET http://local:9200/catalog/products/1?pretty=true//Ans**
- curl-XGET http://localhost:9200/catalog/products/1?type=json
- curl -XGET http://localhost:9200/catalog/products/1?format-type=json
- curl -XGET http://localhost:9200/catalog/products/? Format-type=json

27. What is the purpose of a Kibana index pattern?

- **To define how data is mapped to Elasticsearch//Ans**
- To create custom visualizations
- To define search queries
- To define index settings

28. A _____ is a runnable task that Jenkins controls to achieve a required objective.

- plugin
- **job //Ans**

- stage
- pipeline

29. APIs for dynamically registering servlets and filters can be called at _____

- **Context initialization time//ANS**
- There is no way to dynamically register servlets and filters
- Context destruction time
- Injected into a servlet context

30. enables you to target an element when it's in a particular state, as if you have added a class for that state to the DOM

- Pseudo-elements
- **Pseudo-classes//Ans**
- attribute selector
- element selector

31. What is the correct way to write a JavaScript array?

- `var txt = new Array(1:"arr",2:"kim",3: "jim")`
- `var txt = new Array:1=("arr") 2=("kim") 3=("jim")`
- **`var txt = new Array("arr ", "kim", "jim") //Ans`**
- `var txt = new Array=" arr ", "kim", "jim"`

32. designed for storage that spans mutiple windows and last beyond the current session ?

- Session Stotage
- **Local Storage //Ans**
- Both A & B
- None of the above

33----- is an attribute directive that updates styles for the containing HTML element and sets one or more style properties, specified as colon-separated key-value pairs.

Options

- **ngStyle//Ans**

- ngClass
- ngSwitch
- ngIf

34. If you have a page of search results and want to highlight the search term, Which HTML tag would you use?

- strong>
- **<mark> //Ans**
-
- <highlight>

35. Which of the following attributes of text box control limits the maximum number of characters allowed?

- size
- length
- **maxlength //Ans**
- text

36. Which of the following is a valid Angular expression?

- **{{2+2}} //Ans**
- (2+2)
- [22]
- ((2+2))

37. _____ tag represents an independent piece of content of a document, such as a blog entry or newspaper article.

- <nav>
- <section>
- dialog 0000

- **<article> //Ans**

38. Which of the following is used to check the online status of the browser?-

- document isOnline
- **navigator. Online //Ans**
- navigator is Online
- window online

39. tag is used to split a page into sections like Introduction, Contact Information and Details

- <header >
- <navbar>
- <footer>
- **<section> //Ans**

40. How do you display hyperlinks without an underline?

- a {text-decoration no underline}
- a (decoration:no underline}
- **a { text-decoration: none;}// Ans**
- a (underline:none}

41. Lets say you write custom error handler and a logging service for your application. These will be a part of

- Feature Module
- **Core Module //Ans**
- Shared Module
- NgModule

42. _____ Is a process of initializing or loading our Angular application.

- Differential Loading

- Compilation
- Bundling and Minification
- **Bootstrapping //Ans**

43. provides a more concise syntax for writing function expressions by removing the "function" and "return" keywords.

- **arrow function //Ans**
- spread operator
- rest operator
- default parameter

44. Which of the following decorator specifies that Angular can use this class in the DI system?

- **@Injectable ()//Ans**
- @Component()
- @Service()
- @Inject()

45. Which of the following is not a feature of TypeScript?

- Cross Platform
- ES6 Features
- Static Type Checking
- **Dynamic Type Checking//Ans**

46. You want to change the style properties such as height, width, color, margin, border, etc., or any other internal properties of the host element in a directive, so you need to access these properties on the host element and assign a value to it. This is done by using_____ decorator.

- @Host
- **@HostBinding//Ans**
- @HostListener
- @Attribute

47. Which element is used in the <HEAD> section on an HTML / XHTMLpage, if we want to use an external style sheet file to decorate the page?

- **<link> //Ans**

- <src>
- <styles
- <CSS>

48. What is the output of the below script?

```
var s = "Mphasis";
s = s.substr(2, 3);
console.log(s);
```

- h
- hasis
- **has //Ans**
- Mphasis

49. _____ creates a top-level FormGroup instance and binds it to a <form> element to track aggregated form value and validation status.

- FormsModule
- **NgForm //Ans**
- ReactiveForm
- FormGroup

50. _____ are a great way to organize an application and extend it with capabilities from external libraries.

- **Modules //Ans**
- Directives
- Services
- Pipes

D) 51. What is the output when you load the above component by writing below code?

```
<app-welcome></app-welcome>
```

- It displays Welcome to as name property is not specified while loading the component
- **You will get a blank screen //ANS**

- It displays Welcome to Angular as the default value is assigned to name property
- **Error**

52. Which method of document object returns all the element of specified tag name?

- **getElementsBy TagName() //Ans**
- getElementById)
- getElementsByClassName()
- getElemBy TagName()

53. Which of the following Angular lifecycle hook is used to perform complex initializations shortly after construction and to set up the component after Angular sets the input properties?

- **ngOnInit() //Ans**
- NgAfterInit()
- ngAfterViewInit()
- ngConstruct()

54. Which of the following options would correctly set the viewport to make your website look good on all devices?

- <meta name="port" content="width=device-width,initial-scale=0" />
- **<meta name="viewport" content="width=device-width, initial-scale=1"> //Ans**
- <meta name="port" content="width=device-width, initial-scale=1" />
- <meta name="viewport" content="width=device-width initial-scale=1" />

55. Which of the following statements are valid?

- Component and Directives can be used interchangeably
- Angular application can be bootstrapped using any module
- **Angular application can be bootstrapped using root module//Answer**
- A directive is a component-with-a-template

56. Which of these statements about the await keyword is TRUE?

- **It can only be used inside an async function //Ans**
- It pauses all code executions
- It can only be used inside functions which return a value.

- It can only be used inside functions which take in an argument

57. The support script.tag defines an alternate content to be displayed to users that have disabled scripts in their browser or have a browser that doesn't

- **<noscript> //Ans**
- <script>
- <notscript>
- <js>

58. Let's say you need to uninstall a pm package called mymaterial-ui that is installed globally. Which of the following command is used to do this?

- npm remove mymaterial-ui
- npm purge -l mymaterial-ui
- **npm uninstall -g mymaterial-ui //Ans**
- npm delete -g mymaterial-ui

59. _____ are classes that add additional behavior to elements in your Angular applications.

Options

- **Directives //Ans**
- Modules
- Services
- Components

60. Which of the below option enables the client script to draw graphics dynamically?

- Panel
- Drawpanel
- **Canvas //Ans**
- Paintbox

D)61. Which one of the following statements will make a div element act like an HTML5 button?

- `<a><div href="faq.html" class="faq"></div>FAQ</a>`
- `<a href="#"><div class="home">Home</div></a> //Ans`
- `<div href="#" class="home">Home</div>`
- `<div class="faq">FAQ</div><a href="faq.html">FAQ</a>`

62. Typescript compiler configuration is stored in file.

- package.json
- tsc
- `tsconfig.json //Ans`
- lint.json

63. If you have a page of search results and want to highlight the search term, Which HTML tag would you use?

- `<strong>`
- `<mark> //Ans`
- `<em>`
- `<highlight>`

64. Choose the appropriate option to represent an ordered list with individual list items contained within opening and closing master tags as the container.

- `<dl><dt></dt><dd></dd></dl>`
- `<ol><li></li></ol> //Ans`
- `<ul><li></li></ul>`
- `<list><item> </item></list>`

65. ensure that the nested function is packaged with its environment (outer function variables)

- `Closures //Ans`
- Objects
- Map
- Functions

66. method returns true if the specified value is present in the Set object.

- **Set prototype.has() //Ans**
- Set prototype.hasValueAvailable()
- Set prototype.isValueAvailable()
- Set prototype.isValueExists()

67. Given an array myArray = [15,18, 21,3,6,9, 12] what is returned by the following operation?

myArray.sort()

Options

- **[12, 15, 18, 21, 3, 6, 9] //Ans**
- [3, 6, 9, 12, 15, 18, 21]
- (12, 9, 6, 3, 21, 18, 15]
- (21, 18, 15, 12, 9, 6, 3]

68. Which of these represents a valid syntax to assign a function to a variable?

Options

- let times Two = (x) return 2*x;
- let times Two= (x) { return 2*x };
- let times Two =function (x) = return 2*x;
- **let times Two= function(x) { return 2*x }; //Ans**

69. Which of the following service is used to communicate with a server over the HTTP protocol, to perform CRUD operations?

- Http
- **HttpClient //Ans**
- HttpClientModule
- HttpService

70. You are analyzing the navigation options currently available for your web site before deciding how to remap the HTML links. Each page will need to have a null link to keep the navigation menu consistent. Identify the code that shows the

correct syntax for a null navigation link for the web page

```
<html><head></head><body><a href="#">home</a><br><a href-  
"about.html">About Us</a><br><a href-"careers.htm|">Careers at  
Diallonics</a><br><a href-"contact.htm|">Contact Us</a><br></body></html>
```

Options

- `<a href-"about.htm|"> About Us</a><br>`
- `<a href-"contact.htm|">Contact Us</a><br>`
- `<a href="careers.htm"> Careers at Diallonics</a><br>`
- **`<a href="#">home</a><br>`//Ans**

71. Let's say you want to implement custom sort feature into your Angular application. This functionality has to be reusable across the project. Which of the following option is best suited for implementing this?

Options

- Component
- Module
- Routing
- **Pipes//Ans**

72. Which property of list specifies the type of list item marker?

Options

- **list-style-type //Ans**
- style-type
- list-style
- type

MCQ(QUESTIONS)

1. Which of the following are the valid properties of Promise Object?

- **state**
- **result**
- readyState
- response

2. Which of the below attributes controls the valid numeric boundaries of a range control?

- **min**
- minbound
- count

- **max**

3. Describe some of the essential objectives of the Font-face property.

- Font-face property allows us to link to the styles that are automatically fetched
- **Font-face property allows us to link to the fonts that are automatically fetched //ans**
- **Font-face property allows us to use multiple @font-face rules to construct font formats //ans**
- **Font-face property allows us to use multiple @font-face rules to construct font families //ans**

4. You are creating a web page that requires some text to span across the width of the screen and other pieces of text to be emphasized within those text blocks. Which statements below correctly explain the rules about block and inline elements that will help you build your web page?

- **You should use block-level elements when you want text or content to go across the width of the browser window or to span on a new line.**
- You should use block-level elements when you want text or content to align horizontally with an image on a web page
- **You should use inline-level elements when you want to align or format text without having it start on another line or span width of a browser window.**
- You should use inline-level elements when using the <p> tag.

5. Which of the following are the valid statements with respect to production builds? *

- **Uses AOT compilation**
- **Files are tree shaken**
- **Source maps are generated**
- **Css are extracted to js files** //ans
- Uses JIT compilation

6. Specify some of the essential new features that were added to HTML5.

- **We can now extend web pages with custom semantics using HTML5** //ans
- **Next generation, bi-directional communication can be facilitated in HTML5 using WebSockets** //ans
- **Canvas was introduced in HTML5 which supports 3-dimensional drawing surfaces** //ans
- Forms 3.0

7. Which of the following are the valid syntax of using ngClass directive to apply CSS classes A,B in Angular?

- `<h1 [ngClass]="A B" >Hello World </h1>`
- **`<h1 [ngClass]="['A','B']" >Hello World </h1>`**
- **`<h1 [ngClass]="{'A': true, 'B':true}" >Hello World </h1>`**
- `<h1 [ngClass]="A,B" >Hello World </h1>`

8. Which of the following are the correct statements?

- **Typescript can be compiled to all major versions of Javascript**
- Typescript can be compiled to only latest versions of Javascript
- **Typescript can be used for cross-browser development**
- Typescript supports only specific browsers like Edge

9. Which of the following are the features of HttpClient service?

- **The ability to request typed response objects.**

- **Streamlined error handling.**
- **Request and response interception.**
- Non Testability

10. Which of the following elements are essential for screen readers to navigate through contents in web

- **Form controls //ans**
- Paragraph controls
- **Skip navigation links //ans**
- Jump navigation links

11. Which of the following are the valid properties of XMLHttpRequest object?

- **readyState**
- **status**
- data
- result

12. Select valid states of Promise Object.

- **Pending**
- **fulfilled**
- **rejected**
- completed

13. The <article></article> and <section></section> tags are new to HTML5. Identify the true statements that describes how each is used in HTML5.

- Article elements or tags in HTML5 can be used as CSS styles that group images on a web page that are separated by chunks of content
- **Article elements or tags in HTML5 can be used as RRS syndication and applied to dynamic content such as forum posts, blog posts, comments, and news stories.**
- The Section tag in HTML5 defines and isolates tables and images that have text wrapped around them.
- **The Section tag in HTML5 defines and isolates sections or logical chunks of content in a document such as an Introduction, header, or footer.**

14. Which of the following functions invoked on a string variable "myString" will replace the first occurrence of the sub-string "WebDevelopment" to "programming"?

- **myString.replace("WebDevelopment", "programming");**
- **myString.replace("/webdevelopment/g", "programming");**
- myString.replace("webdevelopment", "programming");
- myString.replaceFirst("WebDevelopment", "programming");

15. Which of the following is not a predefined pipe? *

- **Upper**
- currency
- date
- **datetime**
- slice

16. Which of the following are valid methods .

- **Reverse()**
- **Sort()**
- Search()
- **Filter()**

17. Which of the following are ES6 features?

- var keyword
- **let and const keyword**
- **Arrow functions.**
- **Template Literals**

18 Which of the following members can be included in the class?

- **Constructor**
- **properties**
- **methods**
- modules.

19. Which are the built-in browser validation features?

- **max**
- **minlength**
- conversion
- currency
- **required**

20. Which of these operations are permitted if you apply Object.freeze() on myObject?

- **Adding a property to a nested object in myObject**
- Adding a new top-level property to myObject
- Modifying a top-level property of myObject
- **Modifying a property of a nested object in myObject**

21. Which of the following are the valid HTML5 input types? (lookout)

- **Color**

- time
- datetime
- number

22. The purpose and function of the ADDRESS and CITE tags has changed since HTML 4.01 and are now more suited to RSS syndication purposes. Identify the true statements that describes how the function of each tag has changed in HTML5.

- The Address tag now defines the contact information of the author or owner of a document allowing you to have one author per web page.
- The Cite tag now defines a citation or article footer.
- **The Cite tag now defines the title of a work such as a song, book, or movie.**
- **The Address tag now defines the contact information of the author or owner of a document or an article allowing you to have multiple authors for a single page.**

23. Which of the following options are correct statements to apply the CSS class 'pink' to the paragraph element? **lookout**

- `<p [ngClass]='pink'> background is pink</p>`
- **`<p [ngClass]='{pink:true}'> background is pink</p>`**
- **`<p [ngClass]='["pink"]'> background is pink</p>`**
- `<p ngClass ={pink:true}>background is pink</p>`

24. Mr. Ram was working on a web development project. According to the styling guidelines provided by the organization, Ram wants to add 40px of spacing between all the paragraphs that appear immediately after the heading. Which of the below code suits the above requirement?

- `p{margin-bottom: 20px;}`
- **`h1{margin-bottom: 40px;}`**
- **`h1 (margin: 0 0 40px 0;}`**
- `p{margin-bottom: 0 20px 0;}`

25. Identify which of the following Pseudo classes can be applied to hyperlinks.

- Cue
- Clicked
- **Hover**
- **Visited**
- **Focus**

26. You want to change the style properties such as height, width, color, margin, border, etc., or any other internal properties of the host element in a directive, so you need to

access these properties on the host element and assign a value to it. This is done by using _____ decorator. **(single answer)**

- @Host
- **@HostBinding**
- @HostListener
- @Attribute

27. Given the following object, which of the following options are valid ways to access its "area" property?

```
var country = {name: "Egypt", area: 1010408, population: 99581200}
```

- **country["area"]**
- **country.area**
- country[area]
- area.country

28. Which of the following attribute directive adds and removes CSS classes on an HTML element? **(single answer)**

- **ngClass**
- ngStyle
- ngCSS
- ng-class

29. Which of the following service is used to communicate with a server over the HTTP protocol, to perform CRUD operations? **(single answer)**

- Http
- **HttpClient**
- HttpClientModule
- HttpService

30. Which of the following are the valid attribute-directives in Angular?

- **ngStyle**
- **ngClass**
- ngSwitch
- NgIf

31. Which of the following Pseudo elements can we use to manage styles on HTML documents?

- **::first-line**
- **::first-letter**
- ::training-letter

- ::training-line

32. Which three programming constructs does JavaScript support?

- Arrays
- Type conversion
- **Iteration/looping**
- **Decisions/select**
- **Sequence**

33. List some of the Kibana features that enable dashboard reusability.

- **Copy**
- Clone
- **Snapshot**
- **Share**

34. Which of the following are benefits of Elasticsearch?

- **Searching**
- **Analytics**
- **Near real time**
- Non scalable

35. Select some of the benefits associated with containerization

- **With containerization we can create multiples copies on several machines**
- `StringBuilder(StringBuilder str)`
- **Containerization encapsulates applications**
- **Containerized applications can be automatically deployed on-premise servers**

36. What are the valid reasons for using container?

- Containers cannot be unit or integration tested
- Containers are fixed and cannot be rolled back
- **Containers by design are scalable**

- Containers are highly portable

37. Which of the following are examples of cross-cutting concerns? *

- Logging
- Security
- Debugging
- Wiring

38. Which of the following access modifiers are supported by TypeScript?

- default.
- private
- public
- protected

39. Select valid states of Promise Object.

Options

- pending
- fulfilled
- rejected
- completed
- hp

40. Which of the following are features of HttpClient service?

- The ability to request typed response objects
- Streamlined error handling
- Request and response interception
- Non Testability

41. Which of the following are valid methods of XMLHttpRequest object?

- send()
- open()
- cancel()
- abort()

42. Choose the disadvantages of Single Page Applications.

- User experience
- Fast and Flexible
- **SEO optimization**
- **JavaScript Dependency in Browser**

43. Which of the following are valid methods of Array object?

- **reverse()**
- **sort()**
- search()
- **filter()**

44. Which of the following scopes are available in TypeScript?

- **Local**
- **Global**
- **Class**
- Default

45. The purpose and function of the ADDRESS and CITE tags has changed since HTML 4.01 and are now more suited to RSS syndication purposes.

Identify the true statements below that describe how the function of each has changed in HTML5.

- The Address tag now defines the contact information of the author or owner of a document allowing you to have one author per web page
- The Cite tag now defines a citation or article footer
- **The Cite tag now defines the title of a work such as a song, book, or movie**
- **The Address tag now defines the contact information of the author or owner of a document or an article allowing you to have multiple authors for a singlepage**

42. Choose the valid statements of let keyword.

Options

- **let variables are blocked scoped//Ans**
- let variables can be re-declared
- **let variables cannot be re-declared//Ans**
- let variables are window Scoped

CODE SNIP

1. Sahar has developed a website. One of his webpage has been designed as below.

```
<html>
<head>
<style>
    Input[type="checkbox"]:indeterminate +label
    {
        color: yellow;
    }
</style>
</head>
<body>
    <form>
        <input type="checkbox" name="emp" id="emp" indeterminate></input>
        <label for="emp">Employee</label>
        <ul>
            <li><input type="checkbox" name="empsal" id="empsal"></input>
                <label for="empsal">Employee Salary</label></li>
            <li><input type="checkbox" name="empdes" id="empdes"></input>
                <label for="empdes">Employee Designation</label></li>
        </ul>
    </form>
</body>
</html>
```

Since there exists a parent-child relationship between the checkboxes.

Sahar has declared the state checkbox emp as indeterminate and included styling to the checkbox.

Identify the problem in the below code.

- Nothing wrong with the code its correct

- We cannot style the labels of checkbox
- We cannot explicitly change the state of the checkbox as indeterminate. It is taken care automatically when there is parent child relation
- **We cannot change the state of checkbox indeterminate using HTML similar to checked or unchecked, it can be done using javascript. //answer**

2. What would be the color of element when the below webpage is rendered on the browser?

```
<html>
<head>
<style>
    ul li{color: red;}
    li{color:green;}
    #cf {color: yellow;}
    .cfe{color:pink;}
</style>
<body>
    <ul>
        <li id="cf" class="cfe">Coffee</li>
    </ul>
</body>
</html>
```

- Red
- Green
- **Yellow //ans**
- Pink

3. What will be the output of the below code?

```
import { PipeTransform } from '@angular/core';
.....
export class LengthPipe implements PipeTransform {
    length (value: string): string{
        return "Length =" + value.length;
    }
}
```

}

- length method should be an no args method
- **The Length Pipe class has not implemented the transform() method of Pipe Transform Interface**
- LengthPipe class should be using @Component decorator to make it working
- The length method should return an numeric value.

4. What will be the output for the following code?

```
var s= "friends forever";
```

```
var x=3;
```

```
do{
```

```
    console.log(s.charAt(x));
```

```
    x++;
```

```
} while (x < 9);
```

- **ends f**
- friends
- forev
- nds fore

5. In the below code snippet, what is the correct statement to be placed at line number 6 to display the courseName?

```
Line 1: import { Component } from '@angular/core';
```

```
Line 2: @Component({
```

```
Line 3:     selector: 'my-app',
```

```
Line 4:     template:
```

```
Line 5:         <h1>
```

```
Line 6:         .....
```

```
Line 7:         </h1>
```

```
Line 8:
```

```
Line 9:     })
```

```
Line 10: export class AppComponent
```

```
Line 11: {
```

```
Line 12:     courseName: string = "Angular";
```

```
Line 13: }
```

- **{{ courseName }}**
- this.courseName

- courseName
- AppComponent.courseName

6. Please refer below

How to print the value of "LastName" for the below code?

```
<div ng-app="" ng-init "firstHamam= 'John' ">
  <p>Input something in the input box;</p>
  <p>Name: <input type="text" ng-model="Last Name"></p>
  <p>You wrote:_____</p>
</div>
```

- LastName
- **{{ LastName }}**
- {LastName}
- {%_LastName% }

7. Please refer below

Consider the code snippet given below and find out which line will result in an error.

```
class Person {
  constructor (name, age) {
    this._pname = name;
    this._page= age;
  }

  display() {
    console.log("name:" + this.pname);
    console.log("age:" + this.page);
  }
  set pname (namePerson) { this._pname = namePerson; }
  get pname () { return this._ pname; }
}

let per = new Person();           // line 8
per.pname = "Rexha";             // line 9
console.log(per.pname);           // line 10
let per1 = new Person("Alex", 10); // line 11
console.log(per1.pname);          //line12
```

- **Line no 8 gives error as no arguments have been passed to the constructor.**
- Line no. 10 gives error as this is not the way to fetch the instance variable of a class.
- Line 10 displays undefined and line 12 displays Alex

- **Line 10 displays Rexha and line 12 displays Alex**

8. Consider the code output given below

```
let productId:
[string, number];
productionId= ["Mob342", 675, 7845, "Tab573"];
```

Which of the following statements are true regarding the above code?

- This code will throw compilation error, due to length restriction.
- **It will throw compilation error if a boolean value is added to the productId array**
- **This code will get executed if two values are passed of declared data type**
- The array which is created above is immutable

9. Predict the output of the below mentioned code snippet.

```
<html>
<body>
    <input type='text' id='text1' onkeyup="console.log('Task 1')"
    onkeydown="console.log('Task 2')" onkeypress="console.log('Task 3')">
</body>
</html>
```

- Cannot handle multiple events on an element at one time
- Task 1 Task 2 Task 3. Because Execution order will match the order in which event handlers are written
- Task 3 Task 2 Task 1. Because First keypress will happen, followed by keydown and lastly key release will happen
- **Task 2 Task 3 Task 1. Because First keydown will happen, followed by keypress and lastly key release will happen**

10. What will be the output for the following code?

```
let person = {
    name: "Joe Camel",
    age: 42,
    status: "dead"
}
console.log(typeof person);
```

- Number
- Array
- Boolean
- **Object**

11. What will be the color of the text "Sausage"?

```
<ul class="shopping-list" id="awesome">
  <li><span>Milk</span></li>
  <li class="favorite" id="must-buy"><span class="highlight">Sausage</span></li>
</ul>
```

```
css:ul {
    color: red;
}
```

```
li {
    color: blue;
}
```

- Red
- **Blue**
- Yellow
- Green

12. What would be the color of the paragraph element when the below HTML code is executed on the browser?

```
<div class="outer">
<p id="outer">Paragraph Element</p>
</div>
<style>
    .outer {
        color: red;
    }
    #inner {
        background-color: blue;
    }
}
```


</style>

- Paragraph element will occur in blue color
- Paragraph element will occur with background in blue color
- **Paragraph element will occur in red color**
- Multiple style rules will result in error in the above code

13. What will be the output for the following code?

```
function add(num1: number, num2?: number) {  
    if (num2) {  
        return num1+num2;  
    }  
    return num1;  
}  
console.log (add (3,5)+add(5)+add (3));
```

- 8+5+3
- **16**
- Runtime Error
- Compilation Error

14. Consider the below code snippet

```
app-routing.module.ts  
{path: 'product/:id', component: ProductDetail}  
product-list.component.html:  
<a {routerLink} = "[ '/product', product.productId]"> {{product.productName}}</a>
```

Select the correct option to read the parameter value in the navigated component in Angular.

- parameter value can be read using RouteParams service
- Use router. navigate() method
- **Parameter value can be read using Activated Route service**
- By using route.redirects() method

15. Please refer below

A web page is designed using the below code.

```
<html>
<head>
<style>
    body (width:200px;height: 150px;border: 1px solid black;)
</style>
</head>
<body>
    <p> Click on the below links to navigate to appropriate pages</p>
    <ul>
        <li><a href="#">Home</a></li>
        <li><a href="#">Contact us</a></li>
        <li><a href="#">About us</a></li>
    </ul>
</body>
</html>
```

If there is a requirement to display the links with images instead of bullets and remove the underline. Which of the following properties can be used?

- list-style-image and text-transform
- list-style-type and text-decoration
- **list-style-image and text-decoration** //ans
- list-style-type and text-transform

16. In the below code snippet, what is the correct statement to be placed at line 6 to display the 'numbers' array as a list?

```
Line 1: import {Component } from '@angular/core';
Line 2: @Component ({
Line 3:     selector: 'my-app',
Line 4:     template:
Line 5:         <ul>
Line 6:             .....
Line 7:         </ul>
Line 8:
Line 9: })
Line 10: export class AppComponent
Line 11: {
Line 12:     numbers: any [] = [1,2,3];
Line 13: }
```

- <li *ngFor = 'numbers'>{{numbers }}

- `<li *ngFor = 'num of numbers'>{{ num }} </li>`
- `<li *ngFor = '#num of numbers'>{{ num }} </li>`
- `<li *ngFor = 'let num of numbers'>{{ num }} </li>`

17. Predict the output for the below code snippet.

```
class Cube {
    width;
    height;
    length;
}
let cube1= new Cube ();
let cube2= new Cube();
cube1.height=1;
cube1.length =2;
cube1.width = 1;
cube2 =cube1;
console.log(cube2.height);
```

- 0
- 1 //answer
- 2
- Compilation Error

18. Consider the code snippet below:

```
function printData<T>(data:T): T{
    return data;
}
let data= printData('Hello Generics');
```

What will be the output of the above code when executed?

- Compilation error saying string type parameter has to be mentioned while invoking the function
- Runtime error
- Executes without any error
- Compilation error saying type parameter mismatch

19. What will be the output for the following code?

```
<html>
<head>
<script>
function User (name, age) {
this.name =name; this.age= age;
}
var user = new User('Daniel', 45);
document.write(user[name] + ' ' + 'is' + ' ' + user[age] + ' ' + 'years old!');
</script>
</head>
<body>
</body>
</html>
```

- **Wrong syntax - Objects cannot be accessed using brackets //ans**
- Accessing Object property using bracket is allowed in JavaScript. So output is: Daniel is 45 years old!
- Error - age is not defined
- undefined is 45 years old

20. What will be the output for the following code?

```
//es6 rest parameter
function fun(...input) {
    let sum = 0;
    for(let i of input) {
        sum+=i;
    }
    return sum;
}
console.log(fun (1,2));
console.log(" " +fun (1,2,3));
console.log(" " +fun (1,2,3,4,5));
```

- **3,6,15**
- 3, undefined

- 3,error
- 3,6,error

21. Consider the below code snippet. What will be the styling of "Paragraph Element"?

```
<div class="outer">
  <p style="color: initial">Paragraph Element</p>
</div>
<style>
  P,.outer {
    color: white;
    background-color: orange;
  }
</style>
```

- Background color in orange and color in white
- Background color in white and color in orange
- **Background color in orange and color in black**
- Background color in black and color in white

21. What will be the output for the following code?

```
<html>
  <script>
    var check ='a';
    switch(check) {
      case 'A': document.write('Lower and Uppercase letters are same for JavaScript');
      case 'a': document.write('Lower and Uppercase letters are different for JavaScript');
      default: document.write('Lower and Uppercase letters cannot be compared in
JavaScript');
    }
  </script>
</html>
```

- Lower and Uppercase letters cannot be compared in JavaScript
- Lower and Uppercase letters are different for JavaScript
- **Lower and Uppercase letters are different for JavaScript Lower and Uppercase letters cannot be compared in JavaScript**

- Lower and Uppercase letters are same for JavaScript

22. choose the correct output given code

```
let num1 = 20;  
let num2= "20";  
console.log(num1==num2);  
console.log (num1===num2);
```

- true;true
- **true;false;**
- false;true;
- false;false;

23. Observe the code snippet given below

```
let movie={ name: "Galaxy of guardians", type: "Hollywood" };  
let movieProperty="rating ";  
movie ["current"+movieProperty] = 5;
```

Choose the right option to access the computed property at line 3.

- console.log(movie,movieProperty);
- **console.log(movie.currentrating);**
- console.log(movie.rating);
- **console.log(movie.currentProperty)**

24. What will be the output for the following code?

```
let colors = ["Red", "Green", "Orange", "Blue"]; colors.splice (3,0, "White", "Black")  
colors.splice (1,2, "Pink"); console.log(colors);
```

- [White', 'Black', 'Pink', 'Blue']
- **['Red', 'Pink', 'White', 'Black', 'Blue']**
- [Red', 'Pink', "White", "Black"];
- Compilation Error

25. The below code snippet has some lines of code missing. As per the requirement mentioned below please complete the lines

Requirement:

1. set the background color yellow for the div element (line 3)
2. hide rating if not present (line 15)

```
Line 1: @Component({
Line 2:   template:
Line 3:     <div>
Line 4:       <h1 [textContent]=' "user is" +person.name'> </h1>
Line 5:       <table>
Line 6:         <tr>
Line 7:           <td [attr.colspan]='1+1'>
Line 8:             <button >Female</button>
Line 9:           </td>
Line 10:          <td [attr.colspan]='1+1'>
Line 11:            <button>Male</button>
Line 12:          </td>
Line 13:        </tr>
Line 14:      <tr>
Line 15:        <td>{{(person.rating)}}</td> //line2
Line 16:        <td>{{(person.address)}}</td>
Line 17:      </tr>
Line 18:    </table>
Line 19:  </div>
Line 20:
Line 21: }}

    export class AppComponent {
      private female={
        name: 'Jenny Wesley',
        gender: 'F',
        rating: 4,
        address: 'Wesley states'
      }
      private male={
        name: 'Ross Green',
        gender: 'M',
        address: 'New York'
      }
      private person=this.female;
    }
    • line3: [style.backgroundColor]='yellow' line 15: <td [hidden]='!person.rating'>
```

- line3: [style.backgroundColor]=' "yellow" ' line 15: <td hidden='!person.rating'>
- **line3: [style.background-color]=' "yellow" ' line15: <td [hidden]='!person.rating'> //ans**
- line3: [style.background-color]=' "yellow" ' line 15: <td hidden='!person.rating'>

26. What will be the output for the following code?

```
function add(n1: number, n2: number): number {
    let total: number=n1+n2;
    return true;
}
console.log(add (5,2));
```

- 7
- 5
- No Output
- **Compilation Error**

27. What will be the output for the following code?

```
let originalString="Digital!";
let updatedString1= originalString.substr(2,3);
let updatedString2=originalString.substring(2,3);
document.write(updatedString1+'<br>');
document.write(updatedString2);
```

- **git g**
- git git
- ggit
- gg

28. What will be the output for the following code?

```
Items = [
{name: "Cake", price: 360},
{name: "Vegetables", price:500},
{name: "Drinks" ,price: 410}
];
let totalExpensive= items.filter(i => i.price >=400);
console.log(totalExpensive.reduce((sum, i) => sum + i.price,0));
```


- 1270
- **910**
- 500
- 410
- 550

29. Which of the below code snippets will give the same result?

```
//1
let x = [1,2,3];
let y= [4,5,6];
const combinedArray 1=[.....x, ..... y];
console.log(combinedArray1)
//2
Let x = [3,4,5,6];
const combinedArray2= [1,2,..... x]
console.log (combinedArray2)
//3
let a= [3,4];
const combinedArray3= [1,2,.....a, 6];
console.log(combinedArray3)
```

- 1,3
- **1,2 //answer**
- 1,2,3
- 3,2

30. What will be the output for the following code?

```
let myArr = [ ];
for (i = 0; i < 3 ; i++) {
    myArr.push(function () {
        console.log(i);
    });
}
for (j=0; j < 3; j++) {
    myArr[j]();
}
```

- 012
- 000
- undefined
- **333 //answer**

31. Predict the output of the below code snippet.

```
interface Employee {
  dept: string;
  name: string;
}
function printEmpDetails(emp: Employee){
  console.log(emp.name + " " + emp.dept);
}
let myobj = { dept: "Testing"};
printEmpDetails (myobj);
```

- Tom Testing
- undefined Testing
- Code gets executed without any issue
- **Code throws compile time error //ans**

32. write the output of following code?

```
<html>
  <body>
    <script>
      var str1 = "John";
      var obj1 = new String("John");

      console.log("Type of str1 : ", typeof str1);
      console.log ("Type of obj1 : ", typeof obj1);
      console.log ("Is str1 == obj1 : ",str1==obj1);
      console.log("Is str1 === obj1 : ",str1===obj1);
    </script>
  </body>
</html>
```

- (A) **Type of str1 : string**
Type of obj1 : object
str1 == obj1 : true
str1 === obj1 : false
- (B) Type of str1 : object
Type of obj1 : object
str1 == obj1 : true
str1 === obj1 : true
- (C) Type of str1 : string
Type of obj1 : object
str1 == obj1 : true
str1 === obj1 : true
- (D) Type of str1 : string
Type of obj1 : object
str1 == obj1 : false
str1 === obj1 : false

33. Consider the below code

```
type Color= "green" | "yellow";
type Quantity= "five" | "six";
```

```
// type Itemtwo ="five item" | "six item" | "green item" | "yellow item"
type ItemOne= `${Quantity | Colour} item` ;
```

The code snippet is an example of

- String literals
- **Template Literals**
- Object Literals
- Directives

34. What will be the output when AppComponent is rendered on the browser?

app.component.ts:

```
import { Component } from '@angular/core';
@Component({
```

```

        selector: 'app-root',
        templateUrl: './app.component.html',
        template: 'Welcome to {{course}}'
    })
    export class AppComponent {
        salutation = "Hello";
        course = "Angular";
    }
    app.component.html:
    {{salutation}} {{course}}

```

- Compilation error due to use of inline as well as external template
- **Hello Angular**
- Welcome to Angular
- Hello Angular Welcome to Angular

35. Chris has created a highlight directive as shown below. what the colour will be applied to page if enters yellow in the input field?

```

//highlight.directive.ts
import {Directive, ElementRef, Input} from '@angular/core';
@Directive({
    selector: ['highlight']
})
export class HighlightDirective {
    constructor(el:ElementRef) {
        el.nativeElement.style.backgroundColor="blue";
    }
}
//app.component.ts
template:

```

```

<label>Provide color to be applied on the page</label>
<input type='text' [(ngModel)]='colorName'>
<div highlight>Hello </div>

```

- color red will be applied to the div element
- color yellow will be applied to the div element
- yellow color will not be applied as there is no binding of model colorName and highlight

- **color blue will be applied to the div element**

36. You are analyzing the navigation options currently available for your web site before deciding how to remap the HTML links. Each page will need to have a null link to keep the navigation menu consistent. Identify the code that shows the correct syntax for a null navigation link for the web page.

```
<head>
</head>
<body>
<a href="#">home</a><br>
<a href="about.html">About Us</a><br>
<a href="careers.htm|">Careers at Diallonics</a><br>
<a href="contact.htm|">Contact Us</a><br>
</body><html>
```

Options

- `sa href="about.htm|"> About Us</a><br>`
- `<a href="contact.htm|">Contact Us</a><br>`
- `sa href="careers.htm"> Careers at Diallonics</a><br>`
- **`<a href="#">home</a><br>`**

37. What is the output of below script?

```
let colors = ['Red', 'Yellow'];
let newColors = [...colors, 'Green', colors];
console.log(newColors);
```

Options

- **`['Red', 'Yellow', 'Green', ['Red', 'Yellow']]`//Answer**
- `['Red', 'Yellow']`
- `['Red', 'Yellow', 'Green']`
- Error

38. What is the output of the below script?

```
<script type="text/javascript">
var s = "9123456 or 80000?";
var pattern = /d{4}/;
```

```
var output = s.match(pattern);
document. write(output);
< /script>
```

- **9123 //answer**
- 91234
- 80000
- 8000

39. Study the given CSS3 code excerpt that is formatting three different divs on an HTML page.

Select the DIV option whose font size will be set to a specific value

<Styles >

```
#div1 {
background-color: #b6ff00;
font-family: Arial; width: 300px;
height: 700px;
}
#div2 {
background-color: #22f00; font-family: Helvetica;
width: 300px;
height: 400px;
}
#div3 {
background-color: #b6ffa00; font-family: Times new roman;
font-size: 20pt;
width: 100px;
height: 200px;
}
</style>
```

- <div id="div 4"> this is DIV 4</div>
- **<div id="div 3"> this is DIV 3</div>**
- <div id="div 1"> this is DIV 1</div>
- <div id="div 2">this is DIV 2</div>

40. What does the below snippet result in?

```
function aaa() {
return (lookout)
{
Test:1
};}
alert(typeof aaa());
}
```

- Options
- Function
- Number
- Object
- **Undefined**

41. What is the output of below script?

```
const person = {name: "Sarah", country: "Nigeria", job: "Developer"};
person.name="John", console.log(person);
```

Options

- **(name: 'John', country: 'Nigeria', job: 'Developer')**
- (name: 'Sarah', country: 'Nigeria', job: 'Developer')
- Error
- Person

42. What is the output of the following Script?

```
function Car(make, model, year) {
  this.make make;
  this.model = model,
  this.year = year,
}
const auto = new Car('Honda', 'Accord', 1998);
console.log(auto instanceof Car);
console.log(auto instanceof Object);
```

- **True True**
- false false
- True false
- Error

43.

```
type Color="green" | "yellow":
type Quantity = "five"/"x":
#type Item Two = "five item" | "six Item" | "green item" | "yellow item "
type ItemOne = $(Quantity | Color) Item
```

The above code is an example of

- **String literals**
- **Template Literals**
- Object Literals
- Directives

44. What is the output of the below script?

```
let amnumber(10,20,30,40)
```

```
fonet val of am
```

```
console.logival
```

Cenons

- **10 20:30:40**
- 0123
- 10:20
- 30:40

45. What is the output of below script?

```
function sum(x, y, z) {
```

```
return x + y + Z;
```

```
}
```

```
const numbers = [1, 2, 3];
```

```
console.log(sum(...numbers));
```

Options

- **6 //answer**
- 1
- 5
- 7

46. What is the color of the text Sausage?

```
<ul class="shopping-list" id="awesome" <i><span>Milk</span></li>
```

```
<i class-"favorite" id-"must-buy"><span class="highlight">
```

```
Sausage</span></li></ul>
```

```
css.ul {
```

```
color: red;
```

```
}
```

```
li{
```

```
color: blue;
```

```
}
```

options

- Red
- **Blue //answer**
- error
- orange

47. What is the output of the following script?

```
if(i==1) {
```

(lookout)

```
let name='Prakash Patil':
```



```
console.log(name);
```

Options

- Prakash Patil
- **undefined**
- null
- Error name is not defined

48. Please refer below

What would be the color of link1 and link2 when the below webpage rendered on the browser or the first time?

```
<html>
<head>
<style>
    #link1:link {color:maroon;}
    #link2:link {color: yellow;}
    a:visited{color:red;}
    a:hover{color:pink;}
</style>
</head>
<body>
    <a id="link1" href="1.html">Link1</a><br><br>
    <a class="link2" href = "2.html" >Link2</a>
</body>
</html>
```

- **Link1 will be rendered in maroon color and link2 will be rendered in blue color**
- Link1 will be rendered in maroon color and link2 will be rendered in red color
- Link1 will be rendered in maroon color and link2 will be rendered in yellow color
- Console error as anchor tag is not used for styling.

49. Please refer below

Observe the below code:

```
const rollNumber = 1; //----- line 1
var firstName= "Aaliya"; //-----line 2
let course ="JavaScript"; //-----line 3
rollNumber= 2; //-----line4
console.log("Hi "+firstName+", your roll number is "+rollNumber+. Hope you have knowledge on Html5");
```

Which line in the code is likely to throw an error?

- line 2

- line 1
- line 3
- **line 4 //answer**

50. Please refer below

What will be the output of the below code?

```
var line= "Fundamentals of Javascript";
```

```
console.log("Slice (2, 5) ", line. Slice (2, 5));
console.log("Slice (-20,-16) ",line. Slice(-20,-16));
console.log("Slice (18) ",line .slice (18));
```

- (A)

Slice (2,5) : nda

Slice (-20,-16) : Java

Slice (18) : Fundamentals of Ja

- (B)

Slice (2,5) : ipt

Slice (-20,-16) : Java

Slice (18) : vascript

- **©**

Slice (2,5) : nda

Slice (-20,-16) : enta

Slice (18) : vascript

- (D)

Slice (2,5) : ipt

Slice (-20,-16) : Java

Slice (18) : Fundamentals of Ja

51. Multiple options

Please refer below

Given the HTML structure below, which selector will select both the span elements ?

```
<div class="outerDiv">
  <span id="span1" class="innerSpan">
  </span>
```

```
< span id="span2" class="innerSpan">
  </span>
</div>
```

- div#outerDiv span#innerSpan
- **div.outerDiv span.innerSpan //ans**
- div#outerDiv, span.span1 & div#outerDiv, span.span2
- **div.outerDiv span#span1, div.outerDiv span#span2 //ans**

52. Please refer below

What will be the output of the below code?

```
1. function greetUser (isLoggedIn: boolean) {
2.   if (isLoggedIn) {
3.     let message="Welcome to ES6";
4.   }
5.   return message; }
6.   console.log (greetUser (true));
7.   console.log (greetUser (false));
```

- Welcome to ES6 Welcome to ES6
- Welcome to ES6 undefined
- **Error in greetUser()**
- undefined Welcome to ES6

53. Please refer below

What will be the output for the following code?

```
let employee= {
  empId    : 'I1001',
  empName  : {
    firstName: 'John',
    middleName : 'Thomson',
    lastName   : 'Mark'
```

```

    },
    empStatus : 'Active'
};
let (firstName, middleName , lastName) = employee. empName;
console.log(middleName);

```

- **John //ans**
- Thomson
- Mark
- Compilation Error

54. Please refer below

What will be the output of the best code?

```

lat birth_date=new Date("1993-12-11");
console.log("Boolean of null is "Boolean (" "));
console.log("Boolean of ' 'is ", Boolean(""));
consele, log("Variable date cast to Boolean is "Boolean (birth_date])

```

- (A)

Compilation error saying date type cannot be converted to boolean

- (B)

Boolean of null is 0

Boolean of ' ' is 0

Variable date cast to Boolean is true

- (C)

Boolean of null is 0

Boolean of ' ' is 0

Variable date cast to Boolean is false

- (D)

Boolean of null is false

Boolean of ' ' is false

Variable date cast to Boolean is true

55. Geneva have coded the below web page.

```
<html>
<head>
<style>
    h1 {color:blue ;}
    h1,p{color: yellow;}
</style>
</head>
<body>
    <p>1</p>
    <h1>2</h1>
    <p style="color:green">3</p>
    <h1>4</h1>
</body>
</html>
```

What will be the color of 1,2,3, and 4 when rendered

- 1:yellow, 2:yellow, 3:yellow, 4:yellow
- 1:green, 2:yellow, 3:green, 4: yellow
- 1:yellow, 2:green, 3:green, 4: yellow
- **1: yellow, 2: yellow, 3:green, 4:yellow**

56. Which of the following statements are true regarding the code provided?

```
<suite name= "Test Suite" verbose ="1">
  <test name "My Math Test Using TestNG XML">
    <classes>
      <class name="com . mycompany. service FirstTest"/>
    </classes>
  </test>
  <test name="My Echo Test Using Testng xml">
    <classes>
      <class name="com.mycompany. service. SecondTest"/>
    </classes>
  </test>
</suite>
```

- Two tests are defined but only the second test will run Successfully
- Two tests are defined but only the first Test will Run successfully
- Two tests are defined but the case contains errors
- **Two tests are defined and the code will run Successfully**

1. which of the following statements Angular component consists of? a. component class that handles data and functionality. b. An HTML template that determines the UI. c. Component-specific styles that define the look and feel. d. Angular Routing module
2. Which component life cycle hook is called immediately before Angular destroys the directive or component? a. ngOnFinalize() b. ngAfterDestory() c. ngOnDestroy() d. ngBeforeDestroy()
3. Select the new HTML5 layout containers. a. <aside> b. <boxitle> c. <header> d. <footnote>
4. What is a proper syntax to construct a module? a. var variable_name = angular.module("app_name", []); b. var variable_name = angular.module("app_name"); c. var variable_name = angular.module(); d. var variable_name = new module();
5. Given the HTML structure below, which selector will select both the span elements ? <pre><div class="outerDiv"></pre> <pre></pre> <pre></pre> <pre> </pre> <pre></div></pre> div#outerDiv span#innerSpan div.outerDiv span.innerSpan div#outerDiv, span.sparn1 & div#outerDiv, span.span2 div.outerDiv span#span1, div.outerDiv span#span2
6. Which of the following are the valid methods of XMLHttpRequest object? send() open() cancel() abort()

7. Which are the built-in browser validation features?

max
minlength
conversion
currency
required

8. Rahul has completed the development of a website. Rahul needs to add "note" keyword before each paragraph in one of his and Rahul has forgotten to end headings with "full-stop". How can Rahul achieve these requirements?

- a. Using ::before and after ::pseudo class
- b. Using :: first-letter and ::after pseudo class
- c. Using:: before for first requirement and "full-stop" can be added manually in webpage
- d. Rahul cannot achieve this requirement.

9. `const rollNumber = 1;`
`var firstName = "Aaliya";`
`let course = "JavaScript";`
`rollNumber= 2;`
`console.log("Hi "+firstName+", your roll number is "+rollNumber+". Hope you have knowledge on HTML5");`
Which line in the code is likely to throw an error?

line 2

line 1

line 3

line 4

10. Which option of Angular CLI indicates to use inline templates for the project.

--inline

--inline-templates

--inline-temp

--inline-template

11. In the below code snippet, what is the correct statement to be placed at line number 6 to display the courseName?

Line 1: `import { Component } from '@angular/core';`

Line 2: `@Component({` Line 3:

Line 4:

`selector: 'my-app',`

`template:`

Line 5:

<h1>

1

Line 6:

Line 7: </h1>

Line 8:

Line 9: })

Line 10: export class AppComponent

Line 11: {

Line 12: courseName: string "Angular";

Line 13:)

{{ courseName }}

this.courseName

courseName

AppComponent.courseName

12. What will be the output of the below code?

let originalstring="Digital!";

let updatedString1=originalstring.substr(2,3); let updatedString2=originalstring.substring(2,3);
document.write(updatedString1+'
'); document.write(updatedString2);

git g

git git

ggit

gg

13. Which of the following are the features of HttpClient service?

The ability to request typed response objects.

Streamlined error handling.

Request and response interception.

Non Testability

14. Which of the following are ES6 features?

var keyword

let and const keyword

Arrow functions.

Template Literals.

15. Which of the following options are not a predefined pipe?

upper

currency

date

datetime

slice

16. What will be the output of the below code?

```
function sayHi {person: string} {  
  return "Hello,"+ person;  
}  
var customer :string[] = ["Tom", "Harry", "Jack"]; document.body.innerHTML = sayHi(customer);
```

Hello Tom

Hello

Hello Tom Harry Jack

Compilation Error

17. You are creating a web page that requires some text to span across the width of the screen and other pieces of text to be emphasized within those text blocks. Which statements below correctly explain the rules about block and inline elements that will help you build your web page?

You should use block-level elements when you want text or content to go across the width of the browser window or to start on a new line.

You should use block-level elements when you want text or content to align horizontally with an image on a web page

You should use inline-level elements when you want to align or format text without having it start on another line or s the width of a browser window.

You should use inline-level elements when using the <p> tag

18. Rank the elements of the CSS box model from most inner to outer layer.

Padding

Border

Content

Margin

Margin
Border
Padding
Content

Content
Padding
Border
Margin

Content
Border
Padding
Margin

19. What will be the output for the following code?

```
let colors = ['Red', 'Yellow']; let newColors = [...colors, 'Green', colors]; console.log(newColors);
```

['Red', 'Yellow', 'Green', 'Red', 'Yellow']

['Red', 'Yellow']

'Red', 'Yellow', 'Green']

Error

20. Which of the following members can be included in the class?

constructor

properties

methods

modules

21. Which of the following are the valid properties of Promise Object?

state

result

readyState

response

22. Identify which of the following Pseudo classes can be applied to hyperlinks

Cue

Clicked

Hover

Visited

Focus

23. What would be the color of Sample1, Sample2, and paragraph when the below web page is rendered on the browser)

```
<html>

<head>

<style>
h1#id1
{
color:red;
}
p#id1
{
color:blue;
}
</style>

</head>

<body>

<h1 id="id1">Sample1</h1>

<h2 id="id1">Sample2</h1>

<p id="id1">Paragraph</p>

</body>

</html>
```

Red, red and blue respectively

Red, black and blue respectively

Red, black and black respectively

Red, blue and blue respectively

24. Which of the following Pseudo elements can we use to manage styles on HTML documents?

::first-line

::first-letter

::training-letter

::training-line

25.

Consider the following requirement:

When you hover your mouse over the host element (tag), the color of the tag should change. In addition, when the mouse is moved away, the color of the tag should change to his default color. To do this, you need to handle events raised on the host element in the directive class. Which decorator is used to implement this functionality?

@Host

@HostBinding

@HostListener

@Attribute

26. Consider the below given code snippet:

app-routing.module.ts

(path: 'product/:id', component: ProductDetail)

product-list.component.html:

<a [routerLink] = ['/product', product.productId]"> {{product.productilame))</

Select the correct option to read the parameter value in the navigated component in Angular

Parameter value can be read using RouteParams service

Use router navigate() method

Parameter value can be read using ActivatedRoute service

By using route redirects() method

27. Which of the following are the correct statements?

Typescript can be compiled to all major versions of Javascript

Typescript can be compiled to only latest versions of Javascript

Typescript can be used for cross-browser development

Typescript supports only specific browsers like Edge

28. Which of the following are the valid syntax of using ngClass directive to apply CSS classes A B in Angular?

<h1 [ngClass]="A B" >Hello World </h1>

<h1 [ngClass]="['A','B']" >Hello World </h1>

`<h1 [ngClass]='{'A': true, 'B':true}'>Hello World </h1>`

`<h1 [ngClass]="A,B" >Hello World </h1>`

29. The purpose and function of the ADDRESS and CITE tags has changed since HTML 4.01 and are now more suited to RSS syndication purposes Identify the true statements that describes how the function of each tag has changed in HTML5

The Address tag now delimits the contact information of the author or owner of a document allowing you to have one author per web page

The Cite tag now defines a citation or article footer

The Cite tag now defines the title of a work such as a song book, or movie

The Address tag now delimits the contact information of the author or owner of a document or an article allowing you to have multiple authors for a single page

30. Test rest parameter

```
function fun(...input) {  
  let sum = 0;  
  for (let i of input) {  
    sum += i;  
  }  
  return sum;  
}  
  
console.log(fun(1, 2));  
console.log(" " + fun(1, 2, 3));  
console.log(" " + fun(1, 2, 3, 4, 5));
```

a) 3,6,15

b) 3, undefined

c) 3,error

d) 3,6,error

31. In the below code snippet, what is the correct statement to be placed at line 6 to display the 'numbers' array as a key

Line 1: import (Component) from '@angular/core/

Line 2: Component ((

Line 3: selector: 'my-app'

Line 4: template:

Line 5:

Line 6:

Line 7:

Line 8:

Line 9:"

Line 10: export class AppComponent

Line 11: {

Line 12: numbers: any [] = [1,2,3];

Line 13:)

ab

<li *ngFor = 'numbers'>{{ numbers }}

<li *ngFor = 'num of numbers'>{{ num }}

<li "ngFor = '#num of numbers'>{{ num }}

<li "ngFor = 'let num of numbers'>{{ num }}

32. What will be the output for the following code?

```
let employee = {
```

```
  empId : 'I1001',
```

```
  empName : {
```

```
    firstName : 'John',
```

```
    middleName: 'Thomson',
```

```
    lastName : 'Mark'
```

```
  },
```

```
  empStatus : 'Active'
```

```
let {firstName : middleName , lastName} = (middleName); = employee.empName;
```

```
console.log(middleName);
```

John

Thomson

Mark

Compilation Error

33. What will be the output of the below code?

```
import (PipeTransform) from '@angular/core';
```

```
export class Length Pipe implements PipeTransform { length (value: string): string{ return "Length
="+value.length;
}
}
```

length method should be an no args method

The Length Pipe class has not implemented the transform() method of Pipe Transform Interface

Length Pipe class should be using @Component decorator to make it working

The length method should return an numeric value

34. let prices = new Array (67,23,90,67,12,73); prices.push (110); prices.unshift(43);
prices.pop(); prices.unshift(89); prices.shift(); console.log(prices);

[43, 67, 23, 90, 67, 12, 73]

[67, 23, 90, 67, 12, 89]

[67, 23, 90, 67, 12, 73,89]

[110, 67, 23, 90, 67, 12, 89]

35.

Match each element description option to its HTML5 element

A. <main>

B. <aside>

C. <figure>

D. <nav>

A:Defines self-contained content

B:Defines content other than the page content

C:Defines navigation links

D:Defines the main content of a document

☒ A:Defines the main content of a document

B:Defines content other than the page content

C:Defines self-contained content

D:Defines navigation links.....

☒ A:Defines the main content of a document

B:Defines self-contained content

C:Defines content other than the page content

D:Defines navigation links

☒ A:Defines self-contained content

B:Defines the main content of a document

C: Defines content other than the page content

D: Defines navigation links

36. Keith wants to apply margin property to an element with top-margin value of 30px, bottom-margin value of 20px and left and right margin value of 25px. Which among the following will help her in achieving the requirement?

margin: 25px 30px 20px;
margin: 20px 30px 25px;
margin: 20px 25px 30px;
margin: 30px 25px 20px;

37. Which of the following are the valid properties of XMLHttpRequest object?

readyState
status

38. Which of these operations are permitted if you apply Object.freeze() on myObject?(not sure)

Adding a property to a nested object in myObject
Modifying a property of a nested object in myObject

39. Which of the following are the valid HTML5 input types?

Colors
Time
datetime
number

40. Which of the following option describes the setTimeout() function in JavaScript?

It delays the execution of a specified function by the given amount of time

41. Which of the following will map the name of an input parameter "studentinfo" to a field named "students"?

@Input('studentinfo') students

42. means that if we declare a variable (declared but not initialized) at the end of a function, the runtime will hoist it to the top and we will not have any error if we would have used that variable before being declared.

Variable hoisting

43. Which of the following is true about border property?

Border will not appear if border style is not specified

44. Ryan is trying to create an email account. During the registration process, as soon as he focuses on the name field and enters the character to fill his name, he gets a message - 'This name is taken'. We understand that the website is using JavaScript Event-handling to respond to the data entered by Ryan. Which event do you think is triggering the JavaScript code for the account to check the value entered by the user?

onkeyup

45. Which of the following options would correctly set the viewport to make your website look good on all devices?

<meta name="viewport" content="width=device-width, initial-scale=1" />

46. What is the correct statement to bind required and maxlength validators to name field in a reactive form?

name: ['': [Validators.required, Validators.maxLength(10)]]

47. Which CSS code snippet sets the background top vertically and center horizontally in the container and repeat the background vertically?

background: url('bg.jpg) repeat-y center 0px;

48. Which three programming constructs does JavaScript support?

Arrays
Type conversion

Iteration/loopin
49. Lets say you want to capitalize the first letter of each word and transform the rest of the word to lower case. Here the words are delimited by whitespace characters like space, tab, or line-feed character. Which Pipe do you use to achieve this ? TitleCasePipe
50. Which of the following service is used to communicate with a server over the HTTP protocol, to perform CRUD operations? Ans HttpClient
51. Which of the following are the valid attribute-directives in Angular? ngStyle ngClass
52. Which of the following are the valid methods of Array object? reverse() sort() filter()
53. Which object holds key-value pairs and remembers the original insertion order of the keys? Map
54. Assume you have an array myArray = [3, 4, 5, 6, 7]. How could you use that to produce another array timesTwo = [6, 8, 10, 12, 14] ? var times Two = myArray.map(num => num*2)
55. Rest parameters are used when number of parameters are not known
56. Write a statement with <p> element to initialize two variables (empid and salary). <p ng-app=" ng-init="empid='E1001'; salary=7000">
57. Which of the below attributes controls the valid numeric boundaries of a range control? Min Max
58. Which of the following JavaScript function executes asynchronously? Ans async functionName);
59. Which function is used to encode a Uniform Resource Identifier (URI) by replacing each instance of certain characters by one, two or three escape sequences representing the UTF-8 encoding of the character? encodeURIComponent()
60. Which of the following Pseudo elements can we use to manage styles on HTML documents? ::first-line ::first-letter
61. Which of the following attribute directive adds and removes CSS classes on an HTML element? ngClass
62. Which of the following members can be included in the class? Constructor properties methods
63. Given an array myArray = [15,18, 21,3,6,9, 12] what is returned by the following operation? myArray.sort() Options [12, 15, 18, 21, 3, 6, 9] [3, 6, 9, 12, 15, 18, 21] (12, 9, 6, 3, 21, 18, 15]

- (21, 18, 15, 12, 9, 6, 3]

64. Which of the following functions invoked on a string variable "myString" will replace the first occurrence of the sub-string "WebDevelopment" to "programming"?

`myString.replace("WebDevelopment", "programming");`

`myString.replace(webdevelopment/i, "programming");`

`myString.replace("webdevelopment", "programming");`

`myString.replace First("WebDevelopment", "programming");`

65.

Geneva have coded the below web page.

`<html>`

`<head>`

`<style>`

`h1 {color: blue;}`

`h1,p{color: yellow;}`

`</style>`

`</head>`

`<body>`

`<p>1</p>`

`<h1>2</h1>`

`<p style="color:green">3</p>`

`<h1>4</h1>`

`</body>`

`</html>`

What will be the color of 1,2,3, and 4 when rendered on browser?

1:yellow, 2:yellow, 3: yellow,4:yellow

1:green, 2:yellow, 3:green, 4:yellow

1:yellow, 2:green, 3:green, 4: yellow

1:yellow, 2:yellow, 3:green, 4: yellow

66. Predict the output of the below code snippet.

```
class Demo {  
  n :number= 9;  
  
  check () {  
    if(this.n=9) (  
      let n:number= 10;  
      console.log(this.n);  
    }  
  }  
}  
let demoObject=new Demo ();  
  
demoObject.check();
```

9

10

11

Compilation error

67. The _____ tag defines an alternate content to be displayed to users that have disabled scripts in their browser or have a browser that doesn't support script.

<noscript>

<script>

<notscript>

<js>

68. Which of the following option will print the details of Bag and Unit for the below code?

```
let myBag = {  
  name : "Gucci",  
  type : "Hand Bag",  
  price : 2300,  
  
  mfgUnit :  
  {  
    country : "Canada",  
    quantity : 100,  
  }  
}  
  
function printBagDetails()  
{
```

```

console.log("Name :", his.name);

console.log("Type :", this.type);

console.log("Price :", this.price);
}

function printUnitDetails()
{

console.log("Country : “ ,this.country);
console.log("Quantity: “, this.quantity);
}

```

Output

```

Name: Gucci
Type: Hand Bag
Price: 2300
Country: Canada
Quantity: 100

```

69. Describe some of the essential objectives of the Font-face property.

Font-face property allows us to use multiple @font-face rules to construct font families

Font-face property allows us to link to the styles that are automatically fetched

Font-face property allows us to use multiple @font-face rules to construct font formats

Font-face property allows us to link to the fonts that are automatically fetched

70. Specify some of the essential new features that were added to HTML5.

We can now extend web pages with custom semantics using HTML5

Next generation, bi-directional communication can be facilitated in HTML5 using WebSockets

Canvas was introduced in HTML5 which supports 3-dimensional drawing surfaces

Forms 3.0

71. Typescript compiler configuration is stored in-----file

package.json

tsc.json

tsconfig.json

lint.json

72. Which of the following are the valid statements with respect to HTt builds?(not sure)

Uses AOT compilation

Files are tree shaken

Source maps are generated

Css are extracted to js files

Uses JIT compilation

73. Consider the code snippet given below and find out which line will result in an error.

```
class Person {  
  
  constructor (name, age) {  
    this._pname = name;  
    this._page = age;  
  }  
  display() {  
  
    console.log("name :" + this.pname);  
    console.log("age:" + this.page);  
  }  
  
  set pname (namePerson) { this._pname namePerson; }  
  get pname () { return this._pname;}  
}  
let per = new Person();           // line 8  
per.pname = "Rexha";             // line 9  
console.log(per.pname);           //line 10  
let perl = new Person("Alex", 10); // line 11  
  
console.log(perl.pname):          // line 12
```

Line no 8 gives error as no arguments have been passed to the constructor

Line no. 10 gives error as this is not the way to fetch the instance variable of a class

Line 10 displays undefined and line 12 displays Alex

Line 10 displays Rexha and line 12 displays Alex

74. Which of the following options are correct statements to apply the CSS class 'pink' to the paragraph element?

<p [ngClass]='pink'> background is pink</p>

<p [ngClass]='{pink:true}'> background is pink</p>

<p [ngClass]='["pink"]'> background is pink</p>

<p [ngClass] (pink:true)>background is pink</p>

75. Which of the following is a valid command to create a component in Angular?

ng generate component <component-name>

ng create component <component-name>
ng create <component-name>
ng generate <component-name>

76. What will be the output of the below code?

```
let productDetails : {string, number};  
productDetails = ["Samsung Galaxy J7", 16,8, "Samsung Galaxy J5", 32];  
for (let i=0;i< productDetails.length;i++){  
  console.log (productDetails[i]);
```

Samsung Galaxy J7 16 Samsung Galaxy J5 32

Samsung Galaxy J7 16 8 Samsung Galaxy J5 32

Samsung Galaxy J7 Samsung Galaxy J5 16 8

Compilation Error

77. The <article></article> and <section> </section> tags are new to HTML5. Identify the true statements that describes how each is used in HTML5

- a. Article elements or tags in HTML5 can be used as CSS styles that group images on a web page that are separated by chunks of content
- b. Article elements or tags in HTML5 can be used as RRS syndication and applied to dynamic content such as forum posts, blog posts, comments, and news stories
- c. The Section tag in HTML5 defines and isolates tables and images that have food wrapped around them
- d. The Section tag in HTML5 defines and isolates sections or logical chunks of content in a document such as an Introduction, header, or footer

78. Select valid states of Promise Object

pending

fulfilled

rejected

completed

79. Please refer below

Chris has created a highlight directive as shown below. What color will be applied to the page if he enters yellow in

```
// highlight.directive.ts  
import { Directive, ElementRef, Input } from '@angular/core';
```

```
@Directive({  
  selector: '[highlight]' // Corrected the selector to a string  
})
```

```
export class HighlightDirective {
```

```
constructor(el: ElementRef) {  
  el.nativeElement.style.backgroundColor = 'blue'; // Corrected variable name to el  
}  
}
```

//app. component.

template:

<label>Provide color to be applied on the page</label>

<input type='text' [(ngModel)]='colorName'>

<div highlight>Hello </div>

- a. color red will be applied to the div element
- b. color yellow will be applied to the div element
- c. yellow color will not be applied as there is no binding of model colorName and highlight
- d. color blue will be applied to the div element

1. Which data type is returned by the DataProvider method in TestNG?

Object[]
Object
Object[] [] []
Object[][]

2.

Which statements are valid implementations of Implicit and Fluent wait classes?

(A)
Wait<WebDriver> wait =
new FluentWait<WebDriver> (driver) .
withTimeout(Duration.ofSeconds(30)) .
pollingEvery(Duration.ofMillis(500)) .
ignoring(NoSuchElementException.class);
(B)
Driver.manage().implicitlyWait().timeouts(10, TimeUnit. SECONDS);
(C)
Wait<WebDriver> wait =
new WedDriverWait<WebDriver>(driver) .
withTimeout(Duration.ofSeconds (30)) .
pollingEvery(Duration.ofMillis (500)) .
ignoring (NoElementException.class);
(D)
driver.manage (). timeouts().implicitlyWait (10, TimeUnit. SECONDS);

3.

. Which type of given code is really helpful in selecting two or more items in a list box or text area?

Note: All the options of dropdown are in td tag

(A)
List<WebElement> options =
select.findElements (By.tagName("td"));
Action multipleSelect =
builder.keyDown (Keys.CONTROL) .
click (options.get(2)).click(options.get(4)).
click (options.get(6)).build();
multipleSelect.perform();
(B)
List<WebElement> options =
select.findElement(By.tagName("td"));
Action multipleSelect =
builder.keyDown(Keys.CONTROL).click(options.get(2)) .
click(options.get(4))
click(options.get(6)).build();
multipleSelect.perform();
(C)
Select dropdown=new Select (driver.findElement(By.tagName("td")));
dropdown.selectByVisibleText("item1");

```
dropdown.selectByVisibleText("item2");  
dropdown.selectByVisibleText("item3");
```

(D)

```
Select dropdown = new Select (driver.findElements(By.tagName("td")));  
dropdown.selectByVisibleText("item1");  
dropdown.selectByVisibleText ("item2");  
dropdown.selectByVisibleText("item3");
```

4.

Which of the following @Test attribute creates dependency in TestNG?

dependsOnClasses

dependsOnMethods

dependsOnGroups

dependsOn Tests

5.

Which Kubernetes feature automatically scales the number of Pods in a deployment, replica set, or stateful set based on specific metrics like observed CPU utilization?

- a. Application Load Balancing
- b. Bin Packing Autoscaling
- c. Horizontal Pod Autoscaling
- d. Vertical Pod Autoscaling

6.

Refer the code below and choose the output of the following code.

```
Actions builder = new Actions (driver);  
Action actions = builder.moveToElement (addButton).click().build();  
actions.perform();
```

- a) Mouse over on the addButton Element.
- b) click on the addButton Element.
- c) The element is clicked by keyboard Actions.
- d) Mouse over on the addButton and click on Element

7.

Which of the following are the features of Docker Swarm?

- a) Cluster management
- b) Decentralized design
- c) Multi-host networking
- d) Non Scalability

8. What requirements that must be met in order to successfully install Docker Desktop on a Windows 10 Enterprise system?

- a) TFTP Client
- b) NTP Server
- c) Hyper-V backend
- d) WSL 2 feature

9. Specify some of the methods that we can use to handle keyboard events.

- a) contextClick
- b) keyDown
- c) sendKeys

| |
|--|
| d) doubleClick |
| <p>10. After applying a Docker Swarm service configuration, you need to run a Linux command in a container to verify the configuration was applied. Which base command should you use?</p> <p>a) docker container exec</p> <p>b) docker service verify</p> <p>c) docker container verify</p> <p>d) docker service exec</p> |
| <p>11. Which AWS storage class is used for long-term archiving?</p> <p>a) Standard</p> <p>b) Intelligent tiering</p> <p>c) Glacier</p> <p>d) IA</p> |
| <p>12. Select some of the benefits associated with containerization.</p> <p>a) With containerization we can create multiples copies on several machines</p> <p>b) StringBuilder(StringBuilder str)</p> <p>c) Containerization encapsulates applications</p> <p>d) Containerized applications can be automatically deployed on on-premise servers</p> |
| <p>13. You need to view the hostnames of Docker Swarm cluster computers. Which command will achieve this?</p> <p>a) docker swarm ls</p> <p>b) docker cluster nodes ls</p> <p>c) docker node ls</p> <p>d) docker swarm nodes ls</p> |
| <p>14. Which CLI command is used to deploy an EC2 instance?</p> <p>a) aws create-instances</p> <p>b) aws ec2 create-instances</p> <p>c) aws run-instances</p> <p>d) aws ec2 run-instances</p> |
| <p>15. Which of the following are parameters associated with multithread in testng.xml?</p> <p>a) multi-thread</p> <p>b) pool-size</p> <p>c) thread-count</p> <p>d) parallel</p> |
| <p>16. Which of the following are TestNG annotations?</p> <p>a) @Listeners</p> <p>b) @Source</p> <p>c) @AfterGroups</p> <p>d) @Connection</p> |
| <p>17. Identify the valid commands that we can use to build images from Dockerfile.(multi)</p> <p>a) sudo docker build image</p> <p>b) sudo build</p> <p>c) sudo docker create</p> <p>d) sudo docker build</p> |
| 18. What is the output of given code snippet? |

```

import org.testng.annotations.Test;
public class DependsOnTest
{
    @Test
    public void OpenBrowser() {
        System.out.println("Opening The Browser");
    }
    @Test (dependsOnMethods = { "SignIn", "OpenBrowser" })
    public void LogOut() {
        System.out.println("Logging Out");
    }
    @Test
    public void SignIn() {
        System.out.println("Signing In");
    }
}

```

- a) Error in the order of depend Method names in the list.
- b) Signing In
Opening The Browser
Logging Out
- c) Error!We can include Multiple Dependent Tests.
- d) Opening The Browser
Signing In
Logging Out

19. You need to ensure uploaded files to an S3 bucket are encrypted. What should you do?

- a) Import a PKI certificate to KMS
- b) Import a PKI certificate to ACM
- c) Enable encryption in the upload properties section
- d) Do nothing; encryption cannot be set during uploads

20. Which type of storage service is Amazon S3?

- a) Block
- b) Object
- c) Simple
- d) Secure

21. Which environment variables need to be set in the docker-compose.yaml file?
multi

- a) MYSQL_ROOT_PASSWORD
- b) MYSQL_DB_HOST
- c) WORDPRESS_DB_PATH
- d) WORDPRESS_DB_USER

22. A _____ window is a secondary browser window that appears on top of the main browser window.

- a) Simple Alert
- b) pop-up
- c) Confirmation Alert

| |
|--|
| d) Prompt Alert |
| <p>23. Choose the ways to disable the test cases in TestNG. multi</p> <p>a) Disable the test case in a TestNG Annotation.</p> <p>b) Disable the test case in a @Test annotation</p> <p>c) Disable the test case using @Skip tag.</p> <p>d) Disable the test case in the XML file.</p> |
| <p>24. List some of the features of Kibana. multi</p> <p>a) Delete data</p> <p>b) Data discovery</p> <p>c) Dashboard</p> <p>d) Ingest data</p> |
| <p>25. Which of the following visualizations can be created and managed using Kibana? multi</p> <p>a) Split pane</p> <p>b) Vertical bar</p> <p>c) Dashboards</p> <p>d) Pie charts</p> |
| <p>26. Which of the following commands can we use to set up a private Docker Hub repository?multi</p> <p>a) docker push localhost:5000/ubuntu</p> <p>b) docker run -d -p 5000:5000 --restart always --name registry registry:2</p> <p>c) docker info</p> <p>d) docker pull registry</p> |
| <p>27. _____ handles the current window that uniquely identifies it within this driver instance.</p> <p>a) driver.getWindowHandle();</p> <p>b) driver.getWindowHandles();</p> <p>c) driver.getWindows();</p> <p>d) driver.getWindow();</p> |
| <p>28. What is the purpose of Kibana Beats?</p> <p>a) To collect and ship data to Elasticsearch</p> <p>b) To create custom visualizations</p> <p>c) To manage Kibana settings</p> <p>d) To monitor Elasticsearch performance</p> |
| <p>29. Which wait has scope through out the driver instance?</p> <p>a) Implicit wait</p> <p>b) Explicit wait</p> <p>c) Fluent Wait</p> <p>d) ThreadWait</p> |
| <p>30. Point out the correct statements. multi</p> <p>a) Amazon Elastic Cloud is a system for creating virtual disks(volume)</p> <p>b) SimpleDB interoperates with both Amazon EC2 and Amazon S3</p> <p>c) EC3 is an Analytics as a Service provider</p> <p>d) EC3 is not an Analytics as a Service provider</p> |
| <p>31. What will happen if you use the following code in your script?</p> <pre>driver.manage ().timeouts().setScriptTimeout(-10, SECONDS);</pre> |

a) The script will be allowed to run indefinitely b) Sets the amount of time to wait for an asynchronous script to finish execution before throwing an error c) Sets the amount of time to wait for an synchronous script to finish execution before throwing an error d) The script will wait for 10 seconds
32. The default Elasticsearch log level is a) Error b) Trace c) Warn d) Info
33. An _____ is where end users access services located at AWS. a) Region b) Availability Zone c) Edge Location d) AWS Local Zones
34. _____ is a tool for defining & running multi-container Docker applications. a) Docker Compose b) Docker File c) Docker Volume d) Docker Network
35. which of the following are used to select a drop down? multi a) By value b) By Visible Text c) By Index d) By attribute
36. You plan on using the CLI to deploy a Windows EC2 instance. How can you specify that you want a Windows Server 2022 instance? a) Use the -os parameter b) Specify a Windows Server 2022 AMI id c) Use --windows parameter d) Deploy the instance using Fleet Manager
37. Which of the following command is used to switch between frames? a) driver.switch To().frame() b) driver.switch().frame() c) driver.switch To() d) driver.transition().frame()
38. Which Docker command shows the containers in a Swarm that are running on Swarm nodes? a) docker service rm b) docker service ps c) docker service scale d) docker service create
39. By default _____ of types is enabled in Elasticsearch. a) dynamic mapping b) static mapping

c) complex mapping d) URL
40. Which of the following is not an annotation used in TestNG? a) @BeforeSuite b) @BeforeClass c) @AfterTest d) @AfterException
41. Which code snippet would work to create a list of links in a page? (A) <pre>ArrayList links = driver.findElements (By. tagName ("a")); for(WebElement link : links) { System.out.print(link.getAttribute("href")); } </pre> (B) <pre>List<String> links driver.findElements (By.tagName("a")); for (WebElement link : links) { System.out.print(link.getAttribute("href")); } </pre>
42. Which of the following are features of ELK stack? multi a) Corporate branding b) Visualization c) Querying d) Monetization
43. Select the benefits of EC2. multi a) Elastic b) Expensive c) Reliable d) Not flexible e) Secure f) un-controlled
44. AWS has the concept of a _____, which is a physical location around the world where we cluster da a) Region b) Availability Zone c) Edge Location d) AWS Local Zones
45. Choose the correct Syntax to disable the test using the TestNG @Test enable parameter. a) @Test(enable=false) b) @Test(enable=false} c) @Test(enabled=false) d) @Test(enabled=false}
46. _____ helps to gain powerful insights about data

- a) Elasticsearch
- b) Kibana
- c) Logstash
- d) beats

47. Which of the following is not a Kibana visualization option for time-based data?

- a) Timelion
- b) Vega
- c) Line chart
- d) Geo heatmap

48. Choose the advantages of Selenium and Jenkins integration.(multiple)

- a) Jenkins can schedule your tests to run at specific times//ans
- b) With the help of Jenkins, you can save the history of the test cases and also generate periodic test reports//ans
- c) Jenkins also supports Maven and testing of applications with continuous integration.//ans
- d) Reduce the execution time//ans

49.

Specify some of the essential features of Docker.(multiple)

- a) Simplifies infrastructure automation//ans
- b) Simplifies configuration automation//ans
- c) Helps isolate infrastructure from configuration//ans(check)
- d) Provides tools to simplify DevOps with images//ans

50.

What is the output of the following code?

```
public class Tutorial Example {  
    Test (groups = "group")  
    public void testMethodiForGroupA() {  
        System.out.println("Running test method of group");  
        @Test(groups = "groupA")  
        public void testMethod2ForGroupA() {  
            System.out.println("Running test method of group");  
            Assert.fail();  
            @Test(dependsOnGroups = "group")  
            public void dependent TestOnGroupA() (  
                System.out.println("Running the dependent test");
```



```

    }
    }
    a.
    testMethod1ForGroupA-Pass
    testMethod2ForGroupA-Fail
    dependent TestOnGroupA-Pass
    b.
    test Method1ForGroupA-Pass
    test Method2ForGroupA-Fail dependent
    TestOnGroupA-Fail

    c.
    test Method1ForGroupA-Pass
    testMethod2ForGroupA-Fail
    dependent TestOnGroupA-Skipped
    d.
    testMethod1ForGroupA-Pass
    test Method2ForGroupA-Pass
    dependent TestOnGroupA-Pass

```

51.

Select all the distributions used by AWS CloudFront.(multiple)

a) Web//ans

b) FTP

c) RTMP//ans

d) SMTP

e) POP3

52.

```
<suite name="Test Suite" verbose="1">
```

```

    <test name="My Math Test Using TestNG XML">
        <classes>
            <class name="com.mycompany.service.FirstTest"/>
        </classes>
    </test>

    <test name="My Echo Test Using TestNG XML">
        <classes>
            <class name="com.mycompany.service.SecondTest"/>
        </classes>

```

</test>

</suite>

Option:

- a) Two tests are defined but only the second test will run successfully
- b) Two tests are defined but only the first test will run successfully
- c) Two tests are defined but the code contains errors//ans
- d) Two tests are defined and the code will run successfully

53.

Which of the following is a correct relationship between Regions, Availability Zones, and Edge Locations?

- a) Data centers contain regions
- b) Regions contain Availability Zones.//ans
- c) Availability Zones contain edge locations
- d) Edge locations contain regions

54.

.....global wait which means the wait is applicable for the entire driver.

- a) driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);//ans
- b) WebElement DynamicElement = (new WebDriverWait(driver, 10)).until(ExpectedConditions.presenceOfElementLocated(By.id("DynamicElement")));
- c) Wait wait = new FluentWait(driver) with Timeout(60, SECONDS) pollingEvery(10, SECONDS).ignoring(NoSuchElementException)
- d) Thread.Sleep(1000);

55.

Recall the benefits for automated testing provided by Selenium.(multiple)

a) Runs parallel tests on a single machine//ans

b) Works with all popular programming languages//ans

c) Runs tests under multiple browsers//ans

d) Provides simple to use test management features//ans

56.

What are the attributes of @Test in TestNG?(Multiple)

a) Enabled//ans

b) Name

c) Order

d)Groups

57.

Amazon S3 stores data as objects within resources called?

Auto Scaling

EC2 Instances

Buckets//ans

Amazon RDS

58.

Which file contains the commands to build a Docker image?

dockerfile//ans

docker-compose.yml

requirements.txt

app.py

59.

A Docker service named payroll 1 is currently deployed to a two-node swarm. You add an additional swarm node and would like a service replica hosted on the new node. Which two methods can achieve this?(multiple)

a) Use-replicas global with the docker service command

b) `docker service create-- replicas 3 payroll1//ans`

c) `docker service scale payroll1 = 3//ans`

d) Use-mode global with the docker service command

60.

Which commands can we use to set up custom bridge networks that can be used by Docker?(multiple)

a) bridge configure bro

b) `sudo ip link set dev br0 up//ans`

c) `ip addr add bridge br0`

d) `sudo ip link set dev bridge0 up//ans`

61.

You have disabled the "Block all public access" option in an S3 bucket hosting a static website yet website continues to receive "Access Denied" messages. Why is this happening?

a) Users must authenticate with an IAM account

b) The site certificate has expired

c) `S3 objects permissions are overriding bucket permissions`

d) A bucket policy was not configured//ans

62.

The current directory contains a Dockerfile that will be used to create an image containing Linux, Apache, MySQL and PHP. The image should be labelled as lampapp1. Which command achieves this?

a) `docker build -t lampapp1//ans`

b) `docker build - lampapp1`

c) `docker image new -l lampapp1.`

d) `docker create -t lampapp1.`

63.

Identify the valid selenium webdriver wait.(multiple)

a) Explicitly Wait//ans

b) ThreadWait

c) FluentWait//ans

d) Implicitly Wait//ans

64.

Which CLI commands can be used to create a new S3 Bucket?(multiple)

a) aws s3api make-bucket

b) aws s3 cb

c) aws s3api create-bucket//ans

d) aws s3 mb//ans

65.

Amazon S3 stores data as objects within resources called?

a) Auto Scaling

b) EC2 Instances

c) Buckets//ans

d) Amazon RDS

66.

Which of the following is not valid locator in Selenium?

a) Id//ans

b) CssSelector

c) ElementName

d) TagName

67.

Which states are you primarily interested while using a declarative cluster management approach?(multiple)

- a) Current state//abs
- b) Baseline state
- c) Default state
- d) Desired state//ans

68.

What are valid reasons for using containers?(multiple)

- a) Containers cannot be unit or integration tested
- b) Containers are fixed and cannot be rolled back
- c) Containers by design are scalable//ans
- d) Containers are highly portable//ans

69.

Which systems can Beats ship data to?

- a) Logstash//Ans
- b) Apache
- c) Kibana //Ans
- d) Elasticsearch//Ans

70.

Which of the following command is used to return all container details?

- a) docker ps
- b) docker containers
- c) docker ps -q
- d) docker ps -a //ans

71.

Which of the following is an edge-storage or content-delivery system that caches data in different physical locations?

- a) Amazon Associates Web Services
- b) Amazon Relational Database Service
- c) Amazon CloudFront//Ans

d) Amazon SimpleDB

72.

Which of the following provides API to ship the data from source, configure input options and implement logging?

- a) X-Pack
- b) Beats
- c) Kibana
- d) Logstash

73.

Which of the below are valid commands to refresh browser(driver is instance of WebDriver class)?

- a) `driver.navigate().refresh();`//Answer
- b) `driver.switch To().main().refresh();`
- c) `driver.switchTo().main(URL).refresh();`
- d) `driver.findElement(By.xpath("//body")).sendKeys(Keys. F5);`

74.

Select the Get command which fetches the inner text of the element that you specify in Web driver Selenium.

- a) `get in Text()`
- b) `getinner Text()`
- c) `getText()`//Ans
- d) `get inner_ Text()`

75.

What are valid reasons for using containers?

- A) Containers cannot be unit or integration tested
- B) Containers are fixed and cannot be rolled back
- C) Containers by design are scalable//ans
- D) Containers are highly portable//ans

76.

Select the steps involved in scrolling on web pages using Selenium WebDriver.

Call the `executeScript()` method of `JavaScriptExecutor`

Create an instance of Selenium WebDriver

Create an instance of JavaScriptExecutor

Locate the element by calling the method findElementById()

77.

What components are installed along with the Docker Desktop when we install Docker on Windows 10?

- a) Kubernetes
- b) Docker compose//Answer
- c) Docker assemble
- d) Docker machine//Answer

78.

Which technique is used in Web driver Selenium to check whether a specific checkbox, radio button, or option in a drop-down box is selected?

- a) isSelected()//Answer
- b) selected()
- c) textSelected()
- d) isSelect()

79.

is the central application in the AWS portfolio.

- a) Amazon Simple Queue Service
- b) Amazon Elastic Compute Cloud//Answer
- c) Amazon Simple Notification Service
- d) Amazon Relational Database Service

80.

Adding or creating a document into a type within an index of Elasticsearch is called operation.

- a) Indexing//Answer
- b) deleting
- c) updating
- d) getting

81.

Which port is used for Docker Swarm cluster management?

- a) 2388
- b) 7946
- c) 7950
- d) 2377//Answer

82.

Which of the following are complex datatypes that are supported by Elasticsearch?

- a) Array//Answer
- b) Object//Answer
- c) Nested//Answer
- d) Number

83.

What is the abstract class available in Selenium?

- a) WebDriver//Answer
- b) By
- c) WebElement
- d) getText()

84.

What is the role of IAM?

- a) Allows you to set permission, roles, and users.//Answer
- b) Allows you to stores data as objects.
- c) Allows you to manage users and their access.//Answer
- d) It allows only specific users to use specific services.//Answer

85.

List some of the prominent technological use cases of Docker.

- a) Infrastructure isolation//Answer
- b) Configuration simplification//Answer
- c) Single tenancy
- d) Rapid deployment//Answer

86.

What is the purpose of a Kibana index pattern?

- a) To define how data is mapped to Elasticsearch//Answer
- b) To create custom visualizations
- c) To define search queries
- d) To define index settings

87.

Which of the following command format the response of request in JSON format?

- a) curl -XGET http://localhost:9200/catalog/products/1? pretty=true//Answer
- b) curl -XGET http://localhost:9200/catalog/products/1 ?format=json
- c) curl -XGET http://localhost:9200/catalog/products/ 1 ?type=json
- d) curl -XGET http://localhost:9200/catalog/products/1 ?format-type=son

88. Which capability of X-Pack helps to configure fields and document level security? a) Graph b) Reporting c) Alerting d) Security//Answer
89. Applications and services that run on a distributed network using virtualized resources is known as b) Parallel computing c) Soft computing d) Distributed computing e) Cloud computing//Answer
90. Which Docker network treats each container as its own physical network device? a) Overlay b) Ipvlan c) Host d) Bridge e) Macvlan//Answer
91. Select the method which is used to do drag and drop in actionsClass. a) dragAndDrop(Source, offset) b) dragAndDrop(Xoffset,yOffset) c) dragAndDrop(Source, target)//Answer d) dragandMove To(source, target)
92. What is the use of Amazon S3? a) Auto Next b) Hosting websites c) Storing and retrieving data//Answer d) Managing servers e) Monitoring application performance
93. _____ is a visualization tool for analyzing time-series data and allows executing advanced mathematical calculations. b) Dashboard

- c) DevTools
- d) **Timelion**//Answer
- e) Plugins

94.

Which axis name is associated with the selection of the parents and grandparents in selenium?

- b) Following
- c) **Ancestor**//Answer
- d) Parent
- e) Preceding

95.

Amazon Web Services falls into which of the following cloud-computing category?

- a) Platform as a Service
- b) Software as a Service
- c) **Infrastructure as a Service**//Answer
- d) Back-end as a Service

96.

Which of the following command is used to destroy the instance of WebDriver?

- a) **driver. Quit()**//Answer
- b) driver.close()
- c) driver.abandon()
- d) driver.clear()

97.

) Which of the following is the correct syntax for implicit wait of 5 Secs (d is the instance of Webdriver)?

- a) **d.manage().timeouts() .implicitlyWait(5, TimeUnit.SECONDS)**//Answer
- b) d.timeouts() .manage ().implicitly Wait(5, TimeUnit SECONDS)
- c) d.timeouts() implicitly Wait(5, TimeUnit. SECONDS)
- d) d.timeouts() webDriverWait.implicitlyWait(5, TimeUnit.SECONDS)

98.

Which one of the following runs on Xen Hypervisor?

- a) Azure
- b) **AWS EC2**//Answer
- c) AWS EC3
- d) AWS RDS

99.

Most analysis tools depend upon

- a) **structured data**//Answer
- b) unstructured data
- c) graph data

d) complex data

100.

What is the return type of driver getCurrentUri()?

- a) List
- b) Character
- c) String[]
- d) String//Answer

101.

Which of the following request fetches all documents from all indices from the cluster in Kibana?

- a) GET / search//Answer
- b) GET / searchall
- c) GET / all
- d) GET / documents

102.

Which of the following are keywords of Docker?

- a) build
- b) ship//Answer
- c) run//Answer
- d) deploy

103.

Why does the AWS Console show "Global" when working with S3 buckets?

- a) S3 Bucket names must be globally unique//Answer
- b) S3 Buckets use the AWS global backbone network for uploads
- c) S3 Buckets are replicated globally
- d) S3 Buckets are not tied to a specific region

104.

Which of the following annotation supports data-driven testing?

- a) @DataProvider//Answer
- b) @DataSet
- c) @ Provider

105.

Which of the following are Elasticsearch APIs?

- Document APIs
- Search APIs
- Indices APIs
- Divide APIs

106.

Which of the following is the default TestNG report output folder? testoutput test-output test testresult
114. What is default port of Elasticsearch? 9200 5601 8080 8081
115. What benefit is derived from using the AWS Storage Gateway? Ability to migrate on-premises VMs to AWS Reduced network latency when accessing S3 content Ability to migrate VMs from other cloud providers to AWS Reduced network latency when accessing on-premises content
116. What are the benefits of TestNG? Ability to use annotation Ability to test in parallel Advanced testing interface A wide range of programming language support
117. You need to verify that a Docker secret is available in a Linux-based container. Which Linux file system should you check? /run/secrets /tmp/secrets /tmp/security /var/run

118.

Which of the following statement is wrong about Amazon S3?

- a) Amazon S3 provides large quantities of reliable storage that is highly protected
- b) Amazon S3 is highly available//Answer
- c) Amazon S3 is highly reliable
- d) Amazon S3 is an object storage service

119.

What does the following code snippet do?

```
Actions act = new Actions (driver); // where driver is WebDriver type
act.moveToElement(WebEiement).perform();
act.contextClick () perform ();
```

- a) Drag and Drop
- b) Left click
- c) Right click//Answer
- d) Moves the selection

120.

Which of the following component adds security, monitoring, alerting, reporting and graph capabilities to Elastic Stack?

- a) Logstash
- b) Kibana
- c) X-Pack//Answer
- d) beats

121.

Which file contains the commands to build a Docker image?

- a) dockerfile
- b) docker-compose.yml//Answer
- c) requirements.txt
- d) app.py

122.

Which of the following are valid beats built by elastic team?

- a) Packetbeat//Answer
- b) Filebeat//Answer
- c) Heartbeat//Answer
- d) Microbeat

123.

What are the different types of instances?

- a) General purpose//Answer
- b) Compute Optimized//Answer
- c) Storage Optimized//Answer
- d) Hardware Optimized
- e) Software Optimized

124.

Which file is the main node configuration file for Elasticsearch?

- a) elasticsearch.yml//Answer
- b) elasticsearch.config
- c) jvm.options
- d) log4j2.properties

125.

Which WebDriver method should be used to switch the focus to a browser window, a frame, or an alert?

- a) switchTo()//Answer
- b) setFocus()
- c) setTo()
- d) applyFocus()

126.

Which of the following annotation will be invoked before the execution of each test method?

- a) @BeforeMethod//Answer
- b) @BeforeClass
- c) @Before
- d) @Before Test

127.

Which of the following are the advantages of AWS?

- a) Flexibility//Answer
- b) Cost-effectiveness//Answer
- c) Scalability//Answer
- d) Portability

128.

Which command is used to show the Docker version information in an easy to read layout?

- a) docker version//Answer
- b) docker v

- c) docker details
- d) docker show version

129.

Which of the following command is used to get the details of Docker?

- a) **docker info//Answer**
- b) docker help
- c) docker details

130.

List some of the features of Kibana that extend the capability of Kibana dashboards.(lookout)

- a) **Export dashboards//Answer**
- b) **Manage indexes//Answer**
- c) Delete visualizations
- d) **Generate PDF reports without using pluginsocker show//Answer**

131.

Which of the following statements are true about Docker containers?

- a) **Containers are lightweight//Answer**
- b) **Fewer resources are used//Answer**
- c) **Booting of containers is very fast//Answer**
- d) More resources are used

132.

Choose the testing type that is not supported by Selenium.

- a) Re-testing
- b) Regression Testing
- c) Functional Testing
- d) **Volume Testing//Answer**

133.

Which Docker command is used to display detailed information on one or more networks?

- a) docker network metadata
- b) **docker network inspect//Answer**
- c) docker network details
- d) docker network meta

134.

The Elasticsearch default communication port is

- a) **9300/tcp//Answer**
- b) 3000/ip
- c) 9300/utp
- d) 3000/top

135. _____ is the packaging of software code with all it's necessary components like libraries, frameworks, and other dependencies own "container OneN(not perfect que check it) a) Containerization//Answer b) Virtualization c) Orchestration d) Shipping
136. Which of the following method is used to reject the alert depending on the alert type? a) cancel() b) clear() c) dismiss()//Answer d) reject()
137. Which of the following is not an aws cloud platform service? a) Analytics b) Networking//Answer c) Mobile Servicos d) Internet of Things
138. Which of the following options are correct with respect to creating Docker containers? a) I can build my own image and use it for creating a Docker container.//Answer b) I can pull the image from Docker hub and then create a Docker Container.//Answer c) I can create a Docker container without Docker image. d) I can create more than one container using same Docker Image//Answer
139. Where does the Docker daemon stores the Docker images? a) Docker Client b) Docker Host c) Docker Registry//Answer d) Docker Hub
140. Specify some of the steps that we need to define while writing test cases to automate the process of double-click operation

Call `action.doubleClick(ele).build().perform()` to invoke double-click

Use Actions `action = new Actions(driver)` to define Action

Call the `action.click(click)` method twice

Set `doubleClick` Event handler to `True`

141.

Select the options which are true about `getWindowHandles()`.

It will return window handles for all opened windows

It can have duplicate values

It will return window handles for current opened window

It can't have duplicate values

142.

You have deployed a Docker service across a three-node swarm. Which command shows the service container ID on the?

`docker ps`

`docker ps show`

`docker service ls`

`docker service show`

143.

. Which of the following are benefits of Elasticsearch?

Searching

Analytics

Near real time

Non scalable

144.

List some of the considerations that we have to keep in mind when installing Kibana.

Elasticsearch and Kibana must be of different versions

Elasticsearch should already be installed to successfully start Kibana

Logstash should already be installed before installing Kibana

Elasticsearch and Kibana must be of the same version

145.

Which of the following is the deployment model?

Public

Private

Hybrid

Corporate

Standard

146.

Which of the following is not a major type of cloud computing usage?

Platform as a Service

Software as a Service

Hardware as a Service

Infrastructure as a Service

147.

Which port does SSH use?

25

23

22

3389

148.

What is the return type of getWindowHandles method in webdriver?

Set<String>

Set<int>

List<String>

List<Int>

149.

. ____ are the preferred mechanism for persisting data generated by and used by Docker containers

Docker Volumes

Docker File

Docker Compose

Docker Network

150.

. ____ Is a real-time, distributed search, and analytics engine

1. Elasticsearch

2. Kibana

3. Logstash

4. ELK

151.

Which of the following statements are true about TestNG grouping?
(Multiple Choice)

1. Users can specify groups containing other groups

2. Users can group multiple test methods into a named group

3. Users can specify which method to exclude from group

4. Users can use @BeforeClass to execute pre-condition before running any group methods

152.

Select the Dockerfile commands that we can use to manage images.(Multiple Choice)

1. RUN

2. VOLUME

3. ENTRY

4. MAINTAIN

153.

____ command is used to scale the services.

1. `docker-compose up -d --scale service_name=4`
2. `docker-compose up -d --increase service_name=4`
3. `docker-compose up -d scale e service_name=4`
4. `docker-compose up -d service_name=4`

154.

Which statements are valid when we want to delay execution and locate elements using XPath?

(Multiple Choice)

1. `thread.sleep(long millis)`
2. `findElement(By.xpath) (string id)).click()`
3. `findElement(By.xpath)(string xpathExpression)).click()`
4. `sleep.thread(long millis)`

155.

Which of the following perform transformations on data?

1. Filter plugins
2. Logging Plugin
3. Search Plugin
4. Monitor Plugin

156.

Which of the following are features of Docker?(Multi Choice)

1. Docker has the ability to reduce the size of development by providing a smaller footprint of the operating system via containers.
2. With containers, it becomes easier for teams across different units, such as development, QA and Operations to work seamlessly across applications
3. You can deploy Docker containers anywhere, on any physical and virtual machines and even on the cloud
4. Docker Containers are not easily scalable

157.

You have enabled a CloudFront distribution for an existing Elastic Beanstalk web application. What should you instruct the web app developers to change on the app?

1. Origin URL references
2. The PKI certificate
3. TCP timeout settings
4. The caching TTL

158.

Which of the following framework will have the below disadvantages?

1. Require coding skills
2. It takes more time to set up the framework and test data

1. Linear Scripting Framework
2. Data-Driven Framework
3. Modular Testing Framework
4. Behavior-driven Testing Framework

159.

What is the output of the following code?

```
import org.testng.annotations.Test;
public class TestNG (
@Test (priority = 1)
public void b_method(){
    System.out.println("This is B method");
}
@Test (priority = 1)
public void a_method() {
    System.out.println("This is A method");
}
@Test
public void d_method() {
    System.out.println("This is D Method");
}
@Test
public void c_method() {
    System.out.println("This is C Method");
}
}
```

1.This is B method
This is A method
This is D method
This is C method

2.This is D method
This is C method
This is B method
This is A method

3.This is C method
This is D method
This is B method
This is A method

4.This is A method
This is B method
This is C method
This is D method

160.

Identify the valid Docker commands that we can use to manage containers.(Multi Ans)

1. docker container import
2. docker container attach
3. docker container export
4. docker container associate

161.

Select the code which helps you to locale the element that is having type as input and id as 'q'.

1. driver.findElement(By.xpath("//input[@id='q']"))
2. driver.findElement(By.XPath("//input[id='q']"))
3. driver.findElement(By.XPATH("//input[@id=q]"))
4. driver.findElement(By.xpath("//id='q'))

162.

Select some of the critical wait types that are provided by Selenium WebDriver. (Multi Ans)

1. Thread wait
2. Implicit wait
3. Fluent wait
4. Wait explicit

163.

Identify the network drivers that we can use to set up container networking in Docker. (Multi Ans)

1. Macvlan
2. Route
3. Overlay
4. Localhost

164. .

- 1.

165.

Identify the valid statements that we can use to create a Wait instance and apply conditions while locating web elements. (Multi Ans)

1. `wait.until(ExpectedConditions.visibilityOf(driver.findElement(By.xpath(string xpathExpression)))`
2. `wait.until(ExpectedConditions(driver.findElement(By.xpath(string xpathExpression)))`
3. `WebDriverWait wait = new WebDriverWait(driver, 120)`
4. `WebDriverWait wait = new WebDriverWait(driver, 120)`

1.

166.

Which method is used when you want to verify whether a certain check box, radio button or option in a drop-down box is selected in Web driver Selenium?

1. `Is_Selected()`
2. `isSelected()`
3. `isSelected()`
4. `is_Select()`

167.

Choose the correct code to verify the page title.

```
1.
public class TestTitle {
    @Test
    public void myTest (WebDriver d) {

        d.get("https://www.google.com/");
        String ActualTitle=d.getTitle();
        String ExpectedTitle="Google";
        Assert.assertEquals (ExpectedTitle, ActualTitle) ;
    }
}
```

```
2.
public class TestTitle(
    @Test
    public void myTeat (WebDriver d) {

        d.get("https://www.google.com/")
        String ActualTitle=d.getsTitle();
```



```
String ExpectedTitle="Google";
Assert.assertEquals (ExpectedTitle, ActualTitle);
}
}
```

3.

```
public class TestTitle{
@Test
public void myTest (WebDriver d) {

d.get("https://www.google.com/");
String ActualTitle=d.getPageTitle();
String Expected Title="Google";
Assert.assertEquals (ExpectedTitle, ActualTitle);
```

4.

```
public class TestTitle {
@Test
public void myTest (WebDriver d) {

d.get("https://www.google.com/");
String ActualTitle=d.getTitle();
String ExpectedTitle="Google";
}
}
```

168.
_____ is the central application in the AWS portfolio.
Ans **EC210.**

169.
Which commands can we use to set up custom bridge networks that can be used by Docker?
Ans **sudo ip link set dev br0 up**
sudo ip link set dev bridge up

170.
What are some of the essential components of Docker?
Ans **Docker engine**
Container runtime

171.
You have deployed a Docker service across a three-node swarm. Which command shows the service container ID on the local Docker host?
Ans **docker ps** docker

172.

Which of the following are benefits of TestNG multithreading?

Ans Reduces execution of testing time

Increase test result accuracy

Test multiply browsers and operation systems simultaneously.

173.

Which of the following specify the hardware or software load over web servers, that improves the efficiency of the server as well as the application?

Ans Load Balancing

174.

Which of the following is not valid locator in Selenium?

Ans Element name

175.

What are the benefits of TestNG?

Ans Ability to use annotation

Ability to test in parallel

176.

breakdown fields values of a document into a stream, and inverted indexes are created and updated using these values, and these stream of values are stored in the document

Ans Tokenizer

177.

Identify the valid selenium webdriver wait.

Ans ExplicitlyWait

FluentWait

ImplicitlyWait

178.

Which of the following options are correct with respect to creating Docker containers?

Ans I can build my own image and use it for creating a Docker container.

I can pull the image from Docker hub and then create a Docker Container.

I can create more than one container using same Docker Image.

179.

List some of the features of Kibana that extend the capability of Kibana dashboards.

Ans Export dashboards

Generate PDF reports without using plugins

180.
Specify some of the steps that we need to define while writing test cases to automate the process of double-click operations. Ans <code>Call action.double Click(ele).build().perform)</code> to invoke double-click <code>Use Actions action = new Actions(driver) to define Action</code>
181.
Which of the following is not a major type of cloud computing usage? <code>Ans Hardware as a Service</code>
182.
What are the attributes of @Test in TestNG? <code>Ans Enabled Groups</code>
183.
You are installing Docker on Ubuntu 20.04LTS. Which command downloads the latest version of the community edition of Docker Engine, Docker CLI, and containerd? Ans <code>sudo apt-get install -y docker-ce docker-ce-cli containerd.io</code>
184.
Docker compose uses _____ file to configure application services Ans <code>docker-compose.yml</code>
185.
What is the return type of getWindowHandles method in webdriver? Ans <code>set string</code>
186.
How do you determine the Administrator password for a Windows EC2 instance? Ans <code>Refer to the password you specified when launching the instance</code>
187.
_____ is a static wait and it is not a good way to use in scripts as it will wait without condition. Ans <code>Thread Sleep(1000);</code>
188.
You need to verify that a Docker secret is available in a Linux-based container: Which Linux file system should you check? <code>Ans /run/secrets</code>
189.
Which method selects the option at the given index?

```
selectedByIndex()  
selectByIndex()  
selectByIndexValue()  
selectIndex()
```

1. In the case of stored procedures, which statements are true regarding IN parameters?

IN Parameter is the default parameter

IN Parameter value should be passed to the procedure for processing

The same parameter will serve as input and output

The IN parameter is a variable whose value will change in the invoked procedure.

Give the correct option for above

2. A servlet sets a session scoped attribute product with an instance of com.example.product and forward to a jsp. Which of the following options output the name of the product in the response

Package com.example

```
public class Product
```

```
{
```

```
private String name;
```

```
private double prices;
```

```
Public Product() {
```

```
}
```

```
public String getName()
```

```
{
```

```
return name;
```

```
}
```

```
public void setName (String name) {
```

```
this.name= name;
```

```
}
```

```
public double getPrice() {
```

```
return price;
```

```
}
```

```
public void setPrice(double price)
```

```
{
```

```
this.price =price;
```

```
}
```

```
}
```

OPTION

A.<%= product.getName()%>

B.<jsp:useBean id="product" class="com.example.Product" /> <%= product.getName() %>

C.<jsp:useBean class="com.example.Product" id="product" scope="page"> <%= product.getName()%></jsp:useBean>

D.<jsp:useBean id="product"

type="com.example.Product"scope="page"/><%=product.getName()%>

E.<jsp:useBean id="product" type="com.example.product"> <%=product.getName()%>

3. Please refer below

What will be the output of the following code snippet?

```
<%page language="java"%>

<html>

<head[</head>

<Body>

<java.util.Date now =new java.util.Date(); %>


<H1><%= now.getHours() %>: <%= now.getMinutes()%>:<now.getSeconds() %></H1>
</body>
</html>
```

It will display the date as mm:dd:yy

It will display the date and time

It will display time as hh/mm/ss

It will display the current time as hh:mm:ss

What is output for this

4. Multiple Choice (Select 1 out of 4 options, for the question below)

How will you begin a database transaction?

Asking a Transaction object to you Connection, and calling the method begin() on it.

Asking a Transaction object to you Connection, and setting the autoCommit property of the Transaction to false.

Calling the method begin Transaction() on the Connection Object

Setting the autoCommit property of the Connection to false, and execute an statement in the database

5.

Assume that we have hwo entity classes created for validating the Citizen's voter details. As per the rules, a citizen is supposed to have only one voter identity with Which among the following mappings is true with respect to One to One Mapping?

Option D.

@Entity @Table(name = "Citizen")

public class Citizen

{

@Id private int citizenId;

private String citizenName;

@OneToOne (cascade CascadeType ALL)

@JoinColumn (name="citizenVoterId", unique private Voter citizenVoter; true)

```

//getters and setters
}
@Entity @Table(name = "Voter")
public class Voter
{
@Id private int citizenVoter Id;
private String location;
//getters and setters
}

```

6. Which of these can use the context:component-scan XML element for its bean configuration activities Multiple Choice (There may be more than 1 correct answer)

FileSystemXmlApplicationContext

AnnotationConfig WebApplicationContext

AnnotationConfigApplicationContext

ClassPathXmlApplicationContext

7. During initialization, a servlet finds that the database connection it requires is unavailable. As the system designer, you know that the database start up completes several minutes after the web-container initializes the servlet. How should this be handled?

Retry the connection until it is successful, then allow the init() method to complete.

Throw a ServletException

Throw the IOException

Throw an UnavailableException

8. Consider the following code:

```
<html><body>
```

```

<jsp:useBean id="address" class="AddressBean" scope="session" />
state <jsp:getProperty name="address" property="state" />
</body></html>

```

Which of the following options are equivalent to the third line?

```
<% state = address.getState(); %>
```

```
<% out.write("state = "); out.print(address.getState()); %>
```

```
<% out.print("state = "); out.print(address.getstate()); %>
```

```
<% out.print("state = address.getState()); %>
```

```
state = <%= address.getState() %>
```

9. How can we hook initialization or clean up code in Spring?

Using custom init() and destroy() methods

Using interfaces InitializingBean and DisposableBean

Using <bean> tag

It is not possible

10. Evaluate a SELECT statement:

```
SELECT first_name, last_name FROM physician WHERE physician_id NOT IN (SELECT physician_id  
FROM physician WHERE license_no 17852);
```

Which one of the following SELECT statement would achieve the same result?

(A)

```
SELECT first_name, last_name FROM physician WHERE physician_id IN (SELECT physician_id FROM  
physician WHERE license_no=17852);
```

(B)

```
SELECT first_name, last_name FROM physician WHERE physician_id != ALL (SELECT physician_id  
FROM physician WHERE license_no 17852);
```

(C)

```
SELECT first_name, last_name FROM physician_id WHERE license_no <> 17852 AND license_no IS  
NOT NULL;
```

(D)

```
SELECT first_name, last_name FROM physician WHERE physician_id = 17852;
```

11. Multiple Choice (Select 1 out of 4 options, for the question below)

A relation named Students has columns for StudentID and EmailID, both of which are guaranteed to be unique for a given student. Which of the following column specifications is most appropriate?

Choose varchar(255) for EmailID and mark StudentID as a PRIMARY KEY

Designate both StudentID and EmailID as NOT NULL UNIQUE

Designate both StudentID and EmailID as PRIMARY KEYS

Designate the StudentID as PRIMARY KEY, and the EmailID as NOT NULL UNIQUE

12. Please refer below

You are creating a web form with this HTML:

```
<form action="sendorder.jsp">
```

```
<input type="text" name="creditCard"> 13. <input type="text" name="expirationDate">
```

```
<input type="submit">
```

```
</form>
```

Which HTTP method is used when sending this request from the browser?

GET

PUT

POST

SEND

FORM

13. What are the appropriate methods for a Hibernate Session Object?

save()

load()

get()

remove()

destroy()

14. To create a database only if it doesn't exist, which clause is used? A

IF EXISTS

IF NOT EXISTS

CREATE EXISTS

EXISTS IF

15. You need to retrieve the username cookie from an HTTP request. If this cookie does NOT exist, then the c variable will be null Which code snippet must be used to retrieve this cookie object?

Cookie c= request.getCookie("username");

(B)

10. Cookie c= null;

11. Cookie[] cookies request.getCookies();

12. for (int i = 0; i < cookies.length; i++){

13. if (cookies[i].getName().equals("username")){

14. c cookies[i];

15. break:

16 }}

(C)

```
10 Cookie c=null;
```

```
11. for (Iterator i= request.getCookies ():
```

```
12. i.hasNext()) {
```

```
13. Cookie o=(Cookie) i.next();
```

```
14. if (o.getName().equals("username")) {
```

```
16. break;
```

```
17}}
```

```
10.)}}
```

```
D. 10. Cookie c = null;
```

```
11. Cookie[] cookies = request.getCookies();
```

```
12. for (int i = 0; i < cookies.length; i++) {
```

```
13. if (cookies[i].getName().equals("username")) {
```

```
14. c = cookies[i];
```

```
15. break;
```

```
}
```

```
}
```

Give the correct ans

16. What will be the outcome of running the following jsp code?

```
<body>
```

```
<%
```

```
int i = 1 , j = 10 ;
```

```
do {
```

```
if(i > j) {
```

```
break;
```

```
j --;
```

```
} while (++i < 5) ;
```

```
System.out.println("i="+i+" and j= :+j);  
%>
```

```
</body>
```

i= 6 and j = 5

i = 5 and j = 5

i = 6 and j = 4

1=5 and j = 6

17.

Please refer below

Consider the following JSP code snippet:

```
EmployeeBean john = (EmployeeBean)request.getAttribute("john");  
if (john == null)  
{  
    harry =new EmployeeBean(); request.setAttribute("john", john);  
}
```

Which one of the following is an equivalent action code for the above code?

© <jsp:declareBean id = "john" class = "EmployeeBean" scope = "request" />

<jsp:createBean id = "john" class = "EmployeeBean" />

<jsp:useBean id = "john" class = "EmployeeBean" scope= "request" />

<jsp:useBean id = "john" class = "EmployeeBean" />

18. Which one of the following is a valid declaration of a Tag Library in a Deployment Descriptor?
(A)

<taglib>

<uri>http://www.javabeat.net/studyKit</uri>
<location>/studykit.tld</location> </taglib>

(B)

<taglib>

<uri>http://www.javabeat.net/studyKit<uri>
<location>/studyKit.tld</location>

</taglib>

(C)

<taglib>

<taglib-uri>http://www.javabeat.net/studyKit</taglib-uri>

<taglib-location>/studykit.tld</taglib-location> </taglib>

(D)

<taglib>

<uri>http://www.javabeat.net/studykit</uri>

<path>/studykit.tld</path>

</taglib>

18.

Which operator is used with the condition to check for values in the list?

IN

LIKE

OR

AND

19. Consider the following scenario:

An existing JSP page is modified.

The following steps happen when the modified page is requested:

(a) jspInit()

(b) jspDestroy() (c) translation

(d) Load the class

(a) jspService () (f) Create an Instance

(g) Compile the translated code

Which of the following option gives the correct sequence in which the above steps happen?

acg,f,d,e,b

a,b,c,d,e,f,g

cg.df,a,c,a

cg.t,d,a,e,b

20. Multiple Choice (Select 1 out of 4 options, for the question below)

How will you access a servlet context initialization parameter named "company-name"?

`S(init.company-name}`

`S(init['company-name'])}`

`S(initParam("company-name"))}`

`$(contextParam.company-name}`

What will be the output when we request two times to the following servlet?

```
public class TestDemo implements Servlet {
```

```
    public TestDemo() {System.out.println("Servlet Constructor");
```

```
    public void init(ServletConfig config) throws ServletException {System.out.println("Init Method");
    }
```

```
    public void destroy() { System.out.println("Destroy Method");
    }
```

```
    public void service (ServletRequest request, ServletResponse response) throws ServletException,
    IOException {System.out.println("Service Method");
    }
```

```
    public ServletConfig getServletConfig() {return null;
    }
```

```
    public String getServletInfo() {
    return null;
    }
```

Servlet Constructor

Init Method

Service Method

Service Method

Destroy Method

```
package com.test;
```

```
public class MobileService{
    private long mobileNumber;
```

```
    private String model;
```

```
    //default constructor
```

```
    public void MobileService(){}//parameterized constructor
```

```
    public void MobileService (long mobileNumber, String model)
    { this.mobile Number = mobileNumber;
```

```
}
```

```
this.model = model;
```

```
//setter method
```

```
public void setMobileNumber(long mobileNumber) {this.mobileNumber = mobileNumber;  
}
```

```
public void setModel(String model) {this.model = model;  
}}
```

Assume below given Spring configuration:

```
<!--Bean definitions -->
```

```
<bean id="mobileService1" class="com.test. MobileService" />
```

```
<bean id="mobileService2" class="com.test. MobileService" > <constructor-arg  
value="1234567891" /> <constructor-arg value="Nokia" />
```

```
</bean>
```

```
<bean id="mobileService3" class="com.test MobileService"> <property name="mobileNumber"  
value="1234642323" /> <property name="model" value="Nokia" /> </bean>
```

Choose the RIGHT options.

Spring does not allow both constructor and setter injection together in the configuration

fine, it is allowed to use constructor and setter injection together based on need.

Code works works It is mandatory to define default constructor explicitly in this code example as there exists a parameterized constructor

In mobileService3 bean definition, it is optional to specify name attribute in the property tag.

ADVANCE JAVA- JSP

1.Consider the following JSP page sources:

```
<%-- file one.jsp begins here--%>
<%@page import ="java.util.*" contentType="text/html" session="true"%>
<html>
<body>
<%ArrayList al=new ArrayList(0);
al.add("Jack Russell");
al.add("Labrador");
al.add("Great Dane");%>
<@include file="two.jsp"%>
</body>
</html>

<%--file two.jsp begins--%>
<%@page import="java.util.*" contentType="text/html"%>
<table>

<%for (int i=0;i<al.size(0);i++) {%>
<r><td> <%=al.get(i) %></td></tr>
<%}%>

</table>
<%-- two.jsp ends here--%>
```

Which two of the following statements are true about the above two JSP page sources, given that client is requesting one.jsp?

- a) Translation will fail because the import attribute of the page directive is repeated across the page source.
- b) The order of the import and contentType attributes in both JSP page sources is immaterial..**
- c) Removing the Session attribute from one.jsp will make no difference to the generated servlet code**
- d) The local variable al in two.jsp will not be recognized
- e) The import,contentType,and session attributes should appear in separate page directives.

2. Which three of the following statements are NOT TRUE about JSP scripting elements?

- a) It is illegal to embed a <%-- style comment inside another comment.**
- b) It is illegal to embed an expression inside a scriptlet

c) It is legal to embed an expression inside a declaration

d) It is legal to embed an expression inside a directive

e) It is legal to include a declaration at any point in the JSP page source, provided that it appears outside of other elements.

3. Consider the following code:

```
<html><body>
<% x=3; %>
<% int x=5; %>
<%! int x=7; %>
x = <%=x%>, <%=this.x%>
</body></html>
```

Which one of the following will be the correct output of the above code?

a) x = 3, 5

b) x = 3, 7

c) x = 5, 3

d) x = 5, 7

e) Compilation error

4. Which one of the following exceptions may be thrown by the `_jspService()` method?

a) `javax.servlet.ServletException`

b) `javax.servlet.jsp.JSPException`

c) `javax.servlet.ServletException` and `javax.servlet.jsp.JSPException`

d) `javax.servlet.ServletException` and `java.io.IOException`

e) `javax.servlet.jsp.JSPException` and `java.io.IOException`

5. Which one of the following is a correct way to pass a parameter equivalent to the query string `user=rani` at request time to an included component?

a) `<jsp:include page="other.jsp" >`
`<jsp:param paramName="user" paramValue="rani" />`
`</jsp:include>`

b) `<jsp:include page="other.jsp" >`
`<jsp:param name="rani" value="user" />`
`</jsp:include>`

c) `<jsp:include page="other.jsp" >`
`<jsp:param value="rani" name="user" />`
`</jsp:include>`

- d) `<jsp:include page="other.jsp" >`
`<jsp:param param="user" value="rani"/>`
`</jsp:include>`
- e) `<jsp:include page="other.jsp" >`
`<jsp:param user="rani" />`
`</jsp:include>`

6. Consider the following scenario:

Statement A: `<jsp:include/>` standard action inserts the response of a jsp page at runtime.

Statement B: include directive inserts the source of a jsp page at translation time.

Which one of the following is correct with respect to the above?

- a) Statement A is true and B is false
- b) Statement A is false and B is false
- c) Statement A is true and B is true**
- d) Statement A is false and B is true

7. Which two of the following statements are true?

- a) The default value of `isThreadSafe` attribute of page directive is true.**
- b) If `isThreadSafe` attribute of page directive is set to true, then JSP container dispatches request for the page sequentially.
- c) When `isThreadSafe` attribute of page directive is set to true, a thread is created for each request for the page.**
- d) Setting `isThreadSafe` attribute to true for JSP pages, can lead to poor performance.
- e) The default value of `isThreadSafe` attribute is false.

8. Vijay does not want his programmers to use scripting elements in their JSP pages. Select from the following, the one correct deployment descriptor programmers can code to achieve the intended result.

- a) `<jsp-config>`
`<jsp-property-group>`
`<url-pattern>jsp</url-pattern>`
`<scripting-invalid>`
`true`

```
</scription-invalid>
</jsp-property-group>
</jsp-config>
```

b) <jsp-config>

```
<jsp-property-group>
<url-pattern>jsp</url-pattern>
<scripting-invalid>
    false
</scription-invalid>
</jsp-property-group>
</jsp-config>
```

c) <jsp-config>

```
<jsp-property-group>
<ur-pattern>*.jsp</url-pattern>
<scripting-valid>
    true
</scription-valid>
</jsp-property-group>
</jsp-config>
```

d) <jsp-config>

```
<jsp-property-group>
<ur-pattern>*.jsp</url-pattern>
<scripting-invalid>
    true
</scription-invalid></jsp-property-group>
</jsp-config>
```

9. Given a web page that is requested with the query string ?username=yogesh, which two of the following can result in the output in the page of the string yogesh?

a) `${param.username}`

b) `${request.yogesh}`

c) `${param[yogesh]}`

d) `${request["yogesh"]}`

e) `${param["username"]}`

10. What will be the result of running the following jsp file taking into account that the Web server has just been started and this is the first page loaded by the server?

```
<html><body>
<%= request.getSession(false).getId() %>
</body></html>
```

Select one answer from the following.

- a) It will not compile
- b) It will print the session id**
- c) It will produce a NullPointerException as the getSession(false) method's call returns null, cause no session had been created.
- d) It will produce an empty page.

11. Consider the following JSP page source:

- 1) `<%@ page import="java.io.*"%>`
- 2) `<html>`
- 3) `<body>`
- 4) `<%`
- 5) `PrintWriter out=response.getWriter();`
- 6) `out.write("P");`
- 7) `%>`
- 8) `<%out.write("Q");%>`
- 9) `</body>`
- 10) `</html>`

Which two of the following are true statements about the above code?

- a) "P" will be written to the response, followed by "Q"
- b) Only "Q" will be written to the response
- c) The page has a compilation error because of a directive syntax error
- d) In jsp technology it is a bad idea to get hold of the PrintWriter directly from the response**
- e) The page has a compilation error for other reasons**

12. The page directive is used to convey information about the page to JSP container. Which two of the following are legal syntax of page directive?

- a) `<% page info="test page" %>`
- b) `<%@ page info="test page" session="false"%>`**
- c) `<%@ page session="true" %>`**
- d) `<%@ page isErrorPage="errorPage.jsp" %>`
- e) `<%@ page isThreadSafe=true %>`

13. Consider the following scenario:

An existing JSP page is modified.

The following steps happen when the modified page is requested:

- (a) `jspInit()`
- (b) `jspDestroy()`
- (c) translation
- (d) Load the class
- (e) `_jspService()`
- (f) Create an Instance
- (g) Compile the translated code

Which one of the following options gives the correct sequence in which the above steps happen?

- a) a,b,c,d,e,f,g
- b) a,c,g,f,d,e,b
- c) c,g,f,d,a,e,b
- d) c,g,d,f,a,e,a**

14. Consider the following JSP code:

```
<%@ page language="java" %>
<%@ page import="java.util.*" %>
<html>
<body>
<h1>Welcome</h1>
<%! Public void jspInit()
{
System.out.println("First half of jspInit()");
%>
<% new Date()%>
<%! System.out.println("second half of jspInit()");
}
%>
</body>
</html>
```

Which one of the following will be the correct outcome of attempting to access the above JSP for the second time?

- a) Translation error(HTTP response code of 500)
- b) Page not found error(HTTP response code of 404)
- c) Web page returned showing a heading and the current date; two lines of output written to the server log
- d) Web page returned showing "First half of jsplnit()," "Second half of jsplnit()," a heading, and the current date
- e) Web page returned showing a heading and the current date**

15. Consider the following JSP code:

```
<%@page language="java"%>
<html>
<body>

<%! int x;%>
<%! public void jspDestroy()
{
System.out.println("self-destructing");
}%>
<%!
Public void jsplnit()
{
System.out.println(<%= x %>);
}%>
</body>
</html>
```

Which one of the following is a valid reason for the above JSP failing to translate and compile?

- a) Expression embedded in declaration**
- b) Data member x not initialized before use
- c) Local variable x not initialized before use
- d) Placement of jspDestroy() method before jsplnit() method
- e) The page actually compiles and translates without any problem

**16. What would be the output of attempting to access the following JSP page?
(Select two answers)**

```

<%@ page language="java"%>
<html>
<body>

<%! Public void _jspService(HttpServletRequest request, HttpServletResponse
response)
{
out.write("A");
}%>

<%out.write("B");%>

</body>
</html>

```

a) Cannot resolve symbol compilation error

- b) "A" is output to the response
- c) "B" is output to the response
- d) "A" is output to the response before "B"

e) Duplicate method, compilation error

17. Consider the following JSP page:

```

<html>
<body>
<%!public String methodA()
{
return methodB();
}
%>
<%!public String methodB()
{
return methodC();
}
%>
<% public String methodC() //to put <%! public String methodC()
{
return "method c is returned";
}
%>
<h2><%=methodA()%></h2>
</body>
</html>

```

Which one of the following is the result of attempting to access the above JSP page?

- a) "method c is returned" is output to the resulting web page.

b) A translation error occurs

- c) A runtime error occurs.
- d) The text between the <h1></h1> html tags appears, followed by a java stack trace
- e) The web page is blank

18.Fill in the blanks:

____(1)____ belongs to HttpJspPage interface which inturn extends _____(2)_____ which declares methods such as _____(3)_____ and _____(4)_____.

Select one answer from the following.

- a) (1) jspInit() (2) servlet (3) _jspService() (4) jspDestroy()
- b) (1) _jspService() (2) HttpPage (3) jspInit() (4) jspDestroy()
- c) (1) jspInit() (2) JspPage (3) _jspService() (4) jspDestroy()
- d) **(1) _jspService() (2) JspPage (3) jspInit() (4) jspDestroy()**

19. Which one of the following is the correct syntax to include a 1.2 Applet in your output?

- a) <% plugin type="applet" code="com.rocky.TestApplet.class" %>
- b) <jsp:plugin type="applet" code="com. rocky.TestApplet.class"/>**
- c) <%@ page:plugin type="applet" code="com. rocky.TestApplet.class" %>
- d) <jsp:applet code="com. rocky.TestApplet.class"/>
- e) None of the listed options
- f)

20. Which two of the following statements are true?

- a) <%! int i; %> is a variable declaration that has object scope within the servlet**
- b) <% int i = 0; %> is a variable declaration that has local scope within the service method of the servlet.**
- c) <% int i = 0; %> is a variable declaration that has local scope within the doGet method of the servlet.
- d) <%! int i; > a variable declaration that has local scope within the service method of the servlet
- e) <% int i; %> is a variable declaration that is class scope to the service method.

21. Consider the following scenario:

Anil wants to create an web application.

He is planning to use web components for creating the application.

He is planning to use a model 2 architecture.

Which two of the following are valid suggestions with respect to the above?

- a) Use JSP for processing data
- b) Use Servlets for processing data**
- c) Use JSP for displaying response to the user**
- d) Use Servlet for Displaying response to the user
- e) Use xhtml for processing data

22. Raju wants to implement thread-safe JSP page. Which one of the following gives the correct syntax to be used in his JSP page?

- a) `<%@ page isThreadSafe="true"%>`
- b) `<%@ page isThreadSafe="false"%>`**
- c) `<%@page extends="multithreadedModel"%>`
- d) `<%@page implements="singleThreadedModel"%>`

23. Choose one statement from the following that best describes the relationship between JSP and servlets.

- a) Servlets are built on JSP semantics and all servlets are compiled to JSP pages for runtime usage.
- b) JSP and servlets are unrelated technologies.
- c) Servlets and JSP are competing technologies for handling web requests. Servlets are being superseded by JSP, which is preferred. The two technologies are not useful in combination.
- d) JSPs are built on servlet semantics and all JSPs are compiled to servlets for runtime usage.**

24. Which one of the following gives the correct initial contact point for handling a web request in a Page-Centric architecture?

- a) A JSP page.**
- b) A JavaBean.
- c) A servlet.
- d) A session manager.

25. Which three of the following are correct Page Directives?

- a) `<%@ page import="java.io.*,javax.naming.*" %>`**
- b) `<%@ page errorpage=true %>`

c) `<%@page import="java.io.File"%>`

`<%@page import="java.util.*"%>`

d) `<%@include file="/portal/banner.jsp"%>`

e) `<jsp:include page="/portal/header.jsp" />`

26. Which one of the following gives the correct sequence of steps involved in JSP lifecycle?

a) `jspinit()`, `jspdestory()`, `_jspservice()`, translation, compilation, load class, create an instance

b) `jspinit()`, `_jspservice()`, `jspdestory()`, translation, compilation, load class, create an nstance

c) translation, compilation, load class, create an instance, `jspinit()`, `_jspservice()`, `jspdestory()`

d) translation, compilation, create an instance, load class `jspinit()`, `jspdestory()`, `_jspservice()`

27. Which one of the following comments used in JSP page will be displayed in the response to the user?

a) `<!--this is html comment-->`

b) `<%-- this is jsp comment--%>`

c) `<%/* this is java comment */%>`

d) `<% this is jspscriptlet comment%>`

28. Consider the following JSP code:

```
<html><body>
```

```
<%
```

```
Integer count = (Integer) request.getSession(false).getAttribute("count");
```

```
if(count != null )
```

```
{
```

```
    out.println(count);
```

```
}
```

```
else request.getSession(false).setAttribute("count", new Integer(1));
```

```
%>
```

```
Hello!
```

`</body></html>`

Which one of the following will be printed for the very first request to the above page as well as the web application that contains this page?

- a) It will print Hello!
- b) It will print Hello and will set the count attribute in the session.**
- c) It will throw a NullPointerException at request time.
- d) It will not compile.

29. Which one of the following implicit variables should be used by a jsp page to access a page initialization parameter?

- a) pageContext
- b) application
- c) config**
- d) context
- e) page

30. What will be the output of the following JSP code snippet at run-time when it is accessed the third time?

```
<% int test = 0; %>
```

```
<% ++test; %>
```

The value of test is

```
<%= test %>
```

(select one answer)

- a) The value of the 'test' variable will be 2.
- b) The value of the 'test' variable will be 0.
- c) The value of the 'test' variable will be 1.**
- d) The variable 'test' must be declared at global scope, else a run-time error will occur.

31. Which three of the following will compile and execute without error in a JSP page?

a) `<%! public String cat(String word1, String word2){ return word1 + " " + word2; } %>`

b) `<% <!-- out.print("hello"); --> %>`

c) `<% System.out.println("hello"); %>`

d) `<%! System.out.println("hello"); %>`

e) `<%= "Marcus Green" %>`

f)

32. Which three of the following are implicit objects available to every JSP page by default?

- a) session
- b) context
- c) system
- d) out
- e) request

33. Which two of the following statements are true?

- a) The `jspInit` method has access to the `ServletConfig` and `ServletContext` objects.
- b) The `jspInit` method is called before each time `_jspService` is called
- c) The `jspInit` method is called once and only once for a servlet instance
- d) An overridden `jspInit` method will not be called by the container

34. Consider the following JSP code:

```
<html>
<head>
</head>
<body>
  <% for (int i = 0; i < 5 ; i++)
    { %>
      out.print(i);
    <% } %>
</body>
</html>
```

Which one of the following will be printed in the generated response?

- a) This page will not compile.
- b) It will compile but will not generate any output.
- c) It will print 0 to 4.
- d) It will throw an exception at runtime
- e) `out.print(i); out.print(i); out.print(i); out.print(i); out.print(i);`

35. Which one of the following is the proper way to include `java.util` package in your jsp page?

- a) `<page:import package="java.util">`
- b) `<%@ page import="java.util.*" %>`
- c) `<%= import java.util.*; %>`

- d) `<%@ import class="java.util.*" %>`
- e) `<%@ page import="java.util.*"; %>`
- f)

36. Which three of the following are INVALID directives?

- a) `<%@page isELIgnored="false"%>`
- b) `<%@page session="true"%>`
- c) `<%@page contentType="image/music/text"%>`
- d) `<%@include uri="header.jsp"%>`
- e) `<%@taglib prefix="mytags" uri="<%@taglib prefix="mytags" uri="http://www.osborne.com/taglibs/mytags"%>`

37. Consider the following JSP code:

```
<%@page language="java"%>
<html>
<body>

<%! int x=0;%>

<%! public void jspinit()
{
System.out.println(x++);
}%>

<%=x++;%>

<%System.out.println(x);;%>

<%jspInit();;%>
</body>
</html>
```

What is output to the web page on the second access to the same instance of the above JSP?

Select one answer from the following.

- a) 0
- b) 1
- c) 2 //answer
- d) 3
- e) 4

38. Which one of the following methods is defined by the JSP engine?

- a) jsplnit()
- b) _jspService()
- c) _jspService(ServletRequest, ServletResponse)
- d) _jspService(HttpServletRequest, HttpServletResponse)**
- e) jspDestroy()

39. Which two of the following will evaluate to true?

- a) page == this**
- b) pageContext == this
- c) out instanceof ServletOutputStream
- d) application instanceof ServletContext**
- e) out instanceof Exception Class

40. Which one of the following phases is the servlet class corresponding to the JSP page created with respect to JSP page life cycle?

- a) Transformation phase
- b) Translation Phase**
- c) Servlet Phase
- d) Response Phase
- e) Request Phase

41. Consider the following code:

```
<html><body>
<jsp:useBean id="address" class="AddressBean" scope="session" />
state = <jsp:getProperty name="address" property="state" />
</body></html>
```

Which three of the following are equivalent to the third line above?

- a) <% state = address.getState(); %>
- b) <% out.write("state = "); out.print(address.getState()); %>**
- c) <% out.write("state = "); out.print(address.getstate()); %>
- d) <% out.print("state = " + address.getState()); %>**
- e) state = <%= address.getState() %>**
- f)

42. Which three of the following options locate the bean equivalent to the following action `<jsp:useBean id="address" class="AddressBean" scope="request" />?`

- a) `request.getAttribute("address");`
- b) `request.getParameter("address");`
- c) `getServletContext().getRequestAttribute("address");`
- d) `pageContext.getAttribute("address",PageContext.REQUEST_SCOPE);`
- e) `pageContext.getRequest().getAttribute("address");`

43. Consider the following code for `address.jsp`:

```
<html><body>
<jsp:useBean id="address" class="AddressBean" />
<jsp:setProperty name="address" property="city" value="LosAngeles" />
<jsp:setProperty name="address" property="city" />
<jsp:getProperty name="address" property="city" />
</body></html>
```

Also consider the URL -

`http://localhost:8080/guest/address.jsp?city=Chicago&city=Miami`

(Assume that the `city` property is not an indexed property.)

Which one of the following is the output if the above page is accessed via the mentioned URL?

- a) LosAngeles
- b) Chicago
- c) Miami
- d) ChicagoMiami
- e) LosAngelesChicagoMaimi

44. Consider the following code:

```
<html><body>
<%{%>
<jsp:useBean id="address" class="AddressBean" scope="session" />
<%}%>
//1
</body></html>
```

Which one of the following can be placed at line `//1` above to print the value of the `street` property?

- a) `<jsp:getProperty name="address" property="street" />`
- b) `<% out.print(address.getStreet()); %>`

c) `<%= address.getStreet() %>`

d) `<%= ((AddressBean)session.getAttribute("address")).getStreet() %>`

e) None of the listed options

45. Consider the following code:

```
<html><body>
<%{
<jsp:useBean id="address" class="AddressBean" scope="session" />
<%}%>
<jsp:useBean id="address" class="AddressBean" scope="session" />
<jsp:getProperty name="address" property="street" />
</body></html>
```

Which one of the following is true about the above code?

a) It will give translation-time errors.

b) It will give compile-time errors.

c) It may throw runtime exceptions.

d) It will print the value of the street property

46. Given the following code:

```
package com.smilejava;
public class JobBean {
    private String chef,waiter,cleaner;
    public JobBean() {
        chef="jim";
        waiter="bob";
        cleaner="fred";
    }
    public void setChef(String chef){
        this.chef=chef;
    }
    public void setWaiter(String waiter){
        this.waiter=waiter;
    }
    public void setCleaner(String cleaner){
        this.cleaner=cleaner;
    }
    public String getJobs(){
        return chef+waiter+cleaner;
    }
}
```

And the following JSP page that uses the code:

```
<jsp:useBean id="j" class="com.smilejava.JobBean"/>
<jsp:setProperty name="j" property="*" />
<jsp:getProperty name="j" property="jobs" />
<jsp:setProperty name="j" property="waiter" param="a"/>
<jsp:setProperty name="j" property="cleaner" param="b"/>
<jsp:setProperty name="j" property="chef" param="c"/>
<jsp:getProperty name="j" property="jobs" />
```

What will be output by that JSP code when invoked with the query string
?a=tom&b=dick&c=harry

Select one answer from the following.

- a) fredjimbo harrytomdick
- b) jimbofred tom dickharry
- c) harrytomdick jimbofred
- d) jimbofred harrytomdick**

47. Consider the following code contained in a file named this.jsp:

```
<html><body>
<jsp:useBean id="address" class="AddressBean" />
<jsp:setProperty name="address" property="*" />
<%@ include file="that.jsp" %>
</body></html>
```

Which one of the following is true about the AddressBean instance declared in the above code?

- a) The bean instance will not be available in that.jsp
- b) The bean instance may or may not be available in that.jsp, depending on the threading model implemented by that.jsp
- c) The bean instance will be available in that.jsp, and the that.jsp page can print the values of the bean's properties using <jsp:getProperty />.**
- d) The bean instance will be available in that.jsp, and that.jsp page can print the values of the bean's properties using <jsp:getProperty /> only if that.jsp also contains a <jsp:useBean/> declaration identical to the one in this.jsp and before using <jsp:getProperty/>.

48. Which one of the following techniques would NOT correctly put a bean into application scope?

a) `<jsp:useBean id="app1" class="webcert.ch.exp.AddressBean" scope="application"/>`

b) `<% addressBean ab2=new AddressBean();
application.setAttribute("app2",ab2);%>`

c) `<%AddressBean ab3= new AddressBean();
pageContext.setAttribute("app3",ab3,PageContext.APPLICATION_SCOPE);%>`

d) `<% AddressBean ab4 = new AddressBean();
ServletContext context = getServletContext();
Context.setAttribute("app4",ab4);%>`

e) `<% AddressBean ab5 = new AddressBean();
pageContext.setAttribute("app5",ab5);%>`

49. Which three of the following statements are NOT true about `<jsp:useBean>` standard action attributes?

a) If present, the class attribute must match the object type of your bean

b) **If the type attribute is used, the class attribute must be present**

c) The reference variable used for a bean does not always have the same type as the bean object it refers to

d) **If both are used, class and type attributes must have different values.**

e) **If both are used, class and type attributes must have the same values.**

50. Consider the following scenario:

There is a JSP page source:

```
<jsp:useBean id="name" class="webcar.pack.exam.NameBean"/>  
<jsp:useBean id="address" class="webcar.pack.exam.AddressBean"/>  
<jsp:setProperty name="name" property="name"/>  
<jsp:setProperty name="address" param="*/"/>  
<jsp:getProperty name="name" property="name"/>  
<jsp:getProperty name="address" property="address"/>
```

(Assume NameBean with a "name" property and an AddressBean with an "address" property)

Also consider the URL -

`http://localhost:8080/exam/quest.jsp?name=Raj%20BridgeWate&address=Leeds%20UK`

Which one of the following happens when the above JSP is requested with the mentioned URL?

a) A translation time error occurs

- b) A request time error occurs
- c) "null null" is displayed
- d) "David Bridgewater nul" is displayed
- e) "null Leads UK" is displayed.

**51. You want to use a bean that is stored in com/enthu/GUI.ser file.
Which one of the following statements correctly defines the tag that accesses the bean?**

- a) `<jsp:useBean id="pref" name="com/enthu/GUI.ser" />`
- b) `<jsp:useBean id="pref" name="com.enthu.GUI" />`
- c) `<jsp:useBean id="pref" type="com/enthu/GUI.ser" />`
- d) `<jsp:useBean id="pref" beanName="com.enthu.GUI" type="com.enthu.GUI"/>`**
- e) `<jsp:useBean id="pref" class="com.enthu.GUI.ser" />`

52. Which three of the following options correctly declare a useBean tag?

- a) `<jsp:useBean id="user" class="myco.util.User" />`**
- b) `<jsp:useBean id="user" type="myco.interfaces.IUser" />`**
- c) `<jsp:useBean name="user" class="myco.util.User" />`
- d) `<jsp:useBean id="user" beanName="myco.User" class="myco.util.User" />`
- e) `<jsp:useBean id="user" beanName="myco.User" type="myco.interfaces.IUser" />`**

53. Consider the following scenario:

Ajay is using JSP pages to create his application.

He is using useBean tag(to include a student bean with attribute such as name and age) in JSP.

After using the attributes of the bean in page1.jsp, request is forwarded to page2.jsp.
The tag he used in page1 is `<jsp:useBean id="stu" class="student" />`

Is the bean accessible in page2.jsp?

State Yes or No.

- a) Yes

b) No

54. A Java bean with a property color is loaded using the following statement:

```
<jsp:usebean id="fruit" class="Fruit"/>
```

What is the effect of the following statement?

```
<jsp:setproperty name="fruit" property="color"/>
```

Select one answer from the following.

- a) An error gets generated because the value attribute of setAttribute is not defined.
- b) The color attribute is assigned a value null.
- c) The color attribute is assigned a value "".

d) If there is a non-null request parameter with name color, then its value gets assigned to color property of Java Bean fruit.

55. Following are some of the rules framed for writing a java bean:

- a) it must have a public no-arg constructor
- b) it should have public accessor and mutator methods
- c) it should have private member variables
- d) the return type of getter method should be of same type of setter type argument
- e) it must implement serializable interface

Only some of the above rules are correct.

Select one answer from the following that gives the combination of correct rules.

- a) Only a,b,c are correct
- b) Only b,c,d and e are correct
- c) Only b,d and e are correct

d) All of the listed options are correct

56. Which one of the following gets or sets the bean in the ServletContext container object?

- a) `<jsp:useBean id="address" class="AddressBean" />`
- b) `<jsp:useBean id="address" class="AddressBean" scope="application" />`**
- c) `<jsp:useBean id="address" class="AddressBean" scope="servlet" />`
- d) `<jsp:useBean id="address" class="AddressBean" scope="session" />`
- e) None of the listed options

57. When `<jsp:useBean id="objectName" scope="scopeAttribute" class="ClassName" />` is processed, the JSP engine first searches for the object of the class with the same id and scope. If found, the preexisting bean is used; otherwise, a new bean is created. State True or False.

a) True

b) False

58. Consider the following JSP code snippet:

```
EmployeeBean harry = (EmployeeBean)request.getAttribute("harry");
if (harry == null)
{
    harry = new EmployeeBean();
    request.setAttribute("harry", harry);
}
```

Which one of the following is an equivalent action code for the above code?

a) `<jsp:declareBean id = "harry" class = "EmployeeBean" scope = "request" />`

b) `<jsp:createBean id = "harry" class = "EmployeeBean" />`

c) `<jsp:useBean id = "harry" class = "EmployeeBean" scope = "request" />`

d) `<jsp:useBean id = "harry" class = "EmployeeBean" />`

59. A JSP page has the following useBean declaration that references a bean called Person.

The person class has a private field called firstname and an appropriate getter (accessor) method for retrieving the value of that field.

```
<jsp:useBean id="person" type="com.exam.Person" class="com.exam.Person"
scope="session" />
```

Which one of the following is the correct syntax for a getProperty tag to retrieve the value of the instance of Person?

a) `<jsp:getProperty name="person" property="firstname"/>`

b) `<jsp:getProperty name="person" property="getFirstname"/>`

c) `<jsp:getProperty id="person" property="firstname"/>`

d) `<jsp:getProperty name="person" property="firstname" class="com.exam.Person"/>`

e) `<jsp:getProperty name="person" property="firstname" scope="session"/>`

60. Consider the following code:

```
<html><body>  
<jsp:useBean id="address" class="AddressBean" scope="session" />  
state = <jsp:getProperty name="address" property="state" />  
</body></html>
```

Which three of the following are equivalent to the third line above?

- a) `<% state = address.getState(); %>`
- b) `<% out.write("state = "); out.print(address.getState()); %>`
- c) `<% out.write("state = "); out.print(address.getstate()); %>`
- d) `<% out.print("state = " + address.getState()); %>`
- e) `state = <%= address.getState() %>`
- f)

61. Which two of the following statements are true?

- a) A tag may access tags nested within it using the `getChild` method
- b) A tag may access an enclosing tag using the `getParent` method
- c) A tag may access an enclosing tag using the `getAncestor` method
- d) A tag may access an enclosing tag using the `findAncestorWithClass` method

62. Which three of the following are valid return values for `doStartTag()` method for various kinds of tags?

- a) `BodyTag.SKIP_BODY`
- b) `Tag.SKIP`
- c) `Tag.EVAL_BODY_INCLUDE`
- d) `Tag.EVAL_BODY_AGAIN`
- e) `BodyTag.EVAL_BODY_BUFFERED`
- f)

63. Consider the following code:

```
public class MyTagHandler extends TagSupport  
{  
  
    public int doStartTag() throws JspException  
  
    {
```

```

try
{
    //insert code here
}

catch(Exception e){ }

return super.doStartTag();

}

}

```

Which one of the following options, when inserted in the above code, causes the value "hello" to be output?

- a) **JspWriter out = pageContext.getOut();**
out.print("hello");
- b) JspWriter out = pageContext.getWriter();
out.print("hello");
- c) JspWriter out = getPageContext().getWriter();
out.print("hello");
- d) JspWriter out = new JspWriter(pageContext.getWriter());
out.print("hello");
- e) JspWriter out = getPageContext().getOut();
out.print("hello");
- f)

64. Which two of the following are valid custom tag usages, assuming that the correct taglib directives have been set up?

- a) **<mg:claim>amount</mg:claim>**
- b) <claim>amount</claim>
- c) <mg:claim>amount<mg:claim/>
- d) **<mg:claim></mg:claim>**

65. Which two of the following are valid within a JSP page?

- a) **<%@ taglib uri="/richjava_tag" prefix="question" %>**
- b) <%@ taglib uri="/richjava" class="question" %>

c) `<%@ taglib uri="/customtag" type=com.javarich.customtag prefix="question" %>`

d) `<%@ taglib uri="http://java.sun.com/jstl/core_rt" prefix="c" %>`

66. Which one of the following statements is true?

a) TLD files must be stored under the WEB-INF/tld directory

b) TLD files must be stored under the WEB-INF/lib directory

c) TLD files must be stored under the WEB-INF/ directory or a directory under WEB-INF

d) TLD files can be stored anywhere within the application path

67. Which two of the following are valid return values for the doEndTag() of a custom tag class that extends TagSupport?

a) SKIP_BODY

b) EVAL_PAGE

c) SKIP_PAGE

d) EVAL_BODY_INCLUDE

e) EVAL_BODY_AGAIN

68. The following is an entry from Avarice's DateTime Tag-lib Descriptor:

```
<tag>
  <name>currenttime</name>
  <tag-class>org.apache.taglibs.datetime.CurrentTimeTag</tag-class>
  <body-content>empty</body-content>
  <description>Gets the current time in milliseconds since Jan 1, 1970 GMT.</description>
</tag>
```

Which two of the following are true statements about the above tag?

a) This is an example of an empty custom tag.

b) This is a nested tag.

c) This tag has no attributes

d) This tag can surround other JSP code.

69. Which one of the following is a valid method that will not be called on a custom tag handler if the tag is declared as an empty tag?

- a) doStartTag
- b) doBodyTag
- c) doAfterBody**
- d) doEndTag
- e) doBeforeBody
- f)

70. JSP custom tags can only be developed for HTTP protocol. State True or False.

- a) True
- b) False**

71. Consider the following code in a tag handler class that extends TagSupport:

```
public int doStartTag()  
{  
    //1  
}
```

Which one of the following can you use at //1 to get an attribute from the application scope?

- a) `getServletContext().getAttribute("name");`
- b) `getApplication().getAttribute("name");`
- c) `pageContext.getAttribute("name",PageContext.APPLICATION_SCOPE);`**
- d) `bodyContent.getApplicationAttribute("name");`

72. Which two of the following statements are NOT TRUE?

- a) The TagSupport class has an implicit pageContext field
- b) To obtain a PageContext the getPageContext method must be called within a class that extends TagSupport.**
- c) The pageContext field has a getRequest method
- d) The pageContext field has a getOut method.
- e) The pageContext field has a getHeaders method**

73. Consider the following:

- a) t.setPageContext(...);
- b) ATag t = new ATag();
- c) t.setAttribute1(value1);
- d) t.setAttribute2(value2);
- e) t.doStartTag();
- f) t.setParent(...);
- g) t.release();
- h) t.doEndTag();

Which one of the following gives the correct tag handler method invocation sequence?

- a) b,e,a,f,c,d,h,g
- b) b,a,f,c,d,e,h,g**
- c) b,a,e,f,c,d,g,h
- d) b,e,a,f,c,d,g,h

74. Consider the following:

```
<taglib>
<tag>
<name>heading</name>
<tagclass>coreservlets.tags.HeadingTag</tagclass>
<info>Outputs a 1-cell table used as a heading.</info>
<attribute>
<name>bgColor</name>
<required>true</required> <!-- bgColor is required -->
<rtexprvalue>true</rtexprvalue>
</attribute>
<attribute>
<name>color</name>
<required>false</required>
</attribute>
<attribute>
<name>style</name>
<required>true</required>
<rtexprvalue>false</rtexprvalue>
</attribute>
...
</tag>
</taglib>
```

Which two of the following statements can be inferred from the above code?

- a) Attribute value of bgcolour is required for the tag.**

- b) Attribute value of bgcolor will be retrieved from the parent tag.
- c) Attribute value of colour is optional for the tag.**
- d) Attribute value of style required for the tag and will be given by the user.
- e) Taglib tag can have only a maximum of 15 tags.

75. Which two of the following statements are true about taglibs tags?

- a) taglibs tags within a page can have the same prefix so long as the uri value is different
- b) the prefix attribute is mandatory**
- c) the prefix string is arbitrary, there are no reserved strings
- d) taglibs can be deployed within a jar file**
- e) the uri value must point to the location of the taglib file

76. Which two of the following types of objects can be returned by `PageContext.getOut()`?

- a) An object of type `ServletOutputStream`
- b) An object of type `HttpServletOutputStream`
- c) An object of type `JspWriter`**
- d) An object of type `HttpJspWriter`
- e) An object of type `BodyContent`**

77. Given in a single JSP page:

```
<%@taglib prefix='java' uri='myTags'%>
```

```
<%@taglib prefix='JAVA' uri='moreTags'%>
```

Which two of the following statements are true?

- a) The prefix 'java' is reserved.**
- b) The URI 'myTags' must be properly mapped to a TLD file by the web container.**
- c) A translation error occurs because the prefix is considered identical by the web container
- d) For the tag usage `<java:tag1/>`, the tag1 must be unique in the union of tag names in 'myTags' and 'moreTags'.

78. Given a JSP page with the following text:

```
<html>
<body>
<%@taglib prefix="m" tagdir="/WEB-INF/tags"%>
<m:menu>
Calling JSP <% out.print("hello"); %>
</m:menu>
</body>
</html>
```

And a tag file called menu.tag that is accessible in from the tags directory from that JSP page, with the following code:

Tag file

```
<jsp:doBody/>
```

Which one of the following will happen when you attempt to compile and run the code?

- a) Compilation and output of Tag file Calling JSP hello
- b) Compilation and output of Calling JSP hello Tag File
- c) Compile time error, doBody cannot be used within a tag file
- d) Compile time error, a tag file must be declared with the tagfile attribute

e) Compile time error, the body of a call to a tag file cannot contain scripting

79. You are given a tag library that has:

- 1. A tag named getMenu that takes an attribute 'subject' which can be a dynamic value.
- 2. A tag named getHeading that takes an attribute 'report'.

Which three of the following are correct uses of the above library?

a) <myTL:getMenu subject="Finance"/>

b) <myTL:getMenu subject="<myTL:getHeading report=1/>"/>

c) <myTL:getMenu subject='<myTL:getHeading report="1"/>'/>

d) <% String subject="HR";%> <myTL:getMenu subject="<%=subject%>"/>

e) <myTL:getHeading report="2"/>

80. Which three of the following tags are valid tags that can be used in a JSP page?

(You can assume that the tag library is correctly declared in the jsp page with a prefix of "mytags")

a) <mytags:book/>

b) <mytags:book isbn="<%=isbn%>">

c) <mytags:book isbn="\${isbn}"/>

d) <mytags:book isbn="1861979258"/>

e) `<mytags:book isbn="{isbn}">keep smiling!..</mytags:book>`

81. Consider the following use of the tag `<mytags:convert>`:

`<mytags:convert currency="{param.cur}"> <%=amount%> </mytags:convert>`
(Assume that the tag translates and executes correctly)

Which three of the following statements must be true about its setup and use?

a) The taglib declaration has a prefix of "mytags".

b) In the TLD, the tag's body content element has a value of JSP

c) In the TLD, the tag's name element has the value of currency

d) In the TLD, the tag's currency attribute has the required element set to true.

e) In the TLD, the tag's currency attribute has the `rtexprvalue` element set to true.

f)

82. Which two of the following cannot be returned by either of the `getParent()` methods in the JSP class libraries?

a) An instance of a JSTL core library tag handler

b) An instance of `BodyTagSupport`

c) An instance of an HTML tag

d) An instance of `SimpleTagSupport`

e) An instance of XML template tag in a JSP document

83. Which two of the following are valid locations for a tag file?

a) Directly in the context directory

b) Directly in the `/WEB-INF` directory

c) In a JAR file in `/WEB-INF/lib`, under the `/META-INF/tags` directory

d) In `/WEB-INF/tags` or a subdirectory beneath `/WEB-INF/` tags

e) In `/WEB-INF/taglib`

84. Which of the following is a valid declaration of a Tag Library in a Deployment Descriptor?

(select one answer)

a) <taglib>

<uri>http://www.smilejava.net/studyKit</uri>

<location>/studyKit.tld</location>

</taglib>

b) <taglib>

<uri>http://www.smilejava.net/studyKit<uri>

<location>/studyKit.tld</location>

</taglib>

c) <taglib>

<taglib-uri>http://www.smilejava.net/studyKit</taglib-uri>

<taglib-location>/studyKit.tld</taglib-location>

</taglib>

d) <taglib>

<uri>http://www.smilejava.net/studyKit</uri>

<path>/studyKit.tld</path>

</taglib>

85. We can use the directive <%@ page buffer="8kb" %> to specify the size of the buffer when returning EVAL_BODY_BUFFERED from doStartTag(). State True or False.

a) TRUE

b) FALSE

86. Which one of the following elements can you use to import a tag library in a JSP document?

a) <jsp:root>

b) <jsp:taglib>

c) <jsp:directive.taglib>

d) <jsp:taglib.directive>

e) We cannot use custom tag libraries in XML format

87. A tag `testlib:myFirstSimpleTag` extends `SimpleTagSupport`.

Which one of the following should be done so that scriptlet code can be used in the body of this tag?

- a) body-content should be set to JSP in the TLD for this tag.
- b) scripting-enabled should be set to true in the TLD for this tag.
- c) Create a classic tag adapter for this tag that has the body-content of JSP
- d) This cannot be done because it is a Simple tag.**

88. You have a JSP page with a correctly configured Tag file as follows:

```
<%@ taglib prefix="mytag" tagdir="/WEB-INF/tags" %>
```

Within that page you have correctly configured call to that tag file as follows

```
<mytag:partag username="jimbob"/>
```

Which one of the following lines of code within the tag file will output the string "jimbob"?

- a) `${param.username}`
- b) `${attribute.username}`
- c) `${tag.username}`
- d) `${username}`**
- e) `${mytag.username}`

89. Which two of the following statements are correct for a custom tag that can take any number of arbitrary attributes?

- a) The body-content element for the tag in the TLD file must have a value of JSP.
- b) The tag handler must implement the method `setAttribute(String key, String value)`.
- c) The tag element in the TLD file for the tag must have `<dynamic-attributes>true</dynamic-attributes>`.**
- d) The class implementing the tag must implement `javax.servlet.jsp.tagext.DynamicAttributes` interface.**
- e) Dynamic attributes cannot have request time expression values

90. Which one of the following is a valid declaration of a Tag Library in a Deployment Descriptor?

a) <taglib>
 <uri>http://www.javabeat.net/studyKit</uri>
 <location>/studyKit.tld</location>
 </taglib>

b) <taglib>
 <uri>http://www.javabeat.net/studyKit<uri>
 <location>/studyKit.tld</location>
 </taglib>

c) <taglib>
 <taglib-uri>http://www.javabeat.net/studyKit</taglib-uri>
 <taglib-location>/studyKit.tld</taglib-location>
 </taglib>

d) <taglib>
 <uri>http://www.javabeat.net/studyKit</uri>
 <path>/studyKit.tld</path>
 </taglib>

**91. Consider the following code:
JSP document**

```
<html xmlns:mytags="http://www.javarich.com/taglibs/mytags"
xmlns:jsp="http://java.sun.com/JSP/Page"
xmlns:c="http://java.sun.com/jsp/jstl/core">

<jsp:output omit-xml-declaration="true"/>
<jsp:directive.page contentType="text/html"/>
<jsp:directive.page import="java.io.Writer"/>
<head><title>joy</title></head>
<body>
<jsp:scriptlet>Writer myout=pageContext.getOut();</jsp:scriptlet>
<p>
<mytags:jspking>
<jsp:scriptlet.myout.write("Body");</jsp:scriptlet>
</mytags:jspking>
</body>
```

Tag handler code

Import java.io.IOException;

```

import javax.servlet.jsp.JspException;
import javax.servlet.jsp.tagext.*;
Public class JspKing extends BodyTagSupport
{
public int doAfterBody() throws JspException
{
try{
bodyContent.write("legs");
bodyContent.writeOut(bodyContent.getEnclosingWriter());
}catch(IOException e)
{
throw new JspException(e);
}
return Tag.EVAL_PAGE;
}

public void doInitBody()throws JspException
{
try
{
bodyContent.write("Head");
}catch(IOException e)
{
throw new JspException(e);
}
}

}

```

Which one of the following gives the correct output of the above code?

- a) Translation error(source generation)
- b) Translation error(source compilation)
- c) Run-time error
- d) Output of BodyHeadBodyLegs

e) Output of BodyHeadLegs

92. Which three of the following are valid combination of interface and its sequences of method calls according to custom tag life cycle?

- a) Tag:setParent,setPageContext,doStartTag
- b) Tag:SetPageContext,setParent,doStartTag,doEndTag**
- c) Iteration Tag:doStartTag,doAfterbody,doAfterBody,doAfterBody**
- d) BodyTag:doStartTag,setBodycontent,doInitBody,doAfterBody**
- e) BodyTag:doInitBody,setBodycontent,doStartTag,doEndTag

93. Consider the following tag handler code:

```
Package smile.custom.examplecheck;
Import java.io.IOException;
Import javax.servlet.jsp.JspException;
Import javax.servlet.jsp.tagext.Tag;
Import javax.servlet.jsp.tagext.TagSupport;
Public class Smilingcustom extends TagSupport
{
private int data;
int getData()
{

return data;
}
void setData(int data)
{
this.data=data;
}
public int doEndTag() throws JspException
{
try
{
pagecontext.getOut().write(id = ":" + data);
}catch (IOException e)
{
throw new JspException(e);
}
return Tag.EVAL_PAGE;
}

}
```

Which two of the following TLD tag declarations would best fit the above tag handler code?

- a) <tag>
 <name>A</name>
 <tag-class> smile.custom.examplecheck.Smilingcustom </tag-class>

 <body-content>empty</body-content>

 <attribute>
 <name> id</name>

```
<required>false </required>
<rtexprvalue> true</rtexprvalue>
</attribute>
<attribute>
<name> data</name>
<required>false </required>
<rtexprvalue> true</rtexprvalue>
</attribute>
```

```
</tag>
```

b) <tag>

```
<name>B</name>
<tag-class> smile.custom.examplecheck.Smilingcustom </tag-class>
```

```
<body-content>empty</body-content>
```

```
<attribute>
<name>value </name>
<required>false </required>
<rtexprvalue> true</rtexprvalue>
</attribute>
</tag>
```

c) <tag>

```
<name>C</name>
<tag-class> smile.custom.examplecheck.Smilingcustom </tag-class>
```

```
<body-content> empty </body-content>
```

```
<attribute>
<name> data</name>
<required>false </required>
<rtexprvalue> true</rtexprvalue>
</attribute>
</tag>
```

```

d) <tag>
    <name>D</name>
    <tag-class> smile.custom.examplecheck.Smilingcustom </tag-class>

    <body-content> empty </body-content>

    <attribute>
        <name>id </name>
        <required>false </required>
        <rtexprvalue> true</rtexprvalue>
    </attribute>
</tag>
e) <tag>
    <name>E</name>
    <tag-class> smile.custom.examplecheck.Smilingcustom </tag-class>

    <body-content> empty </body-content>

    </tag>

```

94. Consider the following JSP and tag handler code:

JSP Page Source

```

<%@taglib prefix="mytags" uri="http://www.javarich.com/taglibs/mytags"%>
<html>
<head><title>CustomTags</title></head>
<body>
<p><%session.setAttribute("first","first");%>
<mytags:question12/>
${second}
</p>
</body>
</html>

```

Tag Handler code for <mytags:Question12/>

```

Public class Question12 extends TagSupport
{
public int doStartTag()throws JspException
{
Writer out = pageContext.getOut();
Try
{

```

```

out.write(" " +pageContext.getAttribute("first"));
}catch(IOException e)
{
e.printStackTrace();
}
pagecontext.setAttribute("second","second",pageContext.SESSION_SCOPE);
return super.doStartTag();
}
}

```

Which one of the following is the correct result of accessing the above JSP?

- a) first first
- b) second second
- c) first second
- d) first null

e) null second

95. Which one of the following techniques causes the JSP container to skip the rest of the page after processing a custom action implementing the SimpleTag interface?

- a) Returning Tag.SKIP_PAGE from the doEndTag() method
- b) Returning - 1 from the doPage() method
- c) Returning - 1 from the doTag() method

d) Throwing a SkipPageException within the doTag() method

- e) Throwing a JspException from the doPage() method

96. Which two of the following statements are true?

a) To accept an attribute a tag file must include the attribute directive

b) If it is to be wrapped in a jar a, tag file must include a matching tld file

- c) Tag files cannot access parameters from the calling JSP page
- d) Tag files must begin with the tag-file directive

97. Consider the following code for a tag handler:

```

public class MyTagHandler extends TagSupport {
    public int doStartTag() throws JspException {
        setValue("name", "value");
        return EVAL_BODY_INCLUDE;
    }
}

```

Is the above code correct?
State Yes or No.

a) Yes

b) No

98. Which two of the following statements are correct about the method `BodyTagSupport.doInitBody`?

a) It is a method used by the jsp container and the jsp programmer should not override it

b) It can be overridden by the programmer if needed.

c) It can return either `SKIP_BODY` or `EVAL_BODY_INCLUDE`.

d) Its return type is void.

e) This method throws `JspException`

99. Consider the following:

The `doStartTag()` method returns `EVAL_BODY_INCLUDE` one time.
The `doAfterBody()` method returns `EVAL_BODY_AGAIN` five times.

How many times will the `setBodyContent()` method be called?
Select one answer from the following.

a) Zero

b) One

c) Two

d) Five

e) Six

100. Consider the following two statements:

1) All tags have a parent tag, which is null for top-level tags.

2) All tags have a parent tag, and is the innermost containing tag for nested tags.

Which one of the following is valid with respect to the above statements?

a) Both the statements are true.

b) Both the statements are false.

c) Only the second statement is true.

d) Only the first statement is true.

101. Given a JSP page that is requested with the query string:
?username=yogesh

Which one of the following lines of code will result in the output of yogesh?

- a) `${requestScope["username"]}`
- b) `${requestScope.username}`
- c) `${requestScope.username}`
- d) `${param.username}`**
- e) `${parameter.username}`

102. Which of the following are equivalent?

- 1) `<jsp:expression`
- 2) `<%`
- 3) `<%=`
- 4) `<@`

Select the one correct answer from below that provides the right combination.

- a) 1 and 2
- b) 1 and 3**
- c) 1 and 4
- d) None are equal.

103. Attribute flush of jsp include action is a mandatory attribute. State True or False.

- a) TRUE
- b) FALSE**

104. Which two of the following things may happen in the translation phase of a JSP page?

- a) Execution of `_jspService()` method.
- b) Execution of `jspInit()` method.

c) Creation of the servlet class corresponding to the JSP file

d) Instantiation of the servlet class.

e) Loading of the servlet class.

105. Which one of the following is an Equivalent in jsp of printWriter in servlet?

a) jspWriter

b) jspPrintWriter

c) jspOutputStream

d) jspBufferedStream

106. Which two of the following will compile correctly if placed in a JSP file?

a) `<%@ import page="java.util.*,java.io.*" %>`

b) `<%@ import="java.util.*,java.io.*" %>`

c) `<%@ page import="java.util.*,java.io.*" %>`

d) `<jsp:directive.page import="java.util.*"/>`

e) `<jsp:@page language="java">`

107. Which one of the following represents the correct syntax for using the include directive?

a) `<% @include file="menu.jsp" %>`

b) directly in the /WEB-INF directory

c) `<jsp : include page="menu.jsp" />`

d) `<jsp: include file="menu.jsp" />`

108. Which one of the following statements is true?

a) Every JSP page has an exception implicit object

b) It is possible to map HTTP error codes to an error page but not a Java Exception type

c) It is possible to map Java Exception type to an error page but not HTTP error codes

d) Both Java Exceptions and HTTP error codes can be mapped to an error page

109. Which one of the following is not an attribute of the page directive?

a) info

b) buffersize

c) pageEncoding

d) import

110. Which two of the following are considered thread safe?

- a) class variables.
- b) context attributes.
- c) local variables**
- d) request attributes**
- e) Response attributes

111. Consider the following sources for two JSP pages:

Source for a.jsp

```
<% @ page buffer = "none" autoFlush = "true"%>
<jsp:forward page="c.jsp"/>
```

Source for b.jsp:

```
<%@ page buffer = "none" autoFlush = "true" %><jsp:forward page = "c.jsp"/>
```

(Assume that "c.jsp" is available in the same web application directory as a.jsp and b.jsp)

Which two of the following are possible outcomes of requesting each of the above JSP pages in turn?

- a) Neither of the JSP pages translate
- b) a.jsp translates, b.jsp does not translate
- c) b.jsp translates, a.jsp does not translate
- d) Both JSP pages translate**
- e) b.jsp runs successfully, a.jsp does not**

112. Consider the following JSP source:

```
<%@page import="java.util.*,java.text.*"%>
<%!private String returnTimeStamp(Pagecontext pageContext)
{
    DateFormat df = DateFormat.getDateTimeInstance();
    String s = df.format(new Date());
    pageContext.setAttribute("timestamp",s);
    return s;
}%>
```



```
<jsp:forward page = "forwardedTo.jsp"/>
<%=returnTimeStamp(pageContext)%>
<%System.out.println(pageContext.getAttribute("timestamp"));%>
```

(Assume that "forwardTo.jsp" is an emptyfile)

Which one of the following will be the result of requesting the above JSP page?

- a) Translation error
- b) Run-time error
- c) A formatted date appears in the page output
- d) A formatted date appears both in the page output and in te server console.
- e) A blank page is output**

113. Consider the following JSP page:

```
<%@ page language="java" import="java.util.*" pageEncoding="ISO-8859-1"%>
<%
String path = request.getContextPath();
String basePath =
request.getScheme()+"://"+request.getServerName()+":"+request.getServerPort()+path+"
/";
%>

<html xmlns:jsp="http://godaddy.com/JSP/Page" />
<head>
<title>Guess what</title>
</head>

<body>
<%request.setAttribute("a",new Integer(2));
session.setAttribute("b",new Integer(3));
application.setAttribute("c",new Integer(4));
request.setAttribute("d",new Integer(5));
%>>
???
```

Which two of the following expressions when used at the point marked ??? in the above JSP page will output the number 46?

- a) `${pageContext.c * pageContext.d * pageContext.a + pageContext.a * pageContext.b}`
- b) `${applicationScope.c * requestScope.d * requestScope.a + requestScope.a * sessionScope.b}`**

c) $\{(\text{applicationScope.c} * \text{requestScope.d} * \text{requestScope.a}) + (\text{requestScope.a} * \text{sessionScope.b})\}$

d) $\{(\text{pageContext.c} * \text{pageContext.d} * \text{pageContext.a}) + (\text{pageContext.a} * \text{pageContext.b})\}$

e) $\{(\text{application.c} * \text{request.d} * \text{request.a}) + (\text{request.a} * \text{session.b})\}$

114. Of the five JSP page source extracts below, there are two pairs. Each member of the pair gives rise to identical output. Which one of the following is the odd one out?

a) `<% int i , j, k;
i=1;j=2;k=3;%>
<%=i+j /k%>`

b) `<jsp:scriptlet>int i,j,k;
I = 1;j = 2; k = 3;
</jsp:scriptlet>
<jsp:expression> (i+j)/k </jsp:expression>`

c) `<% int I,j,k;
i = 1;j = 2;k = 3;%>
<%=(i+j)/k + “.0”%>`

d) `<%pageContext.setAttribute(“I”,new Integer(1));
pageContext.setAttribute(“j”,new Integer(2));
pageContext.setAttribute(“k”,new Integer(3));
%>${pageScope.i+pageScope.j/pageScope.k}`

e) `<%pageContext.setAttribute(“i”,new Integer(1));
pageContext.setAttribute(“j”,new Integer(2));
pageContext.setAttribute(“k”,new Integer(3));
%>${(pageScope.i+pageScope.j)/pageScope.k}`

115. Consider the following code:

```
1)<html xmlns:jsp=http://java.sun.com/JSP/Page>
2)<jsp:directive.page contentType="text/html"/>
3)<jsp:declaration>
4)public int squared(int value)
5){
6)return value*value;
7)}
8)</jsp:declaration>

9)<jsp:scriptlet>
```

```

10)int value=Integer.parseInt(request.getParameter("number"));
11)int squared=squared(value);
12)out.write(value+"squared is"+squared);
13)if(squared<100)
14){
15)out.write(";try a bigger number.");
16)}
17)</jsp:scriptlet>
18)</html>

```

Which one of the following will be the outcome of accessing the above page, defined as a JSP document, in a web application?

- a) Translation error at line 1
- b) Translation error at line 2
- c) Translation error at line 13**
- d) Run-time error
- e) No errors, with page displaying successfully

116. Given the following JSP files:

```

<!-- file container.jsp -->
<html><body>
<%@ include file="included.jsp" %>
<%@ page errorPage="OneErrorPage.jsp" %>
</body></html>

///////// different file
<!-- file included.jsp -->
<%@ page errorPage="AnotherErrorPage.jsp" %>
<i>Hello</i>
</body></html>

```

Which one of the following is the correct answer with respect to the above?

- a) File container.jsp will compile if the directive page comes before the directive include
- b) File container will compile and when executed it will show:"Hello".
- c) File container.jsp will compile if the errorPage in container.jsp is the same as in file included.jsp.

d) File container.jsp will compile if instead of directive include (<%@ include ...%>) it is used the action include (<jsp:include...>)

117. Consider the following code:

```

<%@ page language="java" %>
<html><body>
<%

```

```
response.getOutputStream().print ("Hello ");  
out.print("World");  
%>  
</body></html>
```

Which one of the following is the correct statement about the above code?

- a) It will print Hello World in the output.
- b) It will generate compile-time errors.
- c) It will throw runtime exceptions**
- d) It will only print Hello.
- e) None of the listed options

118. Consider the following code:

```
<html><body>  
<% int i = 10 ;%>  
<% while(--i>=0) { %>  
out.print(i);  
<% } %>  
</body></html>
```

Which one of the following is the output of the above?

- a) 9876543210
- b) 9
- c) 0
- d) None of the above**

119. Consider the following code written in a servlet:

```
RequestDispatcher rd = request.getRequestDispatcher("world.jsp");  
rd.forward(request, response);
```

Which one of the following is the JSP equivalent of the above code?

- a) <jsp:forward page="world.jsp"/>**
- b) <jsp:action.forward page="world.jsp"/>
- c) <jsp:directive.forward page="world.jsp"/>
- d) <%@ forward file="world.jsp"%>
- e) <%@ forward page="world.jsp"%>

120. Consider the contents of two JSP files:

File 1: this.jsp

```
<html><body><pre>
<jsp:include page="that.jsp" >
<jsp:param name="color" value="red" />
<jsp:param name="color" value="green" />
</jsp:include>
</pre></body></html>
```

File 2: that.jsp

```
<%
String colors[] = request.getParameterValues("color");
for (int i=0; i<colors.length; i++)
{
out.print(colors[i] + " ");
}
%>
```

Also consider the URL - <http://localhost:8080/chapter13/this.jsp?color=blue>

Which one of the following will be the output when accessing the this.jsp file via the above mentioned URL?

- a) blue
- b) red green
- c) red green blue**
- d) blue red green
- e) blue green red

121. Which of the following implicit variables can be used by a JSP page to access a resource and to forward a request to another JSP page?(Select two answers)

- a) pageContext and config
- b) application and config(directly)**
- c) config and pageContext
- d) application for both(directly)**
- e) config for both(directly)

122. Consider the following contents of three JSP files:

File 1: one.jsp

```
<html><body><pre>
<jsp:include page="two.jsp" >
<jsp:param name="color" value="red" />
</jsp:include>
```

```
</pre></body></html>
```

File 2: two.jsp

```
<jsp:include page="three.jsp" >
<jsp:param name="color" value="green" />
</jsp:include>
```

File 3: three.jsp

```
<%= request.getParameter("color") %>
```

Also consider the URL - <http://localhost:8080/chapter13/one.jsp?color=blue>

Which one of the following will be the output of accessing the one.jsp file via the above mentioned URL?

- a) red
- b) green**
- c) blue
- d) The answer cannot be determined.

123. Consider the following contents of two JSP files:

File 1: test1.jsp

```
<html><body>
<% pageContext.setAttribute("ninetyNine", new Integer(99)); %>
//1
</body></html>
```

File 2: test2.jsp

The number is <%= pageContext.getAttribute("ninetyNine") %>

Which one of the following, when placed at line //1 in the test1.jsp file, will allow the test2.jsp file to print the value of the attribute when test1.jsp is requested?

- a) <jsp:include page="test2.jsp" />
- b) <jsp:forward page="test2.jsp" />
- c) <%@ include file="test2.jsp" %>**
- d) None of the listed options because objects placed in pageContext have the page scope and cannot be shared with other components.

124. Consider the following JSP line:

```
<!-- Today is <%= new java.util.Date() %>.Hava a nice day -->
```

Which one of the following will be the result of the above JSP line if the user takes a look at the page source code?

- a) It will not compile
- b) The user cannot see the comment
- c) The user will see the same jsp line as shown
- d) The user will see the comment with the current date because the expression is evaluated at runtime**

125. Given the following line of code:

```
<%! public int getMax() { return 1000; }%>
```

Which one of the following is the correct XML equivalent for the above?

- a) `<jsp:directive> public int getMax() { return 1000; } </jsp:directive>`
- b) `<jsp:declaration> public int getMax() { return 1000; } </jsp:declaration>`**
- c) `<jsp:directive public int getMax() { return 1000; } />`
- d) `<jsp:scriptlet> public int getMax() { return 1000; } </jsp:scriptlet>`
- e) None of the listed options

126. You want the copyright information contained in /html/copyright.html to be added to the html content that your servlet generates. Which of the following code fragments occurring in the doGet() method as below would achieve this objective?

1.
this.getServletContext().getRequestDispatcher("../html/copyright.html").include(request, response);
2.
this.getServletContext().getRequestDispatcher("/html/copyright.html").include(request, response);
3. request.getRequestDispatcher("../html/copyright.html").include(request, response);
4. request.include("../html/copyright.html", request, response);
5.
this.getServletContext().getRequestDispatcher("/html/copyright.html").forward(request, response);
6. request.getRequestDispatcher("/html/copyright.html").forward(request, response);

Select one answer from the following that has the right combination of required code fragments..

- a) 1, 2 and 3
- b) 1 and 4

- c) 4 and 5
- d) 2 and 3**
- e) only 3

127. Consider the following code:

```
<jsp:useBean id="user" class="com.richBean.UserBean" />
<jsp:setProperty name="user" property="firstName" value="yogesh" />
<%@ include file="menu.jsp" %>
```

Which one of the following statements is true with respect to the above code?

- a) The code will not compile, the attribute to include should be page not file
- b) The page will compile but attributes of user will not be available within menu.jsp
- c) The page will compile and it will be possible to access user within menu.jsp without any additional code**
- d) The page will compile and user will be available within menu.jsp if the same useBean tag is declared

128. Which two of the following techniques will cause JSP page source to be treated as a JSP document by the JSP container?

- a) Setting the <is-xml> subelement of <jsp-config> to a value of true
- b) Using a .jspx extension with a version 2.4 deployment descriptor**
- c) Using a .xml extension with a version 2.4 deployment descriptor
- d) Using <jsp:root> as the root element of your source**
- e) Writing your page source in XML syntax.

129. Consider the following JSP page extract:

```
<html xmlns:jsp="http://java.sun.com/JSP/Page">
<jsp:directive.page contentType="text/html"/>
<body>
<jsp:??><![CDATA[<img src=""]></jsp:??>
<jsp:expression >session.getAttribute("theImage")</jsp:expression>
<jsp:??><![CDATA["/>]]></jsp:??>
</body>
</html>
```


Which one of the following tags will successfully complete the above JSP page extract, at the points marked `<jsp:??>` and `</jsp:??>`?

- a) `<jsp:param>` and `</jsp:param>`
- b) `<jsp:element>` and `</jsp:element>`
- c) `<jsp:img>` and `</jsp:img>`
- d) `<jsp:output>` and `</jsp:output>`
- e) `<jsp:text>` and `</jsp:text>`**

130. Consider the following scenario:

Source `params.jsp`:

```
<jsp:include page="included.jsp">
<jsp:param name="x" value="4"/>
<jsp:param name="x" value="5"/>
<jsp:param name="z" value="6"/>
</jsp:include>
${param.x}
<%=request.getParameter("y")%>
```

source for `included.jsp`

```
${param.x}
${param.y}
<%String[]x=request.getParameterValues("x");
for(int i=0;i<x.length;i++)
{
out.write(x[i]);
}
%>
```

The HTTP request is in this form:

`http://localhost:8080/dynamic/params.jsp?x=1&y=2&z=3`

Which one of the following is the correct outcome of making the HTTP GET request as shown above to `params.jsp`?

- a) 1 2 45 4 6
- b) 1 2 145 4 6
- c) 4 6 451 4 6
- d) 4 6 451 1 2**

131. Consider the following two lines of code in the same JSP page.

1. `<%! int sum = 0; %>`

...//other valid code.

2. <% int sum = 0; %

Which two of the following statements are correct regarding the above code?

- a) Both cannot occur in the same JSP file
- b) JSP file will compile properly even if one of them is removed**
- c) 1. is a declaration and 2 is an expression
- d) 1. is a directive and 2 is a scriptlet
- e) 1 is a declaration and 2 is a scriptlet**

132. Consider the following JSP code:

```
<%! java.util.Locale class1=request.getLocale(); %>
```

```
<%= class %>
```

Which one of the following will be printed out if the above code is executed?

- a) It will not compile.**
- b) It will print the default client's Web browser Locale
- c) It will print the default web server Locale.
- d) It will the first language specified in the Accept-Language request's header.

133. Which three of the following statements are true?

- a) the context implicit object of JSP pages has a getAttribute method
- b) The session implicit object of JSP pages have a getAttributeNames method**
- c) The getAttributeNames method of the JSP request implicit object returns a String array
- d) The getAttributeNames method of the JSP request implicit object returns an enumeration**
- e) The getAttribute method of the JSP request implicit object has a return type of Object**

134. Consider the following JSP code:

```
<%
```

```
String s = "sandfield";
```

```
RequestDispatcher rd = request.getRequestDispatcher("menu.jsp");
```

```
rd.forward(request,response);
```

```
out.print(s);
```

```
%>
```

Which one of the following will happen when you attempt to compile and run the above JSP code?

- a) Compilation error, the getRequestDispatcher parameter must begin with a forward slash (/)
- b) Compilation and output that includes sandfield
- c) Compilation and output that does not include sandfield**
- d) Compilation but runtime error

135. Consider the following JSP source:

JSP page source:

```
<!--Source for smileone.jsp -->
<p> Parameter A has values
<%for(int j=0;j<request.getParameterValues("A").length;j++){%>
${paramValues.A[j]},<%}%></p>
```

Also consider URL for accessing smileone.jsp as:

http://localhost:8080/example24/smileone.jsp?A=1&A=2

Which one of the following is the correct consequence of accessing above JSP page with the URL shown?

- a) Translation error
- b) Run-time error
- c) Output of:"parameter A has values 1,2,"
- d) Output of:"parameter A has values ,,"**
- e) Output of:"parameter A has values 1,,"

136. Consider the contents of two JSP files:

File 1: this.jsp

```
<html><body><pre>
<jsp:include page="that.jsp" >
<jsp:param name="color" value="red" />
<jsp:param name="color" value="green" />
</jsp:include>
</pre></body></html>
```

File 2: that.jsp

```
<%
String colors[] = request.getParameterValues("color");
for (int i=0; i<colors.length; i++)
{
out.print(colors[i] + " ");
}
%>
```

Which one of the following will be the output of accessing the this.jsp file via the URL - <http://localhost:8080/chapter13/this.jsp?color=blue?>

- a) blue
- b) red green
- c) red green blue**
- d) blue red green
- e) blue green red

137. Consider the contents of the following two JSP files:

File 1: test1.jsp

```
<html><body>
<% String message = "Hello"; %>
//1 Insert Lines Of Code here.
The message is <%= message %>
</body></html>
```

File 2: test2.jsp

```
<% message = message + " world!"; %>
```

Which one of the following lines can be inserted at //1 in test1.jsp so that it prints "The message is Hello world!" when requested?

- a) <%@ include page="test2.jsp" %>
- b) <%@ include file="test2.jsp" />**
- c) <jsp:include page="test2.jsp" />
- d) <jsp:include file="test2.jsp" />

138. Which two of the following XML syntax format tags do not have an equivalent in JSP syntax format?

- a) <jsp:directive.page/>
- b) <jsp:directive.include/>
- c) <jsp:text></jsp:text>**
- d) <jsp:root></jsp:root>**
- e) <jsp:param/>

139. In your report.jsp page, you want to include the output of the "customer.jsp" page. But this page requires an additional parameter "custid", which is not present in the request.

Which one of the following code snippets do you need to add?

- a) `<jsp:include page="customer.jsp" custid="1234"/>`
- b) `<jsp:include page="customer.jsp">`
 `<jsp:param>`
 `<name>custid</name>`
 `<value>1234</value>`
 `</jsp:param>`
 `</jsp:include>`
- c) `<jsp:include page="customer.jsp" param-name="custid" param-value="1234"/>`
- d) `<jsp:include page="customer.jsp">`
 `<jsp:param name="custid" value="1234"/>`
 `</jsp:include>`

140. Consider the following:

```
<% int x = 3; %>
<%! int x = 5; %>
<%! int y = 6; %>
The sum of x and y is <%=x+y%>
```

Which one of the following is the correct output of above code?

- a) 9
- b) 11
- c) error variable already declared
- d) 6

ADVANCE JAVA

1. Given the HttpServlet method:

`doGet(HttpServletRequest, HttpServletResponse)`

Which one of the following statement is true?

- a) It returns servlet version information.
- b) It returns servlet context information.
- c) It services an HTTP GET request.**
- d) It services an HTTP OPTIONS request.

2. Which one of the following directory stores the deployment descriptor (web.xml) for a Web application located in the myweb directory?

- a) myweb/WEB-INF**
- b) myweb/WEB-INF/lib
- c) myweb/WEB-INF/catalog
- d) myweb/WEB-INF/classes

3. Which one of the following statement will insert static text from the tip.jsp file into a JSP page?

- a) `<% include file='tip.jsp' %>`
- b) `<%! include file='tip.jsp' %>`
- c) `<%@ include file='tip.jsp' %>`**
- d) `<%= include file='tip.jsp' %>`

4. Which one of the following is true of the setProperty standard action in JSP?

- a) The developer must supply the exact name and value of the property to be set.
- b) The developer cannot calculate the value of the property to be set at runtime.
- c) The developer can use a wild card to set multiple properties using a single action.**
- d) The name of the property to be set must always match a request parameter name.

5. Where should the persistent fields of a CMP bean be defined?

- a) as private instance variables in the bean
- b) in the deployment descriptor**
- c) as public instance variables in the bean
- d) as private class variables in the bean

6. What is the correct HTML for making a hyperlink?

- a) `<a href="http://www.cognizant.com">cognizant</a>`
- b) `<a>http://www.cognizant.com</a>`**
- c) `<a url="http://www.cognizant.com">cognizant.com</a>`
- d) `<a name="http://www.cognizant.com">cognizant.com</a>`

7. How can you make an e-mail link?

- a) `<a href="mailto:xxx@yyy">`**
- b) `<mail>xxx@yyy</mail>`
- c) `<mail href="xxx@yyy">`
- d) `<a href="xxx@yyy">`

8. How can you open a link in a new browser window?

- a) `<a href="url" new>`
- b) `<a href="url" target="new">`
- c) `<a href="url">`
- d) `<a href="url" target="_blank">`**

9. Given:

- 1. `<html>`
- 2. `<body>`
- 3.
- 4. `<% x = x + 50; %>`
- 5. `Set it to: <%= x %>`
- 6. `</body>`
- 7. `</html>`

What should be inserted on line 3 to allow compilation to succeed, and allow the output of "Set it to: 94"?

- a) `<% int x = 44 %>`
- b) `<%@ int x = 44 %>`
- c) `<%! int x = 44; %>`**
- d) `<%! int x = 44; !%>`

10. Given:

`<%= bean1.getSize() %>`

What will produce the same results?

- a) `<jsp:getProperty bean1.size />`
- b) `<jsp:getProperty "bean1.size" />`
- c) `<jsp:getProperty name=bean1 property=size />`
- d) `<jsp:getProperty name="bean1" property="size" />`**

11. Which one of the following will print out 100 on the rendered page?

- a) `<%= Math.pow(10,2) %>`**
- b) `<% Math.pow(10,2); %>`
- c) `<jsp:print>Math.pow(10,2)</jsp:print>`
- d) `<jsp:evaluate>Math.pow(10,2)</jsp:evaluate>`

**12. `Public Customer SEcustomer extends SessionBean()`
In the above definition, Scustomer is a/an**

- a) Home interface
- b) Remote Interface
- c) Session Bean**
- d) Entity Bean

13. With one of the following in respect to servlet is true

- a) Servlet cannot access other Java objects on the server in which it resides
- b) Servlet resides in the client
- c) Servlet is singlethreaded by default
- d) Servlet is a request/response mechanism**

14. In a two tier architecture which one the following is true

- a) Difficult to deploy
- b) Difficult to modify code and redistribute after initial deployment**
- c) Cannot access the database
- d) Only a limited validation can be done

15. Which one of the following statement is true?

- a) All XML elements must have a closing tag**
- b) All XML documents must have a DTD
- c) All XML elements must be lower case

- d) All XML elements must have a root element

16. Which one of the following should NOT be used to share data between servlets in a distributed web application?

a) Attributes of ServletContext

- b) Enterprise Java Beans
- c) Attributes of HttpSession
- d) Database

17. Which one of following packaging options of enterprise beans are recommended for most J2EE applications?

- a) Package each enterprise bean for an application in its own EJB module
- b) Package all enterprise beans for an application in one EJB module

c) Package all related enterprise beans for an application in one EJB module

- d) Package all related enterprise beans for an application in multiple EJB module

18. Which one of the following method cannot be called by a stateful session bean using bean-managed transactions?

- a) getUserTransaction()
- b) getStatus()
- c) **getRollbackOnly()**
- d) Rollbackonly()

19. Which one of the following zodiac_sign elements could be valid?

- a) <zodiac_sign type=animal>Leo</zodiac_sign>
- b) <zodiac_sign type="animal">Aries</zodiac_sign>**
- c) <zodiac_sign type=animal name=Taurus/>
- d) <zodiac_sign "life class"="animal">Cancer<zodiac_sign>

20. Which one of the following XML fragments are well-formed?

- 1) <?xml version='1.0'?>
 <!-- Author: me, version: 1.01 -->
- 2) <!-- Author: me, version: 1.01 -->
 <?xml version='1.0'?>

a) 1 only

- b) 2 only

- c) Both 1,2
- d) Neither 1 nor 2

21. Which one of the following statements is true regarding MyServlet?

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
public class MyServlet extends HttpServlet implements SingleThreadModel {
    String myName;
    public void service(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        response.setContentType("text/plain");
        PrintWriter out = res.getWriter();
        myName = req.getParameter("name");
        sayHello(out);
        out.close();
    }
    public void sayHello(PrintWriter out) {
        out.println("Hello " + myName);
    }
}
```

- a) MyServlet is thread safe.**
- b) MyServlet is not thread safe because myName is an instance variable.
- c) MyServlet is not thread safe because MyServlet implements SingleThreadModel.
- d) MyServlet is an apartment threaded model

22. Which one of the following statement is true?

- 1.XML elements must be properly nested
- 2.XML documents must have a root tag
- 3. XML tags are case sensitive

- a) 1 only
- b) 2 only
- c) 3 only

d) All 1,2,3 are true

23. Given that the following web.xml entry specifies a ServletContext initialization parameter for a web application:

```
<context-param>
  <param-name>com.jspervletcookbook.BoundObj</param-name>
  <param-value>com.jspervletcookbook.BoundObj</param-value>
</context-param>
```

Which one of the following servlet code accesses this initialization parameter?

a) String boundObjectName =

getServletContext().getInitParameter("com.jspservletcookbook.Bound Obj");

- b) `com.jspservletcookbook.BoundObj boundObject = (com.jspservletcookbook.BoundObj) getServletContext().getAttribute("com.jspservletcookbook.BoundObj");`
- c) `String boundObjectName = request.getParameter("com.jspservletcookbook.BoundObj");`
- d) `com.jspservletcookbook.BoundObj boundObject = (com.jspservletcookbook.BoundObj) getSession().getAttribute("com.jspservletcookbook.BoundObj");`

24. Which one of the following HttpSession method stores an object like a shopping cart in a session?

- a) `setId()`
- b) `setObject(String objectName)`
- c) `setSessionObject(String objectName)`

d) `setAttribute(String attributeName, Object value)`

25. Which one of the following statement about locating or using a session bean's home interface is true?

- a) Acquiring an InitialContext is required only for remote clients.
- b) After it is acquired by a client, a home interface can be used multiple times.**
- c) The InitialContext must be narrowed before it can be used.
- d) The client can acquire a handle for the bean's local home interface.

26. Which one of the following is true about the client's view of exceptions received from an EJB?

- a) Receiving an application exception communicates to the client that the bean will no longer be accessible.
- b) The client of a CMP bean can determine if the transaction has been rolled back by calling the `getStatus()` method of the `java.transaction.UserTransaction` interface.
- c) If the client receives an `EJBException`, the client must discontinue the transaction.
- d) If the client receives the `java.rmi.RemoteException` exception, the client will not typically know whether the method has completed.**

27. A CMP entity bean called Customer has the persistent fields `customerID`, `name`, `address`, and `phone`. The `customerID` field is the primary key. When `create` is called, it is

passed the arguments customerId, name, address, and phone. All are of type String. How would the ejbCreate method populate the address persistent field in the bean?

a) setAddress(address)

b) this.address = address

c) address = address

d) getPrimaryKey().setAddress(address)

28.If a JSP page overrides the jsplnit() method using a declaration JSP tag, which one of the following phase of the JSP page life-cycle generates the overridden method in the servlet?

a) JSP page compilation

b) call jsplnit()

c) call jspDestroy()

d) call _jspService()

29. Consider the following code-snippet in the deployment-descriptor from the web-application CD. To which one of the following rule will the following request URI map: /CD/door/file.hall?

<!-- Rule 1 -->

```
<servlet-mapping>
  <servlet-name>VauxHallServlet</servlet-name>
  <url-pattern>/hall/*</url-pattern>
</servlet-mapping>
```

<!-- Rule 2 -->

```
<servlet-mapping>
  <servlet-name>VauxHallServlet</servlet-name>
  <url-pattern>/hall/hall/*</url-pattern>
</servlet-mapping>
```

<!-- Rule 3 -->

```
<servlet-mapping>
  <servlet-name>VauxHallServlet</servlet-name>
  <url-pattern>/door/*</url-pattern>
</servlet-mapping>
```

<!-- Rule 4 -->

```
<servlet-mapping>
  <servlet-name>VauxHallServlet</servlet-name>
  <url-pattern>*.hall</url-pattern>
</servlet-mapping>
```

- a) Rule 1
- b) Rule 2
- c) Rule 3**
- d) Rule 4

30. Which one of the following statements are true about the given method declaration? MyEJBObject createWithName(String name) throws CreateException, RemoteException

- a) It is declared in the local home interface of a stateful session bean.
- b) It is declared in the local home interface of a stateless session bean.
- c) It is declared in the remote home interface of a stateless session bean
- d) It is declared in the remote home interface of a stateful session bean**

31. A JSP page called test.jsp is passed a parameter name in the URL using http://localhost/test.jsp?name="John". The test.jsp contains the following code.

```
<%! String myName=request.getParameter();%>
<% String test= "welcome" + myName; %>
<%= test%>
```

- a) The program prints "Welcome John"
- b) The program gives a syntax error because of the statement**

```
<%! String myName=request.getParameter();%>
```

- c) The program gives a syntax error because of the statement

```
<% String test= "welcome" + myName; %>
```
- d) The program gives a syntax error because of the statement

```
<%= test%>
```

32. What gets printed when the following JSP code is invoked in a browser. Select one correct answer.

```
<%= if(Math.random() < 0.5) { %>
hello
<%= } else { %>
hi
<%= } %>
```

- a) The browser will print either hello or hi based upon the return value of random.
- b) The string hello will always get printed.

c) The string hi will always get printed.

d) The JSP file will not compile.

33. What gets printed when the following is compiled. Select one correct answer.

```
<% int y = 0; %>
```

```
<% int z = 0; %>
```

```
<% for(int x=0;x<3;x++) { %>
```

```
<% z++;++y;%>
```

```
<% }%>
```

```
<% if(z<y) {%>
```

```
<%= z%>
```

```
<% } else {%>
```

```
<%= z - 1%>
```

```
<% }%>
```

a) 0

b) 1

c) 2

d) The program generates compilation error.

34. Given this text from a valid web.xml file for a web application to be deployed at www.myorg.org:

```
<servlet>
```

```
  <servlet-name>shopping</servlet-name>
```

```
  <servlet-class>org.myorg.ShoppingServlet</servlet-class>
```

```
  <init-param>
```

```
    <param-name>year</param-name>
```

```
    <param-value>2004</param-value>
```

```
  </init-param>
```

```
</servlet>
```

```
<servlet-mapping>
```

```
  <servlet-name>shopping</servlet-name>
```

```
  <url-pattern>/shop/*</url-pattern>
```

```
</servlet-mapping>
```

Which one of the following is true?

a) Requests for the URL http://www.myorg.org/shopping will be sent to the org.myorg.ShoppingServlet servlet.

b) It would be invalid to add another <servlet> element that also included <servlet-name>shopping</servlet-name>.

- c) Each time the servlet named "shopping" is used to process a request, it will be passed a parameter named "year" with a value of "2004."
- d) The <servlet-mapping> element should be nested within the <servlet> element.

35. Which one of the following are true when comparing two session objects with different homes?

- a) Using the isIdentical() method, stateless session beans will always return true.
- b) Using the isIdentical() method, stateless and stateful session beans will always return false.**
- c) The isIdentical() method can be used for only local object references.
- d) Using the isIdentical() method, stateful session beans will always return true.

36. Which one of the following method in the EntityContext allows the developer to test the state of the current transaction?

- a) getRollbackOnly()**
- b) getUserTransaction()
- c) getTransactionState()
- d) getCurrentTransaction()

37. What statement do you need to insert at line 4 of the AnimalServlet to pick up all selected animals in the form?

```
<form action="/cd/servlet/com.baboon.cd.servletmodel.AnimalServlet" method="POST">
  <select name="animals" size="3" multiple>
    <option value="TIGER">Tiger</option>
    <option value="ELEPHANT">Elephant</option>
    <option value="LION">Lion</option>
  </select>
  <input type="submit" value="Search">
</form>
```

```
1. public class AnimalServlet extends HttpServlet {
2.     protected void doPost(HttpServletRequest req, HttpServletResponse resp)
3.         throws ServletException, IOException {
4.         // insert code here
5.     }
6. }
```

- a) String animals= req.getParameter("animals");
- b) String[] animals = req.getParameterValues("animals");**
- c) String[] animals = req.getParameterNames("animals");
- d) Syntax Error

38. How is a servlet declared to use the single thread model?

- a) The servlet declares the service method as synchronized.
- b) The servlet implements the interface SingleThreadModel**
- c) Nothing is required, by default a servlet uses the single thread model
- d) The declaration is vendor specific

39. In a distributed thin client architecture which of the following is true

- a) Client has all the application logic
- b) As the application logic reside on the application server, it is easier to modify code as no redistribution is involved**
- c) Can not access the database
- d) Database triggers can not be used as it will make the client fat

40. You want to locate an existing bean object but do not want to create a new instance, if an existing instance is not available. Which one of the following bean attributes will you use in the jsp:useBean action?

- a) class
- b) type**
- c) beanName
- d) class and type

41. Considering the following JSP pages, which one of the following of the statements below are true?

```
//include1.jsp
<html>
<body>
<%@ include file="date.jsp" %>
</body>
</html>
```

```
//include2.jsp
<html>
<body>
<jsp:include page="date.jsp" flush="true" />
</body>
</html>
```



```
// date.jsp
<html>
<body>
<%@ page import="java.util.Date" %>
<%= "The current date is" + new Date() %>
</body>
</html>
```

a) The output will be identical, but the loading time for both JSPs (include1.jsp and include2.jsp) will be different.

- b) The output will be different in two JSPs (include1.jsp and include2.jsp)
- c) The output will be identical, and loading time for both JSPs (include1.jsp and include2.jsp) will be same.
- d) Syntax Error

42. Consider the following session bean home interface located through the Java Naming and Directory Interface (JNDI):

```
Context initialContext = new InitialContext();
ProcessPaymentHome ppHome = (ProcessPaymentHome)
initialContext.lookup("java:comp/env/ejb/pphome");
```

Which one of the following exceptions must be declared in the signatures of the methods contained in the ProcessPaymentHome interface?

- a) RemoteException
- b) FindException
- c) Any application specific exception**
- d) CreateException

43. Assume that you need to write a JSP page that adds numbers from one to ten, and then print the output.

```
<% int sum = 0;
for(j = 0; j < 10; j++) { %>
// XXX --- Add j to sum
<% } %>
// YYY --- Display ths sum
```

Which one of the following statement when placed at the location XXX can be used to compute the sum. Select the one correct statement

- a) `<% sum = sum + j %>`
- b) `<% sum = sum + j; %>`**
- c) `<%= sum = sum + j %>`
- d) `<%= sum = sum + j; %>`

44. You want to use URL rewriting to support client browsers, which do not use cookies. Which one of the following method will you use to attach the session id to a URL that is to be used for the `sendRedirect()` method of the `HttpServletResponse`?

- a) `encodeURL`
- b) `encodeRedirectURL`**
- c) `encodeSessionURL`
- d) `encodeSessionRedirectURL`

45. If the HTTP error 500 is generated by your servlet, you do not want to show the "Internal Server Error" page to the client. Instead, you want a custom error page to be displayed. What is the best way to accomplish this?

- a) Forward the user to the error page using `HttpServletResponse.sendRedirect` method
- b) Forward the user to the error page using `RequestDispatcher.forward` method
- c) Specify the mapping of the error-code 500 and the error page in the deployment descriptor**
- d) It is not possible to accomplish this

46. An EJB is required to perform credit card verification. All necessary information such as card number and amount will be supplied as parameters to a single business method invocation that processes the verification. Assuming that performance is critical, which bean type BEST suits the requirement?

- a) Stateful Session Bean
- b) Stateless Session Bean**
- c) Message Driven Bean
- d) Entity Bean

47. What will be the output of the following Servlet? For example, it can be invoked through a Web Browser with the URL `http://:8080/servlet/HelloServlet`

```
import java.io.*;
import javax.servlet.*;
```

```

import javax.servlet.http.*;
public class HelloServlet extends HttpServlet
{
    public void service(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
    }
    public void doGet(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<html><head><title>The Standard Hello World Servlet</title></head>");
        out.println("<body><b>doGet Hello World!!</b></body></html>");
        out.close();
    }
    public void doPost(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<html><head><title>The Standard Hello World Servlet</title></head>");
        out.println("<body><b>doPost Hello World!!</b></body></html>");
        out.close();
    }
}

```

- a) An HTML page displaying doGet Hello World!!
- b) An HTML page displaying doPost Hello World!!
- c) An error message

d) A blank HTML page

48. Problem: North American Real Estate Associates, a realtor agency would like to convert their inventory listings into XML format. Until now, they used a commercial application to retrieve the information from the database and render it to their clients in the HTML format.

Apart from being able to supply the inventory information to various different clients (web, handheld, wireless devices etc.), the new XML-based system should also allow numerous real-estate agents across the country to send their local inventory listings to a central application. The new listings and/or update of existing listings will arrive by email as an XML document. A processing application is required to be designed that will analyze the incoming XML documents once everyday and will bring up to date the master database.

Question: Which one of the following statements is TRUE with respect to the task of validating the incoming XML document ?

- a) No validation needs to be done since the incoming emails are from trusted partners
- b) Using mixed content modelling for DTDs facilitates maximum flexibility since DTDs can be changed without affecting the validity of existing XML documents

- c) Validating parsers should be used for testing only and should not be used in production for performance reasons
- d) A validating parser is necessary to ensure that incoming documents are well-formed**

49. A servlet needs to acquire a data source through a JNDI naming lookup. Which one of the following is the best place to do this?

- a) Constructor
- b) service method
- c) doGet method
- d) init method**

50. Your application supports multiple client types including HTTP clients. Your business layer is implemented using Enterprise Java Beans. Which one of the following is best suited for maintaining client state?

- a) Stateful session beans**
- b) Entity Beans**
- c) HttpSession attributes**
- d) Cookies**

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Number: 1z0-899
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Time Limit: 120 min
File Version: 14.5

Exam Code: 1Z0-899

Exam Name: Java EE 6 Web Component Developer Certified Expert Exam



1z0-899

QUESTION 1

Given the element from the web application deployment descriptor:

```
<jsp property group>
<url pattern>/main/page1.jsp</url pattern>
<scripting invalid>true</scripting invalid>
</jsp--property--group>
```

And given that /main/page1.jsp contains:

```
<% int i = 12; %>
<b> <%= i %> </b>
```

What is the result?

- A.
- B. 12
- C. The JSP fails to execute.
- D. <% int i = 12 %>
 <%= i %> < b>

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 2

A web application allows the HTML title banner to be set using a context initialization parameter called titlestr.

Which two properly set the title in the scenario? (Choose two)

- A. <title> \$ {titlestr} </title>
- B. <title> \$ {initparam.titlestr}</title>
- C. <title> \$ {param [0]. titlestr} </title>
- D. <title> \$ {paramValues.titleStr} </title>
- E. <title> \$ {iParam ["titleStr"]} </title>
- F. <title> \$ {servletParams.titleStr} </title>
- G. <title> \$ {request.get ("titleStr")} </title>

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 3

Given:

- 11. <%
- 12. request.setAttribute ("vals", new String[] {"1", "2", "3", "4"});
- 13. request.setAttribute ("index", "2");
- 14. %>

15. <% -- insert code here -- %>

Which three EL expressions, inserted at line 15, are valid and evaluate to "3"? (Choose three)

- A. \${vals.2}
- B. \${vals ["2"]} }
- C. \${vals.index}
- D. \${vals[index]} }
- E. \${vals} [index]
- F. \${vals. (vals.index) }
- G. \${vals [vals[index-1]] }

Correct Answer: BDG

Section: (none)

Explanation

Explanation/Reference:

QUESTION 4

Given:

```
11. <% java.util.Map map = new java.util.HashMap();
12.     request.setAttribute("map", map);
13.     map.put("a", "b");
14.     map.put("b", "c");
15.     map.put("c", "d"); %>
16. <%-- insert code here --%>
```

Which three EL expressions, inserted at line 16, are valid and evaluate to "d"? (Choose three)

- A. \${map.c}
- B. \${map.[c]}
- C. \${map["c"]}
- D. \${map.map.b}
- E. \${map.[map.b]}
- F. \${map. (map.b)}

Correct Answer: ACE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 5

You are building a dating service web site. Part of the form to submit a client's profile is a group of radio buttons for the person's hobbies:

- ```
20. <input type = `radio` name = `hobbyEnum` value = `HIKING`> Hiking

21. <input type = `radio` name = `hobbyEnum` value = `SKING`> Sking

22. <input type = `radio` name = `hobbyEnum` value = `SCUBA`> SCUBA

23. <! -- and more options -- >>
```

After the user submits this form, a confirmation screen is displayed with these hobbies listed. Assume that an

application-scoped hobbies, holds a map between the hobby enumerated type and the display name.

Which EL code snippet will display Nth element of the user's selected hobbies?

- A. `${hobbies [hobbyEnum[N]]}`
- B. `${hobbies [paramValues.hobbyEnum[N]]}`
- C. `${hobbies [paramValues @ `hobbyEnum' @N]}`
- D. `${hobbies.get(paramValues.hobbyEnum[N]) }`
- E. `${hobbies [paramValues.hobbyEnum.get(N)] }`

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

### QUESTION 6

Given:

1. `<% int[] nums = {42,420,4200};`
2. `request.setAttribute("foo", nums); %>`
3. `${5 + 3 lt 6}`
4. `S(requestScope["foo"][0] ne 10 div 0>`
5. `${10 div 0}`

What is the result?

- A. true true
- B. false true
- C. false true 0
- D. true true Infinity
- E. false true Infinity
- F. An exception is thrown
- G. Compilation or translation fails

**Correct Answer:** E

**Section:** (none)

**Explanation**

**Explanation/Reference:**

### QUESTION 7

You are building a web application with a scheduling component. On the JSP, you need to show the current date, the date of the previous week, and the date of the next week.

To help you present this information, you have created the following EL functions in the `d' namespace:

- Name : curDate; signature: java.util.utilDate CurrentDate()
- Name : curDate; signature: java.util.utilDate addweek (java.util.Date, int)
- Name: dateString; signature: java.util.String getDataString (jave.util.Data)

Which EL code snippet will generate the string for the previous week?

- A. `${d:dateString(affWeek(curDate(), -1)) }`
- B. `${d:dateString [addWeek[curDate[], - ]] }`



- C. `#{d:dateString [d:addWeek[d:curDate[], - 1]] }`
- D. `#{d:dateString (d:addWeek(d:curDate(), -1)) }`

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 8

Given a header in an HTTP request:

X-Retries: 4

A Which two retrieve the value of the header from a given `HttpServletRequest` request? (Choose two)

- A. `request.getHeader("X-Retries")`
- B. `request.getIntHeader("X-Retries")`
- C. `request.getRequestHeader("X-Retries")`
- D. `request.getHeaders{"X-Retries"}.get(0)`
- E. `request.getRequestHeaders("X-Retries").get(0)`

**Correct Answer:** AB

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 9

Given an `HttpServletRequest` request and `HttpServletResponse`, which sets a cookie "username" with the value "joe" in a servlet?

- A. `request.addCookie("username", "joe")`
- B. `request.setCookie("username", "joe")`
- C. `response.addCookie("username", "joe")`
- D. `request.addHeader(new Cookie("username", "joe"))`
- E. `request.addCookie(new Cookie("username", "joe"))`
- F. `response.addCookie(new Cookie("username", "joe"))`
- G. `response.addHeader(new Cookie("username", "joe"))`

**Correct Answer:** F

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 10

Which annotation enables a servlet to efficiently process requests of type multipart/form-data that involve large files being uploaded by the client?

- A. `@AcceptMultipart`
- B. `@MultiPartConfig`
- C. `@MultiPartFormData`

D. @WebServlet (multipart = true)

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: <http://www.scribd.com/ilinchen2008/d/38764279-Servlet3-0-Specs> (page 22, last paragraph)

#### QUESTION 11

APIs for dynamically registering servlets and filters can be called at:

- A. Context initialization time
- B. Any time
- C. There is no way to dynamically register servlets and filters
- D. Context destruction time
- E. Injected into a servlet context

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 12

ServletRegistration.Dynamic can be used to configure (Choose two)

- A. Filter mappings
- B. add init params
- C. set asyncSupported to true
- D. add Listeners

**Correct Answer:** AB

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: <http://www.softwareengineeringsolutions.com/blogs/2010/08/01/> (3rd paragraph)

#### QUESTION 13

A web application wants to register a new servlet at runtime using one of the `servletContext.addServlet` methods.

In the implementation of which method is it legal to perform this operation?

- A. `AsyncListener.oncomplete`
- B. `Servlet.init`
- C. `ServletContextListener.contextInitialize`
- D. `ServletRequestListener.contextInitialize`

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 14**

You want to create a filter for your web application and your filter will implement

`.servlet.Filter`.

Which two statements are true? (Choose two)

- A. Your filter class must implement an `init` method and a `destroy` method.
- B. Your filter class must also implement `javax.sarvlat.FilterChain`.
- C. When your filter chains to the next filter, it should pass the same arguments it received in its `doFilter` method.
- D. The method that your filter invokes on the object it received that implements `javax servlet.Filterchain` can invoke either another filter or a servlet.
- E. Your filter class must implement a `doFilter` method that takes, among other things, an `HttpServletRequest` object and an `HttpServletResponse` object.

**Correct Answer:** AD

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 15**

You have a simple wpb application that has a single Front Controller servlet that dispatches to JSPs generate a variety of views. Several of these views require further database processing to retrieve the necessary order object using the `orderId` request parameter. To do this additional processing, you pass the request first to a servlet that is mapped to the URL pattern `/WEB INF / retrieveOrder.do`. in the deployment descriptor. This servlet takes two request parameters, the `ordered` and the `jspID` and the `jspURL`. It handles the database calls to retrieve and build the complex order objects and then it dispatches to the `jspURL`.

Which code snippet in the Front Controller servlet dispatches the request to the order retrieval servlet?

- A. `request.setAttribute("orderId", orderIS);`  
`request.setAttribute("jspURL", jspURL);`  
`= context.getRequestDispatcher("/WEB INF / retrieveOrder.do"); view.forward(request, response)`
- B. `request.setAttribute("orderId", orderIS);`  
`request.setAttribute("jspURL", jspURL);`  
`Dispatcher view`  
`= request.getDispatcher("/.WEB INF / retrieveOrder.do"); View.forwardRequest (request, response);`
- C. `String T= "/WEB INF / retrieveOrder.do?orderId = %d&jspURI = %s";` `String url = String.format (T, ordered, jspURL);`  
`= context.getRequestDispatcher (url);`  
`View.forward (request, response) ;`
- D. `String T= "/WEB INF / retrieveOrder.do?orderId = %d&jspURI = %s";` `String url = String.format (T, ordered, jspURL);`  
`= context.getRequestDispatcher (url);`  
`View.forwardRequest (request, response) ;`

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 16**

In `ServletContext.getNamedDispatcher(string arg0)` `arg0` is the

- A. url of a servlet relative to the `ServletRequest`
- B. url of a servlet relative to the `Context`
- C. name of a servlet
- D. absolute url to any servlet in any `Context`

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 17**

When using Servlet asynchronous API if you want to dispatch the request back to a particular url `"/url"` within the same Servlet Context which of the following API would you use?

- A. `AsyncContext.dispatch();`
- B. `AsyncContext.dispatch("/url");`
- C. `AsyncContext.dispatch(servletContext, "/url");`
- D. `AsyncContext.start(runnable);`
- E. `RequestDispatcher.forward("/url");`
- F. `RequestDispatcher.forward(servletContext, "/url");`
- G. `RequestDispatcher.include("/url");`

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: [http://blogs.oracle.com/enterprisetechtips/entry/asynchronous\\_support\\_in\\_servlet\\_3](http://blogs.oracle.com/enterprisetechtips/entry/asynchronous_support_in_servlet_3) (Topic: `AsyncContext` Class, third paragraph)

**QUESTION 18**

Which describes a trigger that causes a call to an `AsyncListener`?

- A. Completion of processing
- B. Receipt of a new request from the same client
- C. Addition of a new filter to the processing stream for this request
- D. Completion of each timer interval

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 19**

Refer to the Exhibit.

```

1. package com.example;
2.
3. public class Product {
4. private String name;
5. private double price;
6.
7. public Product() {
8. this("Default", 0.0);
9. }
10.
11. public Product(String name, double
price) {
12. this.name = name;
13. this.price = price;
14. }
15.
16. public String getName() {
17. return name;
18. }
19.
20. public void setName(String name) {
21. this.name = name;
22. }
23.
24. public double getPrice() {
25. return price;
26. }
27.
28. public void setPrice(double price) {
29. this.price = price;
30. }
31. }

```

A servlet sets a session-scoped attribute product with an instance of com.example.product and forward to a JSP. Which two output the name of the product in the response? (Choose two)

- A. `<%= product.getName() %>`
- B. `<jsp:useBean id="product" class="com.example.Product" /> <%= product.getName() %>`
- C. `<jsp:useBean id="com.example.Product" scope="page"> <%= product.getName() %>`  
`</jsp:useBean>`
- D. `<jsp:useBean id="product" type="com.example.Product" scope="page" />`  
`<%= product.getName() %>`
- E. `<jsp:useBean id="product" type="com.example.Product"> <%= product.getName() %>`  
`</jsp:useBean>`

**Correct Answer:** BD

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 20

Click the Exhibit button.

Given the HTML form:

1. `<html>`

2. <body>
3. <form action="submit.jsp">
4. Name: <input type="text" name="i1"><br>
5. Price: <input type="text" name="i2"><br>
6. <input type="submit">
7. </form>
8. </body>
9. </html>

Assume the product attribute does NOT yet exist in any scope. Which code snippet, in submit.jsp, instantiates an instance of com.example.Product that contains the results of the form submission?

```

1. package com.example;
2.
3. public class Product {
4. private String name;
5. private double price;
6.
7. public Product() {
8. this("Default", 0.0);
9. }
10.
11. public Product(String name, double
price) {
12. this.name = name;
13. this.price = price;
14. }
15.
16. public String getName() {
17. return name;
18. }
19.
20. public void setName(String name) {
21. this.name = name;
22. }
23.
24. public double getPrice() {
25. return price;
26. }
27.
28. public void setPrice(double price) {
29. this.price = price;
30. }
31. }

```

- A. <jsp:useBean id="com.example.Product" />  
<jsp:setProperty name="product" property="" />
- B. <jsp:useBean id="product" class="com.example.Product" /> \${product.name = param.i1}  
\${product.price = param.i2}
- C. <jsp:useBean id="product" class="com.example.Product"> <jsp:setProperty name="product"  
property="name"  
param="i1" />  
<jsp:setProperty name="product" property="price"  
param="i2" />  
</jsp:useBean>
- D. <jsp:useBean id="product" type="com.example.Product"> <jsp:setProperty name="product"  
property="name"  
value="<%= request.getParameter( "i1" ) %>" />

```
<jsp:setProperty name="product" property="price"
value="<%= request.getParameter("i2") %>" />
</jsp:useBean>
```

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 21

You need to create a JavaBean object that is used only within the current JSP page. It must NOT be accessible to any other page including those that this page might import. Which JSP standard action < .in accomplish this goal?

- A. <jsp:useBean id = `pageBean' type = `com.example.MyBean' />
- B. <jsp:useBean id = `pageBean' class = `com.example.MyBean' />
- C. <jsp:makeBean id = `pageBean' type = `com.example.MyBean' />
- D. <jsp:makeBean id = `pageBean' class = `com.example.MyBean' />
- E. <jsp:useBean id = `pageBean' class = `com.example.MyBean' />
- F. <jsp:makeBean id = `pageBean' class = `com.example.MyBean' />

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 22

Given the JavaBean class:

```
public class MyBean {
 private Boolean roasted = false;
 public MyBean () {}
 public Boolean isRoasted () { return roasted }
 public void setRoasted (Boolean roasted) { this.roasted = roasted; }
```

Assume a controller has created an instance of this bean, called setRoasted (true), and inserted the bean into the application scope using the name "myBean".

A JSP contains these two tags:

```
<jsp: useBean id = "aBean" class = "MyBean" scope = "page" \ /> <jsp: getProprty name = "aBean" property =
"roasted" \ />
```

Which is true?

- A. the page will include the output false
- B. the page will include the output
- C. the page will report that the property roasted does not exist
- D. the page will report that the syntax of the useBean tag is incorrect
- E. the page will report that the syntax of the getProperty tag is incorrect

**Correct Answer:** E

**Section:** (none)

## Explanation

### Explanation/Reference:

Reference: <http://www.emacao.gov.mo/documents/18/06/exam.pdf> (4th table from the top)

### QUESTION 23

A developer chooses to avoid using `singleThreadModel` but wants to ensure that data is updated in a thread safe manner.

Which two can support this design goal? (Choose two)

- A. Store the data in a local variable.
- B. Store the data in an instance variable.
- C. Store the data in the `HttpSession` object.
- D. Store the data in the `servletContext` object.
- E. Store the data in the `ServletRequest` object.

**Correct Answer:** AE

**Section:** (none)

## Explanation

### Explanation/Reference:

### QUESTION 24

You have a use in your web application that adds several session-scoped attributes. At the end of the use case, one of these objects, the `manage` attribute, is removed and then it needs to decide which of the other session-scoped attributes to remove.

How can this goal be accomplished?

- A. The object of the manager attribute should implement the `HttpSessionBindingListener` and it should call the `removeAttribute` method on the appropriate session attributes.
- B. The object of the manager attribute should implement the `HttpSessionListener` and it should call the `removeAttribute` method on the appropriate session attributes.
- C. The object of the manager attribute should implement the `HttpSessionBindingListener` and it should call the `deleteAttribute` method on the appropriate session attributes.
- D. The object of the manager attribute should implement the `HttpSessionListener` and it should call the `deleteAttribute` method on the appropriate session attributes.

**Correct Answer:** A

**Section:** (none)

## Explanation

### Explanation/Reference:

### QUESTION 25

The Squeaky Beans Inc. shopping application was initially developed for a non-distributed environment. The company recently purchased the Acme Application Server, which supports distributed `HttpSession` objects. When deploying the application to the server, the deployer marks it as distributable in the web application deployment descriptor to take advantage of this feature. Given this scenario, which two must be true? (Choose two)

- A. The J2EE web container must support migration of objects that implement `Serializable`.
- B. The J2EE web container must use the native JVM Serialization mechanism for distributing `HttpSession` objects.



- C. As per the specification, the J2EE web container ensures that distributed HttpSession objects will be stored in a database.
- D. Storing references to Enterprise JavaBeans components in the HttpSession object might NOT supported by J2EE web contain.

**Correct Answer:** AD

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 26

Given an HttpServletRequest request and an HttpServletResponse response:

```
41. HttpSession session = null;
42. // insert code here
43. if (session == null) {
44. // do something if session does not exist
45. } else {
46. // do something if session exists
47. }
```

To implement the design intent, which statement must be inserted at line 42?

- A. session = response.getSession ();
- B. session = request.getSession ();
- C. session = request.getSession (true);
- D. session = request.getSession (false);
- E. session = request.getSession ("jsessionId");

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 27

Users of your web application have requested that they should be able to set the duration of their sessions. So for example, one user might want a webapp to stay connected for an hour rather than the webapp's default of fifteen minutes; another user might want to stay connected for a whole day.

Furthermore, you have a special login servlet that performs user authentication and retrieves the object from the database. You want to augment this code to set up the user's specified session duration.

Which code snippet in the login servlet will accomplish this goal?

- A. User user = // retrieves the user objects object from the database session.setDurationInterval (user.getSessionDuration());
- B. User user = // retrieves the User objects from the database session.setInactiveInterval (user.getSessionDuration());
- C. User user = // retrieves the user objects object from the database session.setInactiveInterval (user.get.SessionDuration());
- D. User user = // retrieves the user objects object from the database session.setDuration (user.getSessionDuration());

- E. `User user = // retrieves the user objects object from the database session.setMaxInactiveInterval(user.getSessionDuration());`
- F. `User user = // retrieves the user objects object from the database session.setMaxDurationInterval(user.getSessionDuration());`

**Correct Answer:** E

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 28

Which interface must a class implement so that instances of the class are notified after any object added to a session?

- A. `javax.servlet.http.HttpSessionListener`
- B. `javax.servlet.http.HttpSessionValueListener`
- C. `javax.servlet.http.HttpSessionBindingListener`
- D. `javax.servlet.http.HttpSessionAttributeListener`

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 29

Which statement is true about web container session management?

- A. Access to session-scoped attributes is guaranteed to be thread-safe by the web container.
- B. To activate URL rewriting, the developer must use the `HttpServletResponse.setURLRewriting` method.
- C. If the web application uses HTTPS, then the web container may use the data on the HTTPS request stream to identify the client.
- D. The JSESSIONID cookie is stored permanently on the client so that a user may return to the web application and the web container will rejoin that session.

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 30

A method call inside your servlet has thrown an `IOException` resulting from a lost network connection to a necessary resource. The servlet has determined that it will be unable to operate successfully for a few minutes, but expects to be able to resume operation after that period. How should this situation be handled?

- A. The servlet should re-throw the exception to the web container.
- B. The servlet should throw a `ServletException` to the web container.
- C. The servlet should throw an `UnavailableException` to the web container.
- D. The servlet should redirect to an error page that reports the exception message.
- E. The servlet should retry the request until the method returns successfully.

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: [http://jcp.org/aboutJava/communityprocess/first/jsr053/servlet23\\_PFD.pdf](http://jcp.org/aboutJava/communityprocess/first/jsr053/servlet23_PFD.pdf) (page 26, topic: 2.3.3.2, first paragraph)

#### **QUESTION 31**

During initialization, a servlet finds that the database connection it requires is unavailable. As the system designer, you know that the database start up completes several minutes after the web- container initializes the servlet. How should this be handled?

- A. Retry the connection until it is successful, then allow the init() method to complete.
- B. Throw a ServletException
- C. Throw the IOException
- D. Throw an UnavailableException

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: [http://java.sun.com/j2ee/tutorial/1\\_3-fcs/doc/Servlets6.html](http://java.sun.com/j2ee/tutorial/1_3-fcs/doc/Servlets6.html)

#### **QUESTION 32**

A servlet wishes to indicate that it is unable to initialize at the present time, but that the initialization might succeed at some future time. Which is true?

- A. This cannot be expressed. A servlet either initializes correctly or fails.
- B. This expression is not necessary. If a servlet fails to initialize, the container will try again later each time a request is received that attempts to invoke that servlet.
- C. The servlet should delay until it is ready to complete initialization successfully.
- D. The servlet should throw an UnavailableException
- E. The servlet should throw a ServletException

**Correct Answer:** E

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### **QUESTION 33**

You are implementing a model component. You realize that an IOException might arise if you lose connection to the database. How should you address this?

- A. Implement multipathing to provide redundant connectivity to the database, thereby avoiding that risk of connection failure.
- B. Provide an error handler page, and use the page directive in the invoking ISP to redirect to that page if the error arises.
- C. Use the JSTL <c:catch> tag to take control if the exception arises.
- D. Surround the problem area with a try/catch block and implement appropriate recovery or fallback behavior.

**Correct Answer:** C

**Section: (none)**

**Explanation**

**Explanation/Reference:**

Reference: <http://www.ibm.com/developerworks/java/library/j-jstl0318/> (topic: exception handling)

**QUESTION 34**

Which is a valid value for the enum EmptyRoleSemantic in Javax.servlet.annotation.ServletSecurity?

- A. ADMIT
- B. PERMIT
- C. EXCLUDE
- D. DENYALL
- E. ALLOWALL

**Correct Answer: B**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

Reference: <http://tomcat.apache.org/tomcat-7.0-doc/servletapi/javax/servlet/annotation/ServletSecurity.EmptyRoleSemantic.html>

**QUESTION 35**

Given the annotation for MyServlet:

```
@WebServlet("/abc")
@ServletSecurity (value=@HttpConstraint)
public class Myservlet extends HttpServlet {
```

Which two are true? (Choose two)

- A. No protection of user data must be performed by the transport.
- B. All user data must be encrypted by the transport.
- C. Access is to be denied independent of authentication state and identity.
- D. Only authenticated users are to be permitted.
- E. Access is to be permitted independent of authentication state and identity.

**Correct Answer: BC**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

**QUESTION 36**

Which of the following annotations relate to security in a servlet? (i) @WebSecurity

- (ii) @RolesAllowed
- (iii) @WebConstraint
- (iv) @HttpConstraint
- (v) @Servletsecurity

- A. (i) only
- B. (i) and (iii)
- C. (iv) and (v)
- D. (iii) and (v)

E. (ii) and (iv)

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 37**

Which element of a web application deployment descriptor <web-resource-collection> element is used to specify a HTTP method to which the corresponding <security-constraint> will not apply?

- A. <exclude-http-method>
- B. <neglect-http-method>
- C. <http-method-omission>
- D. <http-method-excluded>
- E. <exceptional-http-method>

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: [http://java.sun.com/xml/ns/javaee/web-common\\_3\\_0.xsd](http://java.sun.com/xml/ns/javaee/web-common_3_0.xsd) (search http-method- omission)

**QUESTION 38**

Which two statements are true about the security-related tags in a valid Java EE deployment descriptor? (Choose two)

- A. Every <security-constraint> tag must have at least one <http-method> tag.
- B. A <security-constraint> tag can have many <web-resource-collection> tags.
- C. A given <auth-constraint> tag can apply to only one <web-resource-collection> tag.
- D. A given <web-resource-collection> tag can contain from zero to many <url-pattern> tags.
- E. It is possible to construct a valid <security-constraint> tag such that, for a given resource user roles can access that resource.

**Correct Answer:** BE

**Section:** (none)

**Explanation**

**Explanation/Reference:**

**QUESTION 39**

A cookie may be set to be an HttpOnly cookie. Setting a cookie to be HttpOnly results in (Choose two)

- A. Client to not expose the cookie to client side scripting code
- B. Does not work with https protocol
- C. prevent certain types of cross-site scripting attacks
- D. There is no such thing as an HttpOnly cookie

**Correct Answer:** AC

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: <http://docs.oracle.com/javaee/6/api/javax/servlet/http/Cookie.html>

**QUESTION 40**

A popular Ajax framework and its companion widget library contain several hundreds of files of different types (.js, .css, .html). Your company has mandated that all its web applications use only specific versions of this framework approved by IT and repackaged internally as jar files. Furthermore, web applications should only include the entire jar, without subsetting or modification.

Where should the framework's files be placed inside a jar file to enable this kind of reuse?

- A. under resources
- B. under META-INF/resources
- C. under META-INF/web-contents
- D. under WEB-INF/resources

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: <http://ocpsoft.com/opensource/create-common-facelets-jar/> (check the box with the update)

**QUESTION 41**

Which two actions protect a resource file from direct HTTP access within a web application? (Choose two)

- A. placing it in the /secure directory
- B. placing it in the /WEB-INF directory
- C. placing it in the /META-INF/secure directory
- D. creating a <web-resource> element within the deployment descriptor
- E. creating a <secure-resource> element within the deployment descriptor

**Correct Answer:** BC

**Section:** (none)

**Explanation**

**Explanation/Reference:****QUESTION 42**

In which two locations can library dependencies be defined for a web application? (Choose two)

- A. the web application deployment descriptor
- B. the /META-INF/dependencies.xml file
- C. the /META-INF/MANIFEST.MF manifest file
- D. the /META-INF/MANIFEST.MF manifest of a JAR in the web application classpath

**Correct Answer:** CD

**Section:** (none)

**Explanation**

**Explanation/Reference:****QUESTION 43**

The jquery\_1\_3\_2.jar file contains the JQuery Ajax framework in its META-INF/ resources directory. Where should the jar file be placed inside the web application to ensure the resources it contains are accessible by clients?

- A. WEB-INF/classes
- B. WEB-INF/jar
- C. WEB-INF/lib
- D. WEB-INF/resources

**Correct Answer: C**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

Reference:

[http://www.ibm.com/developerworks/websphere/library/techarticles/0112\\_deboer/deboer.html](http://www.ibm.com/developerworks/websphere/library/techarticles/0112_deboer/deboer.html) (topic: web modules, third paragraph)

#### **QUESTION 44**

Given a jar file packaged with three web fragments with names X, Y and Z respectively.

Which of the following deployment descriptor, web.xml, snippets correspond to the web fragment processing orders of X, Y, Z?

(i) `<absolute-ordering>`  
`<name>X</name>`  
`<name>Y</name>`  
`<name>Z</name>`  
`</absolute-ordering>`

(ii) `<absolute-ordering>`  
`<name>X</name>`  
`<name>Y</name>`  
`</absolute-ordering>`

(iii) `<absolute-ordering>`  
`</others>`  
`<name>Y</name>`  
`<name>Z</name>`  
`</absolute-ordering>`

(iv) `<absolute-ordering>`  
`<name>X</name>`  
`</others>`  
`<name>Z</name>`  
`</absolute-ordering>`

(v) `<absolute-ordering>`  
`<name>X</name>`  
`<name>Y</name>`  
`</others>`  
`</absolute-ordering>`

- A. (i) only
- B. (i) and (ii)
- C. (i) and (iv)
- D. (i), (iii), (iv) and (v)

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Reference: [http://blogs.oracle.com/swchan/entry/servlet\\_3\\_0\\_web\\_fragment](http://blogs.oracle.com/swchan/entry/servlet_3_0_web_fragment)

#### QUESTION 45

Which defines the welcome files in a web application deployment descriptor?

- A. 

```
<welcome>
<welcome-file> /welcome.jsp</welcome-file>
</welcome>
<welcome>
<welcome-file> /index.html</welcome-file>
</welcome>
```
- B. 

```
<welcome-file-list>
<welcome-file> welcome.jsp</welcome-file>
<welcome-file> index.html</welcome-file>
</welcome-file-list>
```
- C. 

```
<welcome>
<welcome-file> welcome.jsp</welcome-file>
</welcome>
<welcome-file> index.html</welcome-file>
</welcome>
```
- D. 

```
<welcome-file-list>
<welcome-file> welcome.jsp</welcome-file>
<welcome-file> /index.html</welcome-file>
</welcome-file-list>
```
- E. 

```
<welcome>
<welcome-file>
<welcome-name> welcome</welcome-name>
<location> welcome.jsp</location>
</welcome-file>
<welcome-file>
<welcome-name> index</welcome-name>
<location> index</location>
</welcome-file>
</welcome>
```

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 46

Which code snippet specifies the metadata for a servlet with:

- (i) url mapping "/mytest"
- (ii) Initial parameter with name "debug" and value "10"
- (iii) Support asynchronous operation?

- A. 

```
@WebServlet("/mytest")
@InitParam(name="debug" value="10")
@Asynchronous
public class TestServlet extends HttpServlet {}
```
- B. 

```
@WebServlet("/myset")
```



- ```
@WebParam(name = "debug" value = "10")
Public class testServlet {}
```
- C. `@WebServlet (urlPatterns = { / "mytest"},
Initparams = {@InitParam (name = "debug", value = "10") }, supportAsync = true) public class TestServlet
extends HttpServlet {}`
- D. `@WebServlet (urlPatterns = { / "mytest"},
Initparams = {@InitParam (name = "debug", value = "10") }, asyncSupported = true) public class TestServlet
extends HttpServlet {}`
- E. `@WebServlet (urlPatterns = { / "mytest"},
Initparam = @WebInitParam (name = "debug", value = "10"), supportAsync = true) public class testServlet
{}`

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 47

Which annotation specifies the metadata as a `javax.servlet.http.HttpSessionListener`?

- A. `@Listener`
- B. `@WebListener`
- C. `@WebSessionListener`
- D. `@HttpSessionListener`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Reference: http://docs.oracle.com/cd/E12840_01/wls/docs103/webapp/annotateservlet.html (table 8-1, third row)

QUESTION 48

You are creating a new JSP page and you need to execute some code that acts when the page is first executed, but only once. Which three are possible mechanisms for performing this initialization code? (Choose three)

- A. In the `init` method.
- B. In the `jspInit` method.
- C. In the constructor of the JSP's Java code.
- D. In a JSP declaration, which includes an initializer block.
- E. In a JSP declaration, which includes a static initializer block.

Correct Answer: BDE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 49

You are creating a web form with this HTML:

11. <form action="sendOrder.jsp">
12. <input type="text" name="creditCard">
13. <input type="text" name="expirationDate">
14. <input type="submit">
15. </form>

Which HTTP method is used when sending this request from the browser?

- A. GET
- B. PUT
- C. POST
- D. SEND
- E. FORM

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 50

Your web application requires the ability to load and remove web files dynamically to the web container's file system? Which two HTTP methods are used to perform these actions? (Choose two)

- A. PUT
- B. POST
- C. SEND
- D. DELETE
- E. REMOVE
- F. DESTROY

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 51

A web application wants to expose to its clients an operation that results in a transaction being performed across several systems, for example the transfer of money from one bank account to another

Which HTTP method should it use?

- A. GET
- B. HEAD
- C. OPTIONS
- D. POST

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Reference: [http://en.wikipedia.org/wiki/POST_\(HTTP\)](http://en.wikipedia.org/wiki/POST_(HTTP))

QUESTION 52

A Client wants to retrieve a representation of a resource, e.g. an XML document, localized according to the preferences of the user. Each user is allowed to specify one or more languages, in order of preference. Such preferences are application independent. How can the Client send language preference Information to the Server using a standard HTTP facility?

- A. In the request body
- B. In a request header
- C. In the request line
- D. In the request URL

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Reference: <http://www.w3.org/International/questions/qa-lang-priorities.en.php> (Check the answer, first paragraph)

QUESTION 53

Which JSTL code snippet can be used to import content from another web resource?

- A. `<c:import url = "foo.jsp"/>`
- B. `<c:import page = "foo.jsp"/>`
- C. `<c:include url = "foo.jsp"/>`
- D. `<c:include page = "foo.jsp"/>`
- E. Importing cannot be done in JSTL. A standard action must be used instead.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 54

In a jsp-centric shopping cart application cart application to move a client's home address of Customer object into the shipping address of the order object. The address data is stored in a value object class called Address with properties for: street address, city, province, country, and postal code.

Which two code snippets can be used to accomplish this goal?

- A. `<c:set var='order' property='shipAddress' value='${client.homeAddress}' />`
- B. `<c:set target='${order}' property='shipAddress' value='${client.homeAddress}' />`
- C. `<jsp:setProperty name='${order}' property='shipAddress' value='${client.homeAddress}' />`
- D. `<c:set var='order' property='shipAddress'>`
`<jsp:getProperty name='client' property='homeAddress' /> </c:store>`
- E. `<c:set target='${order}' property='shipAddress'>`
`<jsp:getProperty name='client' property='homeAddress' /> </c:set>`
- F. `<c:setProperty name='${order}' property='shipAddress'>` `<jsp:getProperty name='client'`
`property='homeAddress' /> </c:setProperty>`

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 55

You are creating a JSP page to display a collection of data. This data can be displayed in several different ways so the architect on your project decided to create a generic servlet that generates a comma delimited string so that various pages can render the data in different ways. This servlet takes on request parameter: objectId. Assume that this servlet is mapped to the URL pattern: /WEB-INF/data.

In the JSP you are creating, you need to split this string into its elements separated by commas and generate an HTML list from the data.

Which JSTL code snippet will accomplish this goal?

- A.

```
<c:import varReader='dataString' url='/WEB-INF/data'>
<c:param name='objectId' value='${currentOID}' />
</c:import>
<ul>
<c:forEach items='${dataString.split(",")}' var='item'> <li>${item}</li>
</c:forEach>
</ul>
```
- B.

```
<c:import varReader='dataString' url='/WEB-INF/data'>
<c:param name='objectId' value='${currentOID}' />
</c:import>
<ul>
<c:forEach items='${dataString}' delims=',' var='item'> <li>${item}</li>
</c:forEach>
</ul>
```
- C.

```
<c:import var='dataString' url='/WEB-INF/data'>
<c:param name='objectId' value='${currentOID}' />
</c:import>
<ul>
<c:forEach items='${dataString.split(",")}' var='item'> <li>${item}</li>
</c:forEach>
</ul>
```
- D.

```
<c:import var='dataString' url='/WEB-INF/data'>
<c:param name='objectId' value='${currentOID}' />
</c:import>
<ul>
<c:forEach items='${dataString}' delims=',' var='item'> <li>${item}</li>
</c:forEach>
</ul>
```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 56

Which JSP standard action can be used to import content from a resource called foo.jsp?

- A. <jsp:import file='foo.jsp' />
- B. <jsp:import page='foo.jsp' />
- C. <jsp:include page='foo.jsp' />
- D. <jsp:include file='foo.jsp' />
- E. <jsp:import>foo.jsp</jsp:import>
- F. <jsp:include>foo.jsp</jsp:include>

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 57

Refer to the Exhibit.

```

1. <?xml version="1.0" encoding="UTF-8" ?>
2.
3. <taglib
xmlns="http://java.sun.com/xml/ns/j2ee"
4.   xmlns:xsi="http://www.w3.org/2001/XMLSchema
a-instance"
5.   xsi:schemaLocation="http://java.sun.com/xml
1/ns/j2ee web-jsptaglibrary_2_0.xsd"
6.   version="2.0">
7.   <tlib-version>1.0</tlib-version>
8.   <short-name>forum</short-name>
9.   <uri>http://example.com/tld/forum</uri>
10.  <tag>
11.    <name>message</name>
12.
<tag-class>com.example.MessageTag</tag-class>
13.
<body-content>scriptless</body-content>
14.   <attribute>
15.     <name>from</name>
16.     <rtexprvalue>true</rtexprvalue>
17.   </attribute>
18.   <attribute>
19.     <name>subject</name>
20.     <required>false</required>
21.     <rtexprvalue>true</rtexprvalue>
22.   </attribute>
23. </tag>
24. </taglib>

```

Assuming the tag library in the exhibit is imported with the prefix forum, which custom tag invocation procedures a transaction error in a jsp page?

- A. <forum:message from="My Name" subject="My Subject" />
- B. <forum:message subject="My Subject">
My message body.
</forum:message>
- C. <forum:message from="My Name" subject="{param.subject}"> \${param.body}
</forum:message>

- D. `<forum:message from="My Name" subject="My Subject"> <%= request.getParameter( "body" ) %>`
`</forum:message>`
- E. `<forum:message from="My Name"`
`subject="<%= request.getParameter( "subject" ) %>"> My message body.`
`</forum:message>`

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 58

Which is a benefit of precompiling a JSP page?

- A. It avoids initialization on the first request.
- B. It provides the ability to debug runtime errors in the application.
- C. It provides better performance on the first request for the JSP page.
- D. It avoids execution of the `_jspService` method on the first request.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 59

A web browser need NOT always perform a complete request for a particular page that it suspects might NOT have changed. The HTTP specification provides a mechanism for the browser to retrieve only a partial response from the web server; this response includes information, such as the Last-Modified date but NOT the body of the page. Which HTTP method will the browser use to retrieve such a partial response?

- A. GET
- B. ASK
- C. SEND
- D. HEAD
- E. TRACE
- F. OPTIONS

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 60

A Client wants to retrieve a representation of a resource, for example an HTML page, allowing a cached version to be returning by an intermediary such a proxy server. Which HTTP method should it use?

- A. GET
- B. PUT

- C. POST
- D. DELETE

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The first version of the HTTP protocol had only one method, namely GET, which would request a page from a server. The response from the server was always an HTML page

QUESTION 61

Which two are true about the JSTL core iteration custom tags? (Choose two.)

- A. It may iterate over arrays, collections, maps, and strings.
- B. The body of the tag may contain EL code, but not scripting code.
- C. When looping over collections, a loop status object may be used in the tag body.
- D. It may iterate over map, but only the key of the mapping may be used in the tag body.
- E. When looping over integers (for example begin= `1` end= `10`), a loop status object may not be used in the tag body.

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 62

Assume a JavaBean com.example.GradedTestBean exists and has two attributes. The attribute name of type java.lang.string and the attribute score is of type java.lang.Integer.

An array of com.example.GradedTestBean objects is exposed to the page in a request-scoped attribute called results. Additionally, an empty java.util.HashMap called resultMap is placed in the page scope.

A JSP page needs to add the first entry in results to resultMap, storing the name attribute of the bean as the key and the score attribute of the bean as the value.

Which code snippet of JSTL code satisfies this requirement?

- A. `${resultMap[results[0].name] = results[0].score}`
- B. `<c:set var="${resultMap}" key="{results[0].name}" value="${results[0].score}" />`
- C. `<c:set var="resultMap" property="${results[0].name}"> ${results[0].value}`
`</c:set>`
- D. `<c:set var="resultMap" property="${results[0].name}"> value="${results[0].score}" />`
- E. `<c:set target="${resultMap}" property="${results[0].name}"> value="${results[0].score}" />`

Correct Answer: E

Section: (none)

Explanation

Explanation/Reference:

QUESTION 63

For manageability purposes, you have been told to add a "count" instance variable to a critical JSP Document so that a JMX MBean can track how frequent this JSP is being invoked. Which JSP code snippet must you use to declare this instance variable in the JSP Document?

- A. `<jsp:declaration>`
 `int count = 0;`
 `</jsp:declaration>`
- B. `<%! int count = 0; %>`
- C. `<jsp:declaration.instance>`
 `int count = 0;`
 `</jsp:declaration.instance>`
- D. `<jsp:scriptlet.declaration>`
 `int count = 0;`
 `</jsp:scriptlet.declaration>`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 64

To take advantage of the capabilities of modern browsers that use web standards, such as XHTML and CSS, your web application is being converted from simple JSP pages to JSP Document format. However, one of your JSPs, /scripts/screenFunctions.jsp, generates a JavaScript file. This file is included in several web forms to create screen-specific validation functions and are included in these pages with the following statement:

```
10. <head>
11.   <script> src='/scripts/screenFunctions.jsp'
12.     language='javascript'
13.     type='application/javascript' > </script>
14. </head>
15. <!--body of the web form -->
```

Which JSP code snippet declares that this JSP Document is a JavaScript file?

- A. `<%@ page contentType='application/javascript' %>`
- B. `<jsp:page contentType='application/javascript' />`
- C. `<jsp:document contentType='application/javascript' />`
- D. `<jsp:directive.page contentType='application/javascript' />`
- E. No declaration is needed because the web form XHTML page already declares the MIME type of the /scripts/screenFunctions.jsp file in the `<script>` tag.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 65

You are building your own layout mechanism by including dynamic content for the page's header and footer sections. The footer is always static, but the header generates the `<title>` tag that requests the page name to be

specified dynamically when the header is imported. Which JSP code snippet performs the import of the header content?

- A. `<jsp:include page="/WEB-INF/jsp/header.jsp">
<jsp:param name="pageName" value="Welcome Page" />
</jsp:include>`
- B. `<jsp:import page="/WEB-INF/jsp/header.jsp">
<jsp:param name="pageName" value="Welcome Page" />
</jsp:import>`
- C. `<jsp:include page="/WEB-INF/jsp/header.jsp">
<jsp:attribute name="pageName" value="Welcome Page" /> </jsp:include>`
- D. `<jsp:import page="/WEB-INF/jsp/header.jsp">
<jsp:attribute name="pageName" value="Welcome Page" /> </jsp:import>`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 66

View the exhibit.

```
1. <?xml version="1.0" encoding="UTF-8" ?>
2.
3. <taglib
xmlns="http://java.sun.com/xml/ns/j2ee"
4.   xmlns:xsi="http://www.w3.org/2001/XMLSchema
a-instance"
5.   xsi:schemaLocation="http://java.sun.com/xml
1/ns/j2ee web-jsptaglibrary_2_0.xsd"
6.   version="2.0">
7.   <tlib-version>1.0</tlib-version>
8.   <short-name>ex</short-name>
9.
<uri>http://example.com/tld/example</uri>
10.   <tag>
11.     <name>hello</name>
12.
<tag-class>com.example.HelloTag</tag-class>
13.
<body-content>scriptless</body-content>
14.   </tag>
15. </taglib>
```

Assume the tag library in the exhibit is placed in a web application in the path /WEB-INF/tld/example.tld.

- 1.
- 2. `<ex:hello />`

Which JSP code, inserted at line 1, completes the JSP code to invoke the hello tag?

- A. `<%@ taglib prefix="ex" uri="/WEB-INF/tld" %>`
- B. `<%@ taglib uri="/WEB-INF/tld/example.tld" %>`
- C. `<%@ taglib prefix="ex"`

uri="http://localhost:8080/tld/example.tld" %>

D. <%@ taglib prefix="ex"

uri="http://example.com/tld/example" %>

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 67

You are working on a JSP that is intended to inform users about critical errors in the system. The JSP code is attempting to access the exception that represents the cause of the problem, but your IDE is telling you that the variable does not exist. What should you do to address this problem?

- A. Add a page directive stating that this page is an error handler
- B. Add scriptlet code to create a variable that refer to the exception
- C. Add a <jsp:useBean tag to declare the and access the exception>
- D. Perform the error handling in a servlet rather than in the JSP
- E. Edit the page that caused the error to ensure that it specifies this page as its error handler

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Exception is a JSP implicit variable

The exception variable contains any Exception thrown on the previous JSP page with an errorPage directive that forwards to a page with an isErrorPage directive.

Example:

If you had a JSP (index.jsp) which throws an exception (I have deliberately thrown a NumberFormatException by parsing a String, obviously you wouldn't write a page that does this, its just an example)

```
<%@ page errorPage="error.jsp" %>
```

```
<% Integer.parseInt("foo"); //throws an exception %> This will forward to error.jsp,
```

If error.jsp was

```
<%@ page isErrorPage = "true"%>
```

```
<body>
```

```
<h2>Your application has generated an error</h2>
```

```
<h3>Please check for the error given below</h3>
```

```
<b>Exception:</b><br>
```

```
<font color="red"><%= exception.toString() %></font> </body>
```

Because it has the

```
<%@ page isErrorPage = "true"%>
```

page directive, the implicit variable exception will contain the Exception thrown in the previous jsp

So when you request index.jsp, the Exception will be thrown, and forwarded to error.jsp which will output html like this

```
<body>
```

```
<h2>Your application has generated an error</h2>
```

```
<h3>Please check for the error given below</h3>
```

Exception:

java.lang.NumberFormatException: For input string: "foo" </body>

As @JB Nizet mentions exception is an instanceof Throwable calling exception.getMessage() For input string: "foo" instead of java.lang.NumberFormatException: For input string: "foo"

QUESTION 68

All of your JSPs need to have a link that permits users to email the web master. This web application is licensed to many small businesses, each of which have a different email address for the web master. You have decided to use a context parameter that you specify in the deployment descriptor, like this:

42. <context-param>

43. <param-name>webmasterEmail</param-name>

44. <param-value>master@example.com</param-value>

45. </context-param>

Which JSP code snippet creates this email link?

- A. contact us
- B. contact us
- C. contact us
- D. contact us

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 69

Which EL expression evaluates to the request URI?

- A. \${requestURI}
- B. \${request.URI}
- C. \${request.getURI}
- D. \${request.requestURI}
- E. \${requestScope.requestURI}
- F. \${pageContext.request.requestURI}
- G. \${requestScope.request.requestDRI}

Correct Answer: F

Section: (none)

Explanation

Explanation/Reference:

QUESTION 70

You are building a dating web site. The client's date of birth is collected along with lots of other information. The Person class has a derived method, getAge() :int, which returns the person's age calculated from the date of birth and today's date. In one of your JSPs you need to print a special message to clients within the age group of 25 through 35.

Which two EL code snippets will return true for this condition? (Choose two.)

- A. \${client.age in [25,35]}

- B. `${client.age between [25,35]}`
- C. `${client.age between 25,35}`
- D. `${client.age <= 35 && client.age >= 25}`
- E. `${client.age le 35 and client.age ge 25}`
- F. `${client.age > 35 && client.age < 25}`

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 71

Given an EL function declared with:

```

11. <function>
12.   <name>spin</name>
13.   <function-class>com.example.Spinner</function-class>
14.   <function-signature>
15.     java.lang.String spinlt ()
16.   </function-signature>
17. </function>

```

Which two are true? (Choose two)

- A. The function method must have the signature:
`public String spin()`.
- B. The method must be mapped to the logical name "spin" in the web.xml file.
- C. The function method must have the signature:
`public String spinlt()`.
- D. The function method must have the signature:
`public static String spin()`.
- E. The function method must have the signature:
`public static String spinlt()`.
- F. The function class must be named Spinner, and must be in the package com.example.

Correct Answer: EF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 72

Which EL expression returns true if no session has been established with current client?

- A. `${not(pageContext.session)}`
- B. `${not(requestScope.session)}`
- C. `${requestScope.sessions.this}`
- D. `${sessionScope.empty}`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Note:

* A session is never null. The session is always present in JSP EL, unless you add

```
<%@page session="false" %>
```

* If you'd like to check if the session is new or has already been created, use `HttpSession#isNew()` instead.

```
<c:if test="${not pageContext.session['new']}">
```

```
<p>You've already visited this site before.</p>
```

```
</c:if>
```

```
<c:if test="${pageContext.session['new']}">
```

```
<p>You've just started the session with this request!</p> </c:if>
```

QUESTION 73

Your web application uses a simple architecture in which servlets handle requests and then forward to a JSP using a request dispatcher. You need to pass information calculated in the servlet to the JSP for view generation. This information must NOT be accessible to any other servlet, JSP or session in the webapp. Which two techniques can you use to accomplish this goal? (Choose two)

- A. Add attributes to the session object.
- B. Add attributes on the request object.
- C. Add parameters on the request object.
- D. Use the `pageContext` object to add request attributes.
- E. Add parameters to the JSP's URL when generating the request dispatcher.

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 74

A web application for business expense reporting allows uploading expense receipts. Multiple receipts can be uploaded single step using one HTTP request. The servlet that processes the request has been marked with the `@MultipartConfig` annotation. Which method should the servlet use to access the uploaded files?

- A. `HttpServletRequest.getParts()`
- B. `HttpServletRequest.getData()`
- C. `servletRequest.getParts()`
- D. `servletRequest.getAllParts()`

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The `request.getParts()` method returns collections of all `Part` objects. If you have more than one input of type file, multiple `Part` objects are returned. Since `Part` objects are named, the `getPart(String name)` method can be used to access a particular `Part`. Alternatively, the `getParts()` method, which returns an `Iterable<Part>`, can be used to get an `Iterator` over all the `Part` objects.

QUESTION 75

For which three events can web application event listeners be registered? (Choose three.)

- A. when a session is created
- B. after a servlet is destroyed
- C. when a session has timed out
- D. when a cookie has been created
- E. when a servlet has forwarded a request
- F. when a session attribute value is changed

Correct Answer: ACF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 76

To add a servlet to a context that has not been declared either via annotation or via the descriptor, during context initialization time the following API can be used (Choose three.)

- A. `servletContext.addServlet("myServletName", "MyServlet");`
- B. `servletContext.addServlet( (<Class extends Servlet>)getClass().getClassLoader().getClassLoader().loaderClass("MyServlet"));`
- C. `servletContext.addServlet(myServlet);`
- D. `servletContext.addServlet( ("myServletName", <Class extends Servlet>)gerClass().getClassLoader().loaderClass("myServlet"));`
- E. `servletContext.addServlet("myServletName", MyServlet);`

Correct Answer: ADE

Section: (none)

Explanation

Explanation/Reference:

`addServlet`

`ServletRegistration.Dynamic addServlet(java.lang.String servletName, java.lang.Class<? extends Servlet> servletClass)`

Adds the servlet with the given name and class type to this servlet context. The registered servlet may be further configured via the returned `ServletRegistration` object.

Parameters:

`servletName` - the name of the servlet

`servletClass` - the class object from which the servlet will be instantiated

QUESTION 77

`ServletContextListeners` are invoked in

- A. Random order
- B. `contextInitialized` and `contextDestroyed` are invoked in the order in which they are declared in the `web.xml`
- C. `contextInitialized` method are invoked in the order in which they are declared in the `web.xml` and the `contextDestroyed` method is invoked in the reverse order in which they are declared in the `web.xml`
- D. `contextInitialized` and `contextDestroyed` are invoked in the reverse order of which they are declared in the `web.xml`

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

public interface ServletContextListener
extends java.util.EventListener

Interface for receiving notification events about ServletContext lifecycle changes.

In order to receive these notification events, the implementation class must be either declared in the deployment descriptor of the web application, annotated with `WebListener`, or registered via one of the `addListener` methods defined on `ServletContext`.

Implementations of this interface are invoked at their `contextInitialized(javax.servlet.ServletContextEvent)` method in the order in which they have been declared, and at their `contextDestroyed(javax.servlet.ServletContextEvent)` method in reverse order.

QUESTION 78

A servlet class is injected with a JDBC data source. After injection has occurred, the servlet needs to create a cache out of some of the data in the database, so as to improve responsiveness. Which two methods can host the cache creation code? (Choose two)

- A. `Servlet.init()`
- B. `Servlet.destroy()`
- C. A method annotated with `@Init`
- D. A method annotated with `@PostConstruct`
- E. A method annotated with `@PreDestroy`
- F. A method annotated with `@Resource`

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

A: Because the `Servlet init()` method is invoked when the servlet instance is loaded, it is the perfect location to carry out expensive operations that need only be performed during initialization. By definition, the `init()` method is thread-safe. The results of operations in the `HttpServlet.init()` method can be cached safely in servlet instance variables, which become read-only in the servlet service method.

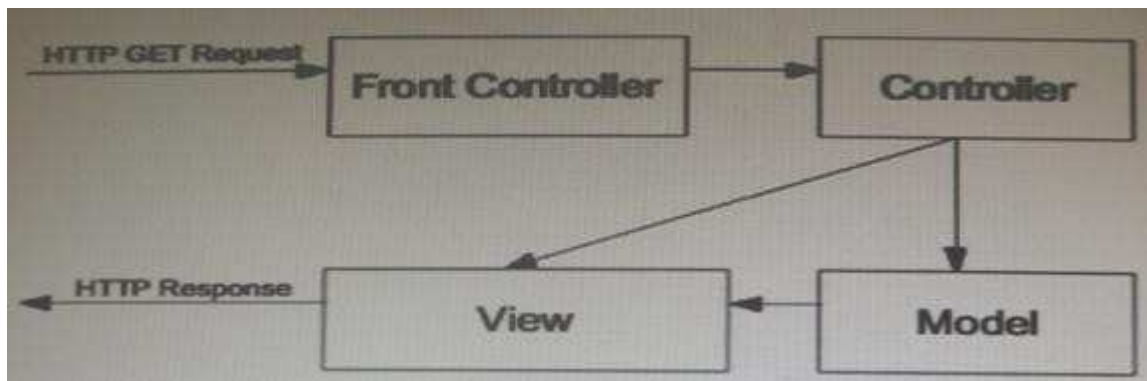
D: Example:

`@PostConstruct`

```
private void init() {  
    cached = (Cached) ctx.lookup(EJB_PATH + Cached.class.getSimpleName());  
}
```

QUESTION 79

View the Exhibit.



Which two technologies would be suitable for use as Front Controller elements? (Choose two)

- A. JSP
- B. Servlet
- C. Filter
- D. POJO
- E. Custom Tag

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

* Servlet only. Works well when:

Output is a binary type. E.g.: an image

There is no output. E.g.: you are doing forwarding or redirection as in Search Engine example.

Format/layout of page is highly variable. E.g.: portal.

* JSP only. Works well when:

Output is mostly character data. E.g.: HTML

Format/layout mostly fixed.

* Combination (MVC architecture). Needed when:

A single request will result in multiple substantially different looking results. You have a large development team with different team members doing the Web development and the business logic.

You perform complicated data processing, but have a relatively fixed layout

Incorrect:

Not D: In computing software, POJO is an acronym for Plain Old Java Object. The name is used to emphasize that a given object is an ordinary Java Object, not a special object. The term "POJO" is mainly used to denote a Java object which does not follow any of the major Java object models, conventions, or frameworks. The term continues the pattern of older terms for technologies that do not use fancy new features.

QUESTION 80

View the Exhibit.


```

// Source Servlet : Source.java
10. public class Source extends HttpServlet {
11.     public void service(HttpServletRequest request,
12.                         HttpServletResponse response)
13.         throws ServletException, IOException {
14.         RequestDispatcher rd =
15.             request.getRequestDispatcher("/dest/Destination");
16.         rd.forward(request, response);
17.     }
18. }

// Filter : ParamAdder.java
12. public class ParamAdder implements Filter {
13.     // ...
23.     public void doFilter(ServletRequest request,
24.                         ServletResponse response,
25.                         FilterChain chain)
26.         throws ServletException, IOException {
27.         request.setAttribute("filterAdded", "addedByFilter");
28.         chain.doFilter(request, response);
29.     }
30.     // ...
50. }

// Destination Servlet Destination.java
10. public class Destination extends HttpServlet {
11.     public void service(HttpServletRequest request,
12.                         HttpServletResponse response)
13.         throws ServletException, IOException {
14.         String filterParam =
15.             (String) request.getAttribute("filterAdded");
16.         response.getWriter().println("filterAdded = "
17.                                     + filterParam);
18.     }
19. }

```

Given the web application deployment descriptor elements:

```

11. <filter>
12. <filter-name>ParamAdder</filter-name>
13. <filter-class>com.example.ParamAdder</filter-class>
14. </filter>
...
31. <filter-mapping>
32. <filter-name>ParamAdder</filter-name>
33. <servlet-name>Destination</servlet-name>
34. </filter-mapping>
...
55. <servlet-mapping>
56. <servlet-name>Destination</servlet-name>
57. <url-pattern>/dest/Destination</url-pattern>
58. </servlet-mapping>

```

What is the result of a client request of the Source servlet with no query string?

- A. The output "filterAdded = null" is written to the response stream.
- B. The output "filterAdded = addedByFilter" is written to the response stream.
- C. An exception is thrown at runtime within the service method of the Source servlet.
- D. An exception is thrown at runtime within the service method of the Destination servlet.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 81

Which three are true about the HttpServletRequestWrapper class? (Choose three.)

- A. The HttpServletRequestWrapper is an example of the Decorator pattern.
- B. The HttpServletRequestWrapper can be used to extend the functionality of a servlet request.
- C. A subclass of HttpServletRequestWrapper CANNOT modify the behavior of the getReader method.
- D. An HttpServletRequestWrapper may be used only by a class implement the javax.servlet.Filter interface.
- E. An HttpServletRequestWrapper CANNOT be used on the request passed to the RequestDispatcher.interface.
- F. An HttpServletRequestWrapper may modify the header of a request within an object implementing the javax.servlet.Filter interface.

Correct Answer: ABF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 82

Given a war file with the following structure

| - WEB-INF/classes/MyServlet.class
| - WEB-INF/lib/wf.jar

Where wf.jar contains a valid web-fragment.xml and the following two classes: MyFilter1.class and MyFiler2.class.

The web-fragment.xml is as follows:

```
<web-fragment metadata-complete="true">
  <filter>
    <filter-name>MyFilter2</filter-name>
    <filter-class>MyFilter2</filter-class>
  </filter>
  <filter-mapping>
    <filter-name>MyFilter2</filter-name>
    <url-pattern>/</url-pattern>
  </filter-mapping>
</web-fragment>
```

The following are some code snippets:

```
(i) @WebServlet("/")
    public class MyServlet extends HttpServlet {}
(ii) @WebFilter(urlPatterns={"/"})
    public class MyFilter1 implements Filter {}
(iii) @WebFilter(urlPatterns={"/abc"})
    public class MyFilter2 implements Filter {}
```

When one access "/" of the above web application, which filters will be executed?

- A. No filters will be executed.
- B. MyFilter1
- C. MyFilter2
- D. MyFilter1 and MyFilter2

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Note:

* <filter-mapping>

This tag specifies a filter name, and either a URL mapping or servlet name, for a filter that has been defined with the <filter> tag.

Multiple <filter-mapping> tags can be specified for a single <filter>, providing different URL patterns. See the <url-pattern> tag for examples.

The <filter-mapping> has two required elements:

<filter-name> - the filter name, as specified in the <filter-name> element of the <filter> tag Either a <url-pattern> or a <servlet-name>.

If a servlet name is specified, the filter will be called whenever the specific servlet is called.

QUESTION 83

Servlet A forwarded a request to servlet B using the forward method of RequestDispatcher. What attribute in B's request object contains the URI of the original request received by servlet A?

- A. REQUEST_URI
- B. javax.servlet.forward.request_uri
- C. javax.servlet.forward.REQUEST_URI
- D. javax.servlet.request_dispatcher.request_uri
- E. javax.servlet.request_dispatcher.REQUEST_URI

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 84

View the Exhibit.

```

1. package com.example;
2.
3. public class Product {
4.     private String name;
5.     private double price;
6.
7.     public Product() {
8.         this( "Default", 0.0 );
9.     }
10.
11.     public Product( String name, double price ) {
12.         this.name = name;
13.         this.price = price;
14.     }
15.
16.     public String getName() {
17.         return name;
18.     }
19.
20.     public void setName(String name) {
21.         this.name = name;
22.     }
23.
24.     public double getPrice() {
25.         return price;
26.     }
27.
28.     public void setPrice(double price) {
29.         this.price = price;
30.     }
31. }

```

Assume the product attribute does NOT yet exist in any scope. Which two create an instance of com.example.Product and initialize the name and price properties to the name and price request parameters? (Choose two)

- A. `<jsp:useBean id="product" class="com.example.Product" /> <jsp:setProperty name="product" property="*" />`
- B. `<jsp:useBean id="product" class="com.example.Product" /> <% product.setName( request.getParameter( "name" ) ); %> <% product.setPrice( request.getParameter( "price" ) ); %>`
- C. `<jsp:useBean id="product" class="com.example.Product" /> <jsp:setProperty name="product" property="name" value="{param.name}" /> <jsp:setProperty name="product" property="price" value="{param.price}" />`
- D. `<jsp:useBean id="product" class="com.example.Product"> <jsp:setProperty name="product" property="name" value="{name}" /> <jsp:setProperty name="product" property="price" value="{price}" /> </jsp:useBean>`

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 85

A session-scoped attribute is stored by a servlet, and then servlet forwards to a JSP page. Which three jsp:useBean attributes must be used to access this attribute in the JSP page? (Choose three.)

- A. id
- B. name
- C. bean
- D. type
- E. scope
- F. beanName

Correct Answer: ADE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 86

View the Exhibit.

```

1. package com.example;
2.
3. public class Product {
4.     private String name;
5.     private double price;
6.
7.     public Product() {
8.         this( "Default", 0.0 );
9.     }
10.
11.     public Product( String name, double
price ) {
12.         this.name = name;
13.         this.price = price;
14.     }
15.
16.     public String getName() {
17.         return name;
18.     }
19.
20.     public void setName(String name) {
21.         this.name = name;
22.     }
23.
24.     public double getPrice() {
25.         return price;
26.     }
27.
28.     public void setPrice(double price) {
29.         this.price = price;
30.     }
31. }

```

Given:

10. <form action='create_product.jsp'>
11. Product Name: <input type= 'text' name='productName'/>

12. Product Price: <input type= 'text' name='productPrice'/>

13. </form>

For a given product instance, which three jsp:setProperty attributes must be used to initialize its properties from the HTML form? (Choose three.)

- A. id
- B. name
- C. type
- D. param
- E. property
- F. reqParam
- G. attribute

Correct Answer: BDE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 87

A developer wants to make a name attribute available to all servlets associated with a particular user, across multiple requests from that user, from the same browser instance. Which two provide this capability from within a tag handler? (Choose two)

- A. `pageContext.setAttribute("name", theValue);`
- B. `pageContext.setAttribute("name", getSession() );`
- C. `pageContext.getRequest().setAttribute("name", theValue);`
- D. `pageContext.getSession().setAttribute("name", theValue);`
- E. `pageContext.setAttribute("name", theValue, PageContext.PAGE_SCOPE);`
- F. `pageContext.setAttribute("name", theValue, PageContext.SESSION_SCOPE);`

Correct Answer: DF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 88

Given that a web application consists of two HttpServlet classes, ServletA and ServletB, and the ServletA.service method:

```
20. String key = "com.example.data";
21. session.setAttribute(key, "Hello");
22. Object value = session.getAttribute(key);
23.
```

Assume session is an HttpSession, and is not referenced anywhere else in ServletA. Which two changes, taken together, ensure that value is equal to "Hello" on line 23? (Choose two)

- A. ensure that the ServletB.service method is synchronized
- B. ensure that the ServletA.service method is synchronized
- C. ensure that ServletB synchronizes on the session object when setting session attributes
- D. enclose lines 21-22 in synchronized block:
`synchronized(this) {`
`session.setAttribute(key, "Hello");`
`value = session.getAttribute(key);`
`}`
- E. enclose lines 21-22 in synchronized block:
`synchronized(session) {`
`session.setAttribute(key, "Hello");`
`value = session.getAttribute(key);`
`}`

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 89

Given an HttpServletRequest request:

```
22. String id = request.getParameter("jsessionId");
23. // insert code here
```

24. `String name = (String) session.getAttribute("name")`

Which three can be placed at line 23 to retrieve an existing `HttpSession` object? (Choose three)

- A. `HttpSession session = request.getSession();`
- B. `HttpSession session = request.getSession(id);`
- C. `HttpSession session = request.getSession(true);`
- D. `HttpSession session = request.getSession(false);`
- E. `HttpSession session = request.getSession("jsessionId");`

Correct Answer: ACD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 90

As a convenience feature, your web pages include an Ajax request every five minutes to a special servlet that monitors the age of the user's session. The client-side JavaScript that handles the Ajax callback displays a message on the screen as the session ages. The Ajax call does NOT pass any cookies, but it passes the session ID in a request parameter called `sessionID`. In addition, assume that your webapp keeps a hasmap of session objects by the ID. Here is a partial implementation of this servlet:

```
10. public class SessionAgeServlet extends HttpServlet {
11. public void service(HttpServletRequest request, HttpServletResponse response) throws IOException {
12. String sessionID = request.getParameter("sessionID");
13. HttpSession session = getSession(sessionID);
14. long age = // your code here
15. response.getWriter().print(age);
16. }
... // more code here
47. }
```

Which code snippet on line 14, will determine the age of the session?

- A. `session.getMaxInactiveInterval();`
- B. `session.getLastAccessed().getTime() session.getCreationTime().getTime();`
- C. `session.getLastAccessedTime().getTime() session.getCreationTime().getTime();`
- D. `session.getLastAccessed() - session.getCreationTime();`
- E. `session.getMaxInactiveInterval() session.getCreationTime();`
- F. `session.getLastAccessedTime() session.getCreationTime();`

Correct Answer: F

Section: (none)

Explanation

Explanation/Reference:

QUESTION 91

Given the definition of `Myobject` and that an instance of `Myobject` is bound as a session attribute:

```
8. package com.example;
9. public class Myobject implements
10. javax.servlet.http.HttpSessionBindingListener {
11. // class body code here
```


12. }

Which is true?

- A. Only a single instance of Myobject may exist within a session
- B. The unbound method of the Myobject instance is called when the session to which it is bound times out
- C. The com.example.MyObject must be declared as a servlet event listener in the web application deployment descriptor
- D. The valueUnbound method of the Myobject instance is called when the session to which it is bound times out

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 92

Your web application requires the adding and deleting of many session attributes during a complex use case. A bug report has come in that indicates that an important session attribute is being deleted too soon and a NullPointerException is being thrown several interactions after the fact. You have decided to create a session event listener that will log when attributes are being deleted so you can track down when the attribute is erroneously being deleted. Which listener class will accomplish this debugging goal?

- A. Create an HttpSessionAttributeListener class and implement the attributeDeleted method and log the attribute name using the getName method on the event object.
- B. Create an HttpSessionAttributeListener class and implement the attributeRemoved method and log the attribute name using the getName method on the event object.
- C. Create an SessionAttributeListener class and implement the attributeRemoved method and log the attribute name using the getAttributeName method on the event object.
- D. Create an SessionAttributeListener class and implement the attributeDeleted method and log the attribute name using the getAttributeName method on the event object.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 93

Which method must be used to encode a URL passed as an argument to HttpServletResponse. sendRedirect when using URL rewriting for session tracking?

- A. ServletResponse.encodeURL
- B. HttpServletResponse.encodeURL
- C. ServletResponse.encodeRedirectURL
- D. HttpServletResponse.encodeRedirectURL

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 94

Which Java expression can be used to check whether the web container is currently configured to track sessions via URL rewriting?

- A. `servletContext.getSessionCookiesConfig().isHttpOnly()`
- B. `servletContext.getSessionCookiesConfig().isSecure()`
- C. `servletContext.getDefaultSessionTrackingModes().contains(SessionTrackingMode.URL)`
- D. `servletContext.getEffectiveSessionTrackingModes().contains (SessionTrackingMode.URL)`

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Code example:

String sessionId = null;

```
if (request.getContext().getEffectiveSessionTrackingModes().contains(SessionTrackingMode.URL)) {
```

```
// Get the session ID if there was one
sessionId = request.getPathParameter(
    SessionConfig.getSessionUriParamName(
        request.getContext()));
if (sessionId != null) {
    request.setRequestedSessionId(sessionId);
    request.setRequestedSessionURL(true);
}
}
```

C: `getDefaultSessionTrackingModes`

`java.util.Set<SessionTrackingMode> getDefaultSessionTrackingModes()` Gets the session tracking modes that are supported by default for this ServletContext.

QUESTION 95

Which security mechanism uses the concept of a realm?

- A. authorization
- B. data integrity
- C. confidentiality
- D. authentication

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 96

Which java code snippet roles "MANAGER" and "EMPLOYEE" in a given application?

- A. `@DeclareRoles({ "MANAGER", "EMPLOYEE" })`
`public class MyServlet extends HttpServlet {}`
- B. `@SecurityRoles({ "MANAGER", "EMPLOYEE" })`
`public class MyServlet extends HttpServlet {}`
- C. `@DeclareRoles("MANAGER")`

```

    @DeclareRoles("EMPLOYEE")
    public class MyServlet extends HttpServlet {}
D. @DeclareRole("MANAGER")
    @DeclareRole("EMPLOYEE")
    public class MyServlet extends HttpServlet {}

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

* The syntax for declaring more than one role is as shown in the following example:

```
@DeclareRoles({"Administrator", "Manager", "Employee"})
```

* @DeclareRoles

This annotation declares the security roles defined by the application.

* javax.annotation.security

Annotation Type DeclareRoles

@Documented

@Retention(value=RUNTIME)

@Target(value=TYPE)

public @interface DeclareRoles

Used by application to declare roles. It can be specified on a class.

QUESTION 97

Given the java code snippet in contextInitialized method of a ServletContextListener:

```

ServletRegistration.Dynamic sr = (ServletRegistration.Dynamic)sc.addServlet ("myServlet", myServletClass);
sr.addMapping("/abc");
sr.setServletSecurityElement(servletSecurityElement); sr.addMapping("/def");

```

Which statement is true?

- A. "/abc" is mapped to "myservlet". The servletSecurityElementy applies to both "/abc" and "/def".
- B. Both "/abc" and "/def" are mapped to "myservlet". The servletSecurityElementy applies to "/abc".
- C. Both "/abc" and "/def" are mapped to "myservlet". The servletSecurityElementy applies to "/def".
- D. Both "/abc" and "/def" are mapped to "myservlet". The servletSecurityElementy applies to both "/abc" and "/def".
- E. Both "/abc" and "/def" are mapped to "myservlet". The servletSecurityElementy applies to "/abc", but the behavior for "/def" is not specified.

Correct Answer: E

Section: (none)

Explanation

Explanation/Reference:

Note:

* addMapping

```
public void addMapping(java.lang.String namespace,
```

```
java.lang.String name,
```

```
java.lang.Class clazz)
```

Defines a direct mapping from a namespace and name to a java class (and vice versa)

QUESTION 98

Given:

```
3. class MyServlet extends HttpServlet {
4. public void doPut(HttpServletRequest req,
HttpServletResponse resp)
throws ServletException, IOException {
5. // servlet code here
...
26. }
27. }
```

If the DD contains a single security constraint associated with MyServlet and its only <http-method> tags and <auth-constraint> tags are:

```
<http-method>GET</http-method>
<http-method>PUT</http-method>
<auth-constrain>Admin</auth-constrain>
```

Which four requests would be allowed by the container? (Choose four)

- A. A user whose role is Admin can perform a PUT.
- B. A user whose role is Admin can perform a GET.
- C. A user whose role is Admin can perform a POST.
- D. A user whose role is Member can perform a PUT.
- E. A user whose role is Member can perform a POST.
- F. A user whose role is Member can perform a GET.

Correct Answer: ABCE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 99

Given this fragment in a servlet:

```
23. if(reg.isUserInRole("Admin")) {
24. // do stuff
25. }
```

And the following fragment from the related Java EE deployment descriptor:

```
812. <security-role-ref>
813. <role-name>Admin</role-name>
814. <role-link>Administrator</role-link>
815. </security-role-ref>
900. <security-role>
901. <role-name>Admin</role-name>
902. <role-name>Administrator</role-name>
903. </security-role>
```

What is the result?

- A. Line 24 can never be reached.
- B. The deployment descriptor is NOT valid.
- C. If line 24 executes, the user's role will be Admin.

- D. If line 24 executes, the user's role will be Administrator.
- E. If line 24 executes, the user's role will NOT be predictable.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 100

Which java code snippet checks whether the user is of the role "MANAGER" for a given HttpServletRequest, httpRequest?

- A. httpRequest.isUserInRole("MANAGER");
- B. httpRequest.isCallerInRole("MANAGER");
- C. httpRequest.isPrincipallInRole("MANAGER");
- D. httpRequest.isAuthenticatedUserInRole("MANAGER");

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

isUserInRole

public boolean isUserInRole(java.lang.String role)

Returns a boolean indicating whether the authenticated user is included in the specified logical "role". Roles and role membership can be defined using deployment descriptors. If the user has not been authenticated, the method returns false.

Parameters:

role - a String specifying the name of the role

Returns:

a boolean indicating whether the user making this request belongs to a given role; false if the user has not been authenticated

Incorrect:

Not B: isCallerInRole is depreciated.

QUESTION 101

A web application uses a cookies to track a client as it navigates through the pages that constitutes the application. Which code snippet can be used by the web application to reduce the chance of a cross-site scripting attack by setting some property of the cookie before returning it to the client?

- A. cookie.setHttpOnly(true)
- B. cookie.setMaxAge(3600)
- C. cookie.setPath("/")
- D. cookie.setSecure(true)

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

When HTTPOnly flag is assigned to a cookie, the browser will restrict the access to such Cookie from Java Script code hence the cookie would only be sent to the subsequent request to server but cannot be accessed using client side script. In such a case even if website is vulnerable to Cross Site Scripting (XSS) attacks, still

the browser would safeguard the data stored into cookies flagged as HTTPOnly.

Incorrect:

Not D: If Secure flag is set for Cookie then it may only be transmitted over secure channel (SSL/HTTPS) ensuring that data is always encrypted while transmitting from client to server.

QUESTION 102

Given the portion of a valid Java EE web application's directory structure:

```
MyApp
|-- Directory1
|   |-- File1.html
|-- META-INF
|   |-- File2.html
|-- WEB-INF
|   |-- File3.html
```

You want to know whether File1.html, File2.html, and/or File3.html is protected from direct access by your web client's browsers. What statement is true?

- A. All three files are directly accessible.
- B. Only File1.html is directly accessible.
- C. Only File2.html is directly accessible.
- D. Only File3.html is directly accessible.
- E. Only File1.html and File2.html are directly accessible.
- F. Only File1.html and File3.html are directly accessible.
- G. Only File2.html and File3.html are directly accessible.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Note:

* WEB-INF is the folder just under the root of a WAR that holds information that you don't want to be accessible to a client via a URL request. Specifically, it holds the web.xml, classes, and lib directories, but you can put anything you want to hide from the client there.

* META-INF is what discriminates a JAR file from a plain ZIP file. It holds the manifest file and may hold other deployment information as needed.

QUESTION 103

Given a web fragment jar file, where should we put the web fragment.xml inside the jar file?

- A. WEB-INF
- B. META-INF
- C. WEB-INF/lib
- D. WEB-INF/classes
- E. META-INF/services

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

* If you're dealing with web applications, /WEB-INF/lib is the portable place to put JARs. This is where web servers servlet containers expect to find an application's jar files.

* The /WEB-INF/classes directory is on the ClassLoader's classpath. This is where .class files are loaded from when the web application is executing. Any JAR files placed in the /WEB-INF/lib directory will also be placed on the ClassLoader's classpath.

QUESTION 104

Given the fragment from Java EE deployment descriptor:

```
341. <error-page>
342. <exception-type>java.lang.Throwable</exception-type> 343. <location>/mainError.jsp</location>
344. </error-page>
345. <error-page>
346. <exception-type>java.lang.ClassCastException</exception-type> 347. <location>/castError.jsp</location>
348. </error-page>
```

If the web application associated with the fragment above throws a ClassCastException.
Which statement is true?

- A. The deployment descriptor is invalid.
- B. The container invokes mainError.jsp
- C. The container invokes castError.jsp
- D. Neither mainError.jsp nor castError.jsp is invoked.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 105

You have built a web application that you license to small businesses. The webapp uses a context parameter, called licenseExtension, which enables certain advanced features based on your client's license package. When a client pays for a specific service, you provide them with a license extension key that they insert into the <context-param> of the deployment descriptor. Not every client will have this context parameter so you need to create a context listener to set up a default value in the licenseExtension parameter.
Which code snippet will accomplish this goal?

- A. You cannot do this because context parameters CANNOT be altered programmatically.
- B.

```
String ext = context.getParameter('licenseExtension'); if ( ext == null ) {
    context.setParameter('licenseExtension' DEFAULT);
}
```
- C.

```
String ext = context.getAttribute('licenseExtension'); if ( ext == null ) {
    context.setAttribute('licenseExtension' DEFAULT);
}
```
- D.

```
String ext = context.getInitParameter('licenseExtension') if ( ext == null ) {
    context.resetInitParameter('licenseExtension' DEFAULT); }
```
- E.

```
String ext = context.getInitParameter('licenseExtension') if ( ext == null ) {
    context.setInitParameter('licenseExtension' DEFAULT); }
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 106

Which of the following are attributes of the annotation `javax.servlet.annotation.WebFilter`?

- (i) Name
- (ii) `servletNames`
- (iii) `urlPatterns`
- (iv) `dispatcherTypes`
- (v) `supportAsync`

- A. (iii) only
- B. (iii) and (iv)
- C. (ii), (iii) and (iv)
- D. (iii), (iv) and (v)
- E. (ii), (iii), (iv) and (v)

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

* (iii) `urlPatterns`

The URL patterns to which the filter applies

* (iv) `dispatcherTypes`

The dispatcher types to which the filter applies

* (not V): `asyncSupported`

Declares whether the filter supports asynchronous operation mode.

Note:

* `javax.servlet.annotation`

Annotation Type `WebFilter`

Annotation used to declare a servlet filter.

This annotation is processed by the container at deployment time, and the corresponding filter applied to the specified URL patterns, servlets, and dispatcher types.

QUESTION 107

Which of following annotations can be used in a servlet class?

- (i) `@javax.annotation.Resource`
- (ii) `@javax.annotation.PreDestroy`
- (iii) `@javax.annotation.security.RunAs`
- (iv) `@javax.annotation.security.RolesAllowed`
- (v) `@javax.servlet.annotation.WebServlet`

- A. (v) only
- B. (i) and (v)
- C. (i), (ii), (iii) and (v)

- D. (i), (ii), (iv) and (v)
- E. (i), (ii), (iii), (iv) and (v)

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

* (i) The javax.annotation.Resource annotation is used to declare a reference to a resource. @Resource can decorate a class, a field, or a method.

* (v) javax.servlet.annotation
Annotation Type@WebServlet

Annotation used to declare a servlet.

This annotation is processed by the container at deployment time, and the corresponding servlet made available at the specified URL patterns.

Incorrect:

* Not (II) : javax.annotation.PreDestroy
Target: Method

QUESTION 108

When using the @WebListener annotation, the class on which the annotation is applied to must also implement at least one of the following interfaces (Choose two):

- A. RequestListener
- B. AttributeListener
- C. ServletContextListener
- D. HttpSessionListener
- E. SessionAttributeListener
- F. AsyncListener

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

@WebListener

The main task of the listener is to listen the particular events and process your own task on that event. For example, if you want to initialize a database connection before your application starts, ServletContextListener will be implemented to do that. Another good example is -when you want to do some task on the creation and destruction of a session. For this purpose you need to implement HttpSessionListener.